

RAS Prelims 2026 · Handwritten Class Notes

Science and Technology

*Chapter-by-chapter notes in coaching-lecture style —
concept first, then the facts, with flow charts, mind
maps, previous-year callouts and current-affairs links.*

ST01 *Physics in Everyday Life*

ST02 *Chemistry in Everyday Life*

ST03 *Biology: Cell, Tissues and the Human Body*

ST04 *Human Health, Diseases and Public Health Programmes*

ST05 *Food and Nutrition*

ST06 *Genetics: Inheritance and Variation*

ST07 *Biotechnology and Genetic Engineering*

ST08 *Nanotechnology and New Materials*

ST09 *Computers, IT and ICT*

ST10 *Space Technology*

ST11 *Defence Technology*

ST12 *Environment, Ecology, EIA, Biodiversity and Sustainable Development*

ST13 *Agriculture, Horticulture, Forestry, Animal Husbandry (Rajasthan focus); S&T*

Policy and Indian Scientists

RPSC RAS & RTS Preliminary Examination · 6 December 2026 · 150 questions · 200 marks · 1/3

negative marking

ST01

Physics in Everyday Life

Science and Technology carries 18-23 questions in every RAS paper and physics supplies 5-7 of them. Units, light and electricity are the three reliable suppliers. Device-to-principle matching and unit-to-quantity matching come up almost every year - build those two tables and you have banked most of the chapter.

Every physics question RPSC asks is really a question about something you have already seen - a pressure cooker whistling, a fuse blowing, a rear-view mirror, a blue sky. The paper never asks you to calculate. It asks you to name the principle behind the everyday thing. So learn it that way: phenomenon first, principle second.

Physics in Everyday Life

Units

- 7 SI base units
- newton, pascal, joule, watt
- Light year, parsec =
- DISTANCE**
- Dioptre = lens power
- 1 hp = 746 W

Fluids

- Pascal - hydraulic brake
- Archimedes - ship, lactometer
- Bernoulli - aerofoil, atomiser
- Surface tension - raindrop

Light (biggest)

- TIR - mirage, diamond, fibre
- Scattering - blue sky, red sunset
- Myopia concave, hypermetropia convex
- Dispersion - rainbow, VIBGYOR

Motion

- 1st = inertia (seat belt)
- 2nd = $F = ma$
- 3rd = rocket, gun recoil
- Impulse - airbag, sand pit

Heat

- Water sp. heat 4.2 J/g/K
- Latent heat - steam burns worse
- Ice floats - density max at 4°C
- Thermos blocks all three modes

Electricity

- $V = IR$, 220 V 50 Hz AC
- Fuse in SERIES in LIVE wire
- Transformer - mutual induction, AC only
- Motor left hand, generator right hand

Gravitation

- g max at poles, zero at centre
- Escape velocity 11.2 km/s
- Geostationary 36,000 km
- Weightlessness = free fall

Sound

- No sound in vacuum
- 20 Hz - 20 kHz audible
- Ultrasound - SONAR, sonography
- Echo needs 17.2 m

Modern physics

- Fission - reactor, U-235
- Fusion - the Sun
- Gamma most penetrating
- Roentgen X-rays, Becquerel radioactivity

Units and measurement - the easiest marks in the paper

Measurement means comparing a quantity with an agreed standard. That standard is the unit. The SI system fixes seven of these as base units and builds everything else out of them. RPSC rarely asks what a unit IS. It asks which quantity a unit belongs to - and it deliberately picks units that sound

like one thing and measure another. Learn the seven, learn the common derived ones, then spend most of your time on the trap list.

The seven SI base units - memorise this whole grid

Base unit	Symbol	Quantity measured
metre	m	Length
kilogram	kg	Mass
second	s	Time
ampere	A	Electric current
kelvin	K	Thermodynamic temperature
mole	mol	Amount of substance
candela	cd	Luminous intensity

Derived units you must be able to match both ways

Unit	Quantity	In base terms / note
newton (N)	Force	kg m/s ²
pascal (Pa)	Pressure, stress	N/m ²
joule (J)	Work, energy, heat	N m
watt (W)	Power	J/s
hertz (Hz)	Frequency	per second
coulomb (C)	Electric charge	A s
volt (V)	Potential difference	J/C
ohm	Resistance	V/A
farad (F)	Capacitance	C/V
tesla (T), weber (Wb)	Magnetic flux density, magnetic flux	-
henry (H)	Inductance	-
diopetre (D)	Power of a lens	1 / focal length in metres

- Pressure units: pascal, bar, torr, atmosphere, mm of mercury - all of these names are pressure.

- $1 \text{ atm} = 1.013 \times 10^5 \text{ Pa} = 760 \text{ torr} = 76 \text{ cm of mercury} = 1.013 \text{ bar}$.
- $1 \text{ horsepower} = 746 \text{ watt}$. 1 kilowatt-hour (the Board of Trade unit on your electricity bill) $= 3.6 \times 10^6 \text{ joule}$.
- Small lengths: angstrom $= 10^{-10} \text{ m}$; nanometre $= 10^{-9} \text{ m}$; micron $= 10^{-6} \text{ m}$; fermi $= 10^{-15} \text{ m}$.
- Scalars have magnitude only - mass, speed, distance, work, energy, power, temperature, density, pressure.
- Vectors have magnitude and direction - displacement, velocity, acceleration, force, momentum, impulse, torque, weight.

TRAP

The unit trap RPSC loves: light year and parsec measure DISTANCE, not time ($1 \text{ parsec} = 3.26 \text{ light years}$). Barn = area. Knot = speed. Decibel = sound level. Dioptre = power of a lens. Poise = viscosity. Chandrasekhar limit = a MASS. And weight is a force, so its unit is the newton, not the kilogram.

ASKED IN RAS

Asked in RAS Pre 2024 - out of bar, pascal, torr and atmosphere, which are units of pressure? Answer: all four. The examiner was betting you would think 'bar' is something else.

Motion, Newton's laws and momentum - explained through the bus you travel in

Newton's three laws are not abstract. Each one describes something that happens to you daily. The first says a body keeps doing whatever it was doing unless a force interferes - that is why you lurch forward when the bus brakes.

The second connects force to the change it produces: $F = ma$, or more properly, force is the rate of change of momentum. The third says forces always come in pairs, equal and opposite, acting on two different bodies.

Law to everyday example - this is how the matching question is set

Law	What it says	Everyday proof
First law (law of inertia)	A body stays at rest or in uniform motion unless an external force acts	Passenger lurching in a braking bus; seat belts; dust flying out when a carpet is beaten; a coin staying put when the card under it is flicked
Second law	Force = mass x acceleration = rate of change of momentum	A cricketer pulls his hands back while catching; heavier vehicles need bigger brakes
Third law	Every action has an equal and opposite reaction	Rocket and jet propulsion; recoil of a gun; walking; swimming; a boat moving back when you step off it

- Momentum = mass x velocity, and it is a vector. Conservation of linear momentum explains gun recoil and rocket propulsion.
- Impulse = force x time = change in momentum. The same change in momentum spread over more time means a smaller force.
- That is exactly why air bags, sand pits at a long jump, foam packing and a cricketer's follow-through reduce injury.
- Inertia depends only on mass - a heavier body has more inertia. Inertia of rest, of motion and of direction.
- Centripetal force keeps a body on a circular path and points towards the centre; centrifugal force is the apparent outward pull felt by the rotating observer, not a real force.
- Work is done only when a force produces displacement along its own direction: work = force x displacement x cosine of the angle.
- A coolie carrying a load on his head across a level platform does ZERO work against gravity - the force is vertical, the motion horizontal.

- Energy is the capacity to do work; it is never destroyed, only converted.
Power is work done per unit time, measured in watt.
- Trick to remember the third law: ROCKET, RIFLE, ROWING, RUNNING. All four R's are action-reaction.

Gravitation, satellites and why astronauts float

Every two bodies in the universe attract each other - that is Newton's universal law of gravitation. On Earth that attraction gives every falling body the same acceleration, g . Mass is how much matter you contain and never changes. Weight is the gravitational force on that mass, so weight changes with place. Get this one distinction right and half the gravitation questions solve themselves.

- $g = 9.8 \text{ m/s}^2$ at sea level. It is MAXIMUM at the poles and MINIMUM at the equator, because the Earth bulges at the equator and spins.
- g decreases as you go up from the surface, and also as you go down into the Earth - it becomes ZERO at the centre of the Earth.
- g on the Moon is about one-sixth of g on Earth. A 60 kg man still has 60 kg of mass on the Moon, but weighs one-sixth as much.
- Escape velocity of the Earth = 11.2 km/s; of the Moon = 2.38 km/s. The Moon's escape velocity is so low that gas molecules leak away - that is why the Moon has no atmosphere.
- Orbital velocity of a near-Earth satellite is about 7.9 km/s. Escape velocity does not depend on the mass or the direction of the body projected.
- Repeatedly set as a statement question: 'g is zero at the centre of the Earth' is TRUE; 'escape velocity depends on the mass of the body' is FALSE.

Satellite orbits - match the orbit to the job

Orbit	Height and period	What it is used for
Geostationary (geosynchronous, equatorial, circular)	About 36,000 km (35,786 km); period 24 hours	Communication, television broadcasting, weather watching - it appears fixed over one point
Polar / sun-synchronous	About 800-900 km	Remote sensing, resource survey, mapping - it crosses each place at the same local time

TRAP

Weightlessness in an orbiting spacecraft is NOT because gravity has vanished. Gravity is very much there - it is what holds the craft in orbit. The astronaut floats because the astronaut and the spacecraft are falling together at the same rate. The same weightlessness is felt inside a lift whose cable snaps.

Pressure and fluids – Pascal, Archimedes, Bernoulli and the pressure cooker

Pressure is force spread over area. Same force, smaller area, more pressure - which is why a sharp knife cuts and a blunt one does not, why a camel's broad feet do not sink into sand, and why school bags are given wide straps. Fluids, meaning liquids and gases, carry pressure in three ways that RPSC asks about constantly. Three names, three sets of devices. Learn them as one block.

The three fluid principles and their devices - the highest-yield table in this section

Principle	In one plain sentence	Devices and effects
Pascal's law	Pressure applied to an enclosed fluid is transmitted equally in all directions	Hydraulic press, hydraulic brakes of a car, hydraulic jack, hydraulic lift

Principle	In one plain sentence	Devices and effects
Archimedes' principle	A body immersed in a fluid is pushed up by a force equal to the weight of the fluid it displaces	Ship, submarine, hydrometer, lactometer (purity of milk); a floating body displaces its own weight
Bernoulli's principle	Where a fluid flows faster, its pressure is lower	Lift on an aeroplane wing (aerofoil), atomiser and spray gun, chimney draught, two boats sailing side by side drawn together, a roof blown off in a storm

- Atmospheric pressure at sea level = 76 cm of mercury = 1.013×10^5 pascal. It is measured by a barometer; Torricelli made the first one.
 - Atmospheric pressure falls with altitude. Lower pressure means a lower boiling point, so water boils below 100 C on a hill and food cooks slowly.
 - A pressure cooker raises the pressure inside, which raises the boiling point of water above 100 C - the food cooks faster because it cooks HOTTER, not because steam is trapped.
 - Nose bleeding at high altitude, breathlessness on mountains and a fountain pen leaking in an aircraft are all low-pressure effects.
 - A sudden fall of the barometer reading warns of a storm; a slow fall means rain; a rise means dry weather.
-
- Surface tension makes a liquid surface behave like a stretched skin. It gives raindrops their spherical shape, lets a greased needle float, and lets insects walk on water.
 - Soap and detergent LOWER the surface tension of water, which is why soapy water spreads into cloth and cleans. Surface tension also falls as temperature rises - hot water washes better.
 - Capillarity is the rise of liquid in a fine tube: oil rises in a lamp wick, ink spreads on blotting paper, water rises in soil and in plant stems.
 - Mercury does NOT wet glass, so mercury is depressed in a capillary tube instead of rising in it.

- Viscosity is a fluid's internal friction. Viscosity of a LIQUID decreases with rise in temperature; viscosity of a GAS increases with temperature. Units: poise (CGS), pascal-second (SI).

ASKED IN RAS

A documented repeat - the working of a hydraulic brake is based on Pascal's law, and 'food cooks faster in a pressure cooker because the boiling point of water rises with pressure'. Both have been set more than once. Do not answer 'because steam carries more heat'.

Heat - specific heat, latent heat and the three ways heat travels

Heat is energy in transit from a hotter body to a colder one; temperature only tells you which way it will flow. Two numbers control almost every heat question. Specific heat is how much heat a substance needs to warm up by one degree. Latent heat is the heat a substance swallows to change its state with no change in temperature at all. Water is exceptional in both, and the exam lives on that exception.

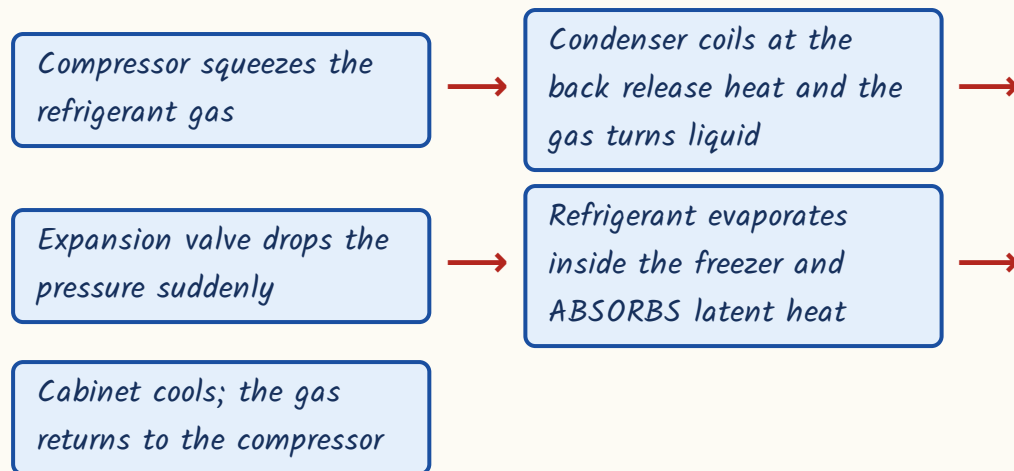
- Specific heat of water = 4.2 J/g/K , the highest among common liquids. That is why water is the coolant in car radiators and why the sea moderates coastal climate.
- Latent heat of fusion of ice = 80 cal/g (336 J/g). Latent heat of vaporisation of water = 540 cal/g (2260 J/g).
- Steam at 100°C burns far worse than boiling water at 100°C because it must first give up that 540 cal/g of latent heat on your skin.
- Anomalous expansion of water: between 0°C and 4°C water CONTRACTS on heating, so its density is maximum at 4°C .
- That is why ice floats and a lake freezes from the surface downwards, leaving liquid water below in which fish survive the winter.

- 0 kelvin = -273.15 C. Minus 40 degrees is the same number on the Celsius and the Fahrenheit scale.
- Evaporation always cools - sweating cools the body, water stays cool in an earthen matka, and the refrigerant cools the fridge, all by absorbing latent heat.
- A thermos flask blocks all three modes at once: the vacuum stops conduction and convection, the silvered walls stop radiation, the cork stopper reduces conduction.
- A bimetallic strip bends because two metals expand unequally - it is the thermostat inside an electric iron, a geyser and a fire alarm.
- Greenhouse effect: short-wavelength sunlight passes in, but the long-wavelength infrared re-radiated by the ground is trapped by carbon dioxide, water vapour and methane.
- Adding salt raises the boiling point of water and lowers its freezing point - which is why salt is spread on icy roads.

The three modes of heat transfer - if the question says vacuum, the answer is radiation

Mode	Needs a medium?	Everyday example
Conduction	Yes - and no bulk movement of matter; chiefly in solids	A metal spoon heating up in hot tea; silver is the best conductor of heat, then copper
Convection	Yes - the fluid itself moves	Land breeze and sea breeze; boiling water; a room heater warming the air; the chimney
Radiation	NO medium needed	Heat from the Sun crossing empty space; a bonfire warming your face

How a refrigerator cools - trace the refrigerant



Sound - why there is silence in space and how SONAR finds the sea floor

Sound is a mechanical wave. It needs particles to pass the vibration along, so it cannot cross a vacuum - there is no sound in space, and an electric bell ringing inside an evacuated jar goes silent. Because the particles of a solid sit closest together, sound travels fastest in solids and slowest in gases.

- Speed of sound in air is about 332 m/s at 0 °C. It increases with temperature and humidity; it does not depend on pressure.
- Order of speed: solids > liquids > gases. Sound travels roughly four times faster in water than in air.
- The audible range for humans is 20 Hz to 20,000 Hz. Above 20,000 Hz is ULTRASONIC; below 20 Hz is INFRASONIC.
- Loudness depends on amplitude (measured in decibel), pitch on frequency, and quality or timbre on the waveform - timbre is how you recognise a voice on the phone.
- An echo is heard only if the reflector is at least about 17.2 m away, because the ear holds a sound for about 0.1 second.

- A stethoscope works by MULTIPLE REFLECTION of sound inside its tube. Sound boards in an auditorium and a megaphone also work by reflection.
- SONAR = Sound Navigation and Ranging. It sends an ultrasonic pulse down and times the echo to find sea depth, shoals of fish, submarines and wrecks.
- Medical uses of ultrasound: sonography of the abdomen and foetus, echocardiography of the heart, and lithotripsy to break kidney stones without surgery.
- Industrial uses of ultrasound: detecting cracks and flaws in metal castings, and cleaning parts a brush cannot reach.
- Bats and dolphins navigate by echolocation. Elephants and whales communicate using infrasound.
- Doppler effect - the apparent change in frequency when source and observer move relative to each other: the rising then falling whistle of a passing train, radar speed guns, and the red shift of receding galaxies.
- Noise limit for a residential zone under India's Noise Pollution (Regulation and Control) Rules, 2000: 55 dB(A) by day and 45 dB(A) by night.

ASKED IN RAS

Asked in RAS Pre 2021 - a stethoscope works on the principle of MULTIPLE REFLECTION of sound. Not refraction, not resonance. The ultrasound and SONAR statement set is a documented repeat cluster as well.

Light and optics - the heaviest scoring section in the whole chapter

Light travels at 3×10^8 m/s in vacuum. Everything you will be asked comes out of four behaviours - it bounces (reflection), it bends when it changes medium (refraction), it splits into colours (dispersion), and it gets thrown

sideways by tiny particles (scattering). Learn which everyday sight belongs to which behaviour. That is the entire section.

- Laws of reflection: the angle of incidence equals the angle of reflection, and the incident ray, the reflected ray and the normal all lie in one plane.
- A plane mirror gives a virtual, erect image of the same size, as far behind the mirror as the object is in front, and laterally inverted - which is why **AMBULANCE** is painted in mirror writing.
- A concave mirror is converging; a convex mirror is diverging and always gives an erect, diminished image with a wide field of view.
- Refraction: light entering a denser medium bends **TOWARDS** the normal and slows down. $\text{Refractive index} = \frac{\text{speed of light in vacuum}}{\text{speed in the medium}}$
- A stick looks bent in water, a coin in a bowl seems to rise, and print looks raised under a glass slab - all plain refraction.

Mirror or lens - match the device to the optic used

Device	Optic used	Why
Rear-view mirror of a vehicle	Convex mirror	Always erect and diminished, gives a wide field of view
Vehicle headlight, torch, searchlight	Concave mirror	Source at the focus throws out a parallel beam
Shaving mirror, dentist's mirror	Concave mirror	Gives an enlarged erect image when the object is inside the focus
Solar cooker and solar furnace	Concave mirror	Concentrates sunlight at the focus
Simple magnifying glass	Convex lens	Enlarged virtual image
Compound microscope	Two convex lenses	Objective plus eyepiece

Device	Optic used	Why
Astronomical telescope	Convex objective of long focal length + convex eyepiece; a reflecting telescope uses a concave mirror instead	Gathers light from distant objects
Periscope	Plane mirrors, or total-reflecting prisms	Bends the line of sight twice
Endoscope	Optical fibres	Carries light around bends inside the body

Now total internal reflection - the single most asked idea in optics. When light tries to go from a denser medium into a rarer one at an angle bigger than the critical angle, it does not escape at all. It is reflected wholly back inside.

Diamond has a very small critical angle, about 24 degrees, so light entering it bounces around many times before it can get out. That is the sparkle.

- Total internal reflection explains the mirage on a hot road, the sparkle of a cut diamond, the shining of a crack in glass, and the working of an optical fibre.
- An optical fibre has a core of higher refractive index and a cladding of lower refractive index - light is trapped in the core by repeated total internal reflection.
- Optical fibre uses: telecommunication and broadband, endoscopy, decorative lamps, and signal transmission immune to electrical interference.
- Atmospheric refraction explains the twinkling of stars, the Sun being seen about two minutes before actual sunrise and after actual sunset, and the oval look of the Sun at the horizon.
- Planets do not twinkle because they are near enough to be extended sources - the flickers from their different points average out.

Defects of vision and their corrective lenses - learn this cold

Defect	What goes wrong	Corrective lens
Myopia (short-sightedness)	Image forms IN FRONT of the retina; distant objects blurred	Concave (diverging) lens

Defect	What goes wrong	Corrective lens
Hypermetropia (long-sightedness)	Image forms <i>BEHIND</i> the retina; near objects blurred	Convex (converging) lens
Presbyopia	Old age - the lens stiffens and loses accommodation, both near and far affected	Bifocal lens
Astigmatism	Uneven curvature of the cornea; lines in one direction blurred	Cylindrical lens
Cataract	The eye lens itself turns cloudy owing to protein changes	No lens helps - corrected by surgery

- Power of a lens in dioptre = 1 divided by the focal length in metres. A convex lens has *POSITIVE* power, a concave lens *NEGATIVE* power.
- The normal adult eye focuses comfortably from 25 cm - the least distance of distinct vision - out to infinity. The ciliary muscles change the lens curvature; that is accommodation.
- The retina carries rods for dim light and cones for colour and bright light; colour blindness is a defect of the cones.
- Dispersion: a prism splits white light into *VIBGYOR* because each colour bends by a different amount. Violet bends *MOST*, red bends *LEAST*.
- A rainbow is refraction plus dispersion plus internal reflection inside raindrops. It always appears in the part of the sky opposite the Sun.
- *LASER* stands for *Light Amplification by Stimulated Emission of Radiation* - its light is monochromatic, coherent and highly directional.

Scattering closes the section and it is the favourite question of the lot. When sunlight meets molecules much smaller than its wavelength, the shorter wavelengths are scattered far more than the longer ones. Blue is scattered most, red least. So the sky you see is scattered blue light. At sunrise and sunset the light travels a long slanting path through the atmosphere, the blue is scattered away entirely, and only red and orange survive to reach your eye.

- Blue sky, red sunrise and sunset, and the reddish look of the Sun at the horizon - all scattering.
- The sky looks BLACK to an astronaut and on the Moon because there is no atmosphere to scatter light.
- Clouds look white because their water droplets are large and scatter all wavelengths about equally.
- Danger signals and traffic stop lights are RED because red scatters the least and travels furthest through fog and dust.
- Electromagnetic spectrum in increasing order of wavelength: gamma rays, X-rays, ultraviolet, visible light, infrared, microwaves, radio waves. The order of frequency is exactly the reverse.
- A television remote control uses INFRARED. A microwave oven uses microwaves, which heat food by making its water molecules rotate.

ASKED IN RAS

Asked in RAS Pre 2023 - match the defect of vision to the corrective lens (myopia to concave, hypermetropia to convex, presbyopia to bifocal, astigmatism to cylindrical). Asked in RAS Pre 2018 - the waves used in a TV remote are infrared. Blue sky, optical fibre and total internal reflection are all documented repeats.

YAAD RAKHO

Yaad rakho: MY-opia is MY-nus - the concave lens has negative power. And for scattering, 'short scatters' - the short wavelength (blue) scatters most, so the sky is blue and the sunset, having lost its blue, is red.

Electricity and magnetism - the fuse, the earth wire and the transformer

Ohm's law is the backbone: at constant temperature the current through a conductor is proportional to the potential difference across it, $V = IR$. Everything in your house is built on top of that one relation plus two safety ideas - a deliberately weak link (the fuse) and a deliberate escape route to the ground (the earth wire).

- Indian domestic supply is alternating current at about 220-230 volt and 50 hertz.
- Direct current flows one way and comes from cells and batteries; alternating current reverses direction fifty times a second and is used for supply because only AC can be stepped up and down by a transformer.
- Household appliances are wired in PARALLEL, so each gets the full voltage and each can be switched independently.
- An ammeter is connected in SERIES and has very low resistance; a voltmeter is connected in PARALLEL and has very high resistance.
- Power = voltage $\times$ current. Electrical energy is billed in kilowatt-hours; $1 \text{ kWh} = 3.6 \times 10^6 \text{ joule}$.
- The resistance of a metal conductor INCREASES with temperature; the resistance of a semiconductor DECREASES with temperature.
- The heating effect of current runs the electric iron, heater, geyser and filament bulb; nichrome is used for heating elements because of its high resistance and high melting point.
- Earthing joins the metal body of an appliance to a thick wire buried in the earth. If the live wire touches the body, the leakage current runs safely to earth instead of through you.

- Traditional Indian wiring colours: red = live, black = neutral, green = earth. The newer IS colour code uses brown for live, light blue for neutral and green-yellow for earth.
- Short circuit - live and neutral touch directly, resistance collapses and current shoots up. Overloading - too many appliances drawing from one line.
- A transformer works on MUTUAL INDUCTION and works only with alternating current - it will do nothing with DC.
- A step-up transformer raises voltage and lowers current; a step-down transformer does the reverse. High-voltage transmission is used because it cuts the heat loss in the wires.

Fuse vs MCB - the exact contrast RPSC sets

Electric fuse	MCB (miniature circuit breaker)
A thin wire of a tin-lead alloy with a low melting point	A switch containing a bimetallic strip plus an electromagnet
Connected in SERIES, in the LIVE wire	Also in the live line, mounted in the distribution board
Melts and breaks the circuit on overload or short circuit	Trips and springs open on overload or short circuit
Must be replaced once it has blown	Simply reset by pushing the lever back up
Slower to act, but cheap	Faster and more precise, and reusable

Device to energy conversion - a standing matching question

Device	Converts	Principle
Electric motor	Electrical energy into mechanical energy	Force on a current-carrying conductor in a magnetic field; Fleming's LEFT-hand rule
Generator or dynamo	Mechanical energy into electrical energy	Faraday's electromagnetic induction; Fleming's RIGHT-hand rule; Lenz's law gives the direction
Photovoltaic (solar) cell	Solar energy DIRECTLY into electrical energy	Photoelectric behaviour of a semiconductor junction

Device	Converts	Principle
Microphone	Sound into electrical energy	-
Loudspeaker	Electrical energy into sound	-
Electric bulb / LED	Electrical energy into light and heat	Heating effect of current; an LED is a semiconductor diode
Battery or dry cell	Chemical energy into electrical energy	-

ASKED IN RAS

Asked in RAS Pre 2016 - a photovoltaic cell converts solar energy **DIRECTLY** into electrical energy, not into heat first. The fuse question - 'a thin wire of low melting point connected in series in the live wire' - is a documented repeat.

Modern physics – radioactivity, fission, fusion, and the master device table

Modern physics in RAS is a naming exercise: who discovered what, which radiation is which, and where fission ends and fusion begins. Radioactivity is the spontaneous breakdown of an unstable nucleus, and it throws out three kinds of thing – alpha, beta and gamma. Understand what each one actually is and the rest of the questions follow.

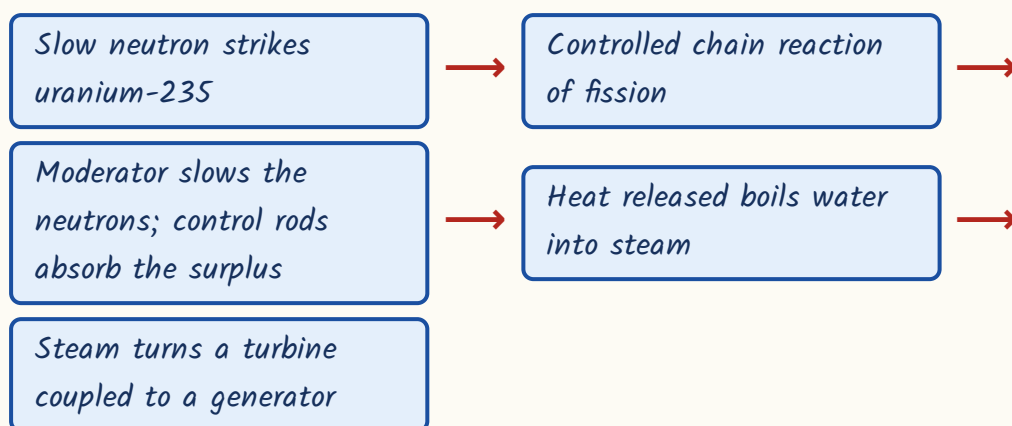
- Alpha particle = a helium nucleus, positively charged, heaviest, least penetrating but most ionising – stopped by a sheet of paper.
- Beta particle = an electron, negatively charged, medium penetration – stopped by a few millimetres of aluminium.

- Gamma ray = an electromagnetic wave, no charge, MOST PENETRATING - needs thick lead or concrete to stop it.
- Half-life is the time in which half of a radioactive sample decays; it is fixed for each isotope and unaffected by temperature or pressure.
- Discoveries to memorise: X-rays - W. C. Roentgen (first Physics Nobel, 1901); radioactivity - Henri Becquerel; the photoelectric effect explained by Albert Einstein (Nobel 1921); the neutron - James Chadwick.
- Superconductivity, in which resistance falls to exactly zero below a critical temperature and the material expels magnetic field (Meissner effect), was found by Kamerlingh Onnes in mercury in 1911.
- Moderator SLOWS neutrons - heavy water (D₂O) or graphite. Control rods ABSORB neutrons - cadmium or boron. Do not swap these two.
- India's Rawatbhata plant in Chittorgarh district, Rajasthan, uses pressurised heavy water reactors.
- The Sun's energy comes from the FUSION of hydrogen into helium - not from burning and not from fission.
- Radioisotope uses: cobalt-60 for cancer radiotherapy, iodine-131 for the thyroid, carbon-14 for dating fossils, sodium-24 for tracing blood flow and pipeline leaks.
- A particle accelerator speeds charged particles up to very high energy and smashes them together - the Large Hadron Collider at CERN is the best known, and it is where the Higgs boson was confirmed in 2012.

Nuclear fission vs nuclear fusion

Fission	Fusion
A heavy nucleus such as uranium-235 splits into lighter nuclei	Light nuclei such as hydrogen join to form a heavier one, helium
Triggered by a slow neutron; can be controlled	Needs enormous temperature and pressure; not yet controllable for power
Runs every nuclear power reactor, and the atom bomb	Powers the Sun and the stars, and the hydrogen bomb
Leaves highly radioactive waste	Produces far less radioactive waste

How a nuclear power station makes electricity



MASTER TABLE – device to principle. If you revise nothing else, revise this

Device or observation	Principle behind it
Hydraulic brake, hydraulic press	Pascal's law
Ship, submarine, lactometer, hydrometer	Archimedes' principle / law of flotation
Aeroplane lift, atomiser, chimney draught	Bernoulli's principle
Pressure cooker	Boiling point rises with pressure
Thermos flask	Blocks conduction, convection and radiation
Refrigerator, sweating, earthen pot	Cooling by evaporation / latent heat
Bimetallic strip in an iron or fire alarm	Unequal thermal expansion
SONAR, sonography, lithotripsy	Reflection of ultrasound
Stethoscope	Multiple reflection of sound

Device or observation	Principle behind it
Mirage, diamond sparkle, optical fibre	Total internal reflection
Blue sky, red sunset	Scattering of light
Rainbow	Refraction + dispersion + internal reflection
Transformer	Mutual induction, AC only
Generator and electric motor	Electromagnetic induction; force on a current in a magnetic field

CURRENT LINK

Current link, as of September 2026: the 2025 Nobel Prize in Physics went to John Clarke, Michel Devoret and John Martinis for discovering macroscopic quantum-mechanical tunnelling and energy quantisation in an electric circuit - the physics that underlies superconducting quantum computers. Expect a straight 'who won it and for what' question; the phrase to hold on to is quantum tunnelling in a circuit.

YAAD RAKHO

Yaad rakho for the radiations: ALPHA is heavy and lazy, paper stops it. BETA is medium, aluminium stops it. GAMMA is the marathon runner, only lead stops it. Penetration order gamma > beta > alpha; ionising power is exactly the reverse.

REVISION RECAP

1. Seven SI base units - metre, kilogram, second, ampere, kelvin, mole, candela. Candela is the one students forget.
2. Light year and parsec are DISTANCE (1 parsec = 3.26 light years); diopetre is lens power; poise is viscosity; knot is speed; barn is area.
3. 1 atm = 76 cm of Hg = 1.013×10^5 Pa = 760 torr. 1 hp = 746 W. 1 kWh = 3.6×10^6 J.

4. First law = inertia (seat belt, carpet dusting). Third law = rocket, gun recoil, walking, swimming. Impulse explains the air bag.
5. g is maximum at the poles, minimum at the equator, zero at the Earth's centre, one-sixth on the Moon. Escape velocity 11.2 km/s.
6. Geostationary satellite - 36,000 km, equatorial, 24-hour period, for communication and weather. Polar orbit at 800-900 km for remote sensing.
7. Weightlessness in orbit is free fall, not the absence of gravity.
8. Pascal - hydraulic brake. Archimedes - ship, lactometer, hydrometer. Bernoulli - aerofoil, atomiser, two boats drawn together.
9. Pressure cooker cooks faster because higher pressure raises the boiling point; on a hill the reverse happens.
10. Water's specific heat 4.2 J/g/K is the highest among common liquids; latent heat of fusion 80 cal/g, of vaporisation 540 cal/g - hence steam burns worse.
11. Water is densest at 4 °C, so ice floats and lakes freeze from the top downwards.
12. Conduction needs contact, convection needs a moving fluid, radiation needs nothing. A thermos defeats all three.
13. No sound in vacuum; fastest in solids; audible 20 Hz to 20 kHz; echo needs 17.2 m; stethoscope = multiple reflection.
14. Ultrasound - SONAR, sonography, echocardiography, lithotripsy, flaw detection. Doppler - train whistle, radar gun, red shift.
15. Total internal reflection - mirage, diamond, optical fibre. Scattering - blue sky, red sunset, black sky in space, red danger signals.
16. Myopia concave, hypermetropia convex, presbyopia bifocal, astigmatism cylindrical, cataract surgery. Convex power positive, concave negative.
17. EM spectrum by increasing wavelength: gamma, X-ray, UV, visible, infrared, microwave, radio. The TV remote uses infrared.
18. Fuse - low melting point wire, in series, in the live wire, replaced after blowing. MCB - trips, then is reset.
19. Transformer works on mutual induction and only on AC. Motor = Fleming's left hand; generator = Fleming's right hand.
20. Fission splits uranium-235 and runs reactors; fusion joins hydrogen and runs the Sun. Moderator slows neutrons, control rods absorb them. Gamma is the most penetrating radiation.

ST02

Chemistry in Everyday Life

Chemistry supplies 5-7 of the 18-23 Science and Technology questions. Alloy composition and common-name-to-formula matching are documented repeats - the alloy list alone has been set as a matching question more than once. Acids with their natural sources, the pH cluster, and the CNG-LPG pair are the other reliable suppliers.

This chapter is a vocabulary test wearing a chemistry coat. Baking soda, washing soda, quicklime, blue vitriol, brass, bronze, german silver - RPSC picks a household name and asks for the formula, the composition or the use. There is very little to understand and a great deal to fix in memory, so these notes are built around three master tables: alloys, ores, and common chemical names.

Chemistry in Everyday Life

Matter

- Solid, liquid, gas + plasma + BEC
- Sublimation - camphor, naphthalene, iodine
- Mercury only liquid metal
- Bromine only liquid non-metal

Atom + table

- Thomson electron, Chadwick neutron
- Mendeleev = atomic mass
- Moseley = atomic number
- Fluorine most electronegative

Acids, bases, pH

- Formic - ant; acetic - vinegar
- Lactic - curd; oxalic - tomato
- pH 7 neutral, blood 7.35-7.45
- Tooth decay below pH 5.5

Corrosion

- Rust = $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$
- Needs oxygen AND moisture
- Galvanisation = zinc
- Kalai / tinning = tin

Alloys (repeat)

- Brass = Cu + Zn
- Bronze = Cu + Sn
- German silver = Cu + Zn + Ni
- Amalgam = anything + mercury

Carbon

- Diamond hardest, four bonds
- Graphite conducts, lubricates
- Fullerene C60 football
- Graphene, carbon nanotubes

Fuels

- CNG = methane
- LPG = butane + propane
- Hydrogen highest calorific value
- Water gas = $\text{CO} + \text{H}_2$

Materials

- Thermoplastic vs thermosetting
- Soap - micelle, fails in hard water
- Cement + gypsum slows setting
- Vulcanisation = rubber + sulphur

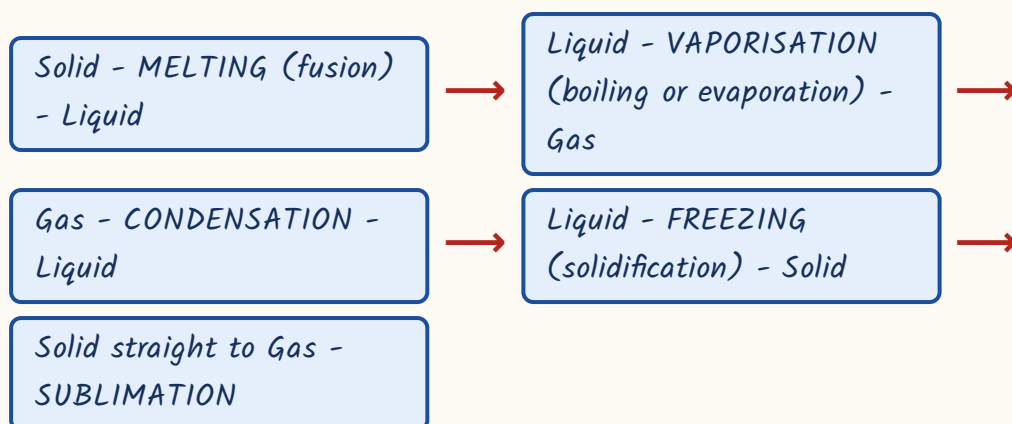
Drugs + food

- Antiseptic living, disinfectant non-living
- Penicillin - Fleming 1928
- Argemone in mustard oil
- Metanil yellow in turmeric

States of matter - and the two extra states the exam adds

Matter exists in three familiar states because of a tug of war between the force holding particles together and the energy making them move. In a solid the force wins, so shape and volume are both fixed. In a liquid the two are balanced, so volume is fixed but shape is not. In a gas the energy wins and neither is fixed. Beyond these three, the exam expects two more names.

Changes of state - learn the name of each arrow



- Sublimation, solid straight to vapour with no liquid stage: camphor, naphthalene balls, iodine, ammonium chloride and dry ice.
- PLASMA is the fourth state - a hot ionised gas. It is what glows inside a fluorescent tube and a neon sign, and what stars are made of.
- The BOSE-EINSTEIN CONDENSATE is the fifth state, formed at temperatures close to absolute zero. Satyendra Nath Bose's work with Einstein gives it the Indian half of its name.
- Evaporation happens at every temperature and only from the surface; boiling happens throughout the liquid at a fixed temperature. Evaporation always causes cooling.
- Mercury is the ONLY metal liquid at room temperature; bromine is the ONLY non-metal liquid at room temperature.
- Sodium and potassium are so reactive that they are stored under kerosene; white phosphorus is stored under water.

TRAP

Do not confuse dry ice with ordinary ice. Dry ice is solid CARBON DIOXIDE - it sublimates, leaves no puddle, and keeps ice cream and vaccines cold. Heavy water, D₂O, is a different thing again: water made with deuterium, used as a moderator in nuclear reactors.

Atomic structure and the periodic table - who found what, and the trends

The atom was pieced together over about forty years, and RPSC asks those discoveries as a matching question. Learn the three particles with their discoverers, then learn the two ways of arranging the elements - Mendeleev's old way and the modern way. That one distinction is the most asked point in this section.

Sub-atomic particles and their discoverers

Particle	Charge	Discovered by
Electron	Negative	J. J. Thomson (1897)
Proton	Positive	Goldstein found the canal rays; Rutherford identified and named the proton
Neutron	No charge	James Chadwick (1932)
The nucleus itself	Positive, tiny, extremely dense	Ernest Rutherford, gold foil experiment

- Atomic number = number of protons. Mass number = protons + neutrons.
- ISOTOPES - same atomic number, different mass number (hydrogen, deuterium, tritium; carbon-12 and carbon-14).
- ISOBARS - same mass number, different atomic number. Do not swap the two words; that swap is the standard trap.
- Bonding in one line: ionic bond = transfer of electrons (common salt); covalent bond = sharing of electrons (water, methane); metallic bond = a sea of free electrons, which is why metals conduct.
- Hydrogen bonding between water molecules gives water its unusually high boiling point and makes ice less dense than liquid water.

Mendeleev arranged the elements in order of increasing atomic MASS and left gaps for elements not yet discovered - eka-aluminium and eka-silicon, later found and named gallium and germanium. But a few pairs stubbornly sat in the wrong order. Moseley fixed that: the modern periodic law says properties are a periodic function of ATOMIC NUMBER. The modern table has 18 groups and 7 periods.

- Group 1 = alkali metals. Group 2 = alkaline earth metals. Group 17 = halogens. Group 18 = noble or inert gases.
- Across a period from left to right: atomic size decreases, ionisation energy and electronegativity increase, metallic character decreases.
- Down a group: atomic size increases, ionisation energy and electronegativity decrease, metallic character increases.
- FLUORINE is the most electronegative element. Caesium and francium are among the most electropositive.
- Metalloids sit on the staircase between metals and non-metals - boron, silicon, germanium, arsenic, antimony, tellurium.
- Graphite is the exception every paper likes: a non-metal that conducts electricity.

Acids, bases, salts and pH - the chemistry of your kitchen and your stomach

An acid gives hydrogen ions in water and tastes sour; a base gives hydroxide ions, tastes bitter and feels soapy. That is all the theory you need. What RPSC actually asks is which acid sits in which everyday thing - the ant that stings you, the curd that turned sour, the tamarind in the chutney. Learn the source list first, then the pH ladder.

Acid to natural source - a documented repeat as a matching question

Acid	Found in
Formic (methanoic) acid	Sting of an ant, a bee and the nettle plant
Acetic acid	Vinegar
Citric acid	Lemon, orange and other citrus fruit
Lactic acid	Curd and sour milk; also builds up in tired muscles
Oxalic acid	Tomato and spinach
Tartaric acid	Tamarind and grapes
Malic acid	Apple
Hydrochloric acid	Gastric juice in the stomach
Butyric acid	Rancid butter
Uric acid	Urine

- Indicators: litmus turns red in acid and blue in base. Methyl orange is red in acid, yellow in base. Phenolphthalein is colourless in acid, pink in base.
- Turmeric paste turns red with a base - which is why a haldi stain on a white shirt turns red the moment soap touches it.
- The pH scale runs 0 to 14. At 25 C, pH 7 is neutral, below 7 acidic, above 7 basic. One pH unit means a tenfold change in acidity.
- Human blood is held tightly at pH 7.35 to 7.45. Tooth decay begins when the pH in the mouth falls below 5.5, which is why toothpaste is mildly basic.
- Rain is called acid rain when its pH drops below about 5.6, caused by oxides of sulphur and nitrogen.
- A bee sting is acidic and is soothed with baking soda; a wasp sting is alkaline and is soothed with vinegar. Keep that pairing straight - it gets reversed in options.

The pH ladder of familiar things - use it for ordering questions

Substance	Approximate pH	Nature
Gastric juice	1 to 3	Strongly acidic
Lemon juice	About 2	Acidic
Vinegar	About 3	Acidic
Milk	About 6.5	Very slightly acidic
Pure water	7	Neutral
Human blood	7.35 - 7.45	Very slightly basic
Baking soda solution	About 9	Basic
Milk of magnesia	About 10	Basic
Sodium hydroxide solution	13 - 14	Strongly basic

- Salts come from an acid plus a base. Sodium chloride is neutral, ammonium chloride is acidic, sodium carbonate is basic.
- Baking soda, NaHCO_3 - an antacid, the raising agent in baking, and the powder in a soda-acid fire extinguisher. Baking powder is baking soda plus a mild edible acid such as tartaric acid.
- Washing soda, $\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$ - used in laundry, in glass and paper making, and to soften hard water.
- Chlor-alkali process: electrolysis of brine gives caustic soda in solution, chlorine at the anode and hydrogen at the cathode.
- Bleaching powder, CaOCl_2 , is made by passing chlorine over dry slaked lime; it bleaches cloth and disinfects drinking water.
- Plaster of Paris, $\text{CaSO}_4 \cdot (1/2)\text{H}_2\text{O}$, is gypsum heated to about 100°C ; it sets hard with water and is used for plaster casts and mouldings.
- Antacid logic: acidity is excess HCl in the stomach, so the cure must be a MILD base - milk of magnesia or baking soda, never a strong base.

ASKED IN RAS

Repeat cluster: 'tooth decay begins when the pH of the mouth falls below 5.5' is set again and again, usually paired with a false statement. The acid-to-source matching - formic with ant, lactic with curd, oxalic with tomato - is also a documented repeat.

Oxidation, reduction and corrosion - why iron rusts and silver goes black

Oxidation used to mean adding oxygen and reduction meant removing it. The wider definition is about electrons: oxidation is loss of electrons, reduction is gain. The two always happen together, which is why the reaction is called redox. Corrosion is simply metal being slowly oxidised by its surroundings, and rusting is the version that costs India the most money.

- Mnemonic: OIL RIG - Oxidation Is Loss of electrons, Reduction Is Gain of electrons.
- Rust is hydrated ferric oxide, $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$. It is brownish and flaky and does not protect the metal underneath it.
- Rusting needs BOTH oxygen and moisture. Iron does not rust in dry air, nor in boiled water free of dissolved air. Salt and acid speed it up, which is why coastal iron rusts fastest.
- Rusting is a chemical change - the rust weighs more than the iron it came from.
- Other corrosion you should be able to name: silver tarnishes BLACK (silver sulphide) in polluted air; copper and bronze turn GREEN (basic copper carbonate); aluminium forms a thin oxide film that actually protects it.
- Reactivity series, most reactive first: K, Na, Ca, Mg, Al, Zn, Fe, Pb, (H), Cu, Hg, Ag, Au, Pt. A metal higher up displaces one lower down from its salt

solution.

- Gold and platinum sit at the bottom, which is why they occur free in nature and do not corrode.

Preventing rust - method and the metal used

Method	What is done	Note
Painting, greasing, oiling	A physical barrier against air and water	Cheapest, but a single scratch breaks it
Galvanisation	Coating iron or steel with ZINC	Zinc is more reactive, so it corrodes first - sacrificial protection; it works even if the coating is scratched
Tinning (kalai)	Coating brass and copper utensils with a thin layer of TIN	Stops copper from contaminating the food; the traditional Indian kalai
Electroplating	Depositing chromium or nickel by electrolysis	Decorative as well as protective
Alloying	Adding chromium to make stainless steel	Chromium forms an invisible, adherent oxide film
Cathodic protection	Bolting blocks of zinc or magnesium to ships and pipelines	The block corrodes instead of the iron

ASKED IN RAS

Asked in RAS Pre 2018 - brass utensils are coated with a thin layer of TIN, the process called kalai, so that copper does not contaminate food. 'Rust is hydrated ferric oxide' is a documented repeat, and galvanisation-with-zinc usually appears alongside it.

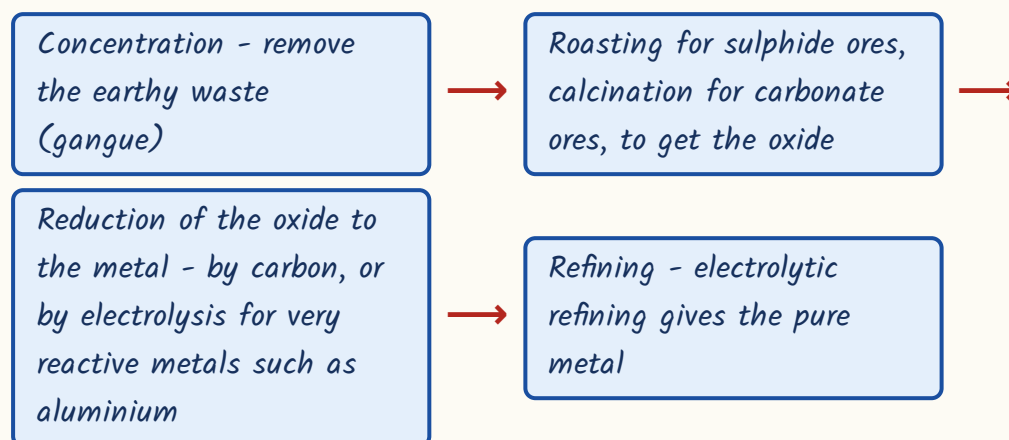
TRAP

Galvanisation is ZINC; tinning or kalai is TIN. The two start the same way in your head and RPSC swaps them deliberately. And remember: stainless steel resists rust because of CHROMIUM, not because of nickel.

Metals, their ores and alloys - the biggest scoring table in this chapter

Metals are almost never found pure. They sit in the ground as oxides, sulphides and carbonates - the ores. Getting the metal out follows the same steps every time, and the pure metal is then usually mixed with another to improve it. That mixture is an alloy, and alloy composition is the most frequently repeated question in the whole chemistry syllabus. Alloying makes a metal harder, more corrosion-resistant, sometimes lower-melting (solder) and sometimes just better looking.

How a metal is extracted from its ore



Ore to metal - straight matching, learn both columns

Ore	Metal	Note
Bauxite	Aluminium	Extracted by electrolysis, aluminium being highly reactive
Haematite (Fe_2O_3), Magnetite (Fe_3O_4)	Iron	Reduced in a blast furnace with coke and limestone
Galena (PbS)	Lead	A sulphide, so it is roasted
Cinnabar (HgS)	Mercury	-

Ore	Metal	Note
Zinc blende (ZnS), Calamine ($ZnCO_3$)	Zinc	Rajasthan's Zawar and Rampura-Agucha are the national centres
Copper pyrites ($CuFeS_2$), Malachite	Copper	Khetri belt in Rajasthan
Pitchblende	Uranium	-
Monazite	Thorium	Kerala's beach sands
Argentite	Silver	-
Rock salt	Sodium	-

ALLOY MASTER TABLE part 1 - the single most repeated matching set

Alloy	Composition	Used for
Brass	Copper + Zinc	Utensils, fittings, musical instruments
Bronze	Copper + Tin	Statues, medals, coins
German silver	Copper + Zinc + Nickel - contains NO silver at all	Utensils and ornamental ware
Stainless steel	Iron + Chromium (about 18%) + Nickel (about 8%) + Carbon	Cutlery, utensils, surgical instruments
Steel	Iron + Carbon in a small percentage	Construction, tools, rails
Duralumin	Aluminium + Copper + Magnesium + Manganese	Aircraft bodies - light and strong
Solder	Lead + Tin, roughly equal parts	Joining electrical wires; it has a low melting point

ALLOY MASTER TABLE part 2

Alloy	Composition	Used for
Gunmetal	Copper + Tin + Zinc	Gears, bearings, marine fittings
Bell metal	Copper + Tin, about 78 : 22	Temple bells and gongs
Nichrome	Nickel + Chromium + Iron	Heating elements in irons, toasters, heaters

Alloy	Composition	Used for
Magnalium	Aluminium + Magnesium	Light structures
Invar	Iron + Nickel	Very low thermal expansion - clock pendulums, measuring tapes
Amalgam	ANY alloy that contains mercury	Dental filling; extraction of gold and silver
Gold jewellery	24 carat is pure gold; 22 carat is about 91.6% gold with copper or silver	Pure gold is too soft to wear, so it is alloyed

ASKED IN RAS

Documented repeats: 'stainless steel is an alloy of iron with chromium and nickel', 'german silver is an alloy of copper, zinc and nickel', and the full alloy matching list with brass, bronze and duralumin. Note also that copper is the base metal of brass, bronze, german silver, gunmetal and bell metal - five of the alloys in the list.

Carbon, its allotropes and the fuels you use at home

Carbon is unique for two reasons. It has four bonds to give (tetravalency) and it can link to itself endlessly (catenation) - which is why there are millions of carbon compounds and only a few hundred thousand of everything else put together. The same pure carbon, arranged differently, gives the hardest substance known and one of the softest. Those different forms are the allotropes, and they are asked every year.

Allotropes of carbon - same element, opposite properties

Allotrope	Structure	Property and use
Diamond	Each carbon bonded to FOUR others in a rigid three-dimensional network	Hardest natural substance; does NOT conduct electricity; cutting, drilling, jewellery

Allotrope	Structure	Property and use
Graphite	Flat layers that slide over each other, with one free electron per atom	Soft and slippery - a lubricant; CONDUCTS electricity - electrodes; pencil lead, mixed with clay
Fullerene (C ₆₀)	Sixty carbon atoms in a hollow football shape - buckminsterfullerene	Nobel Prize in Chemistry 1996 to Curl, Kroto and Smalley
Graphene	A single one-atom-thick layer of graphite	Thinnest and among the strongest materials known; Nobel Prize in Physics 2010 to Geim and Novoselov
Carbon nanotubes	Graphene sheets rolled into cylinders	Extremely strong and conducting; electronics and composites

- Amorphous forms of carbon: coal, coke, charcoal, lamp black. Activated charcoal ADSORBS gases and colouring matter - used in gas masks and water purifiers.
- Hydrocarbons: alkanes have only single bonds (methane, ethane), alkenes have a double bond (ethene), alkynes have a triple bond (ethyne, that is acetylene).
- Calcium carbide with water gives ACETYLENE - used in oxy-acetylene welding and misused for the artificial ripening of fruit, which is banned in India.
- ETHYLENE (ethene) is the natural ripening hormone of fruit - the safe and legal one. Do not confuse it with acetylene.
- Methanol is poisonous and causes blindness and death, which is why hooch tragedies happen. Ethanol is the alcohol in drinks; denatured alcohol is ethanol deliberately made undrinkable.
- Esters give fruits and perfumes their pleasant smell; boiling an ester or a fat with alkali gives soap, and that reaction is saponification.
- Calorific value is the heat released per unit mass of fuel burnt. HYDROGEN has the highest calorific value of any fuel, about 150 kJ per gram.

- A good fuel has a high calorific value, a moderate ignition temperature, little ash and smoke, and is easy to store and carry.
- LPG is stored as a liquid under pressure in the cylinder and leaves as a gas - that is what the L stands for.
- Petroleum refining separates the fractions in order: petroleum gas, petrol, kerosene, diesel, lubricating oil, paraffin wax, bitumen.
- Carbon monoxide is the killer produced by incomplete burning in a closed room - it binds to haemoglobin far more strongly than oxygen does.

Fuel gases and what they are made of

Fuel gas	Chief constituent	Note
CNG (compressed natural gas)	Mainly METHANE	The cleanest of the common vehicle fuels
LPG (liquefied petroleum gas)	Mainly BUTANE and propane	Ethyl mercaptan is added as an odorant so that a leak can be smelt
Biogas or gobar gas	Mainly methane	From anaerobic digestion of dung and waste
Water gas	Carbon monoxide + hydrogen	-
Producer gas	Carbon monoxide + nitrogen	-
Coal gas	Hydrogen + methane + carbon monoxide	-

ASKED IN RAS

Asked in RAS Pre 2021 and repeated - the main constituent of CNG is methane, while LPG is mainly butane and propane. Asked in RAS Pre 2016 - calcium carbide plus water gives acetylene, which is exactly why its use for ripening fruit is banned.

Polymers, plastics, soaps and the everyday materials - cement, glass, fertilisers

A polymer is a long chain built by joining one small unit over and over. Nature makes them - cellulose, starch, protein, natural rubber. Chemistry copies the trick to make plastics and synthetic fibres. The one classification RPSC always wants is whether the plastic can be melted and reshaped, or whether it has set for good.

Thermoplastic vs thermosetting plastic

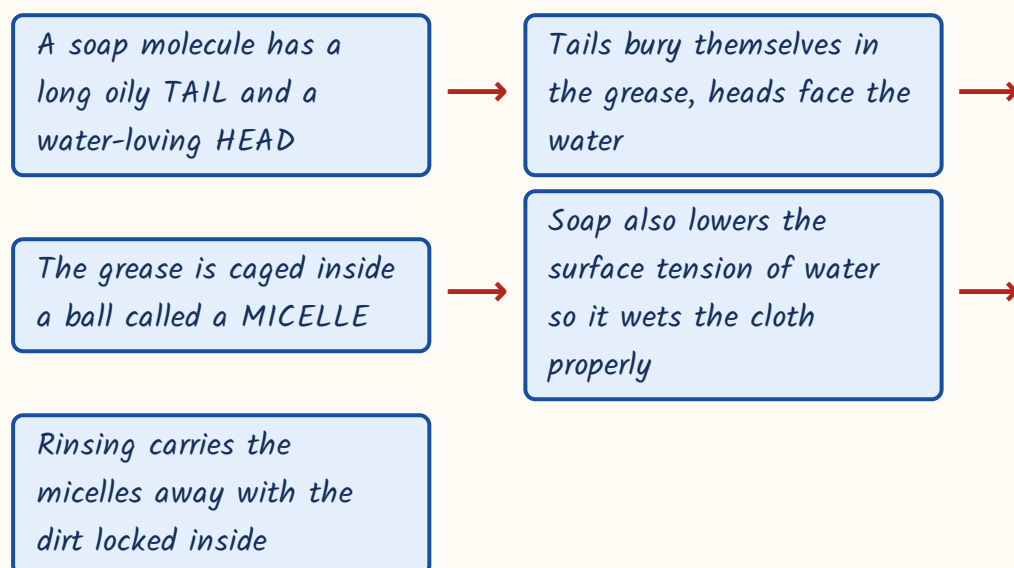
Thermoplastic	Thermosetting plastic
Softens on heating and can be remoulded again and again	Sets permanently once moulded; heating only chars it
Long separate chains held by weak forces	Chains cross-linked into one rigid network
Polythene, PVC, polystyrene, nylon, PET, teflon	Bakelite, melamine
Buckets, bottles, pipes, non-stick coatings	Electrical switches and plugs, crockery, handles of cooking pans
Easier to recycle	Cannot be remelted, so hard to recycle

- Bakelite, made by Leo Baekeland, was the first fully synthetic plastic. Teflon (PTFE) is the non-stick coating on pans.
- Nylon is a synthetic NON-cellulosic fibre - it comes from coal, water and air, not from any plant. Rayon is regenerated cellulose, which is why it is called artificial silk.
- Terylene or Dacron is a polyester; PET, the plastic of drinking-water bottles, belongs to the same family.
- Vulcanisation - heating natural rubber with SULPHUR, discovered by Charles Goodyear - makes rubber harder, stronger and more elastic. Tyres depend on

it.

- India banned identified single-use plastic items from 1 July 2022. Biodegradable plastics such as PLA and PHBV are the substitutes being promoted.
- Cotton, jute, silk and wool are natural fibres; nylon, polyester and acrylic are synthetic.
- Cement is made from limestone and clay. Gypsum, about 2 to 3 per cent, is added to SLOW DOWN the setting - not to speed it up.
- Concrete = cement + sand + gravel + water. Reinforced cement concrete adds steel bars.
- Ordinary glass is mainly silica, SiO_2 . Borosilicate or Pyrex glass resists heat and sudden temperature change. Cobalt oxide colours glass blue; adding lead gives crystal glass.
- Fertilisers: urea, $\text{CO}(\text{NH}_2)_2$, has the highest nitrogen content among solid fertilisers, about 46 per cent. DAP gives nitrogen and phosphorus, muriate of potash is potassium chloride, superphosphate gives phosphorus.
- Explosives: gunpowder = potassium nitrate + charcoal + sulphur; TNT = trinitrotoluene; RDX = cyclonite; dynamite = nitroglycerine soaked in kieselguhr, invented by Alfred Nobel.

How soap actually removes grease - the micelle story



- Soaps are sodium or potassium salts of long-chain fatty acids, made by saponification - boiling oil or fat with alkali. Potassium soaps are the soft ones.
- Soap does NOT lather in hard water: the calcium and magnesium salts in hard water form an insoluble curdy scum and the soap is simply wasted.
- Synthetic detergents are alkyl benzene sulphonates. They work perfectly well in hard water, which is the whole reason they were invented.
- The drawback of detergents is that branched-chain versions resist bacterial breakdown and cause foam in rivers; soaps are biodegradable.
- Temporary hardness is due to bicarbonates and is removed by boiling; permanent hardness is due to sulphates and chlorides and needs washing soda or an ion exchanger.

Batteries, fuel cells, catalysts and the gases around you

A cell turns chemical energy straight into electrical energy. A primary cell such as the dry cell is used once and thrown away; a secondary cell can be recharged by pushing current back through it. A fuel cell is different from both - it is not a store at all, it keeps producing electricity for as long as you keep feeding it fuel.

- Dry cell - a zinc-carbon primary cell; the zinc case is the negative electrode and the carbon rod the positive.
- Lead-acid storage battery - lead, lead dioxide and sulphuric acid; this is the rechargeable battery in cars and inverters.
- Lithium-ion cells power mobile phones, laptops and electric vehicles. The 2019 Nobel Prize in Chemistry went to Goodenough, Whittingham and Yoshino for developing them.
- A fuel cell combines hydrogen and oxygen to give electricity directly, with WATER as the only product - no combustion, no smoke. Fuel cells powered

the Apollo spacecraft.

- Air is about 78 per cent nitrogen and 21 per cent oxygen by volume, with argon and carbon dioxide making up most of the rest.
- Argon, mixed with nitrogen, fills an ordinary incandescent bulb - an inert filling stops the hot filament from burning away.
- Helium fills balloons and airships because it is light and, unlike hydrogen, non-inflammable. It is also in the breathing mixture of deep-sea divers.
- Neon glows red-orange in advertising signs. Ozone, O₃, in the stratosphere absorbs harmful ultraviolet radiation.
- Carbon dioxide is used in fire extinguishers because it is heavier than air and does not support burning; as dry ice it is a refrigerant.
- Hydrogen is the lightest element known; nitrogen provides an inert atmosphere for storing food and for filling tyres.

Industrial process to catalyst - a standing matching question

Process	Product	Catalyst
Haber process	Ammonia	Finely divided IRON
Contact process	Sulphuric acid	Vanadium pentoxide, V ₂ O ₅
Hydrogenation of vegetable oil into vanaspati	Solid fat	NICKEL
Ostwald process	Nitric acid	Platinum

CURRENT LINK

Current link, as of September 2026: the 2025 Nobel Prize in Chemistry went to Susumu Kitagawa, Richard Robson and Omar M. Yaghi for metal-organic frameworks (MOFs) - crystalline materials with enormous internal pore space, being developed to harvest water from desert air, capture carbon dioxide and store hydrogen. Remember the phrase 'metal-organic frameworks' with those three names. Alongside it, India's National Green Hydrogen Mission, approved in January 2023 with an outlay of about Rs 19,744 crore, targets 5 million tonnes of green hydrogen a year by 2030; green hydrogen is made by electrolysis of water using renewable electricity.

Drugs, food adulteration, radioisotopes and the master table of common names

Drugs are asked by class, not by chemistry. You need to know what each class DOES and one or two examples of it. The single most exploited pair is antiseptic against disinfectant - both kill germs, but one goes on you and the other goes on the floor.

- ANALGESIC - relieves pain. Aspirin and paracetamol are ordinary analgesics; morphine, obtained from opium, is a NARCOTIC analgesic.
- ANTIPYRETIC - brings down fever. Paracetamol is both analgesic and antipyretic.
- ANTIBIOTIC - kills or checks micro-organisms inside the body. Penicillin was discovered by Alexander Fleming in 1928.
- ANTACID - neutralises excess acid in the stomach: milk of magnesia and baking soda.
- ANTIHISTAMINE for allergy; TRANQUILLISER for anxiety and sleeplessness; antifertility drugs for conception control.

- Antibiotics do nothing against viruses. Overuse of antibiotics is exactly what breeds antimicrobial resistance.
- Food adulteration to memorise: argemone seed oil in mustard oil, which causes epidemic dropsy; metanil yellow in turmeric and pulses; brick powder in chilli powder; chicory in coffee; melamine in milk; papaya seeds in black pepper; chalk powder in wheat flour.
- Common preservatives: sodium benzoate and sodium metabisulphite, plus the traditional salt, sugar, oil and vinegar.
- FSSAI, the Food Safety and Standards Authority of India, set up under the Food Safety and Standards Act 2006, is the regulator. AGMARK certifies agricultural produce, BIS/ISI industrial goods, Hallmark gold.
- Iodised salt prevents goitre; fortification adds iron, vitamin A and vitamin D to staple foods.
- Aqua regia is 3 parts concentrated hydrochloric acid to 1 part concentrated nitric acid, and it dissolves even gold and platinum - the name means 'royal water' because it attacks the royal metals.

Antiseptic vs disinfectant - the exact distinction that is asked

Antiseptic	Disinfectant
Applied to LIVING tissue - skin, wounds, mouth	Applied to NON-LIVING objects - floors, drains, instruments
Dettol, Savlon, tincture of iodine, boric acid, dilute hydrogen peroxide	Phenol in higher concentration, chlorine, formalin, bleaching powder
Mild, because it must not harm the tissue	Strong, and would harm living tissue
Dilute phenol works as an antiseptic	Concentrated phenol is a disinfectant - same chemical, different strength

Radioisotope to use

Radioisotope	Used for
Carbon-14	Dating of fossils and archaeological material
Cobalt-60	Radiotherapy for cancer

Radioisotope	Used for
Iodine-131	Diagnosis and treatment of thyroid disorders
Phosphorus-32	Agricultural research and leukaemia studies
Sodium-24	Tracing blood circulation and finding leaks in underground pipelines
Uranium-235	Fuel for nuclear fission

MASTER TABLE - common name to formula, part 1

Common name	Chemical name	Formula
Baking soda	Sodium hydrogencarbonate	NaHCO_3
Washing soda	Sodium carbonate decahydrate	$\text{Na}_2\text{CO}_3 \cdot 10\text{H}_2\text{O}$
Caustic soda	Sodium hydroxide	NaOH
Caustic potash	Potassium hydroxide	KOH
Quicklime	Calcium oxide	CaO
Slaked lime	Calcium hydroxide	Ca(OH)_2
Limestone, marble, chalk	Calcium carbonate	CaCO_3
Gypsum	Calcium sulphate dihydrate	$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
Plaster of Paris	Calcium sulphate hemihydrate	$\text{CaSO}_4 \cdot (1/2)\text{H}_2\text{O}$
Bleaching powder	Calcium oxychloride	CaOCl_2
Milk of magnesia	Magnesium hydroxide	Mg(OH)_2
Heavy water	Deuterium oxide	D_2O

MASTER TABLE - common name to formula, part 2

Common name	Chemical name	Formula
Blue vitriol	Copper sulphate pentahydrate	$\text{CuSO}_4 \cdot 5\text{H}_2\text{O}$
Green vitriol	Ferrous sulphate heptahydrate	$\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$
White vitriol	Zinc sulphate heptahydrate	$\text{ZnSO}_4 \cdot 7\text{H}_2\text{O}$
Oil of vitriol	Concentrated sulphuric acid	H_2SO_4
Epsom salt	Magnesium sulphate heptahydrate	$\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
Glauber's salt	Sodium sulphate decahydrate	$\text{Na}_2\text{SO}_4 \cdot 10\text{H}_2\text{O}$

Common name	Chemical name	Formula
Hypo, the photographic fixer	Sodium thiosulphate	$\text{Na}_2\text{S}_2\text{O}_3 \cdot 5\text{H}_2\text{O}$
Dry ice	Solid carbon dioxide	CO_2
Laughing gas	Nitrous oxide	N_2O
Marsh gas	Methane	CH_4
Sal ammoniac	Ammonium chloride	NH_4Cl
Saltpetre	Potassium nitrate	KNO_3
Borax	Sodium tetraborate decahydrate	$\text{Na}_2\text{B}_4\text{O}_7 \cdot 10\text{H}_2\text{O}$
Potash alum	Potassium aluminium sulphate	$\text{K}_2\text{SO}_4 \cdot \text{Al}_2(\text{SO}_4)_3 \cdot 24\text{H}_2\text{O}$

ASKED IN RAS

Asked in RAS Pre 2016 and repeated - the formula of baking soda is NaHCO_3 , and vinegar is a dilute solution of acetic acid. Asked in RAS Pre 2023 - morphine is a narcotic analgesic. Asked in RAS Pre 2018 - nylon is a synthetic non-cellulosic fibre. The common-name-to-formula matching set is a documented repeat, so both tables above are worth a full revision pass.

YAAD RAKHO

Yaad rakho the vitriols by colour: BLUE vitriol is copper, GREEN vitriol is iron (ferrous), WHITE vitriol is zinc, and OIL of vitriol is sulphuric acid itself. Colour to metal - that is the whole question.

REVISION RECAP

1. Sublimation - camphor, naphthalene, iodine, ammonium chloride, dry ice. Plasma is the fourth state, the Bose-Einstein condensate the fifth.
2. Mercury is the only liquid metal, bromine the only liquid non-metal, graphite the non-metal that conducts.
3. Thomson - electron; Chadwick - neutron; Rutherford - the nucleus. Isotopes share atomic number, isobars share mass number.
4. Mendeleev used atomic MASS; the modern periodic law of Moseley uses ATOMIC NUMBER. Fluorine is the most electronegative element.

5. Formic - ant; acetic - vinegar; citric - lemon; lactic - curd; oxalic - tomato; tartaric - tamarind; malic - apple; HCl - gastric juice.
6. pH 7 neutral, blood 7.35-7.45, tooth decay below 5.5, acid rain below about 5.6. Bee sting acidic so use baking soda; wasp sting alkaline so use vinegar.
7. Rust = $\text{Fe}_2\text{O}_3 \cdot x\text{H}_2\text{O}$ and needs oxygen AND moisture. Galvanisation is zinc, tinning or kalai is tin, stainless steel resists rust because of chromium.
8. Brass = Cu + Zn. Bronze = Cu + Sn. German silver = Cu + Zn + Ni with no silver in it. Gunmetal = Cu + Sn + Zn. Bell metal = Cu + Sn about 78:22.
9. Stainless steel = Fe + Cr about 18% + Ni about 8% + C. Duralumin = Al + Cu + Mg + Mn. Solder = Pb + Sn. Amalgam always contains mercury.
10. Bauxite - aluminium; haematite and magnetite - iron; galena - lead; cinnabar - mercury; zinc blende - zinc; copper pyrites - copper; pitchblende - uranium; monazite - thorium.
11. Diamond has four bonds per atom and does not conduct; graphite has sliding layers, one free electron, so it conducts and lubricates. Then fullerene C60, graphene, nanotubes.
12. CNG is methane; LPG is butane and propane with ethyl mercaptan for smell; water gas is $\text{CO} + \text{H}_2$; producer gas is $\text{CO} + \text{N}_2$. Hydrogen has the highest calorific value.
13. Calcium carbide with water gives acetylene, banned for ripening; ethylene is the natural ripening hormone.
14. Thermoplastics remould - polythene, PVC, nylon, PET, teflon. Thermosetting sets for good - bakelite, melamine. Vulcanisation is rubber plus sulphur.
15. Soap fails in hard water and forms scum; detergents work in hard water. Cleaning happens by micelle formation and by lowering surface tension.
16. Gypsum is added to cement to SLOW the setting. Glass is mainly silica; Pyrex is borosilicate. Urea carries about 46 per cent nitrogen.
17. Catalysts: Haber - iron; contact process - V2O5; vanaspati - nickel; Ostwald - platinum.
18. Antiseptic goes on living tissue (Dettol, tincture of iodine); disinfectant on non-living things (phenol, chlorine). Penicillin - Fleming, 1928. Morphine is a narcotic analgesic.
19. Argemone oil in mustard oil causes epidemic dropsy; metanil yellow adulterates turmeric; melamine adulterates milk. FSSAI is the regulator.
20. Carbon-14 dating, cobalt-60 cancer therapy, iodine-131 thyroid, sodium-24 blood flow and pipeline leaks. Aqua regia = 3 parts HCl to 1 part HNO_3 .

ST03

Biology: Cell, Tissues and the Human Body

Science and Technology is 18-23 questions in every RAS paper and biology feeds 5-7 of them. This chapter is the single biggest supplier. Blood groups and the Rh factor are a documented repeat - 2013, 2021 and 2023 - so that one section alone is worth more than the rest of the chapter put together.

This is the chapter about you - the cell you are built from and the nine systems that keep you running. RPSC does not ask biology as reasoning. It asks it as labels: which organelle, which enzyme, which gland, which blood group. So the work here is naming, not theorising, and every label you fix is a mark banked.

Cell, Tissues and the Human Body

The cell

- Hooke 1665, cork
- Schleiden + Schwann + Virchow
- Mitochondrion - powerhouse
- Lysosome - suicide bag
- Plant cell vs animal cell

Cell division

- Mitosis - 2 diploid, somatic
- Meiosis - 4 haploid, gametes
- Crossing over at pachytene
- PMAT sequence

Plant side

- Whittaker 1969, five kingdoms
- Photosynthesis - grana + stroma
- O₂ from photolysis of water
- Auxin, gibberellin, ethylene

Digestion + breathing

- Ptyalin, pepsin, trypsin
- Liver - largest gland
- Absorption in small intestine
- 12 pairs of ribs, tidal volume 500 ml

Heart and blood

- 4 chambers, SA node pacemaker
- 120/80 mm Hg, 72 per minute
- RBC 120 days, spleen graveyard
- Harvey - circulation

Blood groups (repeat)

- Landsteiner 1901
- O universal donor, AB recipient
- Rh from Rhesus monkey, 1940
- Erythroblastosis foetalis, anti-D

Kidney and nerves

- Nephron, 10 lakh per kidney
- Ureotelic, urea in liver
- Cerebrum largest, cerebellum balance
- 12 cranial + 31 spinal nerves

Hormones and bones

- Pituitary - master gland
- Insulin - beta cells
- 206 bones, femur longest, stapes smallest
- 46 chromosomes, XX and XY

The cell - theory, the men who named it, and the organelle table

Start with who said what. Robert Hooke looked at a thin slice of cork in 1665 and called the empty boxes he saw 'cells'. Those were dead cell walls. Anton van Leeuwenhoek was the first to see living cells. Robert Brown found the nucleus in 1831. Then came the cell theory. Learn the three names in order, because RPSC asks the third one most often.

- Cell theory - M.J. Schleiden (1838, plants) and Theodor Schwann (1839, animals).
- Rudolf Virchow (1855) added 'Omnis cellula e cellula' - every cell arises from a pre-existing cell.
- Prokaryote - no true nucleus, no membrane-bound organelles, 70S ribosome (bacteria, blue-green algae).
- Eukaryote - true nucleus with a membrane, organelles present, 80S ribosome in the cytoplasm.
- Plasma membrane is selectively permeable - the fluid mosaic model of Singer and Nicolson (1972).
- The largest human cell is the ovum; the longest is the nerve cell (neuron).

Scientist to discovery - a ready-made match-the-following

Scientist	Discovery / contribution
Robert Hooke	Coined the term 'cell', 1665, from cork
Anton van Leeuwenhoek	First saw living cells
Robert Brown	Nucleus of the cell, 1831
Schleiden and Schwann	Cell theory, 1838-39
Rudolf Virchow	Omnis cellula e cellula, 1855
Camillo Golgi	Golgi apparatus
Christian de Duve	Lysosome
Watson and Crick	Double helix structure of DNA, 1953

Organelle, function and the nickname RPSC hides the question behind

Organelle	Chief function	Nickname / key point
Nucleus	Controls the cell, holds the DNA	Discovered by Robert Brown, 1831
Mitochondrion	ATP through the Krebs cycle and oxidative phosphorylation	Powerhouse of the cell; double membrane, inner one folded into cristae; own circular DNA and 70S ribosomes

Organelle	Chief function	Nickname / key point
Ribosome	Protein synthesis	70S in prokaryotes (50S + 30S), 80S in eukaryotes (60S + 40S)
Endoplasmic reticulum	Rough ER - protein transport; smooth ER - lipid synthesis	The transport channel of the cell
Golgi apparatus	Packaging, secretion, and formation of lysosomes	Named after Camillo Golgi
Lysosome	Intracellular digestion by hydrolytic enzymes at acidic pH	Suicide bag, named by Christian de Duve
Chloroplast	Photosynthesis - light reaction in grana, Calvin cycle in stroma	Kitchen of the cell; chlorophyll holds magnesium
Chromoplast	Gives colour to flowers and fruits	A plastid; plant cells only
Leucoplast	Storage of starch, oil and protein	A colourless plastid
Vacuole	Storage and turgidity	Large and central in the plant cell
Centriole / centrosome	Forms the spindle during cell division	Animal cells; absent in higher plants
Cell wall	Rigidity and protection; made of cellulose	Plants only - fungi have chitin, bacteria peptidoglycan

Plant cell vs animal cell - the exact contrast RPSC sets

Plant cell	Animal cell
Cellulose cell wall outside the membrane	No cell wall, only the plasma membrane
Plastids present - chloroplast, chromoplast, leucoplast	Plastids absent
One large central vacuole	Vacuoles small, many or absent
Centriole absent in higher plants	Centriole and centrosome present
Lysosomes rare	Lysosomes prominent
Reserve food is starch	Reserve food is glycogen
Cytokinesis by a cell plate	Cytokinesis by furrowing of the membrane

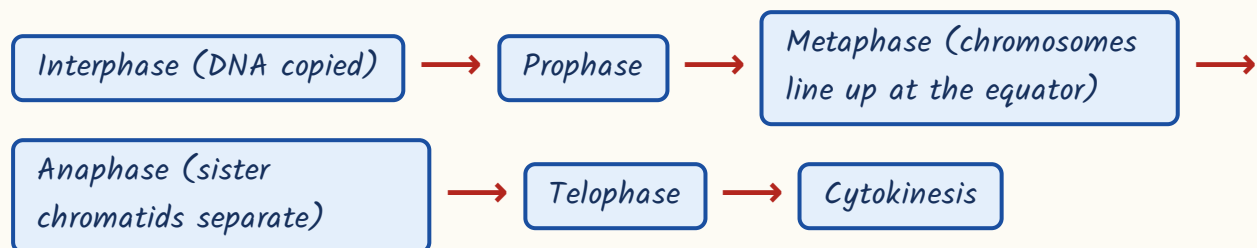
TRAP

Two mismatches RPSC keeps planting. First - 'Ribosome: synthesis of lipids' is **WRONG**; ribosomes make protein, the smooth ER makes lipid. Second - do not say the centriole is absent from all plants; say it is absent from higher plants. Also note chlorophyll contains magnesium while haemoglobin contains iron - that pair is a favourite one-liner.

Cell division, biomolecules and the nucleic acids

Two kinds of division, two entirely different purposes. Mitosis keeps you growing and repairs your wounds - it copies a cell exactly. Meiosis makes eggs and sperm - it halves the chromosome number and shuffles the genes. If you fix that purpose, the rest of the table writes itself.

The sequence of mitosis - asked as an ordering question



Mitosis vs meiosis

Mitosis	Meiosis
Takes place in somatic (body) cells	Takes place in germ cells of the reproductive organs
One division, two daughter cells	Two divisions, four daughter cells
Daughter cells diploid ($2n$) and identical to the parent	Daughter cells haploid (n)
Equational division - chromosome number unchanged	Reductional division - chromosome number halved
No crossing over, so no variation	Crossing over at pachytene of prophase-I creates variation
For growth, repair and healing	For gamete formation - the basis of sexual reproduction

- Crossing over happens in the pachytene sub-stage of prophase-I; the chiasmata become visible later, at diplotene.
- Prophase-I sub-stages in order - leptotene, zygotene, pachytene, diplotene, diakinesis.
- Humans have 46 chromosomes (23 pairs); a gamete carries 23.
- Sex determination - XX is female, XY is male; the sperm decides the sex of the child.

TRAP

'Meiosis takes place in the somatic cells of the body' is a standard false statement in RAS statement-based questions. Meiosis is germ-cell only. And remember mitosis is equational, meiosis-I is the reductional one - not meiosis-II.

Four families of molecules build the body - carbohydrate, protein, fat and nucleic acid - and enzymes run the reactions between them. RPSC asks this at

a very basic level: energy value per gram, what an enzyme does, and the difference between DNA and RNA. Nothing deeper.

- Energy value - carbohydrate 4 kcal/g, protein 4 kcal/g, fat 9 kcal/g. Fat is more than double, and that is the asked fact.
- Carbohydrate is the immediate fuel; protein builds and repairs; fat stores energy and insulates.
- Enzymes are proteins - except ribozymes, which are RNA. They lower the activation energy of a reaction.
- Enzymes are specific, are not consumed in the reaction, and are sensitive to temperature and pH.
- DNA double helix - James Watson and Francis Crick, 1953, using Rosalind Franklin's X-ray work.
- Base pairing - adenine with thymine by two hydrogen bonds, guanine with cytosine by three.
- So G-C rich DNA needs more heat to melt - that is why the number of bonds gets asked.
- RNA has uracil in place of thymine, and ribose sugar in place of deoxyribose.
- DNA is usually double-stranded and lives in the nucleus; RNA is single-stranded and works in the cytoplasm.
- Memory line - 'AT the Two, GC the Three': Adenine-Thymine two bonds, Guanine-Cytosine three.

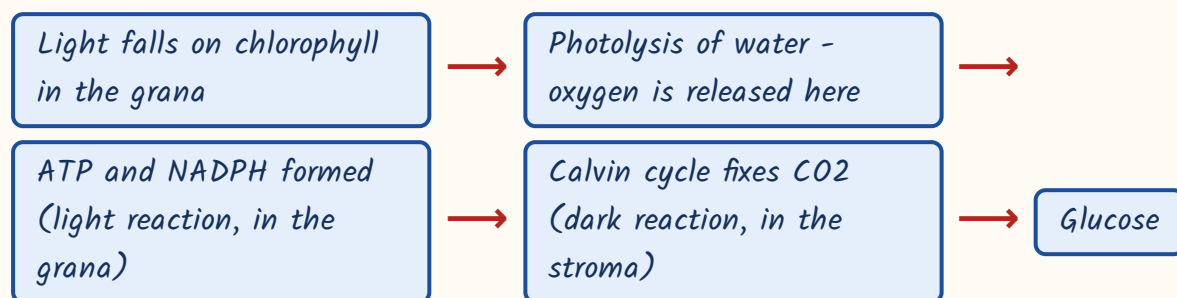
Classification, tissues and how a plant works

Two names carry this whole area. Carolus Linnaeus gave binomial nomenclature - every organism gets a two-word Latin name, genus first with a capital, species second in lower case, the whole thing italicised or underlined. R.H. Whittaker in

1969 divided life into five kingdoms. Learn the year and the five names; that is the question.

- Five kingdoms (Whittaker, 1969) - Monera, Protista, Fungi, Plantae, Animalia.
- Basis of the division - cell structure, body organisation, mode of nutrition and reproduction.
- Monera are prokaryotes (bacteria, cyanobacteria); the rest are eukaryotes.
- Binomial nomenclature - Homo sapiens: Homo is the genus, sapiens the species.
- Plant tissues - meristematic (dividing, at growing tips) and permanent (parenchyma, collenchyma, sclerenchyma; xylem carries water, phloem carries food).
- Animal tissues - four types: epithelial (covering), connective (blood, bone, cartilage), muscular and nervous.

Photosynthesis, step by step



- Overall equation - $6\text{CO}_2 + 6\text{H}_2\text{O}$ gives $\text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$, in sunlight and chlorophyll.
- The oxygen released comes from WATER, not from carbon dioxide - proved with heavy-oxygen labelled water.
- Light reaction in the grana or thylakoid; Calvin or dark reaction in the stroma.
- Chlorophyll contains magnesium at its centre; haemoglobin contains iron.
- Transpiration is loss of water as vapour, chiefly through stomata - it pulls water up the plant.

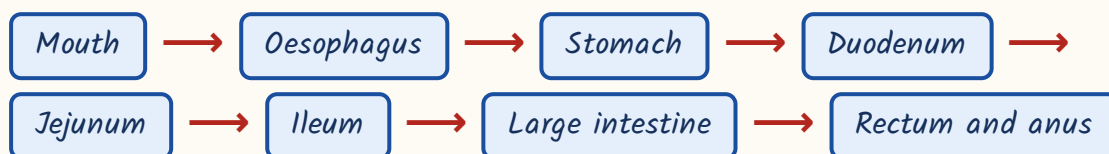
Plant hormones and their principal effect

Hormone	Principal effect
Auxin	Apical dominance, cell elongation, phototropism
Gibberellin	Stem elongation, bolting in rosette plants, seed germination
Cytokinin	Cell division, delays senescence (ageing) of leaves
Abscissic acid	The 'stress hormone' - closes stomata, promotes dormancy
Ethylene	The only gaseous hormone - ripening of fruit and senescence

The digestive system - enzyme, source, substrate

Digestion is a relay of enzymes, and the exam question is always 'which enzyme, from where, acting on what'. Starch is attacked first in the mouth, protein in the stomach, and everything is finished in the small intestine, where absorption also happens through the villi. Bile is the trick answer - it is a secretion, not an enzyme.

The path of swallowed food



Digestive secretions - source and action

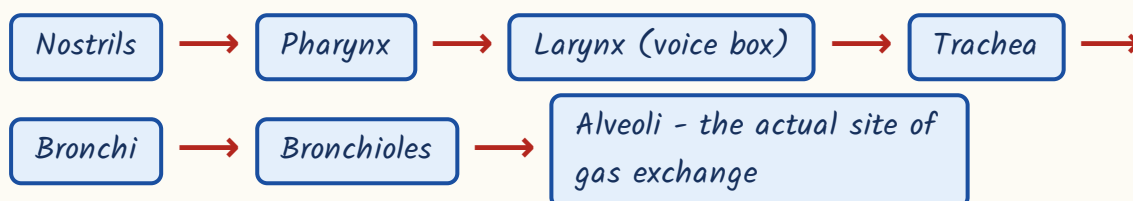
Secretion	Source	Acts on
Ptyalin (salivary amylase)	Salivary glands	Starch to maltose, in the mouth
Pepsin (secreted as pepsinogen)	Gastric glands, needs HCl	Protein to peptones, in the stomach
Rennin	Gastric glands of infants	Curdles the casein of milk
Bile - contains NO enzyme	Liver, stored in the gall bladder	Emulsifies fat, makes the medium alkaline

Secretion	Source	Acts on
Trypsin	Pancreas (as trypsinogen)	Protein and peptones to peptides
Pancreatic amylase and lipase	Pancreas	Starch to maltose; fat to fatty acids and glycerol
Maltase, sucrase, lactase, peptidases	Intestinal juice	Final breakdown before absorption

- Liver is the largest gland in the body, about 1.5 kg - it makes bile, forms urea, stores glycogen, iron and fat-soluble vitamins, and detoxifies.
- The liver does NOT make insulin. That is the pancreas.
- Pancreas is a mixed or heterocrine gland - exocrine (digestive juice) plus endocrine (insulin, glucagon).
- Maximum absorption of food takes place in the small intestine, through finger-like villi that multiply the surface area.
- The appendix is a vestigial organ in humans; the largest and strongest muscle-lined organ of chemical digestion is the stomach.
- HCl of the stomach kills bacteria and gives pepsin the acidic pH it needs.
- Memory line - the three P's in canal order: Ptyalin (mouth), Pepsin (stomach), Pancreas (small intestine).

The respiratory system

The path of air, nose to alveolus



Breathing is mechanics; respiration is chemistry. Air enters through the nose, passes the pharynx, larynx (the voice box), trachea and bronchi, and reaches the

alveoli of the lungs, where the actual gas exchange happens. The muscles that do the pumping are the diaphragm and the intercostals between the ribs.

- Twelve pairs of ribs form the rib cage; the diaphragm plus the intercostal muscles drive breathing.
- Tidal volume - the air moved in one quiet breath - is about 500 ml.
- Normal breathing rate is about 12-16 per minute at rest.
- Oxygen is carried in the blood mainly as oxyhaemoglobin, bound to the iron of haemoglobin in the RBC.
- Carbon dioxide is carried mostly as bicarbonate dissolved in the plasma, not on haemoglobin.
- Alveoli are the site of exchange; the right lung has three lobes, the left has two.
- The medulla oblongata, not the lungs, sets the rhythm of breathing.
- Do not write 'CO₂ is carried as carboxyhaemoglobin' - carboxyhaemoglobin is haemoglobin bound to carbon MONOXIDE, the poison.

The circulatory system - heart, blood and the pump

The human heart has four chambers - two atria on top, two ventricles below - and is made of cardiac muscle, which is striated but involuntary and never tires. It beats on its own because of the sino-atrial node, a patch of tissue in the RIGHT atrium that fires the impulse. That word 'right' is the whole question.

- SA node in the right atrium is the pacemaker; the AV node lies in the inter-atrial septum.
- 'Lub' - the first heart sound - is the closure of the atrio-ventricular valves. 'Dub' is the closure of the semilunar valves.
- Normal blood pressure 120/80 mm Hg; 120 is systolic (contraction), 80 diastolic (relaxation).

- Resting heart rate about 72 beats per minute.
- William Harvey described the circulation of blood, in 1628.
- Blood pressure is measured with a sphygmomanometer; the ECG records the electrical activity of the heart.
- Humans have a closed, double circulation - pulmonary (heart to lungs) and systemic (heart to body).
- Arteries carry blood away from the heart (oxygenated, except the pulmonary artery); veins bring it back (deoxygenated, except the pulmonary vein).

What blood is made of

Component	Normal value	Key point
Plasma	About 55% of blood volume	90-92% water; carries most of the CO ₂ as bicarbonate
RBC (erythrocyte)	4.5-5.5 million per microlitre; lives 120 days	No nucleus in mammals; made in red bone marrow, destroyed in the spleen
WBC (leucocyte)	About 5,000-10,000 per microlitre	Defence; neutrophils are the most numerous, lymphocytes make antibodies
Platelets (thrombocyte)	1.5-4.5 lakh per microlitre	Blood clotting

TRAP

'Mature mammalian red blood corpuscles possess a nucleus' is FALSE and RPSC uses it. Mammalian RBCs lose the nucleus, which leaves more room for haemoglobin; camels and llamas are the odd exceptions with oval RBCs. The spleen is the 'graveyard of RBCs'.

Blood groups and the Rh factor - the documented repeat

Give this section extra time. RPSC has asked blood groups and the Rh factor in 2013, in 2021 and again in 2023 - three papers out of the last four. The logic

is simple once you see it. Your red cell carries an antigen; your plasma carries the antibody against the antigen you do NOT have. A transfusion fails when donor antigen meets recipient antibody and the cells clump.

- The ABO system was discovered by Karl Landsteiner in 1901; he received the Nobel Prize for it in 1930.
- Group A has antigen A on the red cell and anti-B antibody in the plasma. Group B is the mirror image.
- Group AB has both antigens and NO antibody - so it can receive from everyone: the universal recipient.
- Group O has no antigen and both antibodies - so its cells offend nobody: the universal donor.
- Strictly, O-negative is the true universal donor and AB-positive the true universal recipient, once Rh is counted.

ABO system - antigen, antibody and who you can give blood to

Group	Antigen on RBC	Antibody in plasma	Can donate to
A	A	anti-B	A and AB
B	B	anti-A	B and AB
AB	A and B	None	AB only
O	None	anti-A and anti-B	A, B, AB and O - universal donor

The same table read the other way - who you can receive from

Group	Can receive from	Tag
AB	A, B, AB and O	Universal recipient
A	A and O	-
B	B and O	-
O	O only	Universal donor but the fussiest recipient

How ABO is inherited - the genetics behind the group

- ABO is controlled by three alleles of one gene on chromosome 9 - $I-A$, $I-B$ and i .
- $I-A$ and $I-B$ are codominant with each other; i is recessive to both.
- Genotypes - A is $I-A I-A$ or $I-A i$; B is $I-B I-B$ or $I-B i$; AB is $I-A I-B$; O is ii .
- So an A father and a B mother can produce a child of any of the four groups, including O.
- A child of group O needs both parents to carry the i allele - this is the classic parentage question.
- The rare Bombay blood group (hh), first reported from Mumbai in 1952, types as O but cannot receive ordinary O blood.

ASKED IN RAS

Asked in RAS Pre 2023 - which blood group is the universal donor? Answer: O. The same paper asked the statement set on ABO: AB plasma has neither anti-A nor anti-B, O plasma has both, and Landsteiner discovered the system - all three correct.

ASKED IN RAS

Asked in RAS Pre 2013 as a NOT-matched question - 'Blood group AB : anti-A and anti-B antibodies' is the wrong pair. AB plasma has no antibody at all; that is precisely why AB is the universal recipient.

Now the Rh factor. It is a second, separate antigen on the red cell, found by Landsteiner and A.S. Wiener in 1940 in the Rhesus monkey, *Macaca mulatta* - hence the name. If you have it you are Rh-positive; about 93-95 per cent of

Indians are. If you do not, you are Rh-negative, and your body will manufacture anti-Rh antibodies the first time it meets Rh-positive blood.

- First exposure sensitises an Rh-negative person; the SECOND exposure is the one that destroys the red cells.
- So an Rh-negative person cannot safely receive Rh-positive blood 'repeatedly' - that statement is always false.
- Rh incompatibility in pregnancy - an Rh-negative mother carrying an Rh-positive foetus.
- The first pregnancy is usually safe; the danger comes from the second pregnancy onwards.
- The condition in the baby is erythroblastosis foetalis, or haemolytic disease of the newborn.
- Prevention - anti-D immunoglobulin given to the mother, usually within 72 hours of delivery.

ASKED IN RAS

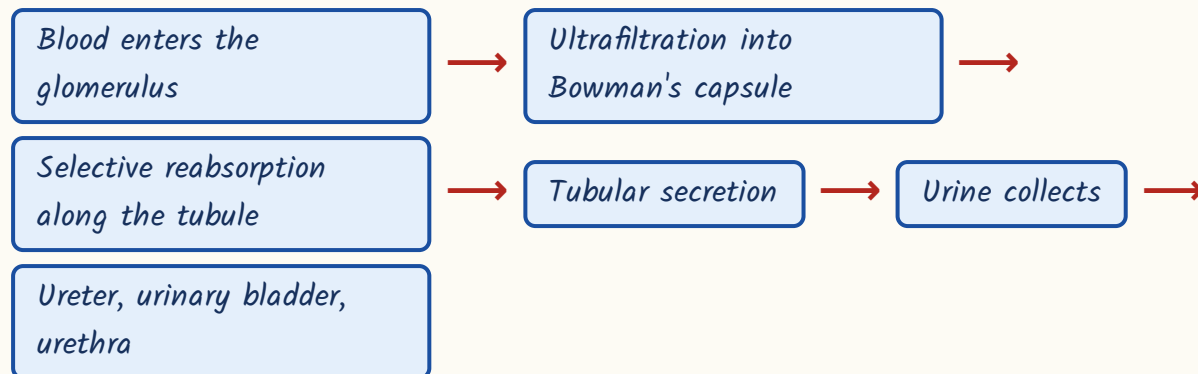
Asked in RAS Pre 2021 - the Rh factor is named after the Rhesus monkey. Asked again in RAS Pre 2023 as statements: erythroblastosis foetalis can occur when an Rh-negative mother carries an Rh-positive foetus (true), anti-D prevents it (true), an Rh-negative person can repeatedly receive Rh-positive blood (false).

YAAD RAKHO

Yaad rakho - 'O gives to all, AB takes from all'. O = zero antigen, so nothing to object to. AB = All antiBodies absent, so nothing to object with. Rh = Rhesus monkey, 1940, Landsteiner and Wiener - and Landsteiner's name therefore attaches to BOTH discoveries.

The excretory system

How urine is made



The kidney filters the blood and the nephron does the filtering. Each kidney holds about ten lakh nephrons. Blood enters a knot of capillaries called the glomerulus, sitting inside a cup called Bowman's capsule - together they make the Malpighian corpuscle, where filtration happens. What the body still wants is reabsorbed further down the tubule; what is left is urine.

- Nephron is the structural and functional unit of the kidney - about 10 lakh per kidney.
- Bowman's capsule plus glomerulus = Malpighian corpuscle, the filtration unit.
- Humans are ureotelic - they excrete urea. Birds and reptiles are uricotelic (uric acid); most fish are ammonotelic.
- Urea is made in the LIVER by the ornithine (urea) cycle and only excreted by the kidney.
- ADH or vasopressin, from the posterior pituitary, controls water reabsorption - less ADH means more, dilute urine.
- Urine is about 95 per cent water; the skin and lungs are accessory excretory organs.
- Dialysis is the artificial substitute when the kidneys fail.

- Planted false statement to watch - 'human beings are uricotelic'. We are UREOtelic; birds are uricotelic.

The nervous system, the brain and the sense organs

The nervous system is made of neurons, the longest cells in the body, and it splits into the central system (brain and spinal cord) and the peripheral system (the nerves). The adult brain weighs about 1300-1400 grams. Learn the brain part-to-function table; RPSC asks nothing else here.

Parts of the brain and what each controls

Part	Controls	Note
Cerebrum	Intelligence, memory, will, voluntary action, speech	The largest part of the brain
Cerebellum	Balance, posture and muscular coordination	Second largest; damage causes loss of balance
Medulla oblongata	Heartbeat, breathing, blood pressure, swallowing	The vital involuntary centre
Hypothalamus	Body temperature, hunger, thirst, sleep	The link between the nervous and endocrine systems
Thalamus	Relay station for sensory impulses	Passes signals up to the cerebrum
Pons	Bridge between brain parts; helps regulate breathing	Part of the hindbrain

- Humans have 12 pairs of cranial nerves and 31 pairs of spinal nerves.
- Reflex action is mediated by the SPINAL CORD through a reflex arc - the brain is informed afterwards.
- The neuron carries the impulse from dendrite to cell body to axon; the gap between two neurons is the synapse.
- Sympathetic nerves prepare the body for emergency; parasympathetic nerves calm it down.

- Eye - rods hold rhodopsin (visual purple) and work in dim light; cones give colour vision.
- The yellow spot or fovea has the maximum visual acuity; the blind spot has no photoreceptors at all.
- Ear ossicles, the three smallest bones - malleus, incus, stapes (hammer, anvil, stirrup).
- The cochlea is for hearing; the three semicircular canals are for balance.
- Tongue detects taste; the nose has olfactory receptors; skin is the largest sense organ.

TRAP

The commonest planted error - 'balance and coordination are controlled by the medulla oblongata'. No. Balance is the CEREBELLUM; the medulla runs heartbeat, breathing and blood pressure. Also, the cochlea is part of the ear, not the brain.

Endocrine glands, and the last sweep - bones, muscles, evolution

Endocrine glands are ductless - they pour hormones straight into the blood. The pituitary sits at the base of the brain and is called the master gland because it controls the others, though it is itself controlled by the hypothalamus. Learn the table in three columns - gland, hormone, and the disorder when it goes wrong. RPSC's favourite trick is a gland-hormone pair that does not match.

Endocrine glands, hormones and disorders

Gland	Hormone	Deficiency / excess
Pituitary (anterior)	Growth hormone	Deficiency - dwarfism; excess in childhood - gigantism; excess in an adult - acromegaly

Gland	Hormone	Deficiency / excess
Pituitary (posterior)	ADH / vasopressin	Deficiency - diabetes insipidus (large volumes of dilute urine)
Thyroid	Thyroxine (needs iodine)	Deficiency - goitre, cretinism in infants, myxoedema in adults; excess - Graves' disease
Thyroid	Calcitonin	Lowers blood calcium - note, calcitonin is thyroid, NOT parathyroid
Parathyroid	Parathormone	Raises blood calcium; deficiency causes tetany
Adrenal cortex	Cortisol and aldosterone	Deficiency - Addison's disease; excess - Cushing's syndrome
Adrenal medulla	Adrenaline and noradrenaline	The emergency or fight-or-flight hormone
Pancreas - beta cells of the islets of Langerhans	Insulin	Deficiency - diabetes mellitus (insulin lowers blood glucose)
Pancreas - alpha cells	Glucagon	Raises blood glucose - antagonistic to insulin
Testis	Testosterone	Male secondary sexual characters
Ovary	Oestrogen and progesterone	Female secondary characters; progesterone maintains pregnancy
Thymus	Thymosin	Maturation of T-lymphocytes; shrinks after puberty
Pineal	Melatonin	Sleep-wake rhythm

- Master gland = pituitary, lying in the sella turcica; its own master is the hypothalamus.
- Diabetes MELLITUS is insulin failure and sugar in the urine; diabetes INSIPIDUS is ADH failure and no sugar at all.
- Adrenaline is the hormone of fright, flight and fight - it raises heart rate, blood sugar and alertness.
- The thyroid needs iodine; that single link ties this chapter to goitre and to salt iodisation.

- An adult has 206 bones; a newborn has about 300, many of which fuse during growth.
- Skull has 22 bones; the femur (thigh) is the longest bone, the stapes of the middle ear the smallest.
- Gluteus maximus of the buttock is the largest muscle; the stapedius of the middle ear the smallest.
- Muscle types - skeletal (striated, voluntary), smooth (unstriated, involuntary), cardiac (striated, involuntary, never fatigues).
- Human evolution sequence - Australopithecus, then Homo habilis (the 'handy man', first tool maker), Homo erectus, Homo neanderthalensis, Homo sapiens.
- Humans carry 46 chromosomes in 23 pairs - 22 pairs of autosomes plus one pair of sex chromosomes, XX in the female and XY in the male.

CURRENT LINK

Current link, as of September 2026 - the Nobel Prize in Physiology or Medicine 2024 went to Victor Ambros and Gary Ruvkun for the discovery of microRNA and its role in gene regulation inside the cell. RAS has asked cell-biology Nobels before, so keep the latest year's medicine Nobel on your revision card.

REVISION RECAP

1. Hooke coined 'cell' (1665, cork); Brown found the nucleus (1831); Schleiden + Schwann gave cell theory, Virchow added 'omnis cellula e cellula'.
2. Mitochondrion = powerhouse, cristae, own circular DNA, 70S ribosomes. Lysosome = suicide bag (de Duve). Chloroplast = kitchen, magnesium in chlorophyll.
3. Ribosome 70S in prokaryotes and in mitochondria/chloroplasts; 80S in eukaryotic cytoplasm.
4. Plant cell only - cellulose wall, plastids, big central vacuole. Animal cell only - centriole, prominent lysosomes.
5. Mitosis = somatic, equational, 2 identical diploid cells. Meiosis = germ cells, reductional, 4 haploid cells, crossing over at pachytene.

6. Fat 9 kcal/g against 4 for carbohydrate and protein. Enzymes are protein (ribozymes excepted) and lower activation energy.
7. DNA - Watson and Crick 1953; A=T two bonds, G-C three; RNA has uracil and ribose.
8. Whittaker 1969 gave five kingdoms; Linnaeus gave binomial nomenclature, genus first. Photosynthesis - light reaction in grana, Calvin cycle in stroma, oxygen from splitting water.
9. Auxin apical dominance, gibberellin bolting, cytokinin cell division, ABA stress and stomatal closure, ethylene ripening.
10. Ptyalin on starch in the mouth, pepsin + HCl on protein in the stomach, trypsin from the pancreas; bile has no enzyme; absorption in the small intestine.
11. Liver is the largest gland - bile, urea, glycogen. Pancreas is mixed. Appendix is vestigial.
12. SA node in the RIGHT atrium is the pacemaker; lub = AV valves, dub = semilunar valves; 120/80 mm Hg; Harvey described circulation.
13. Plasma 55%; RBC 4.5-5.5 million/microlitre, 120 days, anucleate, destroyed in the spleen; platelets 1.5-4.5 lakh.
14. O = universal donor (no antigen, both antibodies); AB = universal recipient (both antigens, no antibody); Landsteiner 1901, Nobel 1930.
15. ABO is three alleles on chromosome 9 - I-A and I-B codominant, i recessive; O child = ii.
16. Rh named after the Rhesus monkey, Landsteiner and Wiener 1940; Rh-negative mother + Rh-positive foetus risks erythroblastosis foetalis from the second pregnancy; anti-D prevents it.
17. Nephron is the kidney's unit (10 lakh each); Malpighian corpuscle = Bowman's capsule + glomerulus; humans are ureotelic; ADH saves water.
18. Cerebrum largest, cerebellum balance, medulla heartbeat and breathing, hypothalamus temperature and thirst; 12 cranial + 31 spinal nerves; reflex through the spinal cord.
19. Pituitary is the master gland - GH gives dwarfism/gigantism/acromegaly; thyroxine needs iodine; insulin from beta cells; adrenaline is the emergency hormone.
20. 206 bones, femur longest, stapes smallest, gluteus maximus largest muscle; 46 chromosomes; Homo habilis was the first tool maker.

ST04

Human Health, Diseases and Public Health Programmes

Science and Technology is 18-23 questions a paper and health takes 4-6 of them. Two facts here are documented identical repeats - Plasmodium falciparum as the malignant form (2021 and 2023) and Trisomy 21 for Down syndrome (2018 and 2021). Vector-borne disease matching appears in 2013, 2021, 2023 and 2024.

Disease, pathogen, vector - that triangle is the most reliably asked thing in the entire Science and Technology paper. RPSC has set a disease-to-organism or disease-to-vector match in every recent paper, and the second half of the chapter, the national health programmes, has become just as regular because RAS loves scheme names, launch years and target dates.

Health, Disease and Health Programmes

Vector-borne

- Malaria - Anopheles
- Dengue/chikungunya/Zika - Aedes
- Filariasis and JE - Culex
- Kala-azar - sandfly
- Plague - rat flea; scrub typhus - mite

Deficiency

- A - night blindness
- B1 beriberi, B3 pellagra
- C scurvy, D rickets
- Iodine goitre, iron anaemia

Programmes

- NHM = NRHM 2005 + NUHM 2013
- Ayushman Bharat 2018, PM-JAY Rs 5 lakh
- NTEP - TB by 2025
- ABDM - 14-digit ABHA
- NCVBDC - malaria by 2030

Bacterial

- TB - Mycobacterium tuberculosis
- Leprosy - M. leprae
- Cholera - Vibrio cholerae
- Typhoid - Salmonella typhi, Widal test

Genetic

- Down - trisomy 21, 47
- Turner 45 XO, Klinefelter 47 XXY
- Haemophilia, colour blindness - X-linked
- Sickle cell, thalassaemia - autosomal recessive

Rajasthan angle

- Right to Health Act - first in India
- Chiranjeevi to Mukhyamantri Ayushman Arogya
- Nishulk Dawa and Janch Yojana
- POSHAN Abhiyaan launched from Jhunjhunu

Viral

- Polio - Sabin OPV, Salk IPV
- Rabies - Pasteur, dog bite
- COVID-19 - SARS-CoV-2, 11 March 2020
- H1N1 swine, H5N1 bird
- Nipah - fruit bat

Immunity

- Active vs passive
- Jenner 1796, smallpox out in 1980
- UIP 1985, 12 diseases
- Mission Indradhanush 25 Dec 2014
- Cold chain 2-8 degrees C

How to read a disease question - the four classes of pathogen

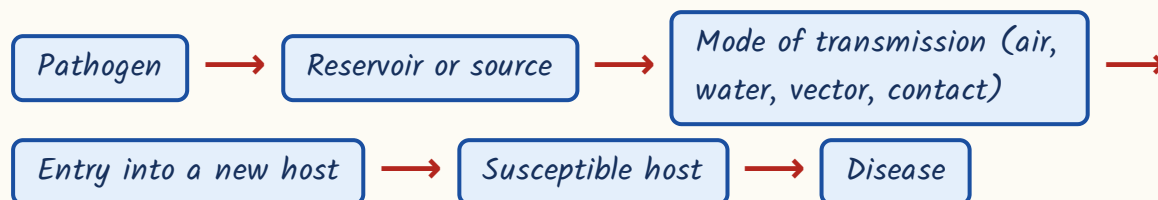
Before memorising anything, sort the diseases into four buckets by the kind of organism that causes them - virus, bacterium, protozoan, worm or fungus. Half the exam's negative-framed questions (which of the following is NOT a viral

disease') are settled by the bucket alone. Then ask the second question: does it need a vector to reach you, or does it come through air, water or contact?

Pathogen class and the diseases that belong to it

Class of pathogen	Diseases
Virus	COVID-19, polio, measles, mumps, chickenpox, rabies, influenza, dengue, chikungunya, hepatitis A-E, AIDS, Nipah, Zika, mpox
Bacterium	Tuberculosis, leprosy, cholera, typhoid, plague, diphtheria, whooping cough, tetanus, pneumonia, scrub typhus
Protozoan	Malaria, kala-azar, amoebic dysentery, sleeping sickness
Worm (helminth)	Lymphatic filariasis (<i>Wuchereria bancrofti</i>), ascariasis, taeniasis
Fungus	Ringworm, athlete's foot, candidiasis

The chain of infection - break any link and the disease stops



The master table - disease, pathogen, vector

This is the centrepiece of the chapter. RPSC asks disease-to-organism or disease-to-vector matching every single year - 2013, 2021, 2023 and 2024 all carried one. Do not read these tables; recite them. Cover the second column and test yourself from the first, then reverse it.

Vector-borne diseases - the highest-yield table in the chapter

Disease	Pathogen	Vector
Malaria	<i>Plasmodium</i> (protozoan)	Female <i>Anopheles</i> mosquito
Dengue	Dengue virus (a flavivirus)	<i>Aedes aegypti</i> - a daytime biter
Chikungunya	Chikungunya virus (alphavirus)	<i>Aedes aegypti</i> and <i>Aedes albopictus</i>
Zika	Zika virus	<i>Aedes aegypti</i>
Yellow fever	Yellow fever virus	<i>Aedes aegypti</i>
Japanese encephalitis	JE virus	<i>Culex tritaeniorhynchus</i> ; the PIG is the amplifying host
Lymphatic filariasis (elephantiasis)	<i>Wuchereria bancrofti</i> (a nematode)	<i>Culex quinquefasciatus</i>
Kala-azar (visceral leishmaniasis)	<i>Leishmania donovani</i> (protozoan)	Female sandfly, <i>Phlebotomus argentipes</i>
Plague	<i>Yersinia pestis</i> (bacterium)	Rat flea, <i>Xenopsylla cheopis</i>
Scrub typhus	<i>Orientia tsutsugamushi</i> (bacterium)	Larval trombiculid mite (chigger); leaves an eschar at the bite
African sleeping sickness	<i>Trypanosoma</i> (protozoan)	Tsetse fly
Kyasanur forest disease	KFD virus	Hard tick, <i>Haemaphysalis</i> - the 'monkey fever' of the Western Ghats

Diseases that need no vector - air, water, food and contact

Disease	Pathogen	How it spreads
Tuberculosis	<i>Mycobacterium tuberculosis</i>	Air-borne droplets
Leprosy (Hansen's disease)	<i>Mycobacterium leprae</i>	Prolonged close contact and droplets
Cholera	<i>Vibrio cholerae</i> (comma-shaped)	Contaminated water and food
Typhoid	<i>Salmonella typhi</i>	Contaminated food and water; Widal test diagnoses it

Disease	Pathogen	How it spreads
Diphtheria	<i>Corynebacterium diphtheriae</i>	Droplets
Whooping cough (pertussis)	<i>Bordetella pertussis</i>	Droplets
Tetanus	<i>Clostridium tetani</i>	Spores from soil entering a wound - NOT water-borne
Pneumonia	<i>Streptococcus pneumoniae</i>	Droplets
Amoebic dysentery	<i>Entamoeba histolytica</i>	Contaminated food and water
Ascariasis	<i>Ascaris lumbricoides</i> (roundworm)	Contaminated food and water
Ringworm	Fungi - <i>Trichophyton</i> , <i>Microsporum</i>	Contact, shared clothing, damp skin
Anthrax	<i>Bacillus anthracis</i>	Contact with infected animals or their products

ASKED IN RAS

Asked in RAS Pre 2024, twice over as matching sets - malaria with *Plasmodium* and *Anopheles*, kala-azar with *Leishmania donovani* and the sandfly, filariasis with *Wuchereria bancrofti* and *Culex*, typhoid with *Salmonella typhi*, dengue with *Aedes aegypti*. If you can reproduce those five rows you have the question.

ASKED IN RAS

Asked in RAS Pre 2024 as a NOT-matched question - 'Kala-azar : tsetse fly' is the wrong pair. Kala-azar is the SANDFLY; the tsetse fly carries African sleeping sickness.

TRAP

The three mosquitoes are where candidates lose the mark. Anopheles = malaria. Aedes = dengue, chikungunya, Zika, yellow fever (day biter, breeds in CLEAN stagnant water). Culex = filariasis and Japanese encephalitis. Also remember plague is bacterial though it has a vector, and scrub typhus is bacterial though it comes from a mite.

Malaria in depth - the single most asked disease

Malaria is caused by a protozoan of the genus Plasmodium and is delivered by the bite of the female Anopheles mosquito - only the female, because only she takes a blood meal. The parasite multiplies first in the liver, then inside the red blood cells, and the cyclical fever with chills comes from the red cells bursting in batches.

The Plasmodium species - the species question is the repeat

Species	Form of malaria	Point to remember
Plasmodium falciparum	Malignant tertian / cerebral malaria	The most fatal form; accounts for almost all malaria deaths
Plasmodium vivax	Benign tertian malaria	The commonest species in India
Plasmodium malariae	Quartan malaria	Fever every fourth day
Plasmodium ovale	Mild tertian malaria	Rare in India

- Ronald Ross demonstrated the malaria parasite inside an Anopheles mosquito at Secunderabad in 1897; Nobel Prize 1902.
- Anopheles is the definitive host of Plasmodium; the human being is the intermediate host.
- Diagnosis - blood smear microscopy or a rapid diagnostic test; treatment includes artemisinin-based combination therapy.

- Control - indoor residual spraying, long-lasting insecticidal nets, larvivorous fish such as *Gambusia*, and source reduction.
- India's target year for malaria elimination is 2030, under the National Center for Vector Borne Diseases Control.
- World Malaria Day is 25 April.

ASKED IN RAS

Asked in RAS Pre 2023, and again in 2021 - which species causes the most severe, malignant form of malaria? Answer: *Plasmodium falciparum*. The 2023 paper also ran it as a statement set where 'P. falciparum causes the benign tertian form' was the planted error, and as an assertion-reason: falciparum infects red cells of every age and the infected cells block cerebral capillaries.

YAAD RAKHO

Yaad rakho - 'Falciparum is Fatal, Vivax is Very common'. F for fatal and for cerebral; V for the vivax that dominates Indian case numbers.

The other vector-borne diseases you must be able to name

After malaria, dengue is the one that reaches the newspapers every monsoon. It is a viral disease - a flavivirus - carried by *Aedes aegypti*, which bites in the daytime and breeds in clean stagnant water inside and around houses: coolers, tyres, flowerpots, overhead tanks. That is why source reduction, not fogging, is the real control measure.

- Dengue - high fever, severe joint and muscle pain ('breakbone fever'), rash, and a characteristic FALL in platelet count (thrombocytopenia).

- Severe forms are dengue haemorrhagic fever and dengue shock syndrome; there is no specific antiviral, treatment is supportive fluids.
- Chikungunya - the same Aedes vector, with crippling joint pain that can persist for months; rarely fatal.
- Zika - Aedes-borne, usually mild in adults, but infection in pregnancy can cause microcephaly in the newborn.
- Japanese encephalitis - Culex tritaeniorhynchus is the vector, the pig is the amplifying host, the ardeid water bird the natural reservoir; JE vaccine is given in endemic districts under UIP.
- Kala-azar - Leishmania donovani through the female sandfly Phlebotomus argentipes; prolonged fever with an enlarged spleen and darkening of the skin, which is what 'kala-azar' means.
- Lymphatic filariasis - the worm blocks the lymph vessels and the limb swells grossly, hence elephantiasis; mass drug administration is the control strategy.

Malaria vs dengue - RPSC sets them against each other

Malaria	Dengue
Protozoan - Plasmodium	Virus - a flavivirus
Female Anopheles, bites at night	Aedes aegypti, bites in the daytime
Breeds in any stagnant water, including dirty	Breeds in CLEAN stored water
Cyclical fever with chills and rigors	High fever with severe joint pain and rash
Platelets usually normal	Platelet count falls - the signature finding
Curable with antimalarial drugs	No specific antiviral; supportive care only

- The NVBDCP directorate is now the National Center for Vector Borne Diseases Control (NCVBDC).
- It covers SIX diseases - malaria, dengue, chikungunya, Japanese encephalitis, kala-azar and lymphatic filariasis.
- Only THREE of them are targeted for elimination - malaria, kala-azar and lymphatic filariasis.

- Dengue, chikungunya and JE are outbreak-prone and are managed by control, not elimination - a favourite statement trap.
- Tuberculosis is not part of this programme at all; it has its own, the NTEP.

Tuberculosis, leprosy and the water-borne classics

Tuberculosis is India's biggest single infectious disease burden and its exam questions are settled by four names: Robert Koch, who identified *Mycobacterium tuberculosis* in 1882; BCG, the live attenuated vaccine given at birth; DOTS, the directly observed treatment; and Ni-kshay, the digital notification portal. World TB Day is 24 March, the day Koch announced the discovery.

- TB is air-borne, spreads by droplets, and most often attacks the lungs (pulmonary TB), though it can hit bone, spine, brain or intestine.
- Diagnosis - sputum microscopy, CBNAAT (a rapid molecular test) and chest X-ray.
- BCG (*Bacillus Calmette-Guerin*) is LIVE ATTENUATED, not killed - it protects mainly against severe childhood TB, miliary TB and TB meningitis.
- Multi-drug resistant TB is resistance to at least isoniazid and rifampicin.
- Leprosy - *Mycobacterium leprae*, discovered by the Norwegian physician G.H.A. Hansen in 1873; treated with multi-drug therapy, free of cost.
- Cholera - *Vibrio cholerae*, profuse painless watery diarrhoea; oral rehydration solution is the life-saver.
- Typhoid - *Salmonella typhi*, step-ladder fever, diagnosed by the Widal test; typhoid vaccine exists but is not in the routine UIP schedule.

Hepatitis A to E - the classic sorting question

Type	How it spreads	Vaccine
Hepatitis A	Faeco-oral - contaminated food and water	Vaccine available

Type	How it spreads	Vaccine
Hepatitis B	Blood, body fluids, mother to child	Vaccine available and part of the UIP, with a birth dose
Hepatitis C	Blood and blood products	No vaccine; curable with modern antivirals
Hepatitis D	Only along with hepatitis B	Prevented by the hepatitis B vaccine
Hepatitis E	Faeco-oral, mainly water	Especially dangerous in pregnancy

TRAP

Split them two and three. A and E travel by WATER (the two vowels at the ends). B, C and D travel by BLOOD. If a question says hepatitis B spreads through contaminated water, it is wrong; if it says hepatitis A has a vaccine in the UIP, that is also wrong - it is hepatitis B that is in the UIP.

The viral diseases - polio, rabies, COVID-19 and the newer ones

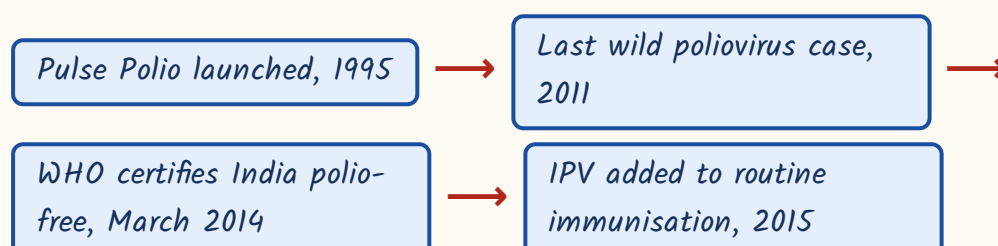
Viruses give the exam its dates. Polio gives you 1955 and 2014, smallpox 1796 and 1980, COVID-19 gives 11 March 2020. Learn the vaccine names attached to each, because the vaccine-to-disease pairing is asked as often as the disease itself.

- Polio - poliovirus. OPV is the LIVE attenuated oral vaccine of Albert Sabin; IPV is the KILLED injectable vaccine of Jonas Salk.
- Pulse Polio Immunisation began in India in 1995; injectable polio vaccine was added to routine immunisation in 2015.
- India was certified polio-free by the WHO in March 2014.
- Smallpox - Edward Jenner used cowpox material in 1796; WHO declared the disease eradicated in 1980, India's last case being in 1975. It is the only

human disease so far eradicated.

- Rabies - a lyssavirus, spread by the bite or lick of an infected animal, most often a dog. Louis Pasteur made the first vaccine, 1885. Once symptoms appear it is virtually 100 per cent fatal, so post-exposure prophylaxis must not be delayed.
- Measles - a virus; measles-rubella (MR) vaccine is given under the UIP at 9-12 months and again at 16-24 months.
- COVID-19 - caused by SARS-CoV-2; the WHO declared it a pandemic on 11 March 2020.
- India's principal vaccines were Covaxin (Bharat Biotech with ICMR-NIV, inactivated) and Covishield (Serum Institute, the Oxford-AstraZeneca viral-vector vaccine).
- Influenza - swine flu is influenza A H1N1 (the 2009 pandemic strain); bird flu is avian influenza H5N1.
- Nipah - the natural reservoir is the fruit bat of the genus *Pteropus*, with pigs as intermediate hosts; India has seen outbreaks in West Bengal and Kerala.
- Mpox - a viral zoonosis of the orthopoxvirus group; the WHO declared it a public health emergency of international concern in August 2024.
- AIDS - HIV, a retrovirus that destroys the CD4 helper T lymphocytes; screened by ELISA and confirmed by Western blot.
- NACO was set up in 1992 and the HIV and AIDS (Prevention and Control) Act was passed in 2017. World AIDS Day is 1 December.

India's polio timeline - a ready-made ordering question



Deficiency diseases, lifestyle diseases and genetic disorders

Not every disease comes from an organism. Some come from what is missing in the plate, some from how we live, and some from the chromosomes we are born with. The deficiency table is pure recall and RPSC has asked it repeatedly as a match-the-following; the genetic table is where Down syndrome sits, and Down syndrome has been asked in identical form in 2018 and 2021.

Deficiency diseases - nutrient to disease

Nutrient	Deficiency disease	Note
Vitamin A (retinol)	Night blindness, xerophthalmia	Fat-soluble; a leading preventable cause of childhood blindness
Vitamin B1 (thiamine)	Beriberi	Polished rice diets are the classic cause
Vitamin B2 (riboflavin)	Cheilosis, angular stomatitis	Cracks at the corners of the mouth
Vitamin B3 (niacin)	Pellagra	The three D's - dermatitis, diarrhoea, dementia
Vitamin B12 (cobalamin)	Pernicious anaemia	The only vitamin containing cobalt
Vitamin C (ascorbic acid)	Scurvy	Bleeding gums, delayed wound healing - collagen fails
Vitamin D (calciferol)	Rickets in children, osteomalacia in adults	The sunshine vitamin
Vitamin K	Delayed blood clotting	Needed for prothrombin
Iodine	Goitre, cretinism in infants	Universal salt iodisation is the answer
Iron	Anaemia	Addressed by Anaemia Mukht Bharat
Calcium	Rickets, osteoporosis, tetany	Works with vitamin D
Fluoride	Deficiency - dental caries; EXCESS - fluorosis	Fluorosis is endemic in parts of Rajasthan; NPPCF is the national programme

Nutrient	Deficiency disease	Note
Protein and energy	Kwashiorkor and marasmus	The two faces of protein-energy malnutrition

- *Non-communicable or lifestyle diseases - diabetes, hypertension, cardiovascular disease, cancer, chronic lung disease and obesity.*
- *They now cause the majority of deaths in India; the national programme for them is NPCDCS, run through district NCD clinics.*
- *Type 1 diabetes is insulin deficiency, type 2 is insulin resistance; obesity is measured by Body Mass Index.*
- *Hypertension is called the 'silent killer' because it usually has no symptoms until damage is done.*
- *World Health Day 7 April, World Cancer Day 4 February, World Blood Donor Day 14 June, Indian Organ Donation Day 3 August.*

Genetic disorders - the basis is the answer

Disorder	Genetic basis	Key point
Down syndrome	Trisomy of chromosome 21 - 47 chromosomes in all	Intellectual disability, characteristic facial features; risk rises with maternal age
Turner syndrome	45, XO	A female with a single X - short stature, sterile
Klinefelter syndrome	47, XXY	A male with an extra X - sterile
Haemophilia	X-linked recessive	Blood fails to clot; the 'royal disease', far commoner in males
Colour blindness	X-linked recessive	Usually red-green; tested by Ishihara charts
Sickle cell anaemia	Autosomal recessive (haemoglobin S)	National Sickle Cell Anaemia Elimination Mission, 2023, targets 2047
Thalassaemia	Autosomal recessive	Defective haemoglobin chains; needs repeated transfusion

Disorder	Genetic basis	Key point
Phenylketonuria	Autosomal recessive	The enzyme phenylalanine hydroxylase is missing

ASKED IN RAS

Asked in RAS Pre 2018 and again, identically, in RAS Pre 2021 - Down syndrome is caused by trisomy of chromosome 21. Learn the number 21 and the resulting total of 47 chromosomes; both have been used as the option.

TRAP

Thalassaemia and sickle cell anaemia are AUTOSOMAL recessive - they are not sex-linked. Only haemophilia and colour blindness are the X-linked pair. RPSC plants 'thalassaemia is a sex-linked disorder' as the false statement.

Immunity, vaccines, the UIP and antimicrobial resistance

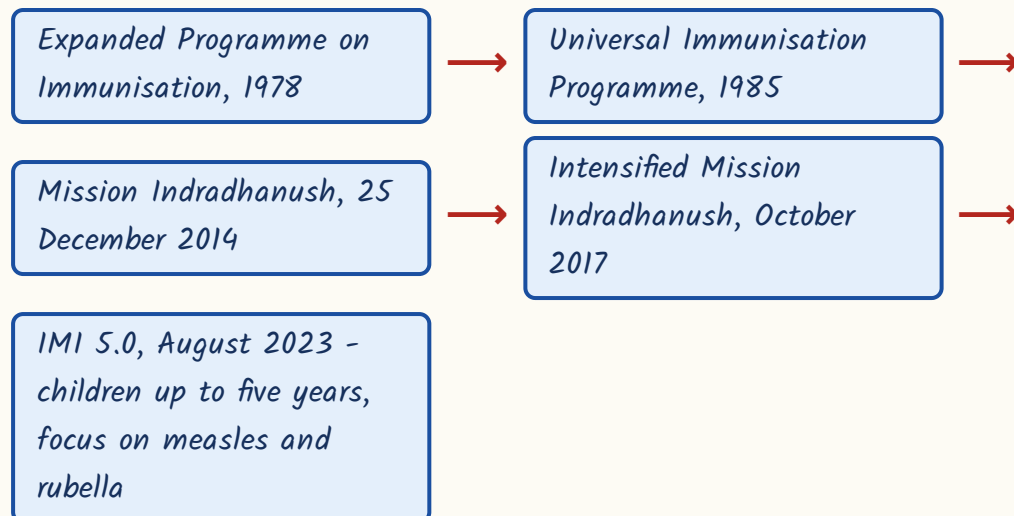
Immunity is either innate - the skin, the stomach acid, the general defence you are born with - or acquired, which is built after exposure. Acquired immunity then splits again: active, where your own body makes the antibodies, and passive, where ready-made antibodies are handed to you. Almost every immunity question in RAS is testing that one split.

Active immunity vs passive immunity

Active immunity	Passive immunity
The body makes its own antibodies	Ready-made antibodies come from outside
Follows an infection or a vaccination	Colostrum, antiserum, anti-D, anti-snake venom
Slow to develop	Acts immediately
Long-lasting, because memory cells are formed	Short-lived, no memory
Example - BCG, OPV, hepatitis B vaccine	Example - mother's milk, anti-rabies serum

- Vaccine types - live attenuated (BCG, OPV, measles), killed or inactivated (IPV, Covaxin), toxoid (tetanus, diphtheria), subunit and recombinant (hepatitis B), and the newer mRNA and viral-vector vaccines.
- BCG is live attenuated; saying 'BCG is a killed vaccine' is a standard planted error.
- The Universal Immunisation Programme was launched in 1985, growing out of the Expanded Programme on Immunisation of 1978.
- The UIP now protects against 12 vaccine-preventable diseases - diphtheria, pertussis, tetanus, polio, measles, rubella, childhood tuberculosis, hepatitis B, Haemophilus influenzae type b infections, rotavirus diarrhoea, pneumococcal pneumonia, and Japanese encephalitis in endemic districts.
- Vaccines are kept in the cold chain at 2 to 8 degrees Celsius; breaking the cold chain destroys the vaccine.
- At birth a child gets BCG, the zero dose of OPV and the birth dose of hepatitis B vaccine.

How India's immunisation drive grew



- Mission Indradhanush aims at full immunisation coverage of at least 90 per cent among children and pregnant women; the name comes from the rainbow's seven colours, the original seven vaccine-preventable diseases.
- Antibiotics work on bacteria only - they do nothing against a virus, so they are useless for a common cold.
- Alexander Fleming discovered penicillin, the first antibiotic, in 1928.
- Antimicrobial resistance grows from over-prescription, self-medication, unfinished courses and heavy antibiotic use in poultry and livestock.
- India adopted a National Action Plan on Antimicrobial Resistance in 2017; the Red Line campaign marks prescription-only antibiotics with a red vertical line on the pack.

National health programmes, the indicators, and Rajasthan's Right to Health

This half of the chapter is pure scheme recall - name, launch year, and the one number attached to it. RAS asks it as matching or as a chronology, and it asks

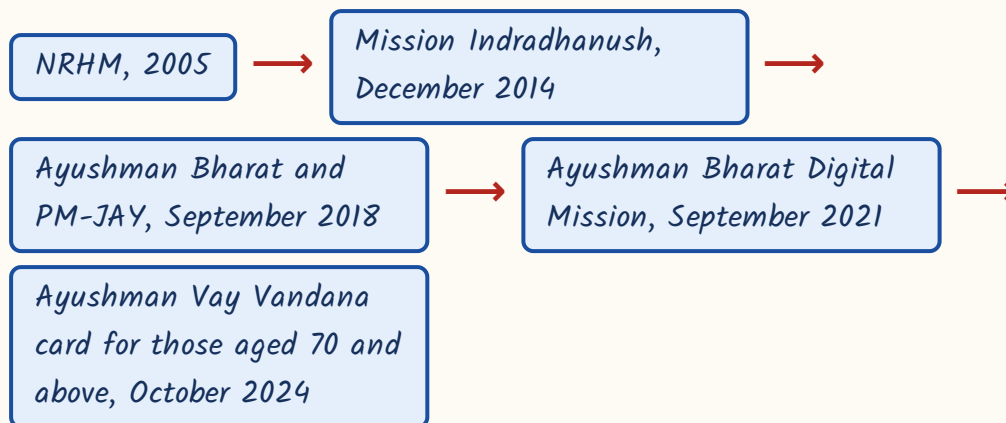
it in current affairs too. Learn the table with the years, then learn the four or five numbers that follow it.

National health programmes - name, year, and what it actually does

Programme	Launched	What it does
National Rural Health Mission	2005	Rural health system; created the ASHA, the community link worker
Janani Suraksha Yojana	2005	Cash assistance for institutional delivery
Janani Shishu Suraksha Karyakram	2011	Free delivery, drugs, diagnostics, diet and transport
National Urban Health Mission	2013	Urban health; NRHM + NUHM together make the National Health Mission
Mission Indradhanush	25 December 2014	Full immunisation coverage of at least 90 per cent
LaQshya	2017	Quality of care in the labour room and maternity operation theatre
Ayushman Bharat	2018	Two arms - Health and Wellness Centres plus PM-JAY
PM-JAY	23 September 2018, from Ranchi	Rs 5 lakh per family per year, secondary and tertiary care, no premium
Ni-kshay Poshan Yojana	April 2018	Nutrition support to TB patients by DBT, raised to Rs 1,000 a month in November 2024
Anaemia Mukta Bharat	2018	The 6x6x6 strategy - six beneficiary groups, six interventions, six institutional mechanisms
POSHAN Abhiyaan	8 March 2018, launched from Jhunjhunu, Rajasthan	Targets stunting, anaemia and low birth weight
National TB Elimination Programme	Renamed from RNTCP in 2020	TB elimination by 2025, five years ahead of the SDG target of 2030

Programme	Launched	What it does
Ayushman Bharat Digital Mission	27 September 2021 (piloted 15 August 2020)	The 14-digit ABHA number, plus the Healthcare Professionals and Health Facility Registries
NCVBDC (from NVBDCP)	Renamed directorate	Six vector-borne diseases; malaria elimination target 2030

Chronology of the big health initiatives - asked as an ordering question



- Health and Wellness Centres were renamed Ayushman Arogya Mandir in November 2023, with the tagline 'Arogyam Parmam Dhanam'.
- PM-JAY is fully government-funded - beneficiaries pay no premium; families were identified from the deprivation criteria of SECC 2011.
- The Ayushman Vay Vandana card, announced in October 2024, extends cover to all citizens aged 70 and above regardless of income.
- The Pradhan Mantri Bhartiya Janaushadhi Pariyojana (generic medicine stores) is a SEPARATE scheme - it is not a component of Ayushman Bharat.
- NOTTO, the National Organ and Tissue Transplant Organisation, sits at New Delhi under the DGHS, with ROTTOs and SOTTOs below it, under the Transplantation of Human Organs Act, 1994, amended in 2011.
- AYUSH covers Ayurveda, Yoga and Naturopathy, Unani, Siddha, Sowa-Rigpa and Homoeopathy; the Ministry was created in November 2014 and the WHO Global Centre for Traditional Medicine is at Jamnagar, Gujarat.

- Other standing programmes - National Programme for Control of Blindness and Visual Impairment, National Mental Health Programme (1982, with the Mental Healthcare Act 2017 and Tele-MANAS), and NPPCF for fluorosis.
- Health indicators, from the SRS Report 2021 released in May 2025 - maternal mortality ratio 93 per lakh live births (2019-21), infant mortality rate 27, under-five mortality rate 31 and neonatal mortality rate 19 per 1,000 live births.
- MMR is counted per lakh live births; IMR, USMR and NMR are per thousand. Mixing up the denominator is the standard error.
- Life expectancy, sex ratio at birth and total fertility rate are the other indicators RAS picks from.

CURRENT LINK

Current link, as of September 2026 - Rajasthan is the FIRST state in India to give its residents a legal right to health. The Rajasthan Right to Health Act was passed by the Assembly on 21 March 2023 and received the Governor's assent on 12 April 2023 (Act No. 7 of 2023). Its core promise is emergency treatment and care at any health-care establishment, public or private, without waiting for prepayment or police clearance. It sets up a State Health Authority and District Health Authorities, with a grievance to be answered within 24 hours and an action-taken report within 30 days. Note also that the state's Chiranjeevi Swasthya Bima Yojana (2021) has been renamed the Mukhyamantri Ayushman Arogya Yojana, with cover raised to Rs 25 lakh per family per year - check the latest state budget for the current figure before the exam.

- Rajasthan's older health flagships still get asked - Mukhyamantri Nishulk Dawa Yojana (free medicines, 2011) and Mukhyamantri Nishulk Janch Yojana (free diagnostics, 2013).
- Nirogi Rajasthan is the state's preventive-health campaign.

- POSHAN Abhiyaan was launched from Jhunjhunu in Rajasthan on 8 March 2018 - a Rajasthan link worth carrying into the current-affairs paper too.

YAAD RAKHO

Yaad rakho the year ladder for the schemes - 2005 NRHM and JSY, 2011 JSSK, 2013 NUHM, 2014 Mission Indradhanush, 2017 LaQshya, 2018 Ayushman Bharat plus Ni-kshay Poshan plus Anaemia Mukht Bharat plus POSHAN, 2020 NTEP, 2021 ABDM. Almost every year has exactly one flagship - and 2018 has four.

REVISION RECAP

1. Malaria - Plasmodium, female Anopheles; falciparum = malignant and cerebral, vivax = commonest in India; Ronald Ross 1897, Nobel 1902.
2. Aedes aegypti (day biter, clean water) carries dengue, chikungunya, Zika and yellow fever; dengue drops the platelet count.
3. Culex carries filariasis (Wuchereria bancrofti) and Japanese encephalitis, whose amplifying host is the pig.
4. Kala-azar = Leishmania donovani + sandfly. Plague = Yersinia pestis + rat flea. Scrub typhus = Orientia + chigger mite. Sleeping sickness = Trypanosoma + tsetse fly.
5. TB - Mycobacterium tuberculosis, Koch 1882, air-borne, BCG (live attenuated), CBNAAT, DOTS, Ni-kshay; World TB Day 24 March.
6. Leprosy = M. leprae (Hansen); cholera = Vibrio cholerae; typhoid = Salmonella typhi, diagnosed by the Widal test.
7. Hepatitis A and E through water; B, C and D through blood; only hepatitis B is in the UIP.
8. Polio - OPV live (Sabin), IPV killed (Salk); Pulse Polio 1995, India polio-free March 2014. Rabies - lyssavirus from a dog bite, Pasteur's vaccine, near-certain death once symptoms start. Smallpox - Jenner 1796, eradicated 1980.
9. COVID-19 = SARS-CoV-2, pandemic declared 11 March 2020; swine flu H1N1, bird flu H5N1; Nipah reservoir is the fruit bat; mpox is an orthopoxvirus.
10. HIV is a retrovirus destroying CD4 T cells - ELISA screens, Western blot confirms; NACO 1992, HIV/AIDS Act 2017.
11. Deficiencies - A night blindness, B1 beriberi, B3 pellagra, B12 pernicious anaemia, C scurvy, D rickets, K clotting, iodine goitre, iron anaemia; excess fluoride gives fluorosis, a Rajasthan problem.

12. Down = trisomy 21 (47 chromosomes); Turner 45 XO; Klinefelter 47 XXY; haemophilia and colour blindness X-linked recessive; sickle cell and thalassaemia autosomal recessive.
13. Active immunity is made by the body and lasts; passive immunity is borrowed and fades. BCG is live, not killed.
14. UIP 1985, from the EPI of 1978, now covers 12 diseases; cold chain 2-8 degrees C; Mission Indradhanush 25 December 2014, IMI 5.0 August 2023.
15. Antibiotics do not touch viruses; Fleming found penicillin in 1928; India's National Action Plan on AMR dates from 2017, plus the Red Line campaign.
16. NHM = NRHM (2005) + NUHM (2013); the ASHA is the NRHM's link worker.
17. Ayushman Bharat 2018 = Ayushman Arogya Mandir (the renamed HWCs, November 2023) + PM-JAY (Rs 5 lakh, launched 23 September 2018 from Ranchi, no premium).
18. NTEP targets TB elimination by 2025; NCVBDC covers six diseases and targets malaria elimination by 2030; ABDM gives the 14-digit ABHA number.
19. SRS 2021 - MMR 93 per lakh, IMR 27, USMR 31, NMR 19 per thousand.
20. Rajasthan passed India's FIRST Right to Health Act - Assembly 21 March 2023, assent 12 April 2023 - guaranteeing emergency care without prepayment; Chiranjeevi is now Mukhyamantri Ayushman Arogya Yojana.

ST05

Food and Nutrition

Science and Technology carries 18-23 questions in every RAS paper. Food and nutrition reliably supplies 2-4 of them, and the vitamin grid alone is worth a question in most papers. The nutrition schemes get counted again in the Current Affairs and Rajasthan blocks, so the return per hour here is among the highest in the syllabus.

This chapter is pure marks. Food and nutrition is the one biology topic where RPSC does not ask you to understand a mechanism - it asks you to match a name to a name. Vitamin to chemical name, vitamin to deficiency disease, adulterant to food item, scheme to launch year. Learn the tables cold and you convert every question in this chapter.

Food and Nutrition

Balanced diet

- Carbs 55-60%, fat 20-30%, protein 10-15%
- Carb 4, protein 4, fat 9, alcohol 7 kcal/g
- Five food groups plus water
- Water = 60-70% of body weight

Vitamins (biggest)

- Fat-soluble: A D E K
- Water-soluble: B-complex and C
- A - night blindness; D - rickets
- B1 beriberi, B3 pellagra, C scurvy
- B12 contains cobalt

Minerals

- Iodine - goitre, cretinism
- Iron - anaemia
- Calcium - rickets, osteoporosis, tetany
- Fluoride - less gives caries, more gives fluorosis

Malnutrition

- Kwashiorkor = protein only, oedema
- Marasmus = protein + calorie, wasting
- Stunting / wasting / underweight
- BMI = kg divided by m squared

Better food

- Fortification = added in processing, +F logo
- Biofortification = bred into the crop
- Dhanashakti bajra, Golden Rice, QPM maize

Food safety

- LTLT, HTST, UHT, irradiation
- Argemone, kesari dal, metanil yellow
- FSSAI 2006, Health Ministry, New Delhi
- Eat Right India, RUCO, trans fat 2%

Programmes

- ICDS 1975
 - NFSA 2013
 - POSHAN Abhiyaan 2018 - Jhunjhunu
 - PM POSHAN 2021
 - Anaemia Mukh Bharat
- 6x6x6

Balanced diet - what the phrase actually means

A balanced diet is not a diet with plenty of everything. It is a diet in which each nutrient comes in the right proportion to the others, in the quantity the body actually needs. Two things must balance at once - the total energy, and the share of that energy contributed by each macronutrient. Get the energy

right but the shares wrong and you still get disease. Food groups are the shortcut: take something from each group and the proportions look after themselves.

- Standard energy split - carbohydrate 55-60%, fat 20-30%, protein 10-15% of total calories.
- Water makes up 60-70% of body weight, the single largest component of the body.
- Nutrients split into macronutrients (carbohydrate, protein, fat) and micronutrients (vitamins, minerals).
- Dietary fibre gives almost no calories but keeps the bowel working - cellulose, hemicellulose, pectin, lignin, gums.

Five food groups and what each one is really for

Group	Examples	Chiefly supplies
Cereals, millets, pulses	Wheat, rice, bajra, ragi, gram	Energy plus plant protein
Vegetables and fruits	Green leafy, roots, citrus	Vitamins, minerals, fibre
Milk and animal foods	Milk, curd, egg, fish, meat	Good-quality protein, calcium, B12
Fats, oils and nuts	Ghee, oil, groundnut	Concentrated energy, fat-soluble vitamins
Sugars	Sugar, jaggery, honey	Energy only, nothing else

TRAP

Dietary fibre is plant material your enzymes cannot digest. Glycogen is animal starch and IS digestible - so glycogen is NOT dietary fibre. RPSC sets this as a 'which one is not a constituent of dietary fibre' question.

Macronutrients and their calorific values

Calorific value is the energy released when one gram of a nutrient is completely oxidised in the body. There are only four numbers in the whole list, and the one students skip - alcohol - is the one the paper likes. Learn all four, in order.

Calorific values - four numbers, learn them cold

Nutrient	Energy per gram	Note
Carbohydrate	4 kcal	The body's first-choice fuel
Protein	4 kcal	Same as carbohydrate - a favourite trap
Fat	9 kcal	Highest, more than double carbohydrate
Alcohol	7 kcal	Second highest, supplies no other nutrient

- Proteins are chains of amino acids; nine are essential for adult humans and must come from food.
- The nine - histidine, isoleucine, leucine, lysine, methionine, phenylalanine, threonine, tryptophan, valine.
- Egg protein is the reference protein, the standard other proteins are scored against.
- Cereal protein is low in lysine, pulse protein is low in methionine - eaten together they complete each other, which is why dal-chawal works.
- Protein is for growth and repair; carbohydrate and fat are for energy. Burning protein for fuel is a waste.

YAAD RAKHO

Descending order of calorific value - Fat 9, Alcohol 7, Carbohydrate 4, Protein 4. Say it as nine-seven-four-four.

Vitamins – the single most examined table in this chapter

Vitamins are organic compounds needed in tiny amounts that the body cannot make in sufficient quantity. The paper picks one of four angles every time – the chemical name, the source, the deficiency disease, or fat-soluble versus water-soluble. So learn all four columns together, never the disease alone. And learn the solubility split first, because that one line answers a whole question by itself.

- Fat-soluble – A, D, E, K. They dissolve in fat, are stored in liver and fat tissue, and an excess can therefore poison you: hypervitaminosis.
- Water-soluble – the whole B-complex and vitamin C. Not stored, the excess leaves in urine, so they must be eaten regularly.
- Because the fat-soluble ones are stored, their deficiency takes months to appear. Water-soluble deficiency shows up quickly.

YAAD RAKHO

Fat-soluble vitamins spell A-DEK. Everything else – all the B's and C – is water-soluble. If tocopherol (vitamin E) turns up in a list of water-soluble vitamins, that is your odd one out.

Fat-soluble vitamins – chemical name, source, deficiency

Vitamin	Chemical name	Good sources	Deficiency
A	Retinol	Carrot, green leafy vegetables, liver, milk	Night blindness, Bitot's spots, xerophthalmia, keratomalacia
D	Calciferol – D2 ergocalciferol, D3 cholecalciferol	Sunlight on skin, fish liver oil, egg yolk	Rickets in children, osteomalacia in adults

Vitamin	Chemical name	Good sources	Deficiency
E	Tocopherol	Vegetable oils, wheat germ, nuts	Antioxidant; deficiency causes sterility in rats
K	Phylloquinone K1, menaquinone K2	Green leafy vegetables; K2 made by gut bacteria	Delayed blood clotting, excess bleeding

- Vitamin D is the one your own skin makes - from 7-dehydrocholesterol, using UV-B in sunlight. Hence 'sunshine vitamin'.
- Vitamin K is needed to synthesise prothrombin, the clotting protein - which is why newborns are given a vitamin K shot.
- Bitot's spots are greyish triangular patches on the white of the eye, a clinical sign of vitamin A deficiency.

Water-soluble vitamins - B-complex and C

Vitamin	Chemical name	Deficiency disease
B1	Thiamine	Beriberi
B2	Riboflavin	Cheilosis, angular stomatitis
B3	Niacin / nicotinic acid	Pellagra - dermatitis, diarrhoea, dementia
B5	Pantothenic acid	Burning feet syndrome
B6	Pyridoxine	Anaemia and convulsions
B7	Biotin	Dermatitis and hair loss
B9	Folic acid	Megaloblastic anaemia; neural tube defects in the foetus
B12	Cyanocobalamin	Pernicious anaemia
C	Ascorbic acid	Scurvy - bleeding gums, delayed wound healing

- Vitamin B12 is the vitamin that contains a metal - cobalt. That single fact is a question on its own.
- B12 comes only from animal sources and microbial fermentation, so strict vegetarians are the risk group.

- Folic acid is given in pregnancy to prevent neural tube defects - the reason it goes into fortified wheat flour.
- Humans, other primates and guinea pigs cannot synthesise vitamin C; most other animals can.

ASKED IN RAS

Vitamin-to-chemical-name and vitamin-to-deficiency matching is a standing feature of the RPSC science block - the question bank for this chapter alone carries four separate matching sets built on it. The four pairs that recur: B1 thiamine - beriberi, B3 niacin - pellagra, B12 cyanocobalamin - pernicious anaemia, C ascorbic acid - scurvy.

TRAP

Pyridoxine (B6) causes anaemia and convulsions, NOT pellagra - pellagra is niacin (B3). The 'not correctly matched' question is built on exactly this swap. Second common swap: riboflavin paired with beriberi, when beriberi belongs to thiamine.

Minerals - fewer names, sharper numbers

Minerals are inorganic, so unlike vitamins they cannot be destroyed by cooking - they only leach into the cooking water. Two angles get asked: which deficiency causes which disease, and the two policy numbers, iodine in salt and fluoride in water. Fluoride is the tricky one, because both too little and too much cause disease.

Minerals and their deficiency effects

Mineral	Main role	Deficiency
Iodine (I)	Thyroid hormone synthesis	Goitre in adults, cretinism in children

Mineral	Main role	Deficiency
Iron (Fe)	Haemoglobin	Anaemia
Calcium (Ca)	Bone, teeth, clotting, muscle contraction	Rickets, osteoporosis, tetany
Zinc (Zn)	Enzymes, immunity, growth	Growth retardation, poor wound healing
Fluoride (F)	Tooth enamel	Too little - dental caries; too much - dental and skeletal fluorosis
Sodium (Na), Potassium (K)	Fluid balance, nerve impulse	Cramps and weakness; excess sodium raises blood pressure

- Under NIDDCP, iodised salt must carry not less than 30 ppm iodine at manufacture and 15 ppm at the consumer end - the gap allows for losses in transit.
- BIS drinking-water standard for fluoride - desirable limit 1.0 mg/L, permissible limit 1.5 mg/L. Rajasthan has serious fluorosis belts, so this is a state-relevant number.
- Iodine deficiency in pregnancy causes cretinism, stunted growth with mental retardation; goitre is the adult sign.
- Double Fortified Salt (DFS) carries both iodine and iron.

ASKED IN RAS

Endemic goitre as a direct consequence of dietary iodine deficiency is a recurring one-liner, and its companion is the 30 ppm / 15 ppm iodised-salt standard. Learn the disease and the number together - the paper has asked both faces.

Malnutrition - PEM, the three indices, anaemia and BMI

Malnutrition does not mean only 'too little food'. It means the wrong balance - and the two classic protein-energy malnutrition conditions differ on exactly that balance. Kwashiorkor is the child getting calories but not protein. Marasmus is the child getting neither. Learn them as a pair, because that is how the question is set.

Kwashiorkor vs Marasmus

Kwashiorkor	Marasmus
Protein deficiency with nearly adequate calories	Protein AND calorie deficiency together
Oedema present - swollen legs and belly	No oedema
Moon face, flaky-paint dermatosis	Extreme wasting, 'old man's face'
Usually after weaning, about 1-3 years	Usually below one year of age
Some subcutaneous fat retained	Fat and muscle both lost
Term coined by Cicely Williams from a Ghanaian word	Name from the Greek for wasting

- Stunting = low height-for-age. It reflects chronic, long-term undernutrition.
- Wasting = low weight-for-height. It reflects acute, recent undernutrition.
- Underweight = low weight-for-age, a composite of the other two.
- NFHS-5 (2019-21) India - stunting 35.5%, wasting 19.3%, underweight 32.1%.
- NFHS-5 anaemia - 67.1% of children aged 6-59 months and 57.0% of women aged 15-49.

Anaemia - WHO haemoglobin cut-offs

Group	Anaemia if haemoglobin below
Adult men	13 g/dL
Non-pregnant women	12 g/dL
Pregnant women	11 g/dL
Children 6-59 months	11 g/dL

- BMI = weight in kilograms divided by the square of height in metres; the unit is kg/m².
- WHO bands - below 18.5 underweight, 18.5-24.9 normal, 25-29.9 overweight, 30 and above obese.
- Asian-Indian cut-offs are lower - 23 and above overweight, 25 and above obese - because Indians carry more body fat at the same BMI.
- India now carries a double burden - undernutrition and obesity together in the same population.

ASKED IN RAS

Kwashiorkor as protein deficiency with oedema is a repeatedly examined statement set. The statement that trips people is the age - kwashiorkor is the post-weaning child, marasmus the infant under one year.

Fortification and biofortification - know the difference

Both raise the nutrient content of food, but at completely different points in the chain. Fortification happens after harvest, in the factory - you add the nutrient to the food, and FSSAI marks such packs with the '+F' logo.

Biofortification happens before harvest, in the field - you breed or grow the nutrient into the crop itself. One word gives it away: 'bio' means it was grown that way - fortification is Factory, biofortification is Bred in the field. Fortified rice now moves through the PDS, ICDS and PM POSHAN. Watch the odd one out in the biofortified list: Golden Rice gets its beta-carotene by genetic engineering, while Dhanashakti and QPM come from ordinary breeding.

Fortification - which staple carries which nutrient

Food vehicle	Nutrient added
Salt	Iodine; iron as well in Double Fortified Salt

Food vehicle	Nutrient added
Wheat flour and rice	Iron, folic acid, vitamin B12
Milk and edible oil	Vitamins A and D

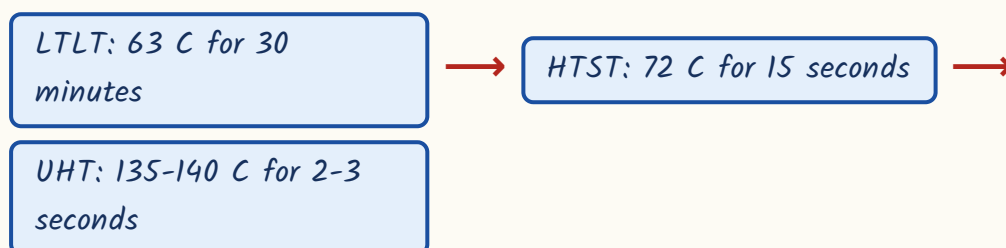
Biofortified varieties worth naming

Variety / crop	Nutrient	Note
Dhanashakti - pearl millet (bajra)	Iron	India's first notified biofortified variety
Golden Rice	Provitamin A (beta-carotene)	Genetically engineered, not conventionally bred
QPM - Quality Protein Maize	Lysine and tryptophan	Fixes the amino-acid gap of ordinary maize

Food preservation, adulteration and FSSAI

Food spoils because microbes and enzymes work on it. Every preservation method therefore does one of three things - kill the microbes, take away the water they need, or change the conditions so they cannot grow. Once you see the list that way it stops being arbitrary. Pasteurisation is the one asked by its numbers, so learn the temperature-time ladder: as the temperature rises, the time falls.

Pasteurisation - hotter means shorter



Preservation methods and the names attached to them

Method	How it works	Name to remember
Canning	Sealed, then heat sterilised	Invented by Nicolas Appert
Irradiation	Cobalt-60 gamma rays kill microbes and stop sprouting	Carries the Radura logo
Lyophilisation	Freeze drying - water removed by sublimation	Instant coffee, vaccines
Dehydration, salting, sugaring	Removes or locks up water	Papad, pickle, murabba
Refrigeration and freezing	Slows microbial growth	Slows spoilage, does not sterilise

- Class I preservatives are natural - salt, sugar, vinegar, spices, honey, edible oil. No legal limit on their use.
- Class II preservatives are chemical - sodium benzoate, sulphur dioxide, sorbic acid, nitrites. These are limited by law.
- Pasteurisation kills pathogens and cuts the microbial load; it does not sterilise, which is why pasteurised milk still needs a fridge.

Adulteration means adding a cheaper or harmful substance to food. RPSC asks it as a matching set and often as a 'not correctly matched' question, so learn the pairs in both directions - food to adulterant, and adulterant back to the disease it causes.

Food - adulterant - effect

Food	Common adulterant	Effect
Mustard oil	Argemone oil	Epidemic dropsy - oedema of the legs, cardiac complications
Arhar and other dals	Kesari dal (<i>Lathyrus sativus</i>), toxin BOAA/ODAP	Lathyrism - crippling paralysis of the lower limbs
Turmeric powder	Metanil yellow, lead chromate	Liver damage, carcinogenic

Food	Common adulterant	Effect
Black pepper	Papaya seeds	Cheating, gastric irritation
Chilli powder	Brick powder	Grit, gastric problems
Milk	Water, starch, detergent, urea	Dilution, gastrointestinal illness

- FSSAI was set up under the Food Safety and Standards Act, 2006, works under the Ministry of Health and Family Welfare and is headquartered in New Delhi.
- The 2006 Act consolidated the earlier laws, chiefly the Prevention of Food Adulteration Act, 1954.
- An FSSAI licence number has 14 digits and must be printed on the pack.
- FSSAI initiatives - Eat Right India (2018), Eat Right Challenge, RUCO (Repurpose Used Cooking Oil), Clean Street Food Hub, BHOG for places of worship.
- Industrial trans fatty acids in foods have been capped at 2% by weight since January 2022.

TRAP

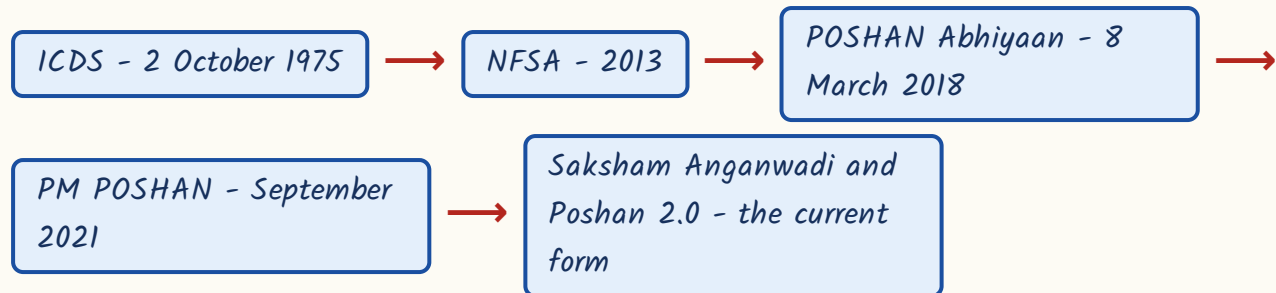
Brick powder adulterates chilli powder, not milk. Milk's usual adulterants are water, starch, detergent and urea. This exact swap is how the 'not correctly matched' option is built.

Nutrition programmes - the scheme grid

These schemes are asked in three different blocks of the paper - Science, Current Affairs and Rajasthan Economy - so the return on learning them is triple. What gets asked is the launch year, the target group and the one

distinguishing feature. Fix the chronology first, because ordering questions are set on it.

Chronology - the order has been asked as an ordering question



The five programmes you must be able to name

Programme	Year	Core feature
ICDS	2 October 1975	Six services through the anganwadi; now runs as Saksham Anganwadi and Poshan 2.0
NFSA	2013	Legal right to foodgrain; covers 75% rural and 50% urban population
POSHAN Abhiyaan (National Nutrition Mission)	8 March 2018	Launched from Jhunjhunu, Rajasthan; targets stunting and anaemia
PM POSHAN (Poshan Shakti Nirman)	September 2021	Replaced the Mid-Day Meal Scheme; covers Bal Vatika and Classes I-VIII
Anaemia Mukh Bharat	2018	6x6x6 strategy - six beneficiary groups, six interventions, six institutional mechanisms

- POSHAN Abhiyaan targets - cut stunting and underweight by 2% a year and anaemia by 3% a year.
- NFSA entitlement - 5 kg of foodgrain per person per month for a priority household, 35 kg per household under Antyodaya Anna Yojana.
- NFSA issue prices - rice Rs 3, wheat Rs 2, coarse grain Rs 1 per kg.
- Jhunjhunu as the POSHAN Abhiyaan launch site is the Rajasthan hook RPSC likes. Remember the district name.

ASKED IN RAS

The launch of POSHAN Abhiyaan from Jhunjhunu on 8 March 2018 is exactly the Rajasthan-linked national-scheme fact RPSC turns into a one-liner. Pair it with ICDS 1975 and PM POSHAN 2021 for the chronological-ordering version.

CURRENT LINK

Current link (as of September 2026): 2023 was the UN International Year of Millets, declared on India's proposal, and millets were rebranded 'Shree Anna'. Millets are still the government's headline nutrition-and-climate crop, and Rajasthan is India's largest producer of bajra - so a millet question can reach you through Science, Current Affairs or Rajasthan Agriculture. Ragi is the richest cereal source of calcium.

REVISION RECAP

1. Calorific values - fat 9, alcohol 7, carbohydrate 4, protein 4 kcal per gram.
2. Energy split of a balanced diet - carbohydrate 55-60%, fat 20-30%, protein 10-15%; water is 60-70% of body weight.
3. Fat-soluble A D E K are stored and can cause hypervitaminosis; B-complex and C are water-soluble and are not stored.
4. A retinol - night blindness and Bitot's spots; D calciferol - rickets and osteomalacia; E tocopherol; K phyloquinone - delayed clotting.
5. B1 thiamine - beriberi; B2 riboflavin - cheilosis; B3 niacin - pellagra; B6 pyridoxine - anaemia and convulsions.
6. B9 folic acid - megaloblastic anaemia and neural tube defects; B12 cyanocobalamin - pernicious anaemia, contains cobalt.
7. Vitamin C is ascorbic acid, deficiency scurvy; humans, primates and guinea pigs cannot make it.
8. Iodine - goitre and cretinism; iron - anaemia; calcium - rickets, osteoporosis, tetany; zinc - growth retardation.
9. Iodised salt 30 ppm at manufacture and 15 ppm at consumer end; fluoride in water 1.0 mg/L desirable, 1.5 mg/L permissible.

10. Kwashiorkor - protein only, oedema, post-weaning child. Marasmus - protein plus calories, wasting, infant under one year.
11. Stunting is height-for-age, wasting is weight-for-height, underweight is weight-for-age.
12. BMI is kg per m²; WHO calls 30 obese, the Asian-Indian cut-off is 25.
13. Fortification is done during processing and carries the +F logo; biofortification is bred or grown into the crop.
14. Dhanashakti is India's first biofortified variety, iron-rich bajra; Golden Rice carries beta-carotene; QPM maize carries lysine and tryptophan.
15. Pasteurisation - LTLT 63 C/30 min, HTST 72 C/15 sec, UHT 135-140 C/2-3 sec; irradiation uses Cobalt-60 and the Radura logo.
16. Argemone oil in mustard oil gives epidemic dropsy; kesari dal in arhar gives lathyrism; metanil yellow adulterates turmeric.
17. FSSAI - Act of 2006, Health Ministry, New Delhi, 14-digit licence, trans fat capped at 2% since January 2022.
18. ICDS 1975, NFSA 2013, POSHAN Abhiyaan 8 March 2018 from Jhunjhunu, PM POSHAN 2021 in place of the Mid-Day Meal Scheme.
19. NFSA gives 5 kg per person per month, 35 kg per AAY household, at Rs 3, Rs 2 and Rs 1 per kg.
20. Anaemia Mukt Bharat runs a 6x6x6 strategy; 2023 was the International Year of Millets, branded Shree Anna.

ST06

Genetics: Inheritance and Variation

Expect 2-4 questions from genetics in the Science block. The three certainties are chromosomal disorders (Down, Turner, Klinefelter), ABO and Rh blood groups, and the Mendelian ratios. Sex-linked disorders - colour blindness and haemophilia - are the fourth regular. Everything else in the chapter is insurance.

Genetics looks frightening because of the crosses. It is not. RPSC almost never asks you to work out a cross - it asks you to recall a ratio, a chromosome number, a disorder name or a scientist. Down syndrome as Trisomy 21 has come up word for word in two different papers. ABO blood groups have come up three times. So this chapter is memory work built on a little understanding, and the understanding takes one sitting.

Genetics

Mendel

- Pisum sativum, 7 contrasting pairs, 1866
- Rediscovered 1900 - de Vries, Correns, Tschermak
- Dominance, Segregation, Independent Assortment
- Bateson coined 'genetics', Johannsen 'gene'

Ratios

- Monohybrid F2 - 3:1 pheno, 1:2:1 geno
- Dihybrid F2 - 9:3:3:1
- Test cross - 1:1 and 1:1:1:1
- Incomplete dominance - 1:2:1 both ways

Blood groups

- 3 alleles IA IB i, chromosome 9
- 6 genotypes, 4 phenotypes
- AB codominant, O universal donor
- Rh -ve mother + Rh +ve foetus = erythroblastosis

Sex and sex-linked

- XX-XY human, XX-XO grasshopper
- ZZ-ZW birds - female heterogametic
- SRY on short arm of Y
- Colour blindness, haemophilia A and B

Chromosomal disorders

- Down = Trisomy 21, karyotype 47
- Edwards 18, Patau 13
- Klinefelter 47 XXY, Turner 45 XO
- Barr bodies = X minus 1

DNA

- Watson and Crick 1953, Franklin's X-ray data
- B-DNA right handed, 10 bp/turn, 3.4 nm pitch
- Chargaff A=T, G=C
- Replication semi-conservative, 5' to 3'

Code and mutation

- 64 codons, 61 sense, 3 stop
- AUG start = methionine
- Point vs frameshift
- de Vries mutation theory, Muller X-rays

Applications

- DNA fingerprinting - Alec Jeffreys 1984
- Lalji Singh, CCMB Hyderabad
- HGP 1990-2003, ~20,000-25,000 genes
- Genome India - 10,000 genomes

Mendel - why a monk with peas got it right

Before Mendel, people believed inheritance was a blending of the parents - mix black and white, get grey. Mendel showed that inheritance is particulate. Traits are carried by discrete units that stay intact, separate cleanly and reappear unchanged in later generations. He got there because of three choices. He chose

the garden pea, which has clear either-or characters and can self-pollinate. He tracked one character at a time. And he counted - he applied statistics to biology, which nobody had done. He published in 1866, was ignored, and was rediscovered in 1900.

- Mendel worked on the garden pea, *Pisum sativum*, and took seven pairs of contrasting characters.
- His paper was published in 1866 and rediscovered in 1900, independently, by Hugo de Vries, Carl Correns and Erich von Tschermak.
- The term 'genetics' was coined by William Bateson; the term 'gene' by Wilhelm Johannsen.
- Mendel is called the Father of Genetics.
- Pea suited him because it is naturally self-pollinating, short-lived, and its characters exist in two sharply distinct forms.

The seven contrasting character pairs Mendel used

Character	Dominant form	Recessive form
Stem height	Tall	Dwarf
Flower colour	Violet	White
Flower position	Axial	Terminal
Pod shape	Inflated	Constricted
Pod colour	Green	Yellow
Seed shape	Round	Wrinkled
Seed colour	Yellow	Green

Mendel's three laws - understand them before you state them

Take the three laws in the order the crosses reveal them. The Law of Dominance comes from the F₁ generation - cross a tall pea with a dwarf pea

and every offspring is tall, so one factor is masking the other. The Law of Segregation comes from the F₂ - the dwarf reappears, which means the dwarf factor was there all along, sitting unchanged inside the tall F₁ plant, and the two factors of a pair separated cleanly when gametes were formed. Each gamete got one and only one. That is why this law is also called the law of purity of gametes. The Law of Independent Assortment comes only from the dihybrid cross - when you track two characters at once, the way one pair assorts has no effect on the other.

- Law of Dominance - in a heterozygote, one allele expresses itself and masks the other.
- Law of Segregation, also called the law of purity of gametes - the two alleles of a pair separate during gamete formation, so a gamete carries only one.
- Segregation is the only one of the three laws that is universally valid, without exception.
- Law of Independent Assortment - alleles of different gene pairs assort independently of each other.
- Dominance and independent assortment both have well-known exceptions; segregation has none.

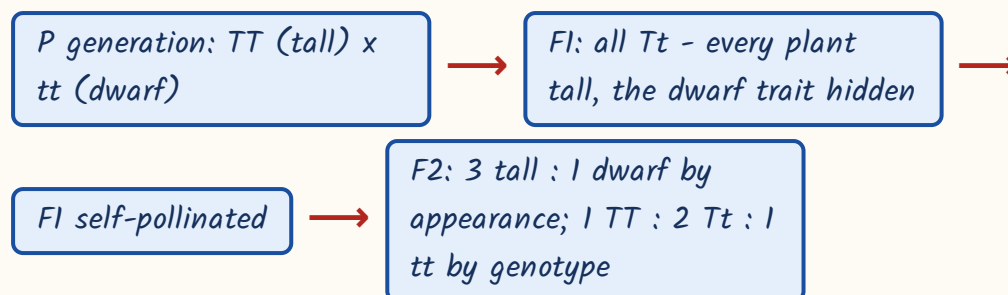
TRAP

Independent assortment holds only for genes on different chromosomes, or far apart on the same one. Genes lying close together on the same chromosome are linked and travel together - which is exactly what T. H. Morgan demonstrated in *Drosophila*. Do not say Mendel's laws have no exceptions.

Monohybrid and dihybrid crosses - the ratios that get asked

A monohybrid cross follows one character; a dihybrid cross follows two. Both start with pure-breeding parents, produce a uniform F₁, and reveal the pattern in the F₂. What the paper wants is the F₂ ratio - phenotypic and genotypic - and it wants the test-cross ratio too. Learn all of them as a block.

How a monohybrid cross runs - tall x dwarf



Every ratio in this section, in one place

Cross	Ratio	Type
Monohybrid F ₂	3 : 1	Phenotypic
Monohybrid F ₂	1 : 2 : 1	Genotypic
Dihybrid F ₂	9 : 3 : 3 : 1	Phenotypic
Monohybrid test cross	1 : 1	Phenotypic and genotypic
Dihybrid test cross	1 : 1 : 1 : 1	Phenotypic and genotypic
Incomplete dominance F ₂	1 : 2 : 1	Phenotypic, which here equals the genotypic

- A test cross means crossing the unknown individual with the homozygous recessive - it is how you find out whether a dominant-looking plant is TT or Tt.
- A back cross is a cross with either parent; a test cross is specifically with the recessive parent.
- In the dihybrid 9:3:3:1, the 9 shows both dominant traits, the 1 shows both recessive, and the two 3s are the new recombinant combinations.
- Phenotype is what you see; genotype is the allele pair actually carried.

TRAP

3:1 is the F₂ phenotypic ratio, 1:2:1 is the F₂ genotypic ratio of the same cross - they are two views of one result, not two different crosses. The paper deliberately asks for one and offers the other.

Beyond Mendel - incomplete dominance and codominance

Mendel's dominance is not the only possibility. Sometimes neither allele fully masks the other and the heterozygote comes out in between - red flower crossed with white gives pink. That is incomplete dominance, and the pink is a blend that does not exist in either parent. Sometimes both alleles express themselves fully and separately, side by side, in the same individual - that is codominance, and the AB blood group is the standard example, because an AB person makes both A antigen and B antigen, not some intermediate antigen. Blend versus both-together is the whole distinction.

Incomplete dominance vs Codominance

Incomplete dominance	Codominance
Heterozygote is intermediate between the two parents	Heterozygote shows both parental traits fully and separately
Neither allele is fully expressed	Both alleles are fully expressed
Example - <i>Mirabilis jalapa</i> (four o'clock plant) and <i>Antirrhinum</i> (snapdragon): red x white gives pink	Example - AB blood group in humans, roan coat in cattle, MN blood group
F ₂ phenotypic ratio 1:2:1, identical to the genotypic ratio	F ₂ phenotypic ratio also 1:2:1, for the same reason
The new phenotype is a blend not seen in either parent	No new blended phenotype - both parental products appear

- Pleiotropy - one gene producing many effects. Examples: phenylketonuria and sickle cell anaemia.
- Polygenic inheritance - many genes producing one continuously varying trait. Examples: human skin colour and height. Wheat kernel colour was studied by Kolreuter.
- Linkage and crossing over were established by T. H. Morgan working on *Drosophila melanogaster*.
- Alfred Sturtevant prepared the first genetic map, using recombination frequency as the measure of distance.

YAAD RAKHO

Incomplete dominance = a mixture, like paint - red and white give pink.

Codominance = both colours visible side by side, like a checked shirt - A and B antigens both present. Whenever the paper says AB blood group, the answer is codominance, not incomplete dominance.

ABO blood groups - the most reliably asked topic in this chapter

The ABO system is the textbook case of multiple alleles - one gene with more than two alternative forms in the population, though any one person still carries only two of them. There are three alleles: $I(A)$, $I(B)$ and i . $I(A)$ and $I(B)$ are codominant with each other and both are dominant over i . Three alleles taken two at a time give six genotypes, and because of the dominance relations those six collapse into only four phenotypes. Learn the grid, then learn the transfusion rule, then learn Rh - the paper has come at this topic from all three sides.

Six genotypes, four phenotypes

Blood group	Possible genotypes	Antigen on RBC	Antibody in plasma
A	$I(A)I(A)$ or $I(A)i$	A	anti-B

Blood group	Possible genotypes	Antigen on RBC	Antibody in plasma
B	$I(B)I(B)$ or $I(B)i$	B	anti-A
AB	$I(A)I(B)$	A and B both	None
O	ii	None	anti-A and anti-B

- The ABO gene lies on chromosome 9.
- AB is the universal recipient (no antibodies in plasma); O is the universal donor (no antigens on the red cells).
- An AB father and an O mother can only have A or B children - never AB and never O. This is the classic cross question.
- Rh incompatibility - an Rh-negative mother carrying an Rh-positive foetus can make anti-Rh antibodies that attack the foetal red cells.
- The result is erythroblastosis foetalis, also called haemolytic disease of the newborn.
- The first child usually escapes because sensitisation happens at delivery; later Rh-positive pregnancies are at risk. Prevention is anti-D immunoglobulin given to the mother.

ASKED IN RAS

Asked in RAS Pre 2023 - the AB blood group is the best-known human example of codominance, and a statement set on ABO inheritance (three alleles, six genotypes, four phenotypes, $I(A)$ and $I(B)$ codominant) was set the same year. RAS Pre 2021 asked the cross itself: an AB man married to an O woman can have children of group A or B only.

ASKED IN RAS

Asked in RAS Pre 2023 - erythroblastosis foetalis arises when an Rh-negative mother carries an Rh-positive foetus. Blood groups and Rh incompatibility have surfaced in 2013, 2021 and 2023, so treat this as a near-certain area.

Sex determination and sex-linked inheritance

Sex determination is about which parent's gamete carries the deciding chromosome. In humans the mother can only give an X, while the father gives either an X or a Y - so the father's sperm decides the sex of the child. The male here is the heterogametic sex, meaning he makes two kinds of gametes. Birds run the opposite arrangement, where the female is heterogametic, and that reversal is the fact the paper tests. Maleness itself comes from one gene: SRY, on the short arm of the Y chromosome.

Sex determination systems - which sex is heterogametic

System	Found in	Heterogametic sex
XX - XY	Humans, Drosophila	Male
XX - XO	Grasshoppers, some insects	Male
ZZ - ZW	Birds, butterflies	FEMALE
Haplodiploidy	Honeybees - drones haploid (16), females diploid (32)	No sex chromosomes; sex depends on ploidy

- Maleness in humans is determined by the SRY gene on the short arm of the Y chromosome.
- The sperm decides the child's sex - the mother contributes only an X.
- Sex-linked genes lie on the sex chromosomes; almost all the disorders you need are X-linked recessive, because the X carries far more genes than the Y.
- A man has only one X, so a single recessive allele on it expresses itself - which is why X-linked recessive disorders are far commoner in men.
- X-linked recessive disorders - red-green colour blindness, haemophilia A, haemophilia B, Duchenne muscular dystrophy, G6PD deficiency.

- Red-green colour blindness affects about 8% of men and about 0.4% of women.
- Haemophilia A is a deficiency of clotting factor VIII; haemophilia B, also called Christmas disease, is a deficiency of factor IX. Queen Victoria was the famous carrier.
- A colour-blind daughter is possible only if the father is colour blind AND the mother is at least a carrier.
- Criss-cross inheritance - the trait passes from an affected father through his carrier daughter to his grandson.

ASKED IN RAS

Asked in RAS Pre 2021 - a statement set on red-green colour blindness confirming that it is X-linked recessive and commoner in males, and a cross question: a carrier woman married to a normal man has one-half of her sons affected by haemophilia. Work that cross once by hand and it will never leave you.

YAAD RAKHO

Sons get their X only from the mother. So for any X-linked recessive disease, look at the mother to explain an affected son, and look at the father to explain an affected daughter. Birds are the reversal - ZZ male, ZW female, so in birds it is the mother who decides.

Chromosomal disorders - Down syndrome gets the full treatment

A normal human body cell has 46 chromosomes - 22 pairs of autosomes plus one pair of sex chromosomes. Arranged by size and centromere position, that display is a karyotype, and the standard arrangement is the Denver system with groups A to G. Chromosomal disorders arise when a chromosome is present in

the wrong number, usually because a pair failed to separate during cell division. That failure is called non-disjunction, and it is the single mechanism behind Down, Turner and Klinefelter alike. Down syndrome is trisomy of chromosome 21 - three copies instead of two, so the total count becomes 47 and not 46. RPSC asked this in identical form in 2018 and again in 2021, which tells you what to do with it: learn the chromosome number, the total count, the person it is named after, and the maternal-age link.

- Down syndrome = trisomy of chromosome 21; karyotype 47, and 47, XX or 47, XY depending on sex.
- First described by Langdon Down; the risk rises sharply with the mother's age.
- Features - short stature, small round head, furrowed protruding tongue, a single palm crease, congenital heart defects, mental retardation.
- It is the commonest chromosomal disorder compatible with survival, and it arises from non-disjunction, not from inheritance of a faulty gene.

Other chromosome-number and structure disorders

Disorder	Chromosome defect
Edwards syndrome	Trisomy 18
Patau syndrome	Trisomy 13
Cri-du-chat syndrome	Deletion of the short arm of chromosome 5
Chronic myeloid leukaemia	Philadelphia chromosome, translocation t(9;22)
Sickle cell anaemia	Not chromosomal - autosomal recessive point mutation

The two sex-chromosome disorders - compare them line by line

Feature	Klinefelter syndrome	Turner syndrome
Karyotype	47, XXY	45, X0
Chromosome count	47	45

Feature	Klinefelter syndrome	Turner syndrome
Sex of the individual	Male, sterile	Female, sterile
Chief features	Tall, gynaecomastia, poor secondary sexual development	Short stature, webbed neck, rudimentary ovaries
Barr bodies	1	0
Cause	Non-disjunction	Non-disjunction

- Number of Barr bodies = number of X chromosomes minus one. Normal female 1, normal male 0, Klinefelter 1, Turner 0.
- Sickle cell anaemia is autosomal recessive - glutamic acid is replaced by valine at the sixth position of the beta-globin chain, a missense point mutation.
- Thalassaemia is a quantitative defect - the globin chain is normal in sequence but made in reduced amounts. Sickle cell is qualitative, thalassaemia is quantitative.

ASKED IN RAS

Asked identically in RAS Pre 2018 and RAS Pre 2021 - Down syndrome is caused by trisomy of chromosome 21, and the affected person has 47 chromosomes. The same papers carried the statement version: described by Langdon Down, incidence rising with maternal age. A fact asked twice in the same form is a fact that will come again.

TRAP

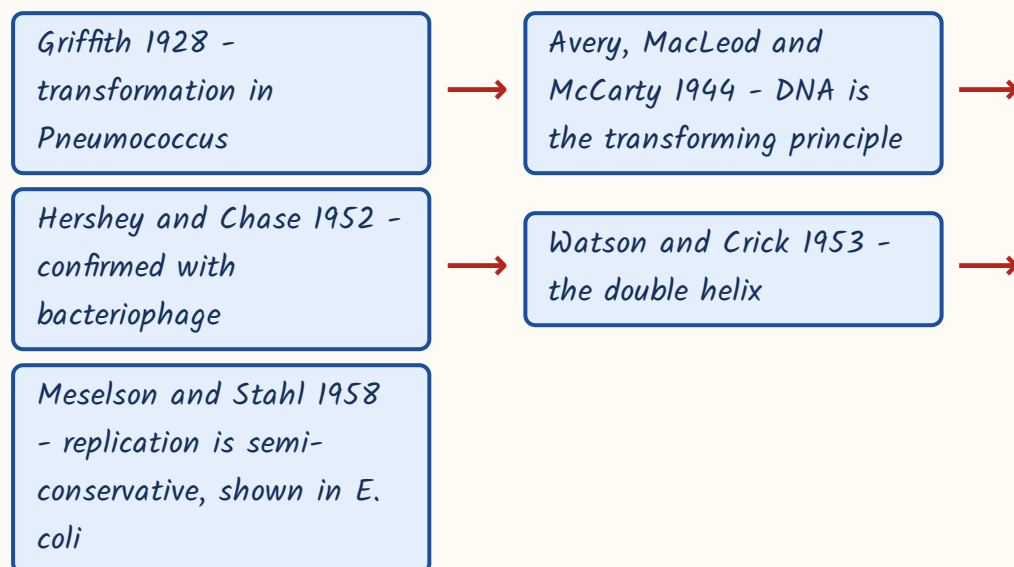
Klinefelter is 47 XXY, a sterile MALE. Turner is 45 XO, a sterile FEMALE. Students reverse them under pressure. Anchor it with the Barr body count - Klinefelter has an extra X, so it has 1 Barr body like a normal female; Turner has only one X, so 0, like a normal male.

DNA - structure, replication and the central dogma

DNA had to be proved to be the genetic material before its structure could matter, and that proof came in stages over twenty-five years. Once Watson and Crick had the double helix in 1953, everything else followed - because the structure itself explains how the molecule copies itself. Two strands, each the exact complement of the other: separate them and each one is a template for rebuilding its partner. Learn the numbers of the helix; the paper asks the numbers.

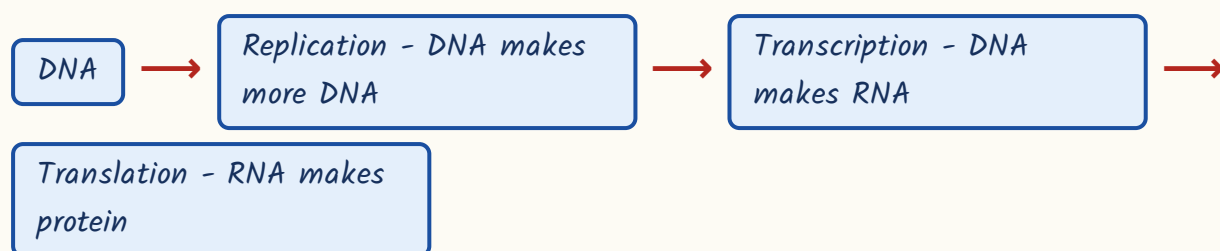
- The double helix was described by Watson and Crick in 1953, using Rosalind Franklin's and Maurice Wilkins' X-ray diffraction data. Nobel Prize 1962.
- B-DNA is right handed, with 10 base pairs per turn, a pitch of 3.4 nm and a rise of 0.34 nm per base pair.
- Chargaff's rule - A equals T and G equals C. A-T is joined by two hydrogen bonds, G-C by three, so G-C rich DNA is more stable.
- The two strands are antiparallel - one runs 5' to 3', the other 3' to 5'.

How DNA was proved to be the genetic material



- Replication is semi-conservative - each daughter molecule keeps one old strand and one new one.
- Synthesis always runs 5' to 3'. The leading strand is made continuously; the lagging strand is made in short Okazaki fragments.
- DNA ligase joins the Okazaki fragments - it is the sealing enzyme.
- Reverse transcription, RNA back to DNA, was discovered by Temin and Baltimore - the exception to the one-way central dogma.

Central dogma - Crick's one-way flow of information



- The genetic code has 64 codons - 61 sense codons and 3 stop codons, UAA, UAG and UGA.
- AUG is the start codon and also codes for methionine. Methionine (AUG) and tryptophan (UGG) are the two amino acids with a single codon each.
- The code is degenerate (more than one codon per amino acid), non-overlapping, commaless and nearly universal.
- The code was deciphered by Nirenberg, Har Gobind Khorana and Holley - Nobel Prize 1968. Khorana is the Indian-origin name to remember.
- The lac operon, an inducible operon, was explained by Jacob and Monod.

Mutation, variation and DNA fingerprinting

A mutation is a permanent change in the DNA sequence, and it is the ultimate source of all new variation - without it natural selection would have nothing to select from. Mutations are classified two ways: by what happens to the sequence, and by what happens to the protein. Adding or removing bases in

numbers not divisible by three shifts the whole reading frame downstream, which is why frameshift mutations are usually far more damaging than a single substitution.

- By sequence change - point or substitution mutation (transition and transversion), and frameshift mutation (insertion or deletion).
- By effect on the protein - silent (no change in amino acid), missense (one amino acid changed, as in sickle cell anaemia) and nonsense (a premature stop codon).
- Physical mutagens - UV light, X-rays, gamma rays. Chemical mutagens - EMS, nitrous acid, 5-bromouracil, acridine dyes.
- Hugo de Vries proposed the mutation theory from *Oenothera lamarckiana* and used the term 'saltation' for large sudden changes.
- H. J. Muller showed that X-rays induce mutations, working on *Drosophila*.
- Hardy-Weinberg principle - $p + q = 1$ and $p^2 + 2pq + q^2 = 1$. It says allele frequencies stay constant in an ideal population.
- The equilibrium needs a large, randomly mating population with no mutation, no migration, no genetic drift and no selection. Any real population violates at least one.
- Founder effect and bottleneck are forms of genetic drift.
- Darwin published 'On the Origin of Species' in 1859, after the HMS Beagle voyage and his study of the Galapagos finches - the classic case of adaptive radiation.
- Alfred Russel Wallace reached the same idea independently; Lamarck had earlier proposed inheritance of acquired characters.
- Industrial melanism in the peppered moth *Biston betularia* is the standard example of directional natural selection.
- Homologous organs indicate divergent evolution; analogous organs indicate convergent evolution.

- DNA fingerprinting was developed by Alec Jeffreys in 1984 and rests on VNTR and STR repeat sequences, which vary enormously between individuals.
- Lalji Singh of CCMB Hyderabad is called the Father of DNA fingerprinting in India.
- Uses - criminal investigation, paternity disputes, identification of disaster victims, wildlife forensics.
- The Human Genome Project ran from 1990 to 2003 and found roughly 20,000-25,000 genes in about 3.2 billion base pairs.

CURRENT LINK

Current link (as of September 2026): the GenomeIndia Project, funded by the Department of Biotechnology, completed whole-genome sequencing of 10,000 Indians, and the dataset was released in January 2025 with the data held at the Indian Biological Data Centre. The point for the exam is the number - 10,000 genomes - and the purpose: India's population carries genetic variation that global reference genomes miss, which matters for disease risk and for drug response.

REVISION RECAP

1. Mendel - *Pisum sativum*, seven contrasting pairs, published 1866, rediscovered 1900 by de Vries, Correns and von Tschermak.
2. Bateson coined 'genetics', Johannsen coined 'gene'; Mendel is the Father of Genetics.
3. Three laws - Dominance, Segregation (purity of gametes, the only universally valid one), Independent Assortment.
4. Monohybrid F2 - 3:1 phenotypic and 1:2:1 genotypic. Dihybrid F2 - 9:3:3:1. Test cross - 1:1 and 1:1:1:1.
5. Incomplete dominance - *Mirabilis jalapa* and *Antirrhinum*, F2 ratio 1:2:1. Codominance - AB blood group, roan cattle, MN group.
6. ABO - three alleles I(A), I(B), i on chromosome 9; six genotypes, four phenotypes; AB universal recipient, O universal donor.
7. Rh-negative mother with an Rh-positive foetus gives erythroblastosis foetalis; prevented by anti-D immunoglobulin.

8. XX-XY human and *Drosophila*, XX-XO grasshopper - male heterogametic. ZZ-ZW birds and butterflies - female heterogametic. Honeybee drones are haploid.
9. SRY on the short arm of the Y makes a male; the sperm decides the child's sex.
10. Colour blindness affects about 8% of men and 0.4% of women; haemophilia A is factor VIII, haemophilia B (Christmas disease) is factor IX.
11. Down syndrome - trisomy 21, total 47 chromosomes, described by Langdon Down, risk rises with maternal age.
12. Edwards trisomy 18, Patau trisomy 13, cri-du-chat deletion on chromosome 5, Philadelphia chromosome t(9;22) in CML.
13. Klinefelter 47 XXY sterile male with gynaecomastia; Turner 45 XO sterile female with webbed neck; both from non-disjunction.
14. Barr bodies = number of X chromosomes minus one - normal female 1, normal male 0, Klinefelter 1, Turner 0.
15. Watson and Crick 1953 from Franklin's X-ray data; B-DNA right handed, 10 bp per turn, pitch 3.4 nm; Chargaff A=T and G=C.
16. Griffith 1928, Avery-MacLeod-McCarty 1944, Hershey-Chase 1952, Meselson-Stahl 1958 for semi-conservative replication.
17. Central dogma DNA to RNA to protein; reverse transcription found by Temin and Baltimore; ligase joins Okazaki fragments.
18. Genetic code - 64 codons, 61 sense and 3 stop (UAA, UAG, UGA); AUG starts and codes methionine; deciphered by Nirenberg, Khorana and Holley.
19. Point and frameshift mutations; de Vries gave the mutation theory from *Oenothera*; Muller showed X-rays induce mutation.
20. DNA fingerprinting - Alec Jeffreys 1984, VNTR/STR based; Lalji Singh of CCMB Hyderabad is its Indian father; HGP 1990-2003, about 20,000-25,000 genes.

ST07

Biotechnology and Genetic Engineering

Biotechnology supplies 2-4 questions in the Science block and feeds Current Affairs on top. Biofuel generations were asked in 2013 and again in 2024. Bt cotton, GEAC and the COVID vaccine platforms are the other recurring anchors. Tools - restriction enzymes, vectors, PCR - are asked as straight one-liners.

Biotechnology is where the science paper and the current-affairs paper overlap. Half of this chapter is settled textbook material - restriction enzymes, PCR, tissue culture - and the other half moves every year: CRISPR therapies, GM mustard in the Supreme Court, ethanol blending targets, new vaccine platforms. Learn the fixed half as tables and keep the moving half dated.

Biotechnology

The toolkit

- Restriction enzyme = molecular scissors
- DNA ligase = molecular glue
- Vector = carrier (plasmid, phage, cosmid)
- Host = *E. coli*, *Agrobacterium*

Techniques

- PCR - Kary Mullis, Taq polymerase
- 94 C denature, 50-55 C anneal, 72 C extend
- Gel electrophoresis - DNA moves to anode
- Southern DNA, Northern RNA, Western protein

Gene editing

- CRISPR-Cas9, guide RNA
- Charpentier and Doudna, Nobel 2020
- FELUDA test - CSIR-IGIB
- Casgevy - first approved CRISPR therapy

GM crops in India

- Bt cotton 2002 - the ONLY approved GM crop
- Bt brinjal - moratorium 2010
- GM mustard DMH-11 - SC split verdict 2024
- GEAC is the apex approver

Plant biotech

- Totipotency, Haberlandt
- Explant - callus - plantlet, MS medium
- Auxin high = roots, cytokinin high = shoots
- Meristem culture = virus-free plants

Biofuels

- 1G edible crops
- 2G lignocellulosic residue - Panipat
- 3G algae
- 4G engineered algae, carbon negative
- E20 achieved 2025

Medicine

- Humulin - first rDNA product
- Dolly 1996, SCNT
- iPSC - Yamanaka and Gurdon 2012
- Gene therapy 1990, ADA deficiency

Institutions

- DBT 1986, BIRAC 2012
- GEAC under MoEFCC
- NBA Chennai, Biological Diversity Act 2002
- BioE3 Policy 2024

What genetic engineering is - and the toolkit it runs on

Biotechnology is the use of living organisms or their parts to make a product or a service. That definition is old - curd, bread and beer are all biotechnology.

What is new is genetic engineering: deliberately cutting a chosen gene out of one organism and putting it into another, so that the second organism now makes something it never made before. To do that you need four things -

scissors to cut the DNA, glue to join it, a carrier to take it inside a cell, and a host cell to multiply it. Learn the four roles first and every technique in this chapter falls into place.

- Recombinant DNA technology was developed by Stanley Cohen and Herbert Boyer in 1972-73.
- Paul Berg made the first recombinant DNA molecule and received the Nobel Prize in Chemistry in 1980.
- Restriction endonucleases are the molecular scissors; DNA ligase is the molecular glue.
- The first restriction enzyme isolated was Hind II. Type II restriction enzymes cut palindromic sequences and are the ones used in genetic engineering.
- The first commercial recombinant DNA product was human insulin - Humulin, 1982-83.

The four tools and what each one does

Tool	Role	Example
Restriction endonuclease	Cuts DNA at a specific sequence	EcoRI, Hind II, BamHI
DNA ligase	Seals the cut ends together	T4 DNA ligase
Vector	Carries the gene into the host cell	pBR322, lambda phage, Ti plasmid
Host cell	Multiplies the recombinant DNA and expresses it	E. coli, yeast, plant or animal cell

Restriction enzymes, vectors and plasmids

A restriction enzyme does not cut DNA at random. It recognises a specific short sequence - usually a palindrome, reading the same on both strands in the 5' to 3' direction - and cuts within it. If the cut is staggered rather than straight

across, the two fragments are left with short single-stranded overhangs. Those are the sticky or cohesive ends, and they are the whole trick of the technology: DNA from two completely different organisms, cut by the same enzyme, will carry complementary overhangs and will pair up. A vector is simply the vehicle that carries the joined DNA into a living cell, and it must satisfy three conditions before it can do the job.

- *EcoRI* is named from *Escherichia coli* strain RY13 - the first letter of the genus, the first two of the species, then the strain.
- *EcoRI* recognises GAATTC and cuts between G and A, leaving sticky ends.
- A cloning vector needs three things - an origin of replication (*ori*) so it can copy itself, and a selectable marker so transformed cells can be picked out.
- The third requirement is unique restriction sites, so the gene can be inserted at a known place.
- Gene transfer into animal cells - microinjection; into plant cells - biolistics or the gene gun, and *Agrobacterium*-mediated transfer.
- Other methods - electroporation, liposome-mediated transfer, and treating bacteria with divalent calcium to make them competent.

Vectors you should be able to name

Vector	What it is	Note
pBR322	Artificial bacterial plasmid	The first widely used plasmid vector
pUC19	Artificial plasmid	Improved cloning vector
Bacteriophage lambda	Virus that infects bacteria	Takes larger inserts than a plasmid
Cosmid	Plasmid plus phage packaging sites	Larger inserts still
BAC and YAC	Bacterial and yeast artificial chromosomes	Very large inserts, used in genome projects
Ti plasmid	Tumour-inducing plasmid of <i>Agrobacterium tumefaciens</i>	The commonest vector for dicot plants

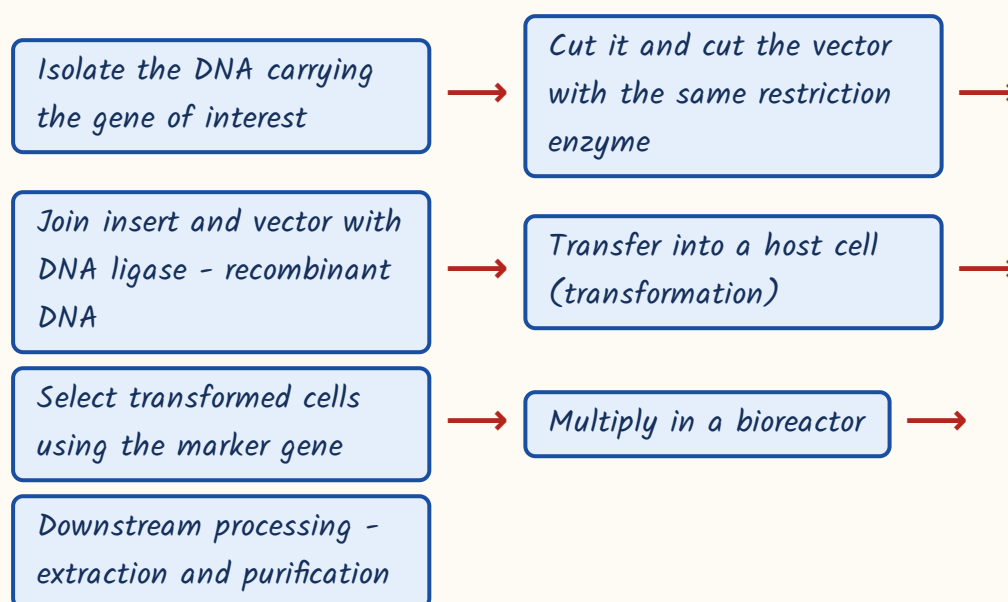
TRAP

Agrobacterium tumefaciens is a plant pathogen - it causes crown gall disease - and yet it is called the natural genetic engineer of plants, because in nature it already transfers its own DNA into the plant genome. Scientists simply disarmed it and loaded their own gene. Do not treat 'pathogen' and 'vector' as contradictory here.

Recombinant DNA technology - the sequence, step by step

This is the process the whole chapter is built on, and it is a sequence, so it can be asked as an ordering question. Say it as a story. You isolate the DNA that carries your gene. You cut it with a restriction enzyme, and cut the vector with the same enzyme so the ends match. You join the two with ligase, which gives you recombinant DNA. You push that into a host cell. The host divides, and every daughter cell carries your gene. Then you select the cells that actually took it up, grow them in a bioreactor, and finally purify the product downstream.

The rDNA sequence - learn it in order

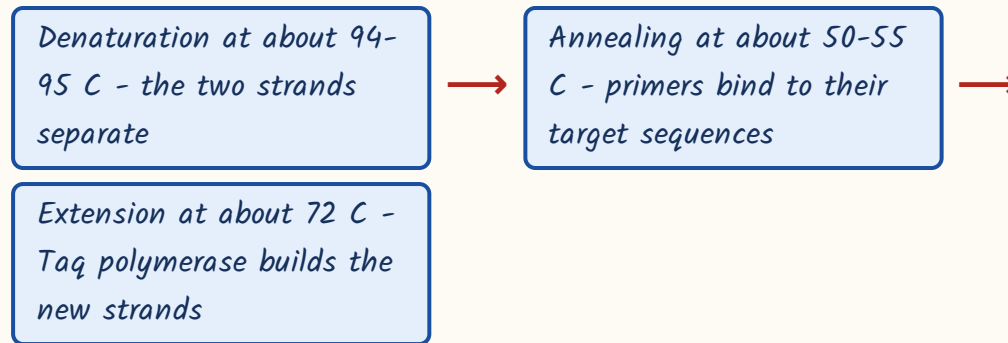


- Bacteria are made 'competent' to take up DNA by treatment with divalent calcium ions followed by a brief heat shock.
 - The selectable marker - classically antibiotic resistance - lets you kill off every cell that did not take up the vector.
 - A bioreactor provides the controlled temperature, pH, oxygen and stirring needed for large-scale culture; the stirred-tank type is the commonest.
 - Downstream processing means separation, purification, formulation and quality control - it is counted as part of the technology, not an afterthought.
-
- Human insulin (Humulin) was the first rDNA product to be marketed, in 1982-83.
 - The A chain (21 amino acids) and the B chain (30 amino acids) were produced separately in *E. coli* and then joined by disulphide bonds.
 - Before this, diabetics used insulin extracted from pig and cattle pancreas, which caused allergic reactions in some patients.
 - Other rDNA products in use - human growth hormone, erythropoietin, clotting factor VIII, hepatitis B vaccine.

PCR and gel electrophoresis - the two workhorse techniques

PCR - the polymerase chain reaction - is a photocopier for DNA. From a trace amount you get millions of copies in a couple of hours, which is why it underlies forensics, diagnosis and every cloning experiment. It works by repeating a three-step temperature cycle, and the enzyme that made it practical is Taq polymerase from a hot-spring bacterium, because an ordinary polymerase would be destroyed by the denaturation heat. Gel electrophoresis is the other half of the pair - it sorts the DNA fragments you now have, by size.

One PCR cycle - repeat it 25-35 times



- PCR was invented by Kary Mullis, who won the Nobel Prize in Chemistry in 1993.
- Taq polymerase comes from *Thermus aquaticus*, a bacterium of hot springs - it survives the 94 C step.
- Each cycle doubles the DNA, so the growth is exponential.
- RT-PCR - reverse transcription PCR - is the gold standard test for RNA viruses, including SARS-CoV-2. It first converts viral RNA into DNA.
- In agarose gel electrophoresis, DNA is negatively charged because of its phosphate backbone, so it moves towards the anode - the positive electrode. Smaller fragments travel further.
- Bands are stained with ethidium bromide and viewed under UV light.

The blotting techniques - a standard matching set

Technique	Molecule detected
Southern blot	DNA
Northern blot	RNA
Western blot	Protein

YAAD RAKHO

Blots: Southern is the only one named after a person, Edwin Southern, and it is for DNA. The others were named as a joke on compass directions. Order them alphabetically and by molecule - DNA, RNA, protein for Southern, Northern, Western. And in electrophoresis remember: DNA is negative, so it runs to the positive anode.

CRISPR-Cas9 and genome editing

CRISPR-Cas9 is not a way of adding a foreign gene. It is a way of editing the organism's own DNA at a chosen spot - correcting a letter, knocking a gene out, switching one off. Bacteria evolved it as an immune memory against viruses: they store snippets of viral DNA between repeated sequences, and use those snippets as a guide to cut that virus if it returns. Scientists borrowed the system. A short guide RNA carries the Cas9 protein to the exact matching sequence, and Cas9 makes the cut. Everything after that is the cell's own repair machinery. Because the edit leaves no foreign gene behind, regulators treat some kinds of editing differently from GM - and India has used that distinction.

- CRISPR stands for Clustered Regularly Interspaced Short Palindromic Repeats; Cas9 is CRISPR-associated protein 9.
- Emmanuelle Charpentier and Jennifer Doudna won the 2020 Nobel Prize in Chemistry for it.
- FELUDA, a CRISPR-Cas9 based paper-strip test for COVID-19, was developed by CSIR-IGIB, New Delhi.
- Casgevy became the first approved CRISPR gene-editing therapy, for sickle cell disease and beta thalassaemia.
- SDN-1 and SDN-2 edits, which introduce no foreign DNA, are exempted from India's GMO rules; SDN-3, which inserts a foreign gene, is not.

Genetic modification (GMO) vs Genome editing

Genetic modification - GMO	Genome editing - CRISPR SDN-1/SDN-2
A foreign gene from another species is inserted	The organism's own DNA is altered; no foreign gene remains
Insertion site is largely random	The change is made at a chosen, precise site
Example - Bt cotton carrying a bacterial cry gene	Example - DRR Dhan 100 'Kamala' rice
Needs full GEAC clearance under the 1989 Rules	SDN-1 and SDN-2 exempted from GMO rules in India
Product is legally a GMO	Product is treated like a conventionally bred variety

CURRENT LINK

Current link (as of September 2026): India released its first genome-edited rice varieties in May 2025 - DRR Dhan 100 'Kamala' from ICAR-IIRR and Pusa DST Rice 1 from ICAR-IARI, both made with SDN-1/SDN-2 editing and therefore outside the GMO rules. This is the fact that shows why the GMO-versus-genome-editing distinction matters in practice, and it is fresh enough to be asked.

Transgenic crops in India and the GEAC route

A transgenic crop carries a gene from another species. India's story here is short and worth learning exactly, because the exam asks the status, not the science. Bt cotton carries cry genes from the soil bacterium *Bacillus thuringiensis*. The bacterium makes a crystal protein that is an inactive protoxin - harmless as it sits. When a bollworm eats it, the alkaline gut of the insect activates the toxin, which punctures the gut lining and kills the larva. The human stomach is acidic,

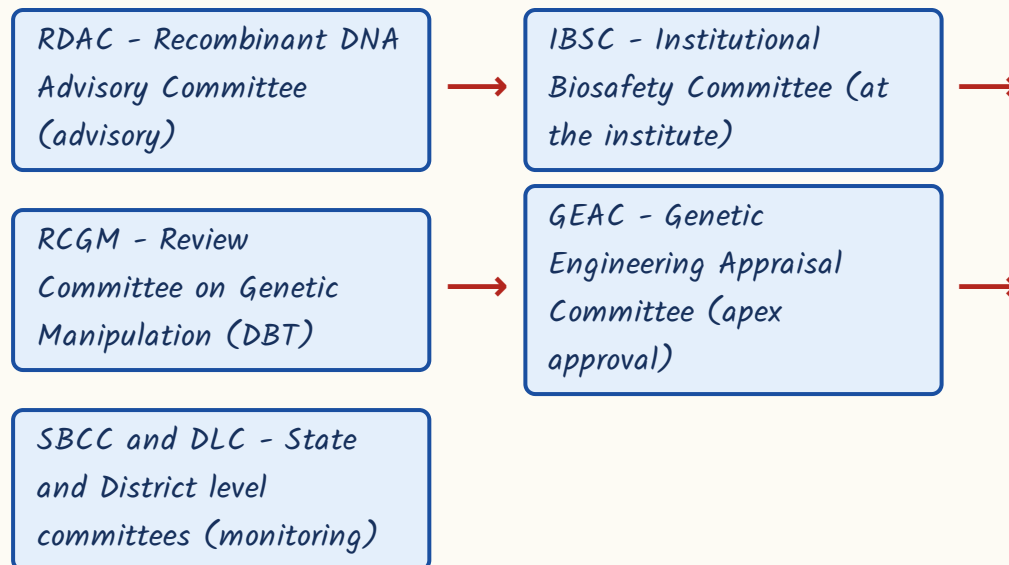
so the protein is not activated in us. That mechanism is the reason Bt is considered selective.

- Bt cotton carries cry genes - cry1Ac and cry2Ab - against bollworms.
- The Bt protein is an inactive protoxin crystal and becomes active only in the alkaline gut of the insect.
- Bollgard-I Bt cotton was approved in India in March 2002.
- Bt cotton remains the **ONLY** GM crop approved for commercial cultivation in India.
- Golden Rice is engineered to make beta-carotene (provitamin A) in the endosperm; Flavr Savr tomato in 1994 was the first genetically modified food crop to be commercialised anywhere.

The three Indian GM crop cases - learn the status, not just the science

Crop	Gene / technology	Indian status
Bt cotton (Bollgard-I)	cry1Ac from <i>Bacillus thuringiensis</i>	Approved March 2002; the only GM crop grown commercially in India
Bt brinjal	cry1Ac, developed by Mahyco	GEAC cleared it in 2009; moratorium imposed February 2010 by Environment Minister Jairam Ramesh. Bangladesh grows it commercially
GM mustard DMH-11	Barnase-barstar-bar system, Deepak Pental's team, University of Delhi	GEAC recommended environmental release October 2022; Supreme Court split verdict July 2024, referred to a larger bench

India's GMO regulatory chain under the 1989 Rules



- *GMOs in India are regulated under the 1989 Rules framed under the Environment (Protection) Act, 1986.*
- *GEAC - the Genetic Engineering Appraisal Committee - sits under the Ministry of Environment, Forest and Climate Change and is the apex body for approving environmental release of GMOs.*
- *The Biological Diversity Act, 2002 created the National Biodiversity Authority at Chennai and the State Biodiversity Boards.*
- *At the local level it created Biodiversity Management Committees, which prepare People's Biodiversity Registers.*
- *The Biological Diversity (Amendment) Act, 2023 decriminalised offences and exempted registered AYUSH practitioners from benefit sharing.*
- *India is a party to the Cartagena Protocol on biosafety and the Nagoya Protocol on access and benefit sharing.*

TRAP

Bt cotton is the only GM crop approved for COMMERCIAL CULTIVATION in India - that is the phrase that matters. Bt brinjal was cleared by GEAC and then stopped by a moratorium; GM mustard was recommended by GEAC but is stuck in court. Cleared, recommended and approved for cultivation are three different things, and the paper uses that difference.

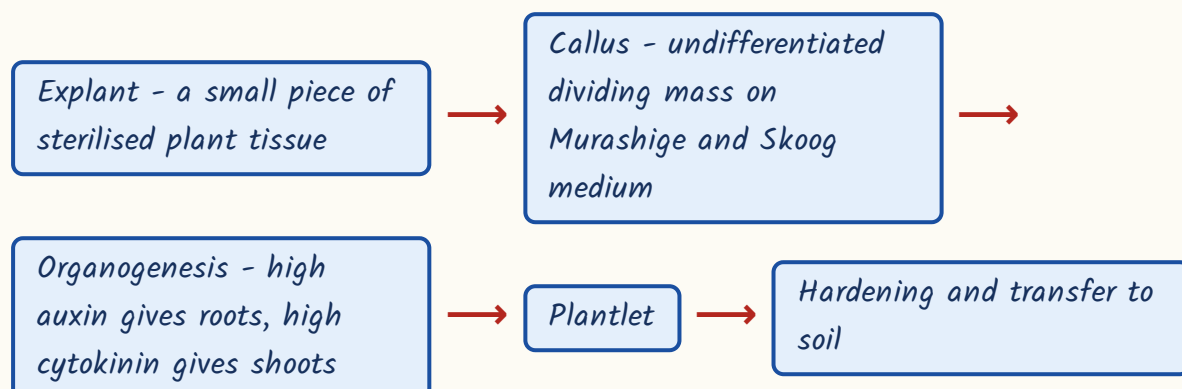
CURRENT LINK

Current link (as of September 2026): the GM mustard DMH-11 case remains unresolved. The Supreme Court's two judges split in July 2024 and the matter went to a larger bench, so no new GM food crop has been released for cultivation. If any movement happens before the exam, the fact to update is only this one line - GEAC recommendation October 2022, split verdict July 2024, status thereafter.

Tissue culture, biofertilisers and industrial biotechnology

Plant tissue culture rests on one property - totipotency. Any living plant cell carries the full genetic instruction set, so a single cell can be coaxed into rebuilding the whole plant. You take a small piece of tissue, the explant, put it on a nutrient medium under sterile conditions, and it first becomes an unorganised mass of dividing cells, the callus. From there, the balance of two hormones decides what grows: more auxin gives roots, more cytokinin gives shoots. That single ratio is the fact the exam asks.

Micropropagation - from a piece of tissue to a whole plant



- G. Haberlandt is the Father of plant tissue culture; the underlying principle is totipotency.
- Murashige and Skoog (MS) medium is the standard culture medium.

- Meristem or shoot-tip culture gives virus-free plants, because the meristem is usually free of virus.
- Anther culture gives haploid plants; protoplast fusion (somatic hybridisation) produced the 'pomato' from potato and tomato.
- Micropropagation produces thousands of genetically identical plants - somaclones - from one parent.
- In nitrogen fixation the enzyme nitrogenase does the work, and leghaemoglobin in the root nodule protects it from oxygen, which would destroy it.
- Bioremediation - *Pseudomonas putida*, the oil-eating 'superbug' engineered by Ananda Mohan Chakrabarty, became the first living organism to be patented, in *Diamond v. Chakrabarty*, 1980.

Biofertilisers and biopesticides - match the organism to the job

Organism	Type	What it does
<i>Rhizobium</i>	Biofertiliser	Symbiotic nitrogen fixation in legume root nodules
<i>Azotobacter</i>	Biofertiliser	Free-living aerobic nitrogen fixer
<i>Azospirillum</i>	Biofertiliser	Associative nitrogen fixer
<i>Anabaena</i> , <i>Nostoc</i> ; <i>Azolla</i> with <i>Anabaena azollae</i>	Biofertiliser	Cyanobacterial nitrogen fixation, used in paddy
<i>Frankia</i>	Biofertiliser	Nitrogen fixation with <i>Casuarina</i>
Mycorrhiza / VAM	Biofertiliser	Improves phosphorus uptake
<i>Bacillus thuringiensis</i> , <i>Trichoderma</i> , <i>Beauveria bassiana</i>	Biopesticide	Bacterial and fungal control of pests and plant disease
NPV (Nucleopolyhedrovirus), <i>Trichogramma</i> , neem azadirachtin	Biopesticide	NPV is highly species specific; <i>Trichogramma</i> is a parasitic wasp

Fermentation products and the microbe behind each

Product	Microorganism
Ethanol	<i>Saccharomyces cerevisiae</i>

Product	Microorganism
Citric acid	<i>Aspergillus niger</i>
Acetic acid	<i>Acetobacter aceti</i>
Penicillin	<i>Penicillium notatum</i> / <i>chrysogenum</i> - Fleming, Chain and Florey, Nobel 1945
Cyclosporin A (immunosuppressant)	<i>Trichoderma polysporum</i>
Statins (cholesterol lowering)	<i>Monascus purpureus</i>
Streptokinase (clot buster)	<i>Streptococcus</i>
Holes in Swiss cheese	<i>Propionibacterium shermanii</i>

Biofuels - the 1G to 4G ladder RPSC keeps coming back to

The generations of biofuel are not about the fuel produced - they are about what the fuel was made FROM. First generation uses food crops, which is why it competes with the food plate. Second generation moves to the parts we do not eat - crop residue, straw, woody biomass, all lignocellulosic. Third generation moves off land entirely and uses algae, which need no farmland. Fourth generation goes one step further and engineers the algae or microbe itself so that the process can pull in more carbon dioxide than the fuel releases - the aim is a carbon-negative fuel. Food, waste, algae, engineered algae. That is the whole ladder.

Biofuel generations - the feedstock is the answer

Generation	Feedstock	Key point
1G - first generation	Edible crops - sugarcane, molasses, corn, vegetable oils	Competes with food and land
2G - second generation	Non-food lignocellulosic biomass and crop residues	Uses waste; India's first 2G plant is at Panipat, Haryana, on paddy straw

Generation	Feedstock	Key point
3G - third generation	Algal biomass	No farmland needed, high yield per area
4G - fourth generation	Genetically or metabolically engineered algae and microbes	Designed for carbon capture; aims at carbon-negative fuel

- The National Policy on Biofuels, 2018, amended in 2022, advanced the 20% ethanol blending target from 2030 to 2025-26. India achieved 20% blending in 2025.
- Pradhan Mantri JI-VAN Yojana supports second-generation ethanol projects.
- SATAT - Sustainable Alternative Towards Affordable Transportation - promotes compressed biogas.
- Biodiesel comes from transesterification of vegetable oils; jatropha was the crop pushed for it in India.

ASKED IN RAS

Asked in RAS Pre 2013 and again in RAS Pre 2024 - first-generation biofuels are those produced from edible food crops such as sugarcane, corn and vegetable oils. A topic asked eleven years apart and then repeated is a topic that is part of the RPSC template.

ASKED IN RAS

Asked in RAS Pre 2013 and 2024 - biofuels from algal biomass are third generation, and fourth-generation biofuels are distinguished by the use of genetically or metabolically engineered algae and microbes aimed at capturing carbon dioxide. If the question names the feedstock, count up the ladder; if it names 'engineered', the answer is 4G.

ASKED IN RAS

Asked in RAS Pre 2024 as a statement set - India's first second-generation ethanol bio-refinery is at Panipat in Haryana and runs on paddy straw, and the 20% blending target was advanced to 2025-26 by the amended National Policy on Biofuels. Learn the place name; that is the part candidates miss.

Cloning, stem cells, gene therapy, vaccines and the Indian set-up

The medical end of biotechnology is a name-and-date section, so treat it as one. Cloning made its point with Dolly - the first mammal produced from an adult body cell, which proved that a fully specialised cell still holds the complete instruction set. Stem cells work from the other direction: cells that have not yet specialised, graded by how many things they can still become. Gene therapy tries to fix the fault at source by supplying a working copy of the gene. And the COVID years handed the exam a ready-made matching set of vaccine platforms, which is now standard material.

- Dolly the sheep, born 5 July 1996 at the Roslin Institute to Ian Wilmut and Keith Campbell, was the first mammal cloned from an adult somatic cell.
- The technique was somatic cell nuclear transfer (SCNT). India's cloned buffalo work was done at NDRI, Karnal.
- Stem cell grades - totipotent is the zygote, which can make the whole organism plus the placenta; pluripotent are embryonic stem cells, which can make any body tissue.
- Multipotent are adult stem cells, with a limited range of possible tissues.
- Shinya Yamanaka shared the 2012 Nobel Prize with John Gurdon for induced pluripotent stem cells - adult cells reprogrammed back to a pluripotent state.

- Umbilical cord blood is the common practical source of stem cells for banking.
- Gene therapy was first attempted in 1990 on a child with adenosine deaminase (ADA) deficiency.
- Somatic gene therapy changes body cells only; germline changes would pass to the next generation and are not permitted.
- RNA interference (RNAi), used for pest resistance in plants, earned Andrew Fire and Craig Mello the 2006 Nobel Prize.
- Monoclonal antibodies are produced by hybridoma technology - Kohler and Milstein, Nobel 1984 - and their names carry the suffix '-mab'.

COVID vaccine platforms - a ready-made matching set

Vaccine	Platform	Maker
Covaxin (BBV152)	Whole-virion inactivated	Bharat Biotech with ICMR-NIV Pune
Covishield	Chimpanzee adenovirus viral vector (Oxford-AstraZeneca)	Serum Institute of India
ZyCoV-D	DNA plasmid - world's first DNA vaccine approved for human use, three needle-free doses	Zydus
GEMCOVAC-19	mRNA - India's first indigenous mRNA vaccine	Gennova
Corbevax	Recombinant protein subunit	Biological E

- The Department of Biotechnology (DBT) was set up in 1986 under the Ministry of Science and Technology.
- BIRAC - Biotechnology Industry Research Assistance Council - was set up in 2012 to support biotech start-ups and small firms.
- The BioE3 Policy - Biotechnology for Economy, Environment and Employment - was approved by the Union Cabinet in August 2024 to promote high-performance biomanufacturing and biofoundries.

- Remember the division of labour: DBT funds and promotes, BIRAC backs start-ups, GEAC approves environmental release, and the National Biodiversity Authority handles access and benefit sharing.

YAAD RAKHO

Fix the four Indian COVID vaccines by platform, because that is how the matching question is set - Covaxin is inactivated, Covishield is a viral vector, ZyCoV-D is DNA and a world first, GEMCOVAC is mRNA and an Indian first. Two 'firsts' in one row is exactly what the paper likes.

REVISION RECAP

- rDNA technology - Cohen and Boyer 1972-73; Paul Berg made the first recombinant DNA molecule, Nobel Chemistry 1980.
- Restriction endonuclease is the molecular scissors, DNA ligase the molecular glue; Hind II was the first restriction enzyme isolated.
- EcoRI comes from *E. coli* strain RY13, cuts GAATTC between G and A and leaves sticky ends.
- A cloning vector needs an origin of replication, a selectable marker and unique restriction sites; pBR322 was the first widely used plasmid vector.
- Ti plasmid of *Agrobacterium tumefaciens*, the crown gall pathogen, is the commonest vector for dicot plants.
- Gene transfer - microinjection for animal cells, gene gun for plant cells, electroporation, calcium treatment for bacteria.
- Human insulin (Humulin, 1982-83) was the first commercial rDNA product - A chain 21 and B chain 30 amino acids joined by disulphide bonds.
- PCR - Kary Mullis, Nobel 1993; denature 94-95 C, anneal 50-55 C, extend 72 C with Taq polymerase from *Thermus aquaticus*.
- In gel electrophoresis DNA moves to the anode, is stained with ethidium bromide and seen under UV. Southern DNA, Northern RNA, Western protein.
- CRISPR-Cas9 comes from a bacterial antiviral defence, is guided by a guide RNA; Charpentier and Doudna, Nobel 2020; FELUDA by CSIR-IGIB; Casgevy the first approved CRISPR therapy.
- Bt cotton carries *cry1Ac* and *cry2Ab*; the protoxin activates only in the insect's alkaline gut; approved March 2002 and still India's only commercially cultivated GM crop.

12. *Bt brinjal* - GEAC cleared 2009, moratorium February 2010 by Jairam Ramesh; Bangladesh grows it. GM mustard DMH-11 - barnase-barstar-bar, GEAC October 2022, Supreme Court split verdict July 2024.
13. GMO chain in India - RDAC, IBSC, RCGM, GEAC, SBCC, DLC, under the 1989 Rules of the Environment (Protection) Act 1986; GEAC sits under MoEFCC.
14. Tissue culture rests on totipotency, Haberlandt is its father; explant to callus to plantlet on MS medium; high auxin gives roots, high cytokinin gives shoots.
15. Meristem culture gives virus-free plants, anther culture gives haploids, protoplast fusion gave the pomato.
16. Biofertilisers - *Rhizobium* symbiotic, *Azotobacter* free-living, *Azospirillum* associative, *Azolla-Anabaena* in paddy, mycorrhiza for phosphorus.
17. Biofuels - 1G edible crops, 2G lignocellulosic residue, 3G algae, 4G engineered algae for carbon capture; India's first 2G plant at Panipat on paddy straw; 20% blending achieved 2025.
18. Dolly 1996 by SCNT at the Roslin Institute; stem cells totipotent, pluripotent, multipotent; iPSC by Yamanaka with Gurdon, Nobel 2012.
19. Gene therapy first attempted 1990 for ADA deficiency; RNAi Nobel 2006 to Fire and Mello; monoclonal antibodies by hybridoma, suffix '-mab'.
20. Covaxin inactivated, Covishield viral vector, ZyCoV-D the world's first human DNA vaccine, GEMCOVAC India's first mRNA vaccine; DBT 1986, BIRAC 2012, BioE3 Policy August 2024.

ST08

Nanotechnology and New Materials

About 2-3 questions a paper out of the 18-23 that Science and Technology carries. The size definition and the quantum dot/well family are the two near-certain suppliers. Materials - Nitinol, Kevlar, aerogel, Meissner effect - come as a matching question. Treat this as a high-return chapter for the effort.

Nanotechnology is a documented repeat in this exam. RPSC asked it in 2016, 2018, 2021 and 2023 - four papers out of the last five. And it keeps asking the same handful of things: the 1 to 100 nanometre size range, the confinement structures (dot, wire, well), the generations, and who did what. Learn those cold and this chapter pays for itself. The second half - composites, superconductors, semiconductors, rare earths - is newer territory that has grown as India's Semiconductor Mission and critical-mineral push moved into the news.

Nanotechnology and New Materials

Size and scale

- $1 \text{ nm} = 10^{-9} \text{ m} = 10 \text{ angstrom}$
- Nano range = 1 to 100 nm
- At least ONE dimension in range
- Hair is 80,000-100,000 nm
- Greek 'nanos' = dwarf

Why nano is different

- Surface area / volume shoots up
- More atoms sit on the surface
- Reactivity and catalysis rise
- Quantum confinement
- Properties become size-dependent

Confinement family

- Quantum well - confined in 1, free in 2
- Quantum wire - confined in 2, free in 1
- Quantum dot - confined in 3, free in 0
- Dot = 'artificial atom', 2-10 nm
- Nobel Chemistry 2023

Carbon nanomaterials

- Fullerene C60 - 0-D, 1985
- Carbon nanotube - 1-D, 1991
- Graphene - 2-D, 2004
- Diamond, graphite - 3-D bulk
- CNT ~100x stronger than steel

Making and seeing

- Top-down: milling, lithography, etching
- Bottom-up: sol-gel, CVD, self-assembly
- SEM and TEM - electron beam
- AFM and STM - scanning probe
- XRD for crystal structure

Generations (Roco)

- 1st - passive nanostructures
- 2nd - active nanostructures
- 3rd - systems of nanosystems
- 4th - molecular nanosystems

India

- Nano Mission, May 2007, DST
- NSTI 2001 came first
- INST Mohali - dedicated institute
- ARCI Hyderabad, CeNS Bengaluru

New materials

- Composite = matrix + reinforcement
- Superconductor - Meissner effect
- Nitinol - shape-memory alloy
- Metamaterial - negative refraction
- Aerogel - 'frozen smoke'

Strategic materials

- Semicon India - Rs 76,000 crore
- ISM under MeitY
- Dholera - first commercial fab
- Rare earths = 15 lanthanides + Sc + Y
- IREL, monazite beach sands

What 'nano' actually means - start with the number

Begin with arithmetic, because that is what RPSC asks. 'Nano' comes from the Greek nanos, meaning dwarf. One nanometre is one-billionth of a metre - 10^{-9} metre. In the older unit that is 10 angstrom. Nanotechnology is the study and manipulation of matter that has at least one dimension in the 1 to 100 nanometre range. That number, 1 to 100 nm, is the single most asked fact in this chapter. Do not paraphrase it. Learn it as it is.

The scale ladder - fix the size in your head

Object	Approximate size
1 nanometre	10^{-9} metre = 10 angstrom
The nanotechnology range	1 to 100 nm
Width of a DNA double helix	about 2 nm
Quantum dot	about 2 to 10 nm
Typical virus	20 to 400 nm
Red blood cell	about 7,000 nm
Human hair thickness	80,000 to 100,000 nm

- Read the definition carefully - at least ONE dimension must lie in 1-100 nm.
- That is why graphene counts: it is one atom thick even if the sheet is centimetres wide.
- A nanoparticle has all three dimensions in the range; a nanowire has two; a thin film has one.
- Nanotechnology existed before the name - gold nanoparticles gave the Roman Lycurgus Cup its ruby-red glow, and Michael Faraday studied colloidal gold in 1857.

ASKED IN RAS

Asked in RAS Pre 2021 - nanotechnology deals with matter having at least one dimension in which size range? Answer: 1 to 100 nm. The topic also appeared in 2016, 2018 and 2023, which makes it a four-paper repeat.

YAAD RAKHO

Chant it: one nanometre = one-billionth of a metre = 10 angstrom, and the working range is one to a hundred. Three numbers, one line.

Why properties change at the nanoscale - the WHY that now gets tested

Gold is yellow and unreactive in a ring. As a nanoparticle it turns ruby-red and becomes a good catalyst. Same element, different behaviour. Two effects explain this, and RPSC has moved from asking the definition to asking the reason. Learn both by name: the surface-area-to-volume ratio, and quantum confinement.

- Reason 1 - surface area to volume ratio. Break a block into smaller and smaller pieces and the total surface area grows enormously while the amount of matter stays the same.
- So a far larger fraction of the atoms sits on the surface, where bonds are unsatisfied and reactions happen.
- Result: much higher chemical reactivity, stronger catalytic action, better adsorption. This is why nano-catalysts and nano-filters work.
- Reason 2 - quantum confinement. Squeeze a particle down to the size of the electron's wave and the allowed energy levels stop being continuous and become discrete.
- Result: optical and electronic properties become size-dependent. Change only the size and you change the colour of the light emitted.

The confinement family - count the dimensions both ways

Structure	Carriers confined in	Carriers free to move in
Bulk crystal	no direction	3 dimensions (3-D)
Quantum well	1 direction	2 dimensions (2-D)
Quantum wire / nanowire	2 directions	1 dimension (1-D)
Quantum dot	all 3 directions	0 dimensions (0-D)

- A quantum dot is a semiconductor nanocrystal of about 2 to 10 nm, so tightly confined that it behaves like an artificial atom.
- Its emission colour is tunable purely by size - bigger dot, redder light; smaller dot, bluer light.
- Nobel Prize in Chemistry 2023 went to Moungi Bawendi, Louis Brus and Alexei Ekimov for discovering and synthesising quantum dots.
- Quantum dots are used in QLED displays and in biological imaging as fluorescent tags.

ASKED IN RAS

Asked in RAS Pre 2018 - a nanostructure with carriers confined in all three spatial dimensions is called? Answer: quantum dot. RAS Pre 2021 returned to the same family twice: once as a statement question on quantum wells and once as a matching question on free dimensions.

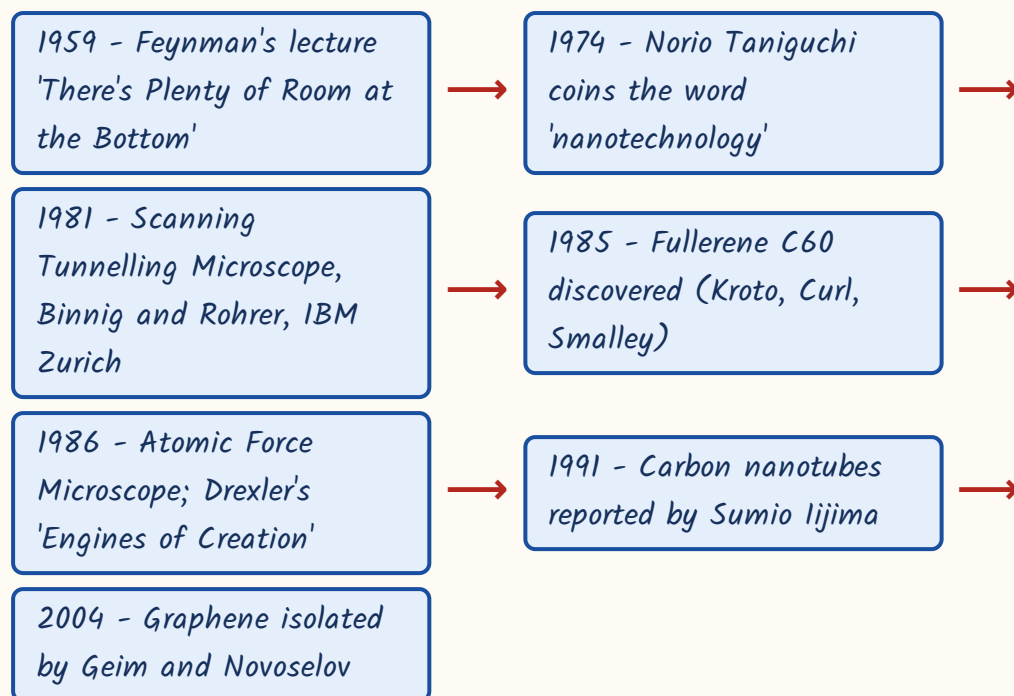
TRAP

Do not swap well and dot. Well = confined in ONE direction. Dot = confined in ALL THREE. The free-dimension count is the mirror image: well 2, wire 1, dot 0. And the colour of a quantum dot DOES depend on its size - a statement saying it does not is the standard wrong option.

The milestones - who did what, in order

This chapter has a clean chronology and RPSC likes ordering questions. Learn the sequence as a story: a physicist imagines it, a Japanese engineer names it, then instruments arrive that can actually see atoms, and then the three carbon forms are discovered one after another - ball, tube, sheet.

The nanotechnology timeline



Nobel Prizes tied to this chapter

Discovery	Who	Prize
Scanning Tunnelling Microscope	Gerd Binnig and Heinrich Rohrer	Physics 1986
Fullerene C60	Harold Kroto, Robert Curl, Richard Smalley	Chemistry 1996
Graphene	Andre Geim and Konstantin Novoselov	Physics 2010
Quantum dots	Bawendi, Brus and Ekimov	Chemistry 2023

- Feynman delivered his lecture on 29 December 1959 at Caltech, to the American Physical Society - the conceptual origin of the field.

- Feynman did not use the word. Taniguchi (Japan) coined 'nanotechnology' in 1974; K. Eric Drexler popularised it in 'Engines of Creation' (1986).
- Graphene was isolated by the adhesive-tape method - peeling layers off graphite until one atomic layer remained.
- Chronological order to remember: Feynman lecture, then C60, then carbon nanotubes, then graphene.

ASKED IN RAS

Asked in RAS Pre 2016 - buckminsterfullerene C60 is made of how many hexagons and pentagons? Answer: 20 hexagons and 12 pentagons. RAS Pre 2023 asked two from this list - who reported carbon nanotubes in 1991 (Iijima) and what the 2010 Physics Nobel was for (graphene).

The nanomaterials - learn them as one table

RPSC asks this as matching: name on the left, description or discoverer on the right. So build the table once and revise it whole. Notice that carbon alone supplies three of the entries, and that the three differ only in how many dimensions they extend in.

Types of nanomaterials

Nanomaterial	What it is	Key point
Nanoparticle	Particle with all dimensions in the nano range	Gold, silver, TiO ₂ , ZnO; carbon black in tyres is a legacy example
Carbon nanotube (CNT)	A rolled-up graphene cylinder	Reported by Iijima, 1991; single-walled and multi-walled
Nanowire	A wire of nanoscale diameter	1-D structure; carriers free in one dimension
Quantum dot	Semiconductor nanocrystal, 2-10 nm	0-D; size-tunable emission; Nobel Chemistry 2023

Nanomaterial	What it is	Key point
Quantum well	Thin layer confining carriers in one direction	2-D; used in laser diodes and LEDs
Fullerene C60	Closed cage of 60 carbon atoms	20 hexagons + 12 pentagons; the 'bucky ball'
Graphene	One-atom-thick honeycomb sheet of sp ² carbon	Isolated 2004; adhesive-tape method
Dendrimer	Repetitively branched, tree-like macromolecule	Drug-delivery carrier; liposomes and nanoshells do the same job

- Carbon by dimensionality: fullerene 0-D, carbon nanotube 1-D, graphene 2-D, diamond and graphite 3-D bulk. Ball - tube - sheet - solid.
- CNTs are about 100 times stronger than steel at roughly one-sixth the weight.
- CNTs are excellent conductors of both heat and electricity.
- CNT electrical behaviour depends on how the sheet is rolled: armchair nanotubes are metallic, zigzag and chiral ones are usually semiconducting.
- So a statement saying 'carbon nanotubes are always insulators' is false - that is a standard wrong option.

TRAP

Nitinol is NOT an allotrope of carbon. It is a nickel-titanium shape-memory alloy. RPSC drops it into a list of diamond, graphite, fullerene and graphene as the odd one out.

Making them and seeing them - synthesis and characterisation

There are only two ways to reach the nanoscale. Either you take a big piece and cut it down, or you take atoms and build up. That is the whole of top-down versus bottom-up. Learn which technique belongs to which side, because the

exam tests exactly that - it puts one technique on the wrong side and asks which statement is correct.

Top-down versus bottom-up synthesis

Top-down	Bottom-up
Start with bulk material and cut it down	Start with atoms and molecules and build up
Sculpting - you remove material	Assembling - you add material
Ball milling, lithography, etching, laser ablation	Sol-gel, chemical vapour deposition, self-assembly, molecular beam epitaxy
Simpler and industrially familiar, but leaves surface defects	Finer control, purer and more uniform structures

TRAP

Chemical vapour deposition is BOTTOM-UP, not top-down. Ball milling is top-down, sol-gel is bottom-up. RPSC's statement question works by flipping exactly one of these three.

Characterisation tools - instrument to principle

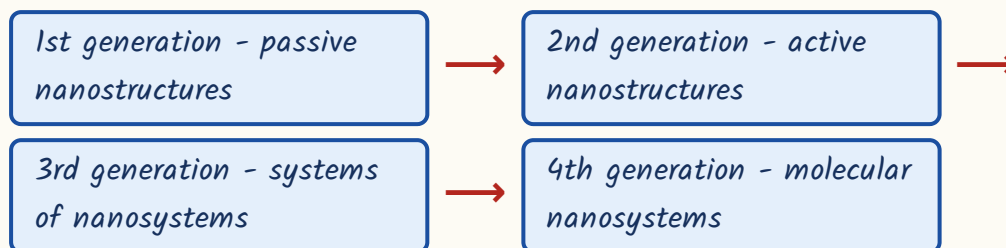
Tool	Working principle	What it gives you
SEM - Scanning Electron Microscope	Scans a focused electron beam across the surface	Surface topography and morphology
TEM - Transmission Electron Microscope	Transmits an electron beam through a very thin specimen	Internal structure and particle size
AFM - Atomic Force Microscope	Measures the force between a sharp tip and the sample surface	3-D surface map; works on non-conducting samples too
STM - Scanning Tunnelling Microscope	Measures the tunnelling current between tip and surface	Atomic-resolution image; sample must conduct
XRD - X-Ray Diffraction	Diffraction of X-rays by the crystal lattice	Crystal structure and phase
DLS - Dynamic Light Scattering	Scattering of light by particles in a liquid	Particle size distribution in suspension

- SEM and TEM are electron microscopes; AFM and STM are scanning probe microscopes. That pairing is the usual matching question.
- The STM (1981) came first and won the 1986 Physics Nobel for Binnig and Rohrer; the AFM followed in 1986.
- Memory hook: the T in STM is tunnelling current, so the sample must conduct. AFM uses Force, so it works on plastics and biological samples as well.

The four generations of nanotechnology

This framework is credited to M.C. Roco and it is pure matching material - RPSC asked it as a four-pair match in 2018. The logic is easy: the technology starts with materials that just sit there, then moves to materials that do something, then to whole systems, and finally to building molecule by molecule.

Generations of nanotechnology (Roco)



- 1st - passive: fixed function. Coatings, nanoparticles in sunscreen and paint, reinforced polymers.
- 2nd - active: the structure changes its behaviour while in use. Targeted drug carriers, adaptive structures, actuators.
- 3rd - systems of nanosystems: many nanocomponents working together, guided assembly, three-dimensional networking.
- 4th - molecular nanosystems: design and assembly molecule by molecule; still largely a research goal.

ASKED IN RAS

Asked in RAS Pre 2018 as a match - first generation to passive nanostructures, second to active nanostructures, third to systems of nanosystems, fourth to molecular nanosystems. Learn the order, not the explanations.

YAAD RAKHO

P-A-S-M: Passive, Active, Systems, Molecular. Four generations, four letters, in order.

Applications, nanotoxicology and India's Nano Mission

Applications come as 'which of these is NOT an application of nanotechnology'. So you do not need depth here - you need coverage, sector by sector, so that nothing on the list looks unfamiliar. Note the one clean negative: generating electricity by nuclear fission has nothing to do with nanotechnology.

Nanotechnology by sector

Sector	Nanotechnology at work
Medicine	Targeted drug delivery using dendrimers, liposomes and nanoshells; nano-scale imaging and diagnostics
Cosmetics	Nano-TiO ₂ and nano-ZnO as transparent UV blockers in sunscreen
Textiles	Stain-resistant and water-repellent finishes; nano-silver as an antimicrobial
Water	Nano-membranes and nano-filters for purification and desalination
Energy	Higher-capacity battery electrodes, better solar cell materials, fuel cell catalysts
Electronics	Ever smaller transistors, quantum-dot displays, flexible electronics
Packaging	Nano-clay barrier films that keep out oxygen and moisture
Agriculture	Nano-fertilisers and controlled-release pesticides
Defence	Lighter and stronger composite armour, sensors, protective coatings

- Nanotoxicology is the study of the harm nanomaterials can do - and it is a fair exam target because policy documents stress it.
- Inhaled ultrafine particles are small enough to cross biological barriers that stop ordinary dust.
- Long, thin carbon nanotubes have raised asbestos-like concerns in animal studies.
- Because of this, risk assessment, labelling and regulation are built into national nanotechnology programmes.

India's programme - dates and the parent department

- The Nano Science and Technology Initiative (NSTI) came first, in 2001.
- The Nano Mission was launched in May 2007 under the Department of Science and Technology - remember DST, not MeitY.
- It concluded on 31 March 2017 and continues as the National Programme on Nano Science and Technology.
- Institute of Nano Science and Technology (INST), Mohali is DST's dedicated autonomous nano institute.
- Other key centres: ARCI, Hyderabad (powder metallurgy and new materials) and CeNS, Bengaluru (nano and soft matter sciences).

New materials - composites, superconductors and smart materials

The second half of this chapter is a vocabulary test. Each material has one defining property and one signature use, and the question is always a match. Do not read around it - just fix the pairs. The one entry worth a second reading is the metamaterial, because its behaviour comes from the engineered structure and not from the chemistry - that wording itself has been made into a statement question.

- A composite = matrix + reinforcement. Two materials combined so that the result behaves better than either alone.
- Carbon-fibre reinforced polymer (CFRP) - carbon fibres in a polymer matrix; used in aircraft and sports goods.
- Kevlar is an aramid fibre - the bullet-proof-vest material.
- Fibreglass is glass fibre in a resin matrix. Reinforced concrete is steel in concrete - the oldest everyday composite.
- The point to remember: a composite's properties differ from those of either constituent taken separately.

Superconductors

- Superconductivity = zero electrical resistance below a critical temperature, PLUS expulsion of magnetic flux from inside the material.
- That expulsion is the Meissner effect - the name RPSC asks for.
- Discovered by Heike Kamerlingh Onnes in mercury in 1911.
- Everyday use: the powerful magnets in MRI scanners; research use: maglev transport and lossless power transmission.

Smart materials, metamaterials and aerogel

Material to property to use - the standard matching set

Material	Type	Signature use
Nitinol (nickel-titanium)	Shape-memory alloy - returns to a remembered shape on heating	Vascular stents, spectacle frames
Quartz, PZT	Piezoelectric - produce a voltage under mechanical stress and vice versa	Sensors, quartz watches, gas lighters
Thermochromic materials	Change colour with temperature	Temperature indicators, novelty coatings
Magnetostrictive materials	Change shape in a magnetic field	Actuators and sonar transducers

Material	Type	Signature use
Aerogel	Extremely low-density solid, popularly 'frozen smoke'	Among the lightest solids; thermal insulation
Metamaterial	Behaviour comes from engineered structure, not chemistry	Negative refractive index; cloaking and superlens research

Semiconductors and rare earths - the strategic materials

This is the newest part of the chapter and the part most likely to be asked as current affairs. Two things to hold: the physics of doping, which is old NCERT material, and India's semiconductor and critical-mineral programmes, which are current. Both are short and both are examinable.

- A semiconductor's conductivity sits between a conductor's and an insulator's; silicon's band gap is about 1.1 eV.
- Doping with a group-15 element such as phosphorus or arsenic adds free electrons - that gives an n-type semiconductor.
- Doping with a group-13 element such as boron or gallium creates holes - that gives a p-type semiconductor.
- Trick: n for negative carriers comes from group 15 (five valence electrons); p for positive holes comes from group 13 (three).
- Silicon carbide (SiC) and gallium nitride (GaN) are wide-band-gap semiconductors, used in high-power and high-temperature electronics.

India's semiconductor push - the facts that get asked

Item	Detail
Programme	Semicon India Programme, approved 2021, outlay Rs 76,000 crore
Who runs it	India Semiconductor Mission (ISM), under the Ministry of Electronics and Information Technology (MeitY)

Item	Detail
First commercial fab	Tata Electronics with Powerchip (PSMC) at Dholera, Gujarat
Assembly and testing units	Micron's ATMP and Kaynes' unit at Sanand, Gujarat; Tata's OSAT plant at Jagiroad, Assam

CURRENT LINK

Current link, as of September 2026: per PIB (1 April 2026) ten projects worth about Rs 1.6 lakh crore stood approved under the India Semiconductor Mission. ISM 2.0 was announced in Budget 2026-27 and cleared by the Union Cabinet at the end of June 2026 with an outlay of roughly Rs 1.25 lakh crore. For the exam, the two safest facts remain the ministry (MeitY) and the first fab location (Dholera).

- Rare earths are 17 elements - the 15 lanthanides plus scandium and yttrium.
- They are not actually rare in the crust; they are rarely found in concentrated, economically workable deposits.
- India's principal source is monazite in beach sands - Kerala, Odisha, Tamil Nadu and Andhra Pradesh.
- IREL (India) Limited is the principal public sector producer of rare earths in India.
- The National Critical Mineral Mission was launched in January 2025 to secure supply of these and other critical minerals.

REVISION RECAP

1. 1 nanometre = 10^{-9} metre = 10 angstrom; nanotechnology = at least one dimension in 1-100 nm. Asked 2016, 2018, 2021, 2023.
2. Two nanoscale effects: surface-area-to-volume ratio (higher reactivity) and quantum confinement (size-dependent optical and electronic behaviour).
3. Confined in 1 = quantum well (free in 2); confined in 2 = quantum wire (free in 1); confined in 3 = quantum dot (free in 0, an artificial atom).

4. Quantum dot: semiconductor nanocrystal 2-10 nm, colour tunable by size, Nobel Chemistry 2023 - Bawendi, Brus, Ekimov.
5. Feynman 1959 lecture, Taniguchi coined the word 1974, Drexler popularised it 1986.
6. Fullerene C₆₀ = 20 hexagons + 12 pentagons, 1985, Kroto-Curl-Smalley, Chemistry Nobel 1996.
7. Carbon nanotubes - Iijima 1991; about 100 times stronger than steel at one-sixth the weight; armchair metallic, zigzag/chiral semiconducting.
8. Graphene - one-atom-thick honeycomb sheet, Geim and Novoselov 2004, Physics Nobel 2010.
9. Carbon by dimension: fullerene 0-D, CNT 1-D, graphene 2-D, diamond and graphite 3-D.
10. Top-down = ball milling, lithography, etching, laser ablation. Bottom-up = sol-gel, CVD, self-assembly, MBE.
11. SEM and TEM are electron microscopes; AFM and STM are scanning probe. STM built 1981 by Binnig and Rohrer, Physics Nobel 1986.
12. Generations: 1st passive, 2nd active, 3rd systems of nanosystems, 4th molecular nanosystems - Roco. Matched in RAS 2018.
13. Dendrimers, liposomes and nanoshells are drug-delivery carriers; nano-silver is antimicrobial; nano-TiO₂ and ZnO block UV.
14. Nanotoxicology: ultrafine particles cross biological barriers; long thin CNTs raise asbestos-like concerns.
15. Nano Mission - May 2007, under DST; NSTI 2001 preceded it; concluded 31 March 2017 and continues as the National Programme. INST Mohali is the dedicated institute.
16. Composite = matrix + reinforcement: CFRP, Kevlar (aramid), fibreglass, reinforced concrete.
17. Superconductivity = zero resistance below the critical temperature + Meissner effect; Kamerlingh Onnes, mercury, 1911.
18. Nitinol = shape-memory alloy (stents); quartz and PZT = piezoelectric; aerogel = 'frozen smoke' insulator; metamaterial = negative refractive index.
19. Semicon India 2021, Rs 76,000 crore; India Semiconductor Mission under MeitY; first commercial fab at Dholera by Tata with PSMC.
20. Rare earths = 15 lanthanides + scandium + yttrium; monazite beach sands; IREL is the PSU producer; National Critical Mineral Mission, January 2025.

ST09

Computers, IT and ICT

Roughly 3-5 questions a paper, and the Rajasthan e-governance part alone has supplied a question in 2016, 2018, 2021 and 2024. Networks, data units and the new technology missions are the other reliable suppliers. Nothing here needs technical depth - it needs exact names, years and ministries.

This is the largest chapter in the Science and Technology block and it has changed character. Ten years ago RPSC asked what a byte is and which key saves a file. It still asks that - but it now also asks about the National Quantum Mission, the IndiaAI Mission, the DPDP Act and Rajasthan's own portals. So you have two jobs: keep the old basics ready for the easy marks, and learn the new missions with their years, outlays and parent ministries.

Computers, IT and ICT

Generations

- I vacuum tube (1946-59)
- II transistor (1959-65)
- III integrated circuit (1965-71)
- IV microprocessor, VLSI (1971-80)
- V AI, ULSI (1980 onwards)

Hardware

- CPU = ALU + Control Unit + registers
- Registers > cache > RAM > secondary
- RAM volatile, ROM not
- EPROM - erased by UV light
- MICR on cheques; plotter is OUTPUT

Data and number systems

- nibble 4 bits, byte 8 bits
- KB 1024 bytes, then MB GB TB PB
- Binary 2, octal 8, decimal 10, hex 16
- ASCII 7-bit, 128 characters
- Unicode is the universal standard

Software

- System vs application software
- Compiler all at once, interpreter line by line
- IGL machine to SGL LISP/Prolog
- FORTRAN 1957 - first high-level
- Codd - relational model, 1970

Networks

- PAN < LAN < MAN < WAN
- Bus, star, ring, mesh, tree, hybrid
- OSI 7 layers, TCP/IP 4
- IPv4 32-bit, IPv6 128-bit
- HTTP 80, HTTPS 443, SMTP 25, DNS 53

Internet and cloud

- ARPANET 1969
- WWW - Berners-Lee, CERN, 1989
- India commercial internet 15 Aug 1995, VSNL
- IaaS, PaaS, SaaS
- MeghRaj - Gov cloud

New technologies

- IoT, big data - five Vs
- Blockchain - distributed, immutable
- AI coined 1956, McCarthy, Dartmouth
- Transformer 2017 - basis of generative AI
- Qubit, superposition, entanglement

India's missions

- IndiaAI Mission 2024, Rs 10,372 cr, MeitY
- National Quantum Mission 2023, Rs 6,003.65 cr, DST
- 5G launched 1 October 2022
- NSM 2015 - PARAM Shivay, PARAM Rudra, AIRAWAT

Cyber security and law

- Virus needs a host, worm spreads itself
- IT Act 2000 - UNCITRAL model law
- Sec 66A struck down - Shreya Singhal 2015
- CERT-In under MeitY; I4C under MHA, 1930
- DPDP Act 2023 - max penalty Rs 250 crore

E-governance

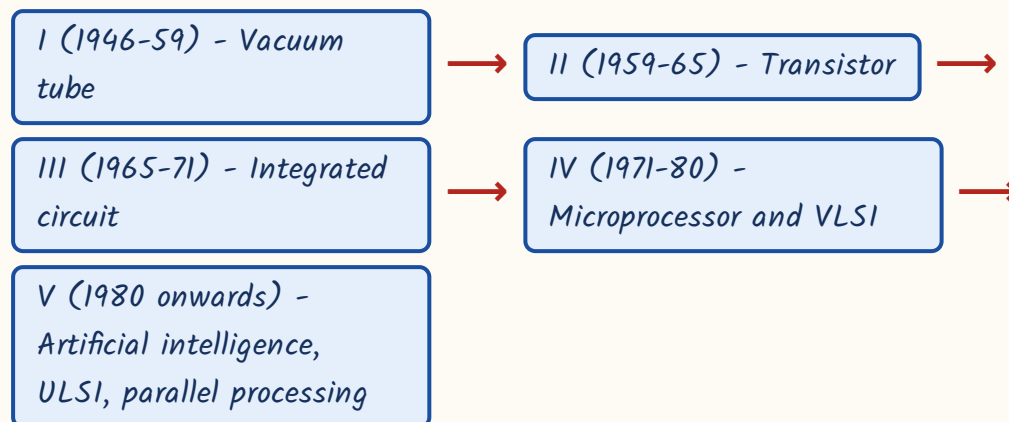
- Digital India - 1 July 2015
- DigiLocker, UMANG, Aadhaar, UPI, ONDC

- Kerala - first digital state, 2016
- e-Mitra, Jan Aadhaar, Raj Kaj, iStart
- DoIT&C is Rajasthan's nodal department

Generations of computers - the matching question you can bank

Every generation is named after the electronic component it was built from. That is the whole logic, and that is exactly how RPSC asks it - generation on the left, component on the right. Learn the five components in order and you have the answer for any arrangement of the pairs.

Five generations by principal component



- Charles Babbage is the 'Father of the Computer' - Difference Engine 1822, Analytical Engine 1837.
- Ada Lovelace is regarded as the world's first programmer.
- ENIAC (1946) was the first general-purpose electronic digital computer.
- UNIVAC-I was the first computer sold commercially.
- EDVAC embodied the stored-programme concept, associated with John von Neumann.

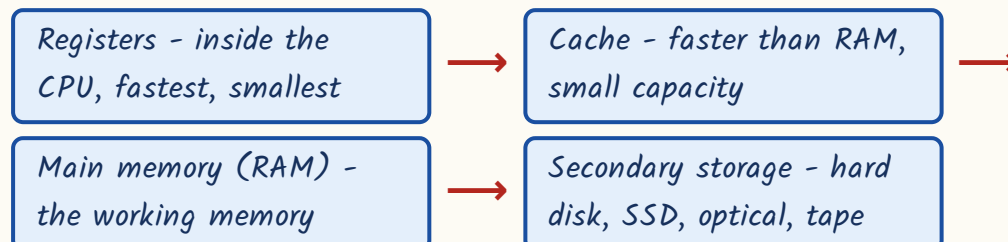
YAAD RAKHO

Say the components as one string: Vacuum - Transistor - IC - Microprocessor - AI. Five words, five generations, in order. The years are secondary.

Hardware - the CPU and the memory hierarchy

The CPU is made of three things: the Arithmetic Logic Unit, which does the arithmetic and the logic; the Control Unit, which fetches and decodes instructions and directs everybody else; and registers, which are tiny holding places inside the processor. Registers are the fastest memory in the machine, not the slowest - RPSC has built a statement question on exactly that reversal.

Memory hierarchy - fastest and smallest at the top



Types of memory - the volatile test

Memory	Volatile?	Key point
RAM	Yes	Working memory; contents lost when power goes off
ROM	No	Retains contents without power; holds firmware
PROM	No	Programmable once, then permanent
EPROM	No	Contents erased by exposure to ultraviolet light
EEPROM	No	Erased electrically
Flash memory	No	A form of EEPROM - pen drives, memory cards, SSDs
Cache	Yes	Faster than main memory but much smaller

- Input devices: keyboard, mouse, scanner, joystick, light pen, barcode reader, OMR, OCR and MICR.
- MICR - Magnetic Ink Character Recognition - is the one used on bank cheques.
- Output devices: monitor, printer, plotter, speaker.
- Printers: dot-matrix is an impact printer; laser and inkjet are non-impact.
- As you go down the hierarchy speed falls and capacity rises - and cost per bit falls.

TRAP

Two reversals RPSC loves. First: a plotter is an OUTPUT device, even though it appears in lists of input devices as the odd one out. Second: the CPU cannot execute instructions lying on a hard disk - they must first be loaded into main memory. A statement saying otherwise is false.

Number systems and data units

Computers store everything as 0 and 1. Number systems are simply different bases for writing the same value, and RPSC asks two things only - the base of a system, and the order of the data units. Both are memory work. Nothing here requires calculation beyond a small binary conversion.

The four number systems

System	Base	Digits used
Binary	2	0, 1
Octal	8	0 to 7
Decimal	10	0 to 9
Hexadecimal	16	0 to 9 and A to F

Data units in ascending order

Unit	Equals
Bit	A single 0 or 1
Nibble	4 bits
Byte	8 bits
Kilobyte (KB)	1024 bytes
Megabyte (MB)	1024 KB
Gigabyte (GB)	1024 MB
Terabyte (TB)	1024 GB
Then PB, EB, ZB, YB	each 1024 times the one before

- A nibble is half a byte - four bits. That single fact is a whole question.
- In hexadecimal, A stands for 10 and F for 15.
- Decimal 25 in binary is 11001. Convert by repeated division by 2 and reading the remainders upwards.
- ASCII is a 7-bit code covering 128 characters; Unicode is the universal standard that covers the world's scripts, including Devanagari.
- Ascending order to memorise: KB, MB, GB, TB, PB, EB, ZB, YB.

Software, operating systems and programming languages

Software splits in two. System software runs the machine - the operating system, compilers, assemblers, interpreters, device drivers. Application software does your work - a word processor, a browser, a payroll package. The operating system is the manager sitting between hardware and everything else: it handles memory, files, processes and devices.

- Operating systems: Windows, Linux, Unix, macOS, Android, iOS.
- Android is built on the Linux kernel.

- Oracle is a database management system, NOT an operating system - the classic odd-one-out option.
- Open source means the source code is freely available to read, modify and redistribute; Linux is the standard example.
- Firmware sits in ROM and starts the machine; a device driver lets the OS talk to a particular piece of hardware.
- MS Office is application software. Shortcuts asked as odd-one-out: Ctrl+C copy, Ctrl+X cut, Ctrl+V paste, Ctrl+Z undo, Ctrl+Y redo, Ctrl+S save, Ctrl+P print, Ctrl+A select all, Ctrl+F find.
- Function keys: F1 help, F2 rename, F5 refresh, Alt+F4 close the window.
- File extensions get asked the same way: .docx word processing, .xlsx spreadsheet, .pptx presentation, .mp3 audio, .jpg image, .pdf portable document.
- A compiler translates the whole programme at one go and reports all the errors together.
- An interpreter translates and executes line by line, stopping at the first error.
- An assembler converts assembly language into machine code.
- DBMS: E.F. Codd proposed the relational model in 1970.
- In a relation, a row is a tuple or record and a column is an attribute or field; the primary key uniquely identifies a row.

Generations of programming languages

Generation	Type	Examples
1GL	Machine language	Binary instructions the CPU runs directly
2GL	Assembly language	Mnemonics such as ADD, MOV
3GL	High-level languages	FORTRAN (1957, the first), COBOL, C, Java, Python
4GL	Query and report languages	SQL
5GL	Artificial intelligence languages	LISP, Prolog

Computer networks – the part RPSC has started leaning on

A network is just computers joined so they can share data and resources. Three things get asked: the classification by geographical span, the topology (the shape of the wiring), and the protocol-to-function matching. Take them in that order and the section is short.

Networks by geographical span – smallest to largest

Network	Span	Typical example
PAN – Personal Area Network	A few metres around one person	A Bluetooth link between phone and earphones
LAN – Local Area Network	One building, office or campus	An office or college network
MAN – Metropolitan Area Network	A city	A city-wide cable or broadband network
WAN – Wide Area Network	A country or the whole world	The internet itself

Network topologies

Topology	Layout	Note
Bus	All nodes share a single backbone cable	Cheapest; failure of the backbone brings the whole network down
Star	Every node connects to a central hub or switch	Easy to manage; failure of the hub kills the network
Ring	Nodes form a closed loop, data moves node to node	A break in the loop disrupts the network
Mesh	Every node connected to every other node	Most fault-tolerant, but needs the most cabling
Tree	Hierarchical, star networks joined to a backbone	Used in large structured setups
Hybrid	Any combination of the above	Most real networks

OSI reference model versus TCP/IP model

OSI reference model	TCP/IP model
7 layers	4 layers
Physical, Data Link, Network, Transport, Session, Presentation, Application	Network Access, Internet, Transport, Application
A theoretical reference framework	The model the internet actually runs on
Layers are strictly separated	Layers are merged and practical

- *Devices by layer: repeater and hub work at the physical layer; switch and bridge at the data link layer; a router at the network layer.*
- *A gateway joins two dissimilar networks. A modem modulates and demodulates - digital to analogue and back.*
- *An IPv4 address is 32 bits, written as four octets in dotted decimal, for example 192.168.1.1.*
- *An IPv6 address is 128 bits - created because IPv4 addresses ran out.*
- *Bandwidth is the data-carrying capacity of a link, measured in bits per second.*

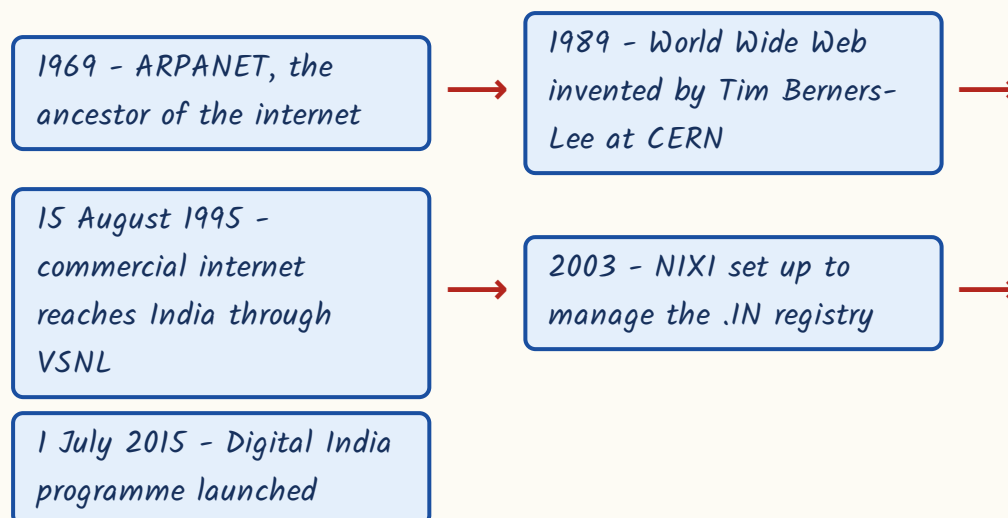
Common protocols and their default ports

Protocol	Function	Default port
HTTP	Transfer of web pages	80
HTTPS	Secure (encrypted) web transfer	443
FTP	File transfer between hosts	20 and 21
SMTP	Sending electronic mail	25
POP3	Retrieving mail from a server	110
DNS	Translating domain names into IP addresses	53
Telnet	Remote login	23
DHCP	Automatic assignment of IP addresses to hosts	-

The internet, the web and cloud computing

Keep the two apart. The internet is the global network of networks - the wires, the routers, the addressing. The World Wide Web is one service running on it, a system of linked pages accessed through a browser. Email, file transfer and video calls also run on the internet without being 'the web'. RPSC has asked the invention of the web separately from the origin of the internet, so hold both dates.

From ARPANET to Digital India



Cloud service models - who manages what

Model	What the provider gives you	What you still manage
IaaS - Infrastructure as a Service	Virtualised servers, storage and network	The operating system and your applications
PaaS - Platform as a Service	A ready development and deployment platform	Only your own application and its data
SaaS - Software as a Service	Finished software delivered over the internet	Nothing - you simply use it

- Deployment models: public cloud, private cloud, hybrid cloud and community cloud.

- *MeghRaj is the cloud computing initiative of the Government of India.*
- *A URL identifies a resource - protocol, domain name and path. A domain name is translated into an IP address by DNS.*
- *Common domain endings: .com commercial, .org organisation, .gov government, .edu education, .in the India country code.*
- *A browser (Chrome, Firefox, Edge) displays pages; a search engine (Google, Bing) finds them. They are not the same thing.*

The new technologies - where RPSC is now moving

This is the growth area of the chapter. The examiner is not testing whether you can build any of these. It is testing whether you know the one-line definition and the Indian mission attached to it - the year, the outlay and the ministry. So learn the table, then learn the missions table, and stop there.

The technologies - definition plus the fact that gets asked

Technology	One-line definition	Exam anchor
<i>Internet of Things (IoT)</i>	<i>A network of physical objects fitted with sensors that collect and exchange data</i>	<i>Objects plus sensors plus connectivity - not just phones and computers</i>
<i>Big data</i>	<i>Data too large, too fast and too varied for ordinary tools to handle</i>	<i>Five Vs - volume, velocity, variety, veracity, value</i>
<i>Blockchain</i>	<i>A decentralised, immutable ledger distributed across many computers</i>	<i>Bitcoin (2009) attributed to 'Satoshi Nakamoto'; RBI began Digital Rupee (CBDC) pilots in 2022</i>
<i>Artificial intelligence</i>	<i>Machines performing tasks that would need human intelligence</i>	<i>Term coined by John McCarthy at the Dartmouth Conference, 1956; Alan Turing proposed the Turing Test in 1950</i>
<i>Machine learning</i>	<i>Systems that learn patterns from data instead of being explicitly programmed</i>	<i>Supervised (labelled data), unsupervised (clustering), reinforcement (rewards and penalties)</i>

Technology	One-line definition	Exam anchor
Generative AI	Models that produce new text, images, audio or code	Built on the transformer architecture, introduced in 2017
Quantum computing	Computing with qubits instead of bits	Superposition and entanglement; a qubit is the unit of quantum information
5G	Fifth-generation mobile telephony - high speed, low latency	Launched in India on 1 October 2022 at the India Mobile Congress

India's missions in these areas - years, outlays, ministries

Learn the parent department for each - RPSC swaps them

Mission or document	Key numbers	Under
National Strategy for AI, '#AIforAll'	2018	NITI Aayog
IndiaAI Mission	Approved 7 March 2024, about Rs 10,372 crore, seven pillars	MeitY
Bhashini	India's language AI platform for Indian languages	MeitY
National Quantum Mission	Approved 19 April 2023, Rs 6,003.65 crore, period 2023-24 to 2030-31; target 50-1000 physical qubits in eight years	DST
NQM thematic hubs	Computing at IISc Bengaluru; communication at IIT Madras; sensing and metrology at IIT Bombay; materials and devices at IIT Delhi	DST
Bharat 6G Vision	Document March 2023; Bharat 6G Alliance July 2023; leadership targeted around 2030	Department of Telecommunications
National Supercomputing Mission	Launched 2015	MeitY and DST, executed through C-DAC Pune and IISc Bengaluru

- PARAM 8000 (1991), built by C-DAC under Dr Vijay Bhatkar, was India's first indigenous supercomputer.
- PARAM Shivay at IIT (BHU) Varanasi, 2019, was the first system installed under the National Supercomputing Mission.
- Three PARAM Rudra supercomputers were dedicated to the nation in September 2024 - at Pune, Delhi and Kolkata.
- AIRAWAT, at C-DAC Pune, ranked 75th in the Top500 list announced at ISC 2023 - India's AI supercomputing platform.
- Pratyush and Mihir are the supercomputers used for weather and climate forecasting.

CURRENT LINK

Current link, as of September 2026: the National Quantum Mission (2023-24 to 2030-31) and the IndiaAI Mission (March 2024) are both mid-run, so the examinable facts are still the approval year, the outlay and the ministry - Rs 6,003.65 crore under DST for quantum, about Rs 10,372 crore under MeitY for AI. Watch for the four NQM hub institutions being asked as a match; that pattern is already visible in mock and state papers.

Cyber security - malware, agencies and the law

Cyber security questions come in three shapes: identify the malware from its behaviour, name the agency and its ministry, or state a fact about the IT Act or the DPDP Act. The behaviour descriptions are the easiest marks in the chapter because each type of malware does exactly one distinctive thing.

Malware - one behaviour each

Type	How it behaves
Virus	Needs a host programme or file to attach to and spread

Type	How it behaves
Worm	Spreads by itself across a network; needs no host
Trojan horse	Disguises itself as a legitimate programme to get installed
Ransomware	Encrypts the victim's files and demands payment for the decryption key
Spyware and keylogger	Silently collect information and record keystrokes
Botnet	A network of compromised machines controlled remotely by an attacker
Adware	Pushes unwanted advertisements onto the system

- *Phishing - a fake email or website that tricks you into giving up passwords or card details. Over a phone call it is vishing; by SMS it is smishing.*
- *A firewall filters traffic between a trusted internal network and the outside world, allowing or blocking it by rule.*
- *Encryption converts readable data into cipher text. Symmetric encryption uses one shared key; asymmetric (public key) encryption uses a key pair.*
- *A digital signature is created with the sender's private key and verified with the sender's public key.*
- *A digital signature gives three assurances: authenticity of the sender, integrity of the message, and non-repudiation.*

Laws and agencies - fix the ministry with the name

Item	Key facts
IT Act, 2000	Based on the UNCITRAL Model Law on Electronic Commerce, 1996; came into force 17 October 2000; amended in 2008
Section 66A	Penalised 'offensive' messages; struck down by the Supreme Court in <i>Shreya Singhal v. Union of India</i> (2015)
CERT-In (2004)	National nodal agency for computer security incidents; under MeitY
NCIIPC	Protects critical information infrastructure; under NTRO, created under Section 70A

Item	Key facts
14C	Indian Cyber Crime Coordination Centre; under the Ministry of Home Affairs; runs the cybercrime portal and helpline 1930
Cyber Swachhta Kendra	Botnet cleaning and malware analysis centre
DPDP Act, 2023	Data Principal, Data Fiduciary, Consent Manager, Significant Data Fiduciary; sets up the Data Protection Board of India; maximum penalty Rs 250 crore; DPDP Rules notified 14 November 2025

TRAP

Two swaps to guard against. The maximum penalty under the DPDP Act 2023 is Rs 250 crore, not Rs 50 crore. And CERT-In sits under MeitY while 14C sits under the Home Ministry - RPSC exchanges the two ministries in options.

E-governance - India and Rajasthan

This is the highest-yield section of the chapter. Rajasthan e-governance has supplied a question in 2016, 2018, 2021 and 2024. The format is almost always name-to-purpose matching, so build the two tables below and revise them as tables, not as prose. Note the launch dates - they are asked directly.

Digital India and its building blocks

Initiative	Year	What it does
Digital India	1 July 2015	Umbrella programme - three vision areas and nine pillars
MyGov	2014	Citizen engagement platform
DigiLocker	2015	Cloud platform for storing and sharing authentic digital documents

Initiative	Year	What it does
UMANG	2017	Single mobile app giving access to many government services
Aadhaar / UIDAI	UIDAI 2009; Aadhaar Act 2016	12-digit unique identity number
UPI	NPCI, 2016	Instant mobile-based payments
ONDC	2021	Open Network for Digital Commerce
BharatNet and Common Service Centres	-	Rural broadband backbone and last-mile service delivery points
SWAYAM, SWAYAM PRABHA, DIKSHA, NPTEL, e-PathShala, PM eVidya	-	ICT in education - online courses, DTH channels, school content

ASKED IN RAS

Asked in RAS Pre 2016 - which state was declared India's first digital state? Answer: Kerala, in February 2016. RPSC thought so highly of this fact that it asked it TWICE in the same 2016 paper.

TRAP

Kerala, not Rajasthan, is India's first digital state. Rajasthan is genuinely strong on e-governance, and the mind slides towards it - which is precisely why the option is there. Do not fall for it.

Rajasthan - DoIT&C is the nodal department

Rajasthan's portals and schemes - pure matching material

Portal or scheme	Purpose
e-Mitra	Electronic delivery of 600-plus government and private services through 80,000-plus kiosks

Portal or scheme	Purpose
Rajasthan SSO	Single sign-on access to state online services
Jan Aadhaar	Launched 18 December 2019 on the principle 'One Number, One Card, One Identity'; subsumed Bhamashah
Bhamashah Yojana	Relaunched 2014; benefits routed by treating the adult woman as head of the family
Raj Kaj	e-Office and service matters of government employees
RAJ NIVESH	Single-window clearance for investors
iStart	Startup registration, incubation and support
RajNET and RajSWAN	State connectivity backbone
Rajasthan Sampark	Grievance redressal; helpline 181
R-CAT, Jaipur	Rajasthan Centre for Advanced Technology - advanced technology training
RKCL's RS-CIT	Basic digital literacy certification course

ASKED IN RAS

Asked in RAS Pre 2018 - what are e-Mitra kiosks primarily meant for?

Answer: electronic delivery of government and private services to citizens.

Asked in RAS Pre 2021 - the principle of Jan Aadhaar Yojana? Answer: 'One Number, One Card, One Identity'.

ASKED IN RAS

Asked in RAS Pre 2024 as a four-pair match - Raj Kaj to e-office and employee service matters, RAJ NIVESH to single-window investment clearance, iStart to startups, Rajasthan SSO to single sign-on. Learn these four pairs word for word; they have now been asked in this exact form.

REVISION RECAP

1. Generations by component: I vacuum tube, II transistor, III integrated circuit, IV microprocessor and VLSI, V AI and ULSI.
2. Babbage - father of the computer; Ada Lovelace - first programmer; ENIAC 1946 first general-purpose electronic digital computer; EDVAC carried the stored-programme idea.
3. CPU = ALU + Control Unit + registers. Registers are the FASTEST memory, not the slowest.
4. Hierarchy: registers, cache, RAM, secondary storage. RAM volatile, ROM not. EPROM erased by UV, EEPROM electrically, flash is a form of EEPROM.
5. MICR is used on bank cheques. Plotter is an output device.
6. Nibble 4 bits, byte 8 bits, 1 KB = 1024 bytes; then MB, GB, TB, PB, EB, ZB, YB. ASCII 7-bit / 128 characters; Unicode universal.
7. Bases: binary 2, octal 8, decimal 10, hexadecimal 16 (0-9 and A-F). Decimal 25 = binary 11001.
8. Compiler translates the whole programme and lists all errors; interpreter goes line by line. FORTRAN (1957) was the first high-level language; SQL is 4GL; LISP and Prolog 5GL.
9. Codd proposed the relational model in 1970 - row is a tuple, column an attribute, primary key identifies the row.
10. Span order PAN < LAN < MAN < WAN. Mesh is the most fault-tolerant topology; bus shares one backbone.
11. OSI has 7 layers, TCP/IP has 4. Repeater/hub physical, switch/bridge data link, router network, gateway joins dissimilar networks.
12. IPv4 = 32 bits, IPv6 = 128 bits. Ports: HTTP 80, HTTPS 443, FTP 20/21, SMTP 25, DNS 53, Telnet 23, POP3 110.
13. ARPANET 1969; WWW by Tim Berners-Lee at CERN in 1989; commercial internet in India 15 August 1995 via VSNL; NIXI 2003 for the .IN registry.
14. Cloud: IaaS infrastructure, PaaS platform, SaaS software. MeghRaj is the Government of India cloud.
15. Big data's five Vs; blockchain is decentralised and immutable; Bitcoin 2009 'Satoshi Nakamoto'; RBI Digital Rupee pilots 2022.
16. AI coined by John McCarthy, Dartmouth 1956; Turing Test 1950; ML is supervised, unsupervised or reinforcement; generative AI rests on the 2017 transformer.
17. IndiaAI Mission - 7 March 2024, about Rs 10,372 crore, MeitY. National Quantum Mission - 19 April 2023, Rs 6,003.65 crore, DST, hubs at IISc, IIT Madras, IIT Bombay, IIT Delhi.
18. 5G launched in India 1 October 2022; Bharat 6G Vision March 2023, Alliance July 2023. PARAM 8000 (1991, Vijay Bhaskar, C-DAC); NSM 2015; PARAM Shivay at IIT-BHU; PARAM Rudra x3 in September 2024; AIRAWAT 75th at ISC 2023.

19. IT Act 2000 on the UNCITRAL model law, in force 17 October 2000, amended 2008; Section 66A struck down in *Shreya Singhal* (2015). CERT-In under MeitY, 14C under MHA with helpline 1930, DPDP Act 2023 maximum penalty Rs 250 crore.
20. Digital India 1 July 2015. Kerala - first digital state, 2016, NOT Rajasthan. Rajasthan: e-Mitra, Jan Aadhaar (18 December 2019), Raj Kaj, RAJ NIVESH, iStart, SSO, DoIT&C as nodal department.

ST10

Space Technology

About 2-4 questions a paper, in every paper since 2013. The launch-vehicle-to-payload match and the ISRO centre-to-location match are near-certain. Add one current-affairs question on the newest mission.

India's space programme is the most reliably asked block in the Science and Technology section. RPSC does not test physics here - it tests names, dates, launch-vehicle-to-satellite matching and firsts. Learn it as a set of tables, not as a story.

Indian Space Programme

Basics

- Orbits - LEO, MEO, GEO, SSO, GTO
- Escape velocity 11.2 km/s
- Multi-stage rockets
- Cryogenic engines

Science missions

- Chandrayaan-1/2/3
- Mangalyaan
- AstroSat, Aditya-L1, XPoSat
- SpaDeX, NISAR

Policy and private

- Space Policy 2023
- IN-SPACe, NSIL, Antrix
- Skyroot, Agnikul, Pixxel
- Project NETRA, space debris

Institutions

- INCOSPAR 1962
- ISRO 15 Aug 1969
- Department of Space 1972
- VSSC, SDSC-SHAR, URSC, SAC, NRSC

Application satellites

- INSAT / GSAT
- IRS, Cartosat, Oceansat, RISAT
- NavIC / IRNSS
- Kalpana-1, EDUSAT

Launch vehicles

- SLV-3, ASLV
- PSLV and its variants
- GSLV Mk-II
- LVM3 / GSLV Mk-III
- SSLV, RLV-TD, HLVM3

Human spaceflight

- Gaganyaan on HLVM3
- TV-D1 crew escape test
- Vyommitra
- Bharatiya Antariksh Station 2035

The physics first - orbits, escape velocity, staging

Start with one idea. A satellite is not held up by anything. It is falling towards the Earth all the time, but it is moving sideways so fast that the Earth curves away beneath it at the same rate. That is an orbit - a permanent, controlled fall. Two speeds follow from this. Orbital velocity is the sideways speed needed to keep circling, about 7.9 km/s just above the atmosphere. Escape velocity is the speed at which you break free of Earth's gravity altogether - 11.2 km/s. Escape velocity does not depend on the mass of the object being launched; it depends on the mass and radius of the Earth.

- Orbital velocity near Earth's surface: about 7.9 km/s. Escape velocity: 11.2 km/s.
- Higher the orbit, slower the satellite and longer the period.
- At 35,786 km the period becomes 24 hours - the satellite then appears fixed over one point.
- Rockets are built in stages so that spent tanks and engines can be thrown away; less mass to accelerate means more speed from the fuel that remains. It is not about drag or heat.
- SLV-3 and ASLV were all-solid. PSLV alternates solid and liquid stages. Cryogenic stages sit on top, where efficiency matters most.

Orbit types - altitude, character, use

Orbit	Altitude / inclination	Character	Typical use
LEO - Low Earth Orbit	160 - 2,000 km	Period about 90 minutes	Earth observation, ISS, Gaganyaan at about 400 km
MEO - Medium Earth Orbit	Above LEO, up to 35,786 km	Period a few hours	Navigation constellations such as GPS at about 20,200 km
GEO - Geostationary	35,786 km, inclination about 0 degrees	Period about 24 hours, appears stationary	INSAT and GSAT communication and weather satellites

Orbit	Altitude / inclination	Character	Typical use
Polar / Sun-synchronous (SSO)	About 600-900 km, inclination about 98 degrees	Crosses a given latitude at the same local solar time every pass	IRS, Cartosat, RISAT, Resourcesat remote sensing
GTO - Geostationary Transfer Orbit	Perigee about 180 km, apogee about 35,786 km	A stretched transfer ellipse, not a parking orbit	Intermediate step before a satellite raises itself to GEO
Halo orbit at L1	About 1.5 million km, Sun-Earth L1	Balanced-gravity point with an uninterrupted view of the Sun	Aditya-L1

Understand why sun-synchronous matters, because RPSC has asked it as reasoning. The satellite passes over the same place at the same local solar time on every visit. So the Sun angle and the shadows are comparable from one pass to the next. That is exactly what you need to compare two images and detect change - a new building, a shrinking lake, a burnt forest. A geostationary satellite sits too far away to do this at high resolution.

YAAD RAKHO

Ladder going upward: LEO - MEO - GEO. Sun-synchronous is a special polar LEO tilted at about 98 degrees. GTO is an ellipse used only to reach GEO. If an option calls GTO a 'circular orbit', it is wrong.

ISRO - the institution, its people and its centres

India's space programme began with a civilian, development-first idea: use space to solve problems on the ground - communication, weather, resource survey - not for prestige. Vikram Sarabhai set that direction. The institutional dates below are asked as straight one-liners, so lock them.

- INCOSPAR set up in 1962 under the Department of Atomic Energy.
- ISRO formed on 15 August 1969, superseding INCOSPAR.
- Department of Space and the Space Commission created in 1972.
- Vikram Sarabhai - father of the Indian space programme.
- Satish Dhawan chaired ISRO from 1972 to 1984; Sriharikota is named after him.
- V. Narayanan took charge as ISRO Chairman in January 2025.
- TERLS at Thumba, Thiruvananthapuram - chosen because the magnetic equator passes over it.
- First sounding rocket from Thumba: a Nike-Apache, launched 21 November 1963.

The centres and where they are

ISRO's principal centres

Centre	Location	What it does
VSSC - Vikram Sarabhai Space Centre	Thiruvananthapuram, Kerala	Design and development of launch vehicles
LPSC - Liquid Propulsion Systems Centre	Valiamala (Thiruvananthapuram) and Bengaluru	Design of liquid propulsion stages
IPRC - ISRO Propulsion Complex	Mahendragiri, Tamil Nadu	Testing of liquid and cryogenic engines
SDSC-SHAR - Satish Dhawan Space Centre	Sriharikota, Andhra Pradesh	India's principal launch port
URSC - U R Rao Satellite Centre	Bengaluru, Karnataka	Building of satellites
SAC - Space Applications Centre	Ahmedabad, Gujarat	Payloads and space applications
NRSC - National Remote Sensing Centre	Hyderabad, Telangana	Reception and dissemination of remote sensing data
MCF - Master Control Facility	Hassan, Karnataka (also Bhopal)	Control of India's geostationary satellites

Centre	Location	What it does
ISTRAC	Bengaluru, Karnataka	Tracking and low-Earth-orbit mission operations
Aeronautical Test Range	Chitradurga, Karnataka	RLV / Pushpak autonomous landing experiments

The supporting bodies - equally examinable

Body	Location	Role
PRL - Physical Research Laboratory	Ahmedabad, Gujarat	Space science research; runs the Udaipur Solar Observatory
NARL - National Atmospheric Research Laboratory	Gadanki, Andhra Pradesh	Atmospheric research
IIST - Indian Institute of Space Science and Technology	Valiamala, Thiruvananthapuram	India's first space university
HSFC - Human Space Flight Centre	Bengaluru, Karnataka	Runs the Gaganyaan human spaceflight programme
IN-SPACE	Ahmedabad, Gujarat	Single-window authorisation and regulation of private space activity
NSIL - NewSpace India Limited	Bengaluru, Karnataka	Commercial arm of ISRO, created 2019

ASKED IN RAS

Asked repeatedly as a centre-to-location match - VSSC Thiruvananthapuram, SAC Ahmedabad, NRSC Hyderabad, URSC Bengaluru, IPRC Mahendragiri. And as a NOT-correctly-matched question: IIST is at Valiamala near Thiruvananthapuram, not Bengaluru.

Rajasthan link. The Udaipur Solar Observatory sits on an island in Fateh Sagar Lake. Water around it keeps the air steady, which reduces image shimmer - that

is why an island was chosen. It was founded in 1976 by Arvind Bhatnagar, has been run by the Physical Research Laboratory, Ahmedabad, since 1981, and its 50 cm Multi Application Solar Telescope (MAST) became operational in 2015.

ASKED IN RAS

Asked in RAS Pre 2018 and again in RAS Pre 2021, in near-identical wording - the Udaipur Solar Observatory is on an island in Fateh Sagar Lake. A repeat this exact is a gift; do not lose it.

The launch vehicles - the table RPSC keeps returning to

A launch vehicle is judged by two numbers: how much it can lift, and to which orbit. A rocket that can put 1,750 kg into a low polar orbit will manage far less into a geostationary transfer orbit, because height costs energy. Read the table with that in mind, and memorise the mission column - that is where the marks are.

Indian launch vehicles - payload capacity and the missions each carried

Vehicle (first flight)	Configuration	Payload capacity	Key missions
SLV-3 (1979)	Four stages, all solid	40 kg to LEO	Rohini RS-1, 18 July 1980; project director A.P.J. Abdul Kalam
ASLV (1987)	Five stages, all solid	150 kg to LEO	SROSS series of experimental satellites
PSLV (1993)	Four stages, solid and liquid alternating	About 1,750 kg to a 600 km sun-synchronous orbit	Chandrayaan-1, Mangalyaan, AstroSat, Aditya-L1, XPoSat, SpaDeX, Cartosat, RISAT

Vehicle (first flight)	Configuration	Payload capacity	Key missions
GSLV Mk-II (2001)	Three stages, indigenous CE-7.5 cryogenic upper stage	About 2,250 kg to GTO	EDUSAT (GSLV-F01), NVS-02 navigation satellite, NISAR (GSLV-F16)
LVM3 / GSLV Mk-III (2014, experimental)	Three stages, CE-20 cryogenic upper stage	4,000 kg to GTO, 8,000 kg to LEO	Chandrayaan-2, Chandrayaan-3, OneWeb constellation, CMS-03, BlueBird Block-2
SSLV (2022)	Three solid stages plus a liquid velocity trimming module	About 500 kg to a 500 km LEO	EOS-08 on SSLV-D3, August 2024
RLV-TD / Pushpak	Winged reusable technology demonstrator	Experimental - no operational payload	HEX-01 in May 2016; LEX autonomous landings at Chitradurga
HLVM3	Human-rated LVM3 with a crew escape system	About 10,000 kg to LEO	Gaganyaan uncrewed and crewed flights

PSLV is called ISRO's workhorse because it flies almost anywhere - polar orbits, transfer orbits, even out to the Moon and Mars. Its flexibility comes from strap-on boosters that can be added or removed, and the variant name simply tells you how many. Learn this; it converts a hard question into an easy one.

PSLV variants - the letters count the strap-ons

Variant	Strap-on boosters	Note
PSLV-G (generic)	6	The original configuration, now retired
PSLV-CA	0 - 'core alone'	Lightest version, used for smaller payloads
PSLV-DL	2	Introduced with PSLV-C44
PSLV-QL	4	Introduced with PSLV-C45
PSLV-XL	6 extended-length boosters	Heaviest version - carried Chandrayaan-1 and Mangalyaan

- PSLV-D1 in 1993 failed; the first fully successful flight was PSLV-D2 in 1994.
- PSLV-C37, 15 February 2017 - world record of 104 satellites in a single flight, primary payload Cartosat-2 series.
- That record was surpassed by SpaceX's Transporter-1 with 143 satellites in January 2021.
- PSLV-C55, April 2023 - launched Singapore's TeLEOS-2 as a dedicated commercial mission for NSIL.
- SSLV-D1, August 2022, failed to reach a stable orbit; SSLV-D2 (February 2023) and SSLV-D3 (August 2024) succeeded.
- SSLV technology has been transferred to HAL; a second launch port for SSLV is coming up at Kulasekarapattinam, Thoothukudi district, Tamil Nadu.
- GSLV-F15 carrying NVS-02, on 29 January 2025, was ISRO's 100th launch from Sriharikota.

PSLV versus LVM3 - the swap RPSC exploits

PSLV	LVM3 / GSLV Mk-III
Four stages, solid and liquid alternating	Three stages, two solid boosters plus a cryogenic upper stage
About 1,750 kg to sun-synchronous orbit	4,000 kg to GTO and 8,000 kg to LEO
Carried Chandrayaan-1 (2008) and Mangalyaan (2013)	Carried Chandrayaan-2 (2019) and Chandrayaan-3 (2023)
Aditya-L1, XPoSat, SpaDeX, AstroSat	OneWeb constellation, CMS-03, BlueBird Block-2
ISRO's workhorse	India's heaviest operational rocket; being human-rated as HLVM3

ASKED IN RAS

Two straight repeats. RAS Pre 2018 - PSLV-C37 launched 104 satellites in February 2017. RAS Pre 2023 - TeLEOS-2 was launched by PSLV-C55. RAS Pre 2023 also gave a mission-to-launch-vehicle match: Chandrayaan-1 to PSLV-C11, Mangalyaan to PSLV-C25, Chandrayaan-3 to LVM3-M4, Aditya-L1 to PSLV-C57. Learn the C-numbers - RPSC puts them in the options.

TRAP

Chandrayaan-1 and Mangalyaan flew on PSLV; Chandrayaan-2 and Chandrayaan-3 flew on GSLV Mk-III / LVM3. Every year somebody swaps these. Second trap in the same family: SLV-3 was all-solid and never had a cryogenic stage - India's first indigenous cryogenic flight was GSLV-D5 in January 2014.

Cryogenic engines and reusability

A cryogenic engine burns liquid hydrogen with liquid oxygen. Both must be held at extremely low temperatures - hydrogen below about minus 253 degrees Celsius. That is the hard part: pumps, seals and tanks that survive those temperatures. The reward is the highest efficiency of any chemical propellant, so cryogenics go in the upper stages. India was denied the technology in the 1990s and built it in-house, which is why the exam keeps asking about it.

- CE-7.5 - indigenous cryogenic engine on GSLV Mk-II.
- CE-20 - the larger cryogenic engine on LVM3.
- GSLV-D5, January 2014 - India's first successful flight with an indigenous cryogenic stage.
- SCE-200 semi-cryogenic engine, burning kerosene with liquid oxygen, is under development.

- RLV-TD is ISRO's winged reusable demonstrator; HEX-01 flew in May 2016 and the Pushpak (LEX) landing experiments were done at the Aeronautical Test Range, Chitradurga, Karnataka.

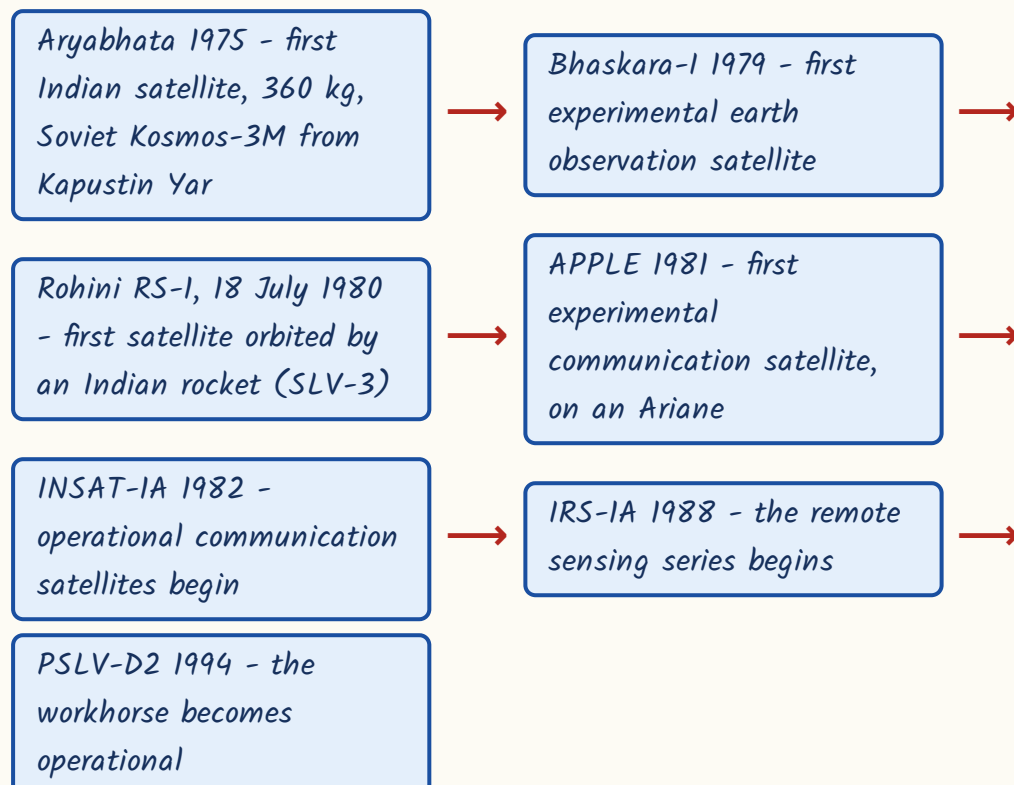
YAAD RAKHO

Cryogenic means liquid hydrogen plus liquid oxygen. Semi-cryogenic means kerosene plus liquid oxygen. Any option that speaks of a 'solid cryogenic engine' is wrong by definition.

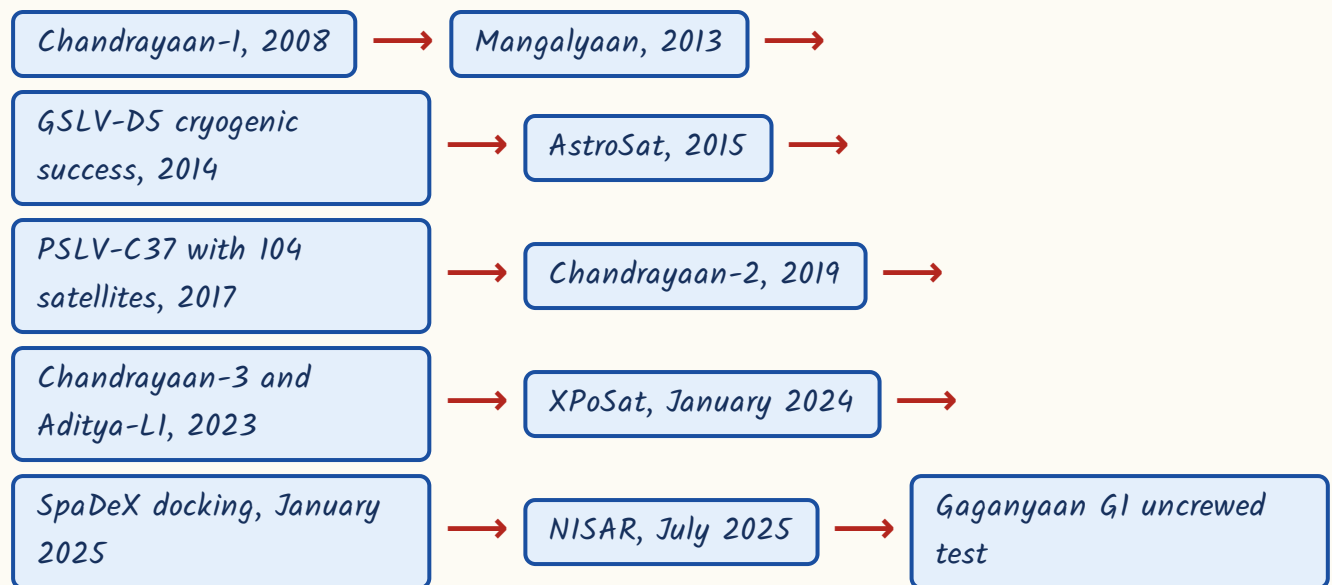
The milestones - lock the chronology

RPSC asks chronological ordering in this chapter. The safe method is to fix a handful of anchor years and place everything else against them: 1975 Aryabhata, 1980 Rohini, 2008 Chandrayaan-1, 2013 Mangalyaan, 2023 Chandrayaan-3.

The foundation years



The modern milestones



- *Aryabhata - India's first satellite, 360 kg, launched 19 April 1975 from Kapustin Yar in the USSR.*
- *Rohini RS-1 - first Indian satellite placed in orbit by an Indian launch vehicle.*
- *APPLE, 1981 - first experimental communication satellite, launched by an Ariane rocket.*
- *Kalpana-1, 2002, originally METSAT - India's first dedicated meteorological satellite, renamed after Kalpana Chawla.*
- *EDUSAT, 2004 on GSLV-F01 - first Indian satellite built exclusively for education.*
- *Remember the trio A-R-A: Aryabhata on a Soviet rocket, Rohini on an Indian rocket, APPLE on a European rocket.*

Deep space and science missions - the highest-yield block

These missions carry the 'firsts' examiners love. For each one hold four things: launch date, launch vehicle, what it achieved, and one distinguishing detail. Chandrayaan-1 (2008) was an orbiter that also dropped a Moon Impact Probe. Its most quoted result did not come from an Indian instrument - NASA's Moon

Mineralogy Mapper, riding on the Indian orbiter, confirmed water molecules on the lunar surface.

- Chandrayaan-1 - October 2008 on PSLV-C11; Moon Impact Probe; confirmed water molecules on the Moon.
- Chandrayaan-2 - 22 July 2019 on GSLV Mk III-M1; the Vikram lander crashed on 7 September 2019; the orbiter still works.
- The Chandrayaan-2 crash site was named Tiranga Point.
- Chandrayaan-3 - 14 July 2023 on LVM3-M4; Vikram soft-landed on 23 August 2023 near the lunar south pole.
- India became the 4th country to soft-land on the Moon and the 1st to land near the lunar south pole.
- The Chandrayaan-3 landing site is Shiv Shakti Point; 23 August is now National Space Day.
- Chandrayaan-3 cost about Rs 615 crore.

ASKED IN RAS

Asked in RAS Pre 2023 - Vikram soft-landed on 23 August 2023. A second standard one-liner: Chandrayaan-1 is chiefly credited with the discovery of water molecules on the lunar surface. Do not attribute that discovery to Chandrayaan-2.

Science missions at a glance

Mission	Launched on	Domain and achievement
Mangalyaan (Mars Orbiter Mission)	5 November 2013, PSLV-C25	Mars orbit insertion 24 September 2014; India the first country to reach Mars orbit on its maiden attempt; cost about Rs 450 crore
AstroSat	2015, PSLV-C30	India's first multi-wavelength space observatory
Aditya-L1	2 September 2023, PSLV-C57	Solar mission; halo orbit around Sun-Earth L1 from 6 January 2024; main payload VELC built by the Indian Institute of Astrophysics

Mission	Launched on	Domain and achievement
XPoSat	1 January 2024, PSLV-C58	World's second X-ray polarimetry mission after NASA's IXPE; payloads POLIX and XSPECT
SpaDeX	30 December 2024, PSLV-C60	SDX01 Chaser and SDX02 Target docked on 16 January 2025 - India the 4th country to dock in space after the USA, Russia and China
NISAR	30 July 2025, GSLV-F16	ISRO-NASA dual-band synthetic aperture radar - NASA L-band plus ISRO S-band, in a sun-synchronous orbit

ASKED IN RAS

Asked in RAS Pre 2016 - Mangalyaan entered Mars orbit on 24 September 2014. Carry both dates: launch 5 November 2013, insertion 24 September 2014. Match the mission to what it looks at - AstroSat many wavelengths, Aditya-L1 the Sun, XPoSat X-ray polarisation, NISAR radar imaging of the Earth, SpaDeX docking.

Application satellites, remote sensing and GIS

This is the part of the programme that pays for itself. Communication satellites carry television, telephony and disaster warnings. Earth observation satellites survey crops, water, forests and coastlines. Navigation satellites give position. RPSC asks these as series-to-purpose matches, so learn the family names and one line each.

The satellite families and what each does

Series	Orbit	Principal application
INSAT and GSAT	Geostationary	Communication, broadcasting and meteorology
IRS - Indian Remote Sensing	Sun-synchronous polar	Earth observation; the series began with IRS-1A in 1988

Series	Orbit	Principal application
Cartosat	Sun-synchronous polar	High-resolution cartographic mapping; Cartosat-3 gives about 0.25 m panchromatic resolution
Resourcesat	Sun-synchronous polar	Land use, agriculture and natural resource survey
Oceansat	Sun-synchronous polar	Ocean colour and ocean surface wind studies
RISAT	Sun-synchronous polar	All-weather synthetic aperture radar imaging - sees through cloud and at night
EOS series	Mostly sun-synchronous	Renamed earth observation satellites, for example EOS-08 on SSLV-D3
NavIC / IRNSS	Geosynchronous and geostationary	Regional navigation - designed as a constellation of 7 satellites
Kalpana-1, 2002	Geostationary	First dedicated Indian meteorological satellite, originally METSAT
EDUSAT, 2004	Geostationary	First Indian satellite exclusively for education

- NavIC, earlier IRNSS, covers India and about 1,500 km beyond India's borders - regional, not global.
- GAGAN is a satellite-based augmentation system built jointly by ISRO and the Airports Authority of India to sharpen GPS accuracy for civil aviation. NavIC and GAGAN are two different systems.
- Remote sensing means recording the electromagnetic radiation an object reflects or emits, without touching it.
- Passive sensors read reflected sunlight or emitted heat; active sensors such as radar send their own signal and read the echo - which is why RISAT works through cloud and at night.
- Spatial resolution is the smallest object distinguishable; spectral resolution is the number and width of bands; temporal resolution is the revisit interval.
- GIS - a Geographic Information System - stores map layers and analyses the relations between them; it is the software that turns imagery into a decision.

- Indian uses: crop forecasting, groundwater prospecting, forest cover mapping, urban planning, disaster damage assessment, and the Bhuvan geoportal.

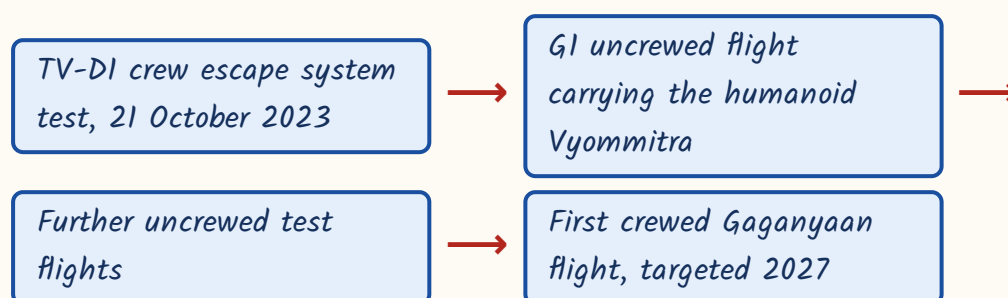
TRAP

Kalpana-1 is meteorological, not navigational - navigation is NavIC / IRNSS. RPSC has used exactly this NOT-correctly-matched pair. Second confusion in the same set: Cartosat is mapping, RISAT is radar, Oceansat is ocean colour.

Gaganyaan and human spaceflight

Gaganyaan is India's crewed spaceflight programme. Three crew, an orbit of about 400 km, a mission of up to seven days, and a safe splashdown. The launcher is HLV M3 - the LVM3 rebuilt to human-rating standards and fitted with a crew escape system that can pull the crew module clear of a failing rocket. Approved outlay about Rs 20,193 crore; the programme is run from the Human Space Flight Centre, Bengaluru.

The Gaganyaan sequence



CURRENT LINK

Current link - as of September 2026 the uncrewed Gaganyaan G1 mission had not yet flown; ISRO has been targeting the second half of 2026. The crew module completed an air-drop test from an Indian Air Force Chinook in April 2026, and Vyommitra was being integrated into the crew module. Expect a question on the launcher name (HLVM3) and on Vyommitra.

Two Indians have been to space. Rakesh Sharma flew aboard Soyuz T-11 in 1984. Group Captain Shubhanshu Shukla flew the Axiom-4 mission to the International Space Station in June 2025 - the second Indian in space and the first Indian to visit the ISS.

- The Union Cabinet cleared a long-term package on 18 September 2024 - learn the three names, costs and target years together.
- Chandrayaan-4 - lunar sample return mission, about Rs 2,104 crore.
- Venus Orbiter Mission (Shukrayaan) - about Rs 1,236 crore, targeted for 2028.
- Bharatiya Antariksh Station - first module by 2028, the station complete by 2035.
- Indian crewed landing on the Moon - target 2040.

Space policy, private players, debris and the world outside

Until 2020, space in India meant ISRO and nothing else. The Indian Space Policy 2023 changed the architecture: ISRO concentrates on research and advanced missions, while private companies build and fly. Three names carry the new structure, and RPSC has already begun asking about them.

Who does what after the 2023 policy

Body	Set up	Role
IN-SPACe	2020, headquartered at Ahmedabad	Single-window autonomous agency that authorises and regulates non-government space activity
NSIL - NewSpace India Limited	2019, Bengaluru	Commercial PSU arm - owns and sells launches and satellite services
Antrix Corporation	1992	The earlier marketing arm of ISRO - not to be confused with NSIL

The private players you should be able to name

Company	Product	Distinction
Skyroot Aerospace	Vikram-S sounding rocket	Mission Prarambh, 18 November 2022 - India's first privately built rocket
Agnikul Cosmos	Agnibaan SOrTeD	May 2024 - the world's first single-piece 3D-printed semi-cryogenic engine
Pixxel	Hyperspectral imaging satellites	Commercial earth observation
Dhruva Space	Satellite orbital deployers	Small satellite platforms and deployment hardware

- Every dead satellite and spent rocket stage stays in orbit; at orbital speed even a fleck of paint can hole a spacecraft, and one collision creates thousands more fragments.
- Project NETRA is ISRO's space object tracking initiative.
- IS40M, at Bengaluru, is the ISRO System for Safe and Sustainable Space Operations Management.
- ISRO's stated goal is debris-free space missions by 2030.

International programmes - the minimum you must know

Agency / programme	Country	Key fact
NASA - Artemis	United States	Artemis-1 flew uncrewed around the Moon in November-December 2022
James Webb Space Telescope	NASA with ESA and CSA	Launched December 2021 on an Ariane-5; operates from the Sun-Earth L2 point
International Space Station	USA, Russia, ESA, Japan, Canada	In low Earth orbit; Shubhanshu Shukla visited on Axiom-4 in 2025
CNSA - Chang'e programme	China	Chang'e-6 returned the first samples from the far side of the Moon in June 2024
Roscosmos	Russia	Launched Aryabhata in 1975 and Rakesh Sharma on Soyuz T-11 in 1984

Agency / programme	Country	Key fact
JAXA	Japan	The Japanese space agency
CNES	France	The French space agency - Canada's is the Canadian Space Agency, CSA
SpaceX - Falcon 9, Starship	United States, private	Transporter-1 set the 143-satellite record in January 2021

TRAP

CNES is French, not Canadian. Antrix is 1992, NSIL is 2019. IN-SPACe regulates and authorises; NSIL sells. These three swaps carry almost every NOT-correctly-matched question in this section.

CURRENT LINK

Current link, and read the wording carefully. CMS-03, launched by LVM3-M5 on 2 November 2025 and weighing about 4,410 kg, was the heaviest satellite launched from Indian soil AT THE TIME. It was surpassed on 24 December 2025, when LVM3-M6 placed the roughly 6,100 kg BlueBird Block-2 satellite in orbit for AST SpaceMobile. So an unqualified option saying CMS-03 is 'the heaviest ever launched from India' is wrong - as of September 2026 the holder is BlueBird Block-2.

REVISION RECAP

1. Escape velocity 11.2 km/s, orbital velocity about 7.9 km/s, GEO at 35,786 km with a 24-hour period.
2. Sun-synchronous polar orbit, inclination about 98 degrees - used for remote sensing because the local solar time stays constant.
3. Staging works by shedding dead mass, not by reducing drag.
4. INCOSPAR 1962, ISRO 15 August 1969, Department of Space 1972; Sarabhai the father, Satish Dhawan chairman 1972-84, V. Narayanan chairman from January 2025.
5. Thumba TERLS - Nike-Apache sounding rocket, 21 November 1963, site chosen for the magnetic equator.

6. VSSC Thiruvananthapuram, SAC Ahmedabad, NRSC Hyderabad, URSC Bengaluru, IPRC Mahendragiri, MCF Hassan, IIST Valiamala.
7. Udaipur Solar Observatory - island in Fateh Sagar Lake, founded 1976 by Arvind Bhatnagar, run by PRL, 50 cm MAST since 2015.
8. Payloads: SLV-3 40 kg, ASLV 150 kg, PSLV about 1,750 kg to SSO, GSLV Mk-II about 2,250 kg to GTO, LVM3 4,000 kg to GTO, SSLV about 500 kg to LEO, HLV3 about 10,000 kg to LEO.
9. PSLV variant names count the strap-ons - CA zero, DL two, QL four, XL six.
10. PSLV-C37 put up 104 satellites on 15 February 2017; beaten by SpaceX Transporter-1 with 143 in January 2021.
11. Launch pairs: Chandrayaan-1 PSLV-C11, Mangalyaan PSLV-C25, AstroSat PSLV-C30, Aditya-L1 PSLV-C57, XPoSat PSLV-C58, SpaDeX PSLV-C60, Chandrayaan-3 LVM3-M4, NISAR GSLV-F16.
12. CE-7.5 on GSLV Mk-II, CE-20 on LVM3; the first indigenous cryogenic success was GSLV-DS in January 2014.
13. Firsts: Aryabhata 1975 Soviet rocket, Rohini RS-1 1980 SLV-3, APPLE 1981 Ariane, Kalpana-1 2002 weather, EDUSAT 2004 education.
14. Chandrayaan-3 landed on 23 August 2023 at Shiv Shakti Point - 4th nation to soft-land, 1st near the lunar south pole; 23 August is National Space Day.
15. Mangalyaan reached Mars orbit on 24 September 2014 at the first attempt, at about Rs 450 crore.
16. SpaDeX docked on 16 January 2025, making India the 4th docking nation; NISAR is the ISRO-NASA dual-band radar mission of July 2025.
17. NavIC is a 7-satellite regional system covering India plus 1,500 km; GAGAN is the ISRO-AAI augmentation system for aviation.
18. Gaganyaan - Rs 20,193 crore, HLV3 launcher, 3 crew, about 400 km, up to 7 days, Vyommitra on the uncrewed flight, crewed flight targeted 2027.
19. Cabinet of 18 September 2024 - Chandrayaan-4 sample return, Venus Orbiter for 2028, Bharatiya Antariksh Station complete by 2035, crewed Moon landing 2040.
20. IN-SPACe authorises from Ahmedabad, NSIL sells since 2019, Antrix was the 1992 marketing arm; Vikram-S was India's first private rocket, November 2022.

STII

Defence Technology

About 2-4 questions a paper. DRDO facts and the exercise-to-country match are the two near-certain items; missile ranges and new commissionings supply the rest.

DRDO and its products have appeared in every RAS paper since 2016 - the founding year, the motto and Lukoskin are documented repeats. Around them sit the missiles with their ranges, the platforms of the three services, the nuclear doctrine, and the joint exercises matched to their partner countries.

India's Defence Technology

DRDO

- Founded 1 January 1958
- Balasaya Mulam Vigyanam
- About 52 laboratories
- Lukoskin, 2-DG, rations
- Defence Laboratory Jodhpur

Missiles

- IGMDP 1983 - Prithvi, Agni, Trishul, Akash, Nag
- BrahMos, Astra, Nirbhay
- QRSAM, VSHORADS, Pinaka, Pralay
- K-series SLBM
- Agni-V MIRV, HSTDV

Air

- Tejas LCA, AMCA
- Rafale, Su-30MKI
- LCH Prachand, Dhruv
- TAPAS-BH, Netra AEW&C

Sea

- INS Vikrant, Vikramaditya
- Arihant, Arighaat SSBN
- Kalvari class, Project 75
- Project 15B and 17A

Land

- Arjun MBT, Zorawar
- ATAGS, Dhanush, K-9 Vajra
- Pinaka MBRL
- AK-203 at Korwa

Nuclear

- Pokhran-I 1974
- Pokhran-II 1998
- Doctrine 2003 - No First Use
- Triad completed 2018
- MTCR, Wassenaar, Australia Group

Structure and industry

- CDS and DMA 2019
- Andaman and Nicobar Command 2001
- Agnipath 2022
- OFB into 7 DPSUs 2021
- Defence corridors UP and Tamil Nadu
- Joint exercises

DRDO - the organisation RPSC asks about every year

DRDO is the research and development wing of the Ministry of Defence. Its job is not to manufacture in bulk - that is for the DPSUs and private industry - but to design systems and hand the technology over. It began small and became the largest research network in the country. These founding facts are

documented repeats: RPSC has asked the year, the motto and one product in paper after paper.

- Founded on 1 January 1958 by merging the Technical Development Establishment, the Directorate of Technical Development and Production, and the Defence Science Organisation.
- Started with 10 laboratories; today about 52.
- Motto: 'Balasya Mulam Vigyanam' (बलस्य मूलं विज्ञानम्) - 'Strength's origin is in Science', drawn from Kautilya's Arthashastra.
- Headquarters: DRDO Bhawan, New Delhi.
- Works under the Department of Defence Research and Development in the Ministry of Defence.
- Its head is Secretary, Department of Defence R&D and Chairman, DRDO.
- Five DRDO Young Scientist Laboratories were opened in 2020 - artificial intelligence, quantum technologies, cognitive technologies, asymmetric technologies and smart materials.
- Aero India, the biennial air show, is held at Yelahanka, Bengaluru.

ASKED IN RAS

Two near-certain repeats. RAS Pre 2024 asked the year DRDO was established - 1958. RAS Pre 2018 asked the motto - Balasya Mulam Vigyanam. If you learn nothing else in this chapter, learn these two.

Now the laboratories. The pattern to hold is city-to-subject, because that is how the match-the-following is framed. Hyderabad means missiles, Bengaluru means aeronautics and radar, Pune means armaments, Visakhapatnam and Kochi mean underwater, Mysuru means food.

DRDO laboratories and their specialities

Laboratory	Location	Area of work
DRDL, RCI and ASL	Hyderabad	Missile design, guidance and advanced systems
ADE - Aeronautical Development Establishment	Bengaluru	Unmanned aerial vehicles - TAPAS, Nishant, Lakshya
GTRE - Gas Turbine Research Establishment	Bengaluru	Aero gas turbine engines - the Kaveri engine
LRDE - Electronics and Radar Development Establishment	Bengaluru	Radars
CABS - Centre for Airborne Systems	Bengaluru	Netra Airborne Early Warning and Control
CVRDE - Combat Vehicles R&D Establishment	Avadi, Chennai	Combat vehicles - the Arjun main battle tank
ARDE and HEMRL	Pune	Armaments and explosives - Pinaka, ATAGS
TBRL - Terminal Ballistics Research Laboratory	Chandigarh	Terminal ballistics and warheads
NPOL - Naval Physical and Oceanographic Laboratory	Kochi	Sonar and underwater sensing
NSTL - Naval Science and Technological Laboratory	Visakhapatnam	Torpedoes - Varunastra
DFRL - Defence Food Research Laboratory	Mysuru	Food technology and combat rations
INMAS, DIPAS and DIPR	Delhi	Nuclear medicine, physiology and psychological research
DRDE	Gwalior	Chemical and biological defence
DIHAR, SASE and DIBER	Leh, Chandigarh and Haldwani	High-altitude agriculture, snow and avalanche study, bio-energy

Rajasthan's own DRDO laboratory is the Defence Laboratory Jodhpur (DLJ). It works on camouflage, desert environment sciences, isotopes and radiation-

detection kits - all of which follow from where it sits. Expect this in a Rajasthan-flavoured science question.

DRDO beyond weapons - the life-science products

DRDO also works on keeping soldiers alive and healthy in extreme places, and those spin-offs reach civilians. This is the corner of the syllabus RPSC likes precisely because candidates skip it. Lukoskin has been asked directly.

- Lukoskin - a polyherbal drug for leucoderma (vitiligo), developed by DIBER, Haldwani, from Himalayan herbs; technology transferred to Aimil Pharmaceuticals.
- 2-deoxy-D-glucose (2-DG) - anti-COVID-19 drug developed by INMAS, Delhi, with Dr Reddy's Laboratories in 2021.
- Oxycare - an oxygen delivery system developed by DEBEL.
- Retort-pouch rations and extreme-climate food products from DFRL, Mysuru.
- Seabuckthorn-based beverages from DIHAR, Leh.
- Varunastra heavyweight torpedo from NSTL, Visakhapatnam - remember the lab, it is a common match.

ASKED IN RAS

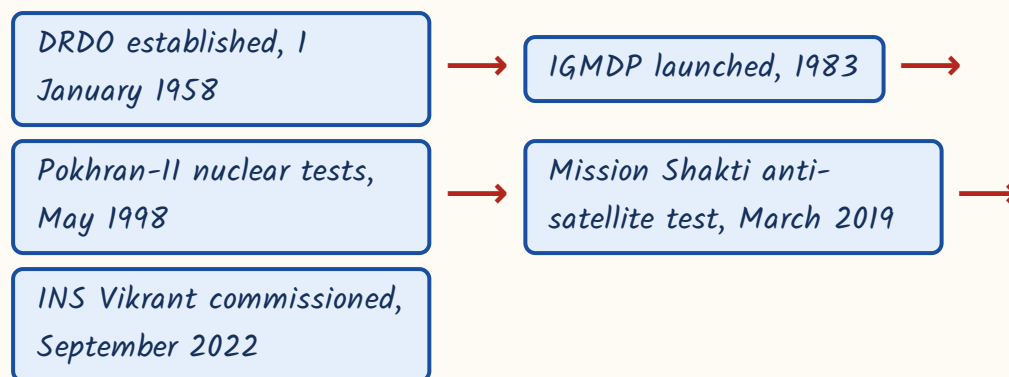
Asked in RAS Pre 2016 - Lukoskin is used in the treatment of leucoderma (vitiligo). Note the lab too: DIBER, Haldwani. RPSC has framed the same fact as both a direct question and a match.

IGMDP and the missile ladder

The Integrated Guided Missile Development Programme was launched in 1983 under A.P.J. Abdul Kalam. The idea was to build five missiles of five different

kinds at once, so that India would acquire the whole set of skills rather than one. It was formally concluded in 2008, its objectives met. Learn the five names as a closed set - the exam's favourite trick is to slip a sixth name into the list.

Defence technology chronology - fix these five anchors



Missiles and their approximate ranges

Missile	Type	Approximate range	Note
Prithvi-I	Surface-to-surface ballistic	150 km	IGMDP; Army version
Prithvi-II	Surface-to-surface ballistic	250 - 350 km	Air Force version
Agni-I	Short-range ballistic	About 700 - 900 km	IGMDP
Agni-II	Intermediate-range ballistic	2,000+ km	
Agni-III	Intermediate-range ballistic	3,000+ km	
Agni-IV	Intermediate-range ballistic	About 4,000 km	
Agni-V	Intercontinental class	5,000+ km	MIRV tested as Mission Divyasthra, 11 March 2024
Agni-Prime	Short to medium range	1,000 - 2,000 km	Canisterised, newer generation

Missile	Type	Approximate range	Note
Trishul	Short-range surface-to-air	About 9 km	IGMDP
Akash	Medium-range surface-to-air	25 - 30 km	Akash-NG about 70 km
Nag	Anti-tank guided missile	About 4 km	Fire-and-forget; IGMDP
QRSAM	Quick-reaction surface-to-air	About 30 km	For mobile formations
VSHORADS / MANPADS	Man-portable air defence	About 6 km	Shoulder-fired against low-flying aircraft
Astra Mk-I	Beyond-visual-range air-to-air	100+ km	India's first indigenous BVR air-to-air missile

ASKED IN RAS

Asked in RAS Pre 2016 - which of these missiles was NOT a part of IGMDP? Answer: BrahMos. The examiner puts BrahMos among four names and waits for you to accept it as one of the five.

ASKED IN RAS

Two more straight repeats from this table. RAS Pre 2021 - Astra is India's first indigenous beyond-visual-range air-to-air missile. RAS Pre 2023 - MANPADS stands for a man-portable air defence system.

YAAD RAKHO

The IGMDP five, as one word to remember - P-A-T-N-A: Prithvi, Agni, Trishul, Nag, Akash. Range ladder for Agni: I about 900 km, II 2,000, III 3,000, IV 4,000, V 5,000 - roughly a thousand more at each step from Agni-II onwards.

TRAP

BrahMos is NOT part of IGMDP - it is a separate India-Russia joint venture of 1998. Also do not confuse the types: Prithvi is surface-to-surface ballistic, Akash is surface-to-air, Nag is anti-tank, Astra is air-to-air. RPSC matches these four against each other almost every time.

Cruise missiles, hypersonics and missile defence

A ballistic missile is thrown - it burns its engine early, arcs high above the atmosphere and falls on the target under gravity. A cruise missile flies - it stays low, keeps its engine running, and steers all the way. That single distinction answers a large family of questions, so hold it clearly.

Ballistic missile versus cruise missile

Ballistic missile	Cruise missile
Powered only in the initial phase, then follows a ballistic arc	Powered and guided throughout the flight
Travels high, largely outside the atmosphere	Flies low, hugging the terrain to avoid radar
Very long range - Agni-V exceeds 5,000 km	Shorter range - BrahMos about 290 km originally, Nirbhay about 1,000 km
Carries larger warheads, mainly strategic	High accuracy, mainly tactical and precision strike
Examples: Prithvi, Agni series, K-series	Examples: BrahMos (supersonic), Nirbhay (subsonic)

The systems beyond IGMDP

System	Category	Key facts
BrahMos	Supersonic cruise missile	India-Russia joint venture of 1998, named after the Brahmaputra and Moskva rivers; Mach 2.8 to 3, the world's fastest operational supersonic cruise missile; range extended beyond the original 290 km after India joined MTCR; the Philippines was the first export customer, contracted 2022, delivered 2024
Nirbhay	Subsonic cruise missile	Range about 1,000 km
Pinaka	Multi-barrel rocket launcher	Developed by ARDE, Pune; Mk-I about 38 km, enhanced about 45 km, guided Pinaka 75+ km; used in the 1999 Kargil conflict
Pralay	Quasi-ballistic surface-to-surface missile	150 - 500 km
K-15 Sagarika	Submarine-launched ballistic missile	About 750 km - the sea leg of the nuclear triad
K-4	Submarine-launched ballistic missile	About 3,500 km, carried by Arihant-class boats
HSTDV	Hypersonic technology demonstrator	Scramjet propulsion, flight tested 7 September 2020 at about Mach 6
LRAShM	Long-range hypersonic missile	Tested from Dr A.P.J. Abdul Kalam Island in November 2024

- Ballistic Missile Defence Phase-I has two interceptors: the exo-atmospheric Prithvi Air Defence, which kills outside the atmosphere, and the endo-atmospheric Advanced Air Defence, which kills inside it.
- Mission Shakti, 27 March 2019, destroyed a satellite at about 283 km, making India the 4th anti-satellite power.
- A scramjet is a supersonic-combustion ramjet - air enters at supersonic speed and the fuel burns without slowing it to subsonic. No moving

compressor, which is why it works only at very high speed.

Air power – aircraft, helicopters and unmanned systems

For the exam, an aircraft question is almost always a platform-to-category match. So fix the category first – fighter, transport, helicopter, UAV, early-warning aircraft – and only then the details of who designs and who builds. Tejas is the standard example of the split: designed by the Aeronautical Development Agency, manufactured by Hindustan Aeronautics Limited.

Aerial platforms – what each one is

Platform	Category	Key facts
Tejas LCA Mk-1 / Mk-1A / Mk-2	Light combat aircraft	Designed by ADA, built by HAL; IOC 2013, FOC 2019; Mk-1A powered by the GE F404; the larger Mk-2 is under development
AMCA	Fifth-generation stealth fighter	India's fifth-generation programme, approved by the Cabinet Committee on Security in March 2024
Su-30MKI	Heavy multirole fighter	Licence-built by HAL at Nashik
Rafale	Multirole fighter	36 aircraft from Dassault; first squadron No. 17 'Golden Arrows' at Ambala, 2020; 26 Rafale-M for the Navy contracted in 2025
C-295	Medium transport aircraft	Assembled at Vadodara by Airbus with Tata
C-17 Globemaster and C-130J	Heavy and special-operations transport	Imported strategic airlift
Dhruv ALH and Rudra	Advanced light helicopter and its armed version	HAL
LCH Prachand	Light combat helicopter	Inducted October 2022; the first unit was raised at Jodhpur
HTT-40	Basic trainer aircraft	HAL

Platform	Category	Key facts
TAPAS-BH	Medium-altitude long-endurance UAV	ADE, Bengaluru; earlier called Rustom-II
Nishant and Lakshya	Tactical UAV and pilotless target aircraft	ADE, Bengaluru
SWiFT / Ghatak	Stealth unmanned combat air vehicle demonstrator	DRDO
Netra AEW&C	Airborne early warning and control	Built by CABS; flies on an Embraer ERJ-145
Phalcon AWACS	Airborne warning and control system	Mounted on an IL-76 airframe

- The MiG-21 was retired from Indian Air Force service in September 2025 after more than six decades.
- Netra is the smaller Indian AEW&C on an Embraer; Phalcon is the larger Israeli-origin AWACS on an IL-76. Do not swap the airframes.
- The Kaveri engine, from GTRE Bengaluru, was intended for Tejas but Tejas flies on the GE F404 instead.

TRAP

Prachand is an attack helicopter, not a trainer - the trainer is HTT-40. Dhruv is a utility helicopter and Rudra its armed version. RPSC's NOT-correctly-matched question in this section is almost always built on one of these three.

Sea power - the Navy's platforms

Naval questions are current-affairs questions in disguise. A ship is asked about in the year it is commissioned, and the hook is always a 'first' - first

indigenously designed carrier, first indigenous diving support vessel, last warship imported from abroad. Learn the ship, the year and the superlative together.

Ships and submarines you must be able to place

Ship	Type	Commissioned	Why it is asked
INS Vikrant	Aircraft carrier	2 September 2022	India's first indigenously designed and built aircraft carrier; Cochin Shipyard; about 45,000 tonnes
INS Vikramaditya	Aircraft carrier	2013	Acquired from Russia - not indigenous
INS Arihant	Nuclear ballistic missile submarine	2016	India's first indigenous SSBN; completed the nuclear triad with its first deterrence patrol in 2018
INS Arighaat	Nuclear ballistic missile submarine	2024	Second indigenous SSBN
INS Vagsheer	Kalvari-class (Scorpene) diesel-electric submarine	January 2025	Sixth and last boat of Project 75
INS Nilgiri	Project 17A stealth frigate	January 2025	Lead ship of the class
INS Surat	Project 15B destroyer	January 2025	Visakhapatnam-class guided missile destroyer
INS Arnala	Anti-submarine shallow water craft	June 2025	First of its class
INS Tamal	Talwar-class frigate	July 2025	The last warship imported from abroad
INS Nistar	Diving support vessel	July 2025	India's first indigenously designed and built diving support vessel
INS Himgiri and INS Udaygiri	Project 17A stealth frigates	26 August 2025	Commissioned together

- ***Project 75 - the six Kalvari-class (Scorpene) conventional submarines, built at Mazagon Dock.***

- Project 15B - Visakhapatnam-class destroyers; Project 17A - Nilgiri-class stealth frigates.
- An SSBN is a nuclear-powered submarine carrying ballistic missiles; a conventional Kalvari boat is diesel-electric and cannot stay submerged as long.

CURRENT LINK

Current link - as of September 2026 the run of indigenous commissionings from 2025 (Nilgiri, Surat, Vagsheer, Arnala, Nistar, Himgiri, Udaygiri) is the freshest material in this chapter. If a naval question appears in RAS 2026, it will most likely be built on one of these 'firsts', or on INS Tamal as the last imported warship.

Land systems - the Army's equipment

Group the Army's equipment by function: tanks, artillery, rockets, small arms. Then attach the developing laboratory or the manufacturer, because that is the second half of every match question here.

Army systems, their category and their maker

System	Category	Developed or made by	Note
Arjun	Main battle tank	CVRDE, Avadi	India's indigenous MBT
Zorawar	Light tank	DRDO with Larsen and Toubro	For high-altitude terrain
ATAGS	155 mm / 52 calibre towed howitzer	ARDE, Pune	Range about 48 km
Dhanush	155 mm / 45 calibre towed howitzer	Ordnance factories, now AWEIL	Called the 'Desi Bofors'
K-9 Vajra-T	Tracked self-propelled gun	L&T with Hanwha	Mobile artillery for desert and plains

System	Category	Developed or made by	Note
M777	Ultra-light howitzer	Imported from BAE Systems	Air-portable, for mountain formations
Pinaka	Multi-barrel rocket launcher	ARDE, Pune	Used in the Kargil conflict, 1999
AK-203	Assault rifle	Indo-Russian joint venture at Korwa, Amethi	Replacing the INSAS rifle

- INSAS was the earlier indigenous rifle family; AK-203 is its replacement, made in India at Korwa in Amethi district, Uttar Pradesh.
- Varunastra (torpedo, NSTL) and Arjun (tank, CVRDE) are the two lab-to-product pairs RPSC has actually used.
- Note the calibre difference: ATAGS is 155 mm / 52 cal, Dhanush is 155 mm / 45 cal. The calibre number after the slash is barrel length in calibres, and a longer barrel means longer range.

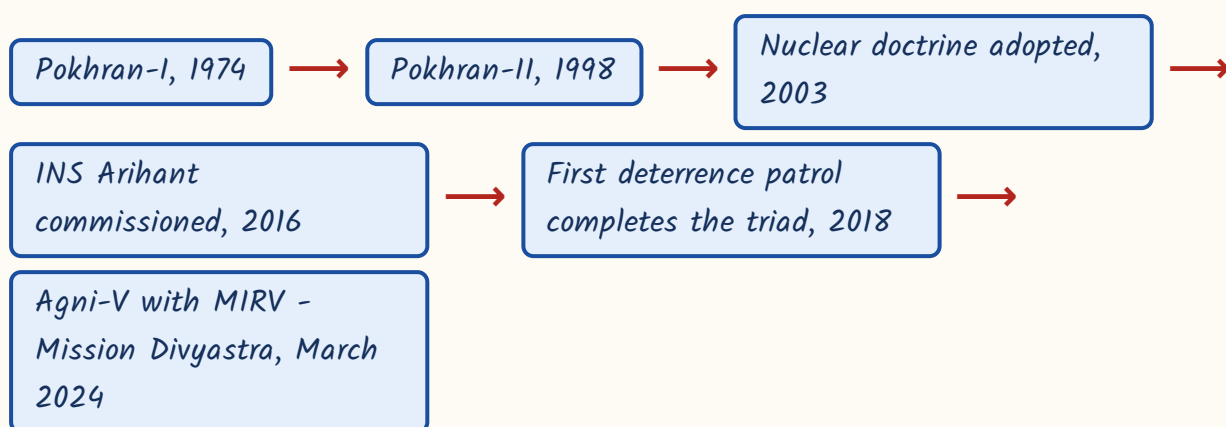
Nuclear India - tests, doctrine, triad and the export-control regimes

India's nuclear story has a Rajasthan address. Both test series were conducted at Pokhran, in Jaisalmer district. Pokhran-I in 1974 was called a 'peaceful nuclear explosion'; Pokhran-II in 1998 was an open declaration of weapon capability, and it brought sanctions that shaped the next two decades of Indian technology policy.

Pokhran-I versus Pokhran-II

Pokhran-I - Smiling Buddha	Pokhran-II - Operation Shakti
18 May 1974	11 and 13 May 1998
A single device, described as a peaceful nuclear explosion	Five devices, including a thermonuclear design
Prime Minister Indira Gandhi	Prime Minister Atal Bihari Vajpayee
Led to the creation of the Nuclear Suppliers Group in 1975	11 May is now observed as National Technology Day
No formal doctrine followed	Doctrine formally adopted in 2003

How the nuclear triad was completed



- The triad means the ability to deliver nuclear weapons from land, air and sea. India's legs are the Agni missiles, aircraft such as the Mirage-2000 and Su-30MKI, and the K-series missiles on Arihant-class submarines.
- Doctrine of 2003 - No First Use; credible minimum deterrence; massive retaliation if attacked; no use against non-nuclear weapon states.
- The Strategic Forces Command, created in 2003, manages the nuclear arsenal.
- Mission Divyastra, 11 March 2024, was the first Indian flight test of Agni-5 with Multiple Independently Targetable Re-entry Vehicles - one missile, several warheads aimed at different targets.

Export-control regimes - what India is in and what it is not

Regime	India's status	Year
MTCR - Missile Technology Control Regime	Member	2016
Wassenaar Arrangement	Member	2017
Australia Group	Member	2018
NSG - Nuclear Suppliers Group	NOT a member	-
NPT - Non-Proliferation Treaty	Not signed	-
CTBT - Comprehensive Test Ban Treaty	Not signed	-

TRAP

India is in three of the four regimes but NOT in the Nuclear Suppliers Group, and has signed neither the NPT nor the CTBT. RPSC's statement-based question here usually plants NSG membership as the false statement. Second confusion: Pokhran-I was 1974 under Indira Gandhi, Pokhran-II 1998 under Vajpayee - National Technology Day, 11 May, commemorates the 1998 tests, not 1974.

Defence production, exports, joint exercises and the tri-services structure

The last decade's story is indigenisation - build at home, then sell abroad. Three structural changes carry it: the ordnance factories were corporatised, two defence corridors were created, and start-ups were brought in through iDEX. The numbers move every year, so carry the latest figure with a date attached.

- The Ordnance Factory Board was dissolved on 1 October 2021 and its units reorganised into 7 new DPSUs - Munitions India, Armoured Vehicles Nigam (AVANI), Advanced Weapons and Equipment India, Troop Comforts, Yantra India, India Optel and Gliders India.

- Two defence industrial corridors were announced in Budget 2018-19 - Uttar Pradesh (Aligarh, Agra, Jhansi, Kanpur, Chitrakoot, Lucknow) and Tamil Nadu (Chennai, Coimbatore, Hosur, Salem, Tiruchirappalli).
- iDEX (2018) engages start-ups and MSMEs through the Defence Innovation Organisation.
- The SRIJAN portal and the Positive Indigenisation Lists support Atmanirbharta by barring the import of listed items after fixed dates.
- Defence exports reached about Rs 23,622 crore in 2024-25, against a target of Rs 50,000 crore by 2029.

CURRENT LINK

Current link - India's defence exports rose to an all-time high of about Rs 38,424 crore in 2025-26, up roughly 63 per cent on the previous year's Rs 23,622 crore, with DPSUs contributing about 54.8 per cent and the private sector about 45.2 per cent (Ministry of Defence figures released April 2026). If RPSC asks a defence-export number in 2026, this is the one to carry - always with the financial year attached.

Now the joint exercises. This is asked almost every year, and it is pure memorisation, so treat it as a vocabulary list. Two tables below; learn the exercise name to the country, and note which service it belongs to, because RPSC sometimes asks that too.

Joint military exercises - army and bilateral (part I)

Exercise	Partner country	Service
Yudh Abhyas	United States of America	Army
Vajra Prahar	United States of America	Special Forces
Indra	Russia	Bilateral, services-wise
Surya Kiran	Nepal	Army
Sampriti	Bangladesh	Army

Exercise	Partner country	Service
<i>Mitra Shakti</i>	<i>Sri Lanka</i>	<i>Army</i>
<i>Ekuverin</i>	<i>Maldives</i>	<i>Army</i>
<i>Dustlik</i>	<i>Uzbekistan</i>	<i>Army</i>
<i>Nomadic Elephant</i>	<i>Mongolia</i>	<i>Army</i>
<i>Al Nagah</i>	<i>Oman</i>	<i>Army</i>
<i>Cyclone</i>	<i>Egypt</i>	<i>Special Forces</i>
<i>Maitree</i>	<i>Thailand</i>	<i>Army</i>
<i>Dharma Guardian</i>	<i>Japan</i>	<i>Army</i>
<i>Harimau Shakti</i>	<i>Malaysia</i>	<i>Army</i>

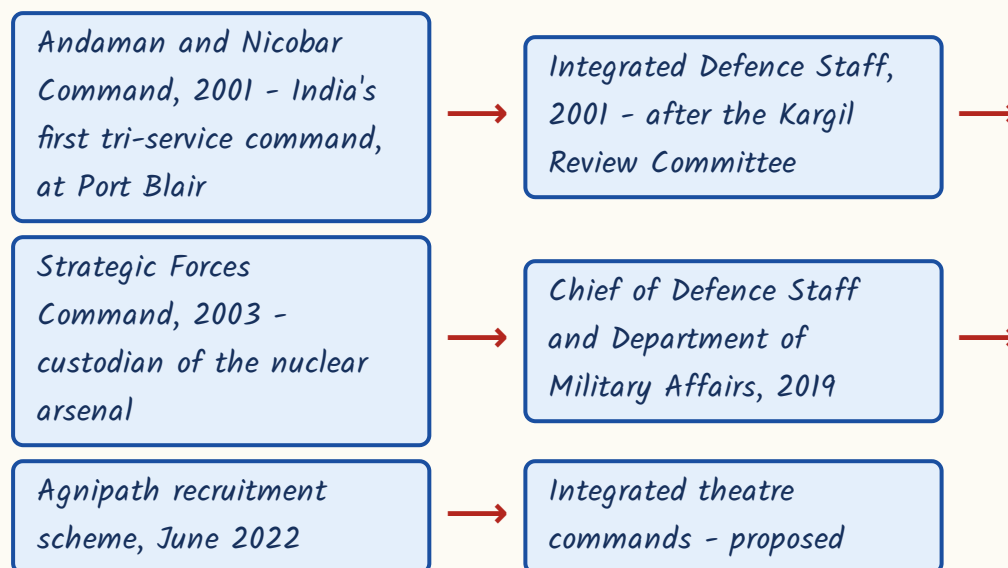
Joint military exercises – naval, air and multilateral (part 2)

Exercise	Partner country	Service
<i>Malabar</i>	<i>USA, Japan and Australia</i>	<i>Navy</i>
<i>Varuna</i>	<i>France</i>	<i>Navy</i>
<i>Garuda</i>	<i>France</i>	<i>Air Force</i>
<i>Shakti</i>	<i>France</i>	<i>Army</i>
<i>Konkan</i>	<i>United Kingdom</i>	<i>Navy</i>
<i>Ajeya Warrior</i>	<i>United Kingdom</i>	<i>Army</i>
<i>SIMBEX</i>	<i>Singapore</i>	<i>Navy</i>
<i>SLINEX</i>	<i>Sri Lanka</i>	<i>Navy</i>
<i>JIMEX</i>	<i>Japan</i>	<i>Navy</i>
<i>AUSINDEX</i>	<i>Australia</i>	<i>Navy</i>
<i>AUSTRA HIND</i>	<i>Australia</i>	<i>Army</i>
<i>Cope India</i>	<i>United States of America</i>	<i>Air Force</i>
<i>Al Mohed Al Hindi</i>	<i>Saudi Arabia</i>	<i>Navy</i>
<i>Samudra Shakti</i>	<i>Indonesia</i>	<i>Navy</i>

TRAP

Four swaps carry almost every question in this section. France has three exercises - Varuna (navy), Garuda (air force) and Shakti (army). Al Nagah is with Oman, not Egypt; the India-Egypt exercise is Cyclone. Malabar is a four-nation naval exercise of India, the USA, Japan and Australia, not a bilateral. And in the structural set: Agnipath (June 2022) is a recruitment scheme, NOT the corporatisation of the ordnance factories - that was 2021; the Integrated Defence Staff dates from 2001, while 2019 is the year of the CDS and the Department of Military Affairs.

Finally, the command structure. India's forces are still organised service by service, but a set of tri-service bodies has been layered on top to make them fight as one. Learn the years - RPSC has asked exactly this as a NOT-correct question.

Building the tri-services structure

- The post of Chief of Defence Staff was created in 2019. General Bipin Rawat became the first CDS on 1 January 2020; General Anil Chauhan has held the post since September 2022.
- The CDS heads the Department of Military Affairs and is the Permanent Chairman of the Chiefs of Staff Committee.

- Agnipath, launched June 2022 - a four-year Agniveer tenure with up to 25 per cent retained in regular service.
- As of September 2026 no integrated theatre command had been formally raised; the CDS submitted a final plan to the Defence Ministry in 2026 and it awaits government clearance.
- Rajasthan and defence: the Pokhran Field Firing Range (nuclear tests, Exercise Bharat Shakti 2024, the IAF's Vayu Shakti firepower demonstration), the Mahajan Field Firing Range near Bikaner, the Chandan range near Jaisalmer, Defence Laboratory Jodhpur, and the Jodhpur air base which hosted a phase of the multinational Exercise Tarang Shakti in 2024.

Central armed police forces - the raising years RPSC asks

Force	Raised	Note
Assam Rifles	1835	The oldest paramilitary force in India
Indo-Tibetan Border Police	1962	Raised after the India-China war
Central Industrial Security Force	1969	Guards industrial and airport installations
Border Security Force	1965	India's largest border guarding force
National Security Guard	1984	Counter-terrorism force - not 1974

REVISION RECAP

1. DRDO - founded 1 January 1958, headquarters DRDO Bhawan New Delhi, about 52 laboratories, motto Balasya Mulam Vigyanam from the Arthashastra.
2. Lab-to-city: Hyderabad missiles, Bengaluru aeronautics and radar, Pune armaments, Kochi sonar, Visakhapatnam torpedoes, Mysuru food, Avadi combat vehicles.
3. Rajasthan's DRDO lab is Defence Laboratory Jodhpur - camouflage, desert sciences, radiation detection.
4. Lukoskin for leucoderma from DIBER Haldwani; 2-DG anti-COVID drug from INMAS Delhi; OxyCare from DEBEL; rations from DFRL Mysuru.
5. IGMDP - launched 1983 under A.P.J. Abdul Kalam, concluded 2008; the five are Prithvi, Agni, Trishul, Akash and Nag. BrahMos is not one of them.
6. Ranges: Prithvi-I 150 km, Prithvi-II 250-350 km, Agni-I about 900 km, Agni-II 2,000+, Agni-III 3,000+, Agni-IV about 4,000, Agni-V 5,000+, Agni-Prime 1,000-2,000 km.

7. Types: Prithvi surface-to-surface, Akash surface-to-air, Nag anti-tank, Astra beyond-visual-range air-to-air, QRSAM quick-reaction SAM.
8. BrahMos - India-Russia joint venture 1998, named after the Brahmaputra and Moskva, Mach 2.8-3; the Philippines was the first export customer.
9. Mission Divyasthra, 11 March 2024 - first Agni-5 test with MIRV. HSTDV, September 2020 - scramjet at about Mach 6. Mission Shakti, March 2019 - anti-satellite kill at about 283 km.
10. Ballistic missile arcs unpowered after boost; cruise missile flies low under power all the way. BMD Phase-I = exo-atmospheric PAD plus endo-atmospheric AAD.
11. Tejas - designed by ADA, built by HAL, IOC 2013 and FOC 2019, Mk-1A on the GE F404. AMCA is the fifth-generation programme cleared by the CCS in March 2024.
12. Netra AEW&C on an Embraer ERJ-145 from CABS; Phalcon AWACS on an IL-76. TAPAS-BH from ADE Bengaluru. Prachand is the light combat helicopter, first unit at Jodhpur.
13. INS Vikrant, September 2022 - first indigenously designed and built carrier; Vikramaditya was bought from Russia in 2013.
14. Arihant 2016 and Arighaat 2024 are the indigenous SSBNs; Vagsheer, January 2025, closed Project 75; INS Nistar is the first indigenous diving support vessel; INS Tamal is the last imported warship.
15. Arjun from CVRDE Avadi, Pinaka and ATAGS from ARDE Pune, Varunastra from NSTL Visakhapatnam, AK-203 from Korwa in Amethi.
16. Pokhran-I 'Smiling Buddha' 18 May 1974, Pokhran-II 'Operation Shakti' 11 and 13 May 1998 - both in Jaisalmer district; 11 May is National Technology Day.
17. Doctrine 2003 - No First Use, credible minimum deterrence, massive retaliation, no use against non-nuclear weapon states; triad completed 2018 by INS Arihant's first deterrence patrol.
18. MTCR 2016, Wassenaar 2017, Australia Group 2018 - but NOT the NSG, and neither NPT nor CTBT signed.
19. OFB dissolved 1 October 2021 into 7 DPSUs; defence corridors in Uttar Pradesh and Tamil Nadu; exports about Rs 38,424 crore in 2025-26 against a Rs 50,000 crore target for 2029.
20. CDS created 2019, Bipin Rawat first from 1 January 2020, Anil Chauhan since September 2022; Andaman and Nicobar Command 2001 was the first tri-service command; Agnipath from June 2022 with four-year tenure and up to 25 per cent retention.

ST12

Environment, Ecology, EIA, Biodiversity and Sustainable Development

Science and Technology is 18-23 questions in every paper and environment alone supplies 4-6 of them. Documented repeats: alpha/beta/gamma diversity (2013, 2016, 2018), greenhouse gases ranked by contribution (2016 and 2018), the acid rain pollutant pair (2016 and 2018 - the identical question twice), Ramsar sites of Rajasthan (2016, 2023), tiger reserves of Rajasthan (2023, 2024).

This is the biggest chapter in Science and Technology and the one RPSC mines hardest. Read it as three stacked layers - first the ecology (how an ecosystem actually runs), then the law (what India has enacted to protect it), then the diplomacy (what the world has agreed on climate). Almost every question is a definition, a year, a rank order or a matching pair, so the marks are there for anyone who memorises cleanly.

Environment & Ecology

Ecosystem

- Tansley 1935
- Grazing vs detritus chain
- 10% law - Lindeman 1942
- Energy pyramid always upright
- Gaseous vs sedimentary cycles

Biodiversity

- Genetic / species / ecosystem
- Alpha - beta - gamma (Whittaker)
- Hotspots - Myers 1988
- India's four hotspots
- IUCN 9 categories; threatened = CR+EN+VU

Conservation

- In-situ vs ex-situ
- Core - buffer - transition
- Project Tiger 1973, NTCA 2006
- Rajasthan - 5 tiger reserves
- Rajasthan - 5 Ramsar sites (2025)

Environmental law

- WPA 1972 - 4 schedules after 2022
- Water 1974, Air 1981
- EPA 1986 - umbrella, Art. 253
- BD Act 2002 + 2023 amendment
- FCA 1980 renamed in 2023
- NGT Act 2010

Pollution

- AQI 2014 - 6 bands, 8 pollutants
- NAAQS 2009, NCAP Jan 2019
- Acid rain = SO₂ + NO_x
- BOD 5 days at 20 deg C
- SUP ban 1 July 2022

Climate change

- CO₂ > CH₄ > CFC > N₂O
- SF₆ highest GWP
- IPCC 1988, UNFCCC 1992
- Kyoto 1997, Paris 2015
- COP30 Belem, COP31 Antalya
- Net-zero 2070; NAPCC 8 missions

EIA & disasters

- Notification 2006 under EPA
- Screening-Scoping-Consultation-Appraisal
- Category A central, B state
- Public hearing by SPCB
- DM Act 2005, Sendai 2015-30

The ecosystem - structure, energy flow and the 10 per cent law

Start with the word itself. An ecosystem is a community of living organisms plus the non-living surroundings they interact with, working as one functional unit. The term was coined by A.G. Tansley in 1935 - that name alone is a one-

mark question. Around it runs a ladder of organisation that RPSC asks you to order: individual, population, community, ecosystem, landscape, biome, biosphere. Learn it upward, from the smallest to the largest.

- Structure of an ecosystem = biotic components (producers, consumers, decomposers) + abiotic components (light, temperature, water, soil, nutrients).
- Functions of an ecosystem = productivity, decomposition, energy flow, nutrient cycling. Four functions, learn all four.
- Producers are autotrophs; consumers are heterotrophs; decomposers such as bacteria and fungi are saprotrophs.
- Gross Primary Productivity minus respiration = Net Primary Productivity. Only NPP is available to the herbivore.
- Energy flow is one-way (unidirectional); nutrients go round in cycles. Never mix the two.

Now the food chain. Two types, and the difference is only in where the chain begins. The grazing food chain starts with a green plant. The detritus food chain starts with dead organic matter. Here is the point students get wrong - in a terrestrial ecosystem far more energy flows through the DETRITUS chain, because most leaves are never eaten alive; they fall and rot. In many aquatic systems the grazing chain dominates. Energy transfer between successive levels then obeys the 10 per cent law, given by Raymond Lindeman in 1942: only about a tenth of the energy at one trophic level reaches the next, the rest being lost as heat in respiration. That is exactly why a food chain rarely runs beyond four or five links.

- Trophic level 1 producers, level 2 herbivores (primary consumers), level 3 secondary consumers, level 4 tertiary consumers.
- 10 per cent law - Raymond Lindeman, 1942. The ecological pyramid concept - Charles Elton.

- Pyramid of energy is ALWAYS upright, in every ecosystem, because of the 10 per cent law.
- Pyramid of numbers can be inverted - one big tree supports many insects and still more parasites.
- Pyramid of biomass is inverted in the open ocean - a small standing crop of phytoplankton supports a larger mass of zooplankton and fish.

TRAP

Two traps live in this one page. First: it is the pyramid of ENERGY that is always upright, not biomass and not numbers. Second: Lindeman gave the 10 per cent law, Elton gave the ecological pyramid, Tansley coined 'ecosystem', Odum wrote the textbook. RPSC has offered all four names in a single option set.

Nutrient cycles, succession and the special species categories

Nutrients move in closed loops called biogeochemical cycles. Split them into two families by where the main reservoir sits. If the reservoir is the atmosphere it is a gaseous cycle; if it is rock and sediment it is a sedimentary cycle. The phosphorus cycle is the perennial favourite because it has no significant gaseous phase at all.

Gaseous cycle vs sedimentary cycle - the standard question

Gaseous cycle	Sedimentary cycle
Reservoir is the atmosphere	Reservoir is the earth's crust - rocks and sediments
Carbon, nitrogen, oxygen	Phosphorus, sulphur
Fast, well buffered, global in reach	Slow, easily disturbed, local in reach
Has a gaseous phase	Phosphorus has NO gaseous phase

Nitrogen fixers matched to their mode - a standing matching set

Organism	Mode of nitrogen fixation
<i>Rhizobium</i>	Symbiotic, in the root nodules of legumes
<i>Frankia</i>	Symbiotic, with <i>Casuarina</i>
<i>Azotobacter</i>	Free-living, aerobic bacterium
<i>Clostridium</i>	Free-living, anaerobic bacterium
<i>Anabaena</i> / <i>Nostoc</i>	Cyanobacteria; <i>Anabaena azollae</i> lives in the water fern <i>Azolla</i>

- Ecological succession - the orderly replacement of one community by another until a stable climax community is reached.
- Primary succession begins on a lifeless surface; secondary succession begins where soil survives, and is much faster.
- Xerarch succession starts on bare rock and its pioneers are LICHENS; hydrarch succession starts in water and its pioneers are phytoplankton.
- Ecotone = the transition zone where two ecosystems meet, such as a mangrove between land and sea.
- Edge effect = the higher species density found in an ecotone; species confined to it are edge species.
- Keystone species - an influence far out of proportion to its numbers (sea otter, fig trees).
- Flagship species - charismatic, used to raise public support (tiger, Great Indian Bustard).
- Umbrella species - needs so much area that protecting it protects a whole community (tiger, elephant).
- Indicator species - its condition reveals environmental quality; lichens indicate clean air.
- Invasive alien species in India - *Lantana camara*, *Parthenium* (congress grass), *Eichhornia* or water hyacinth ('terror of Bengal'), *Prosopis juliflora*.

Biodiversity - levels, the alpha-beta-gamma trio and the hotspots

Biodiversity means the variety of life, measured at three levels. Genetic diversity is variation within a species - the thousands of rice landraces. Species diversity is the number and abundance of species in an area. Ecosystem diversity is the variety of habitats in a region. The word itself was coined by Walter Rosen in 1985 and popularised by E.O. Wilson. On top of that sits R.H. Whittaker's alpha-beta-gamma trio, and this is a documented RPSC repeat. It is a scale story, not a difficulty story. Alpha is diversity INSIDE one habitat. Beta is the CHANGE in species as you move from one habitat to the next along a gradient. Gamma is the TOTAL for the whole landscape. Because gamma is the regional total, it can never be smaller than the alpha of a single habitat within it - RPSC has built a deliberately wrong statement on exactly that point.

Whittaker's three measures of diversity

Measure	What it counts	Scale
Alpha diversity	Species within a single community or habitat	Local, within-habitat
Beta diversity	Turnover in species composition between communities along a gradient	Between habitats
Gamma diversity	Total diversity of a large landscape or geographical region	Regional, the sum total

ASKED IN RAS

Asked in RAS Pre 2018 - the diversity of species found within a single community or habitat is termed? Answer: alpha diversity. Asked again in RAS Pre 2016 as a statement question, where 'alpha diversity of a habitat is always greater than the gamma diversity of its region' was the wrong statement. Add the 2013 hotspot question and this cluster has appeared in three papers - treat it as certain.

A biodiversity hotspot is not simply a rich area; it is a rich area in danger. Norman Myers proposed the idea in 1988 and there are two criteria, both of which must be met together. About 36 hotspots are recognised worldwide and four of them reach into Indian territory.

- Criterion 1 - at least 1,500 species of endemic vascular plants. Criterion 2 - at least 70 per cent of its primary vegetation already lost. Both, together.
- India's four hotspots - the Himalaya, Indo-Burma, the Western Ghats and Sri Lanka, and Sundaland.
- Sundaland enters India ONLY through the Nicobar Islands, not through the mainland.
- Cerrado (Brazil), Caucasus, Madagascar, Mesoamerica and the Cape Floristic Region do NOT extend into India - RAS Pre 2013 asked exactly this, with Cerrado as the answer.
- Rajasthan lies in none of the four; the Aravalli is ecologically vital but is not a hotspot.
- Trick to remember India's four - 'HIWS': Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland.

IUCN Red List and the two styles of conservation

The International Union for Conservation of Nature was founded in 1948 and works out of Gland in Switzerland. Its Red List places every assessed species in one of nine categories. You must be able to run the ladder in order, and you must know which three of them together make up the word 'threatened'.

IUCN Red List - the ladder from safe to gone



- Nine categories in all - the seven above plus Data Deficient and Not Evaluated, which sit outside the risk ladder.
- 'Threatened' = Critically Endangered + Endangered + Vulnerable. Exactly these three, no more, no fewer.
- Extinct in the Wild means it survives only in cultivation, captivity or outside its natural range.
- The Great Indian Bustard (Godawan), Rajasthan's state bird, is Critically Endangered - the single most examinable Red List entry for this state.
- The cheetah became extinct in India in 1952; African cheetahs were reintroduced at Kuno, Madhya Pradesh, from 2022.
- CITES (1973, in force 1975) regulates international trade in species through three appendices - do not confuse it with the IUCN Red List, which only assesses risk.

In-situ vs ex-situ conservation

In-situ (on site)	Ex-situ (off site)
Species protected in its natural habitat	Species maintained outside its natural habitat
National parks, sanctuaries, biosphere reserves, conservation and community reserves, sacred groves	Zoos, botanical gardens, seed and gene banks, cryopreservation of gametes, tissue culture, captive breeding
The whole ecosystem is conserved along with the species	Only the species or its germplasm is conserved
Cheaper, self-sustaining; evolution continues	Costly, but the only route for species already too rare in the wild

- *Biosphere reserve zones - core (legally protected, no human activity), buffer (limited research and education), transition (settlement and sustainable use).*
- *India's first biosphere reserve - Nilgiri, notified in 1986. Eighteen have been notified in all and twelve are on UNESCO's World Network.*
- *Rajasthan has NO biosphere reserve. Learn this negative fact; it is exactly what RPSC asks.*
- *Nokrek - Meghalaya; Simlipal - Odisha; Dihang-Dibang - Arunachal Pradesh; Agasthyamalai - Kerala and Tamil Nadu.*

Protected areas, species projects and Rajasthan's wetlands

India's species projects are simple year-matching questions. Project Tiger came first, in 1973, with nine reserves - and Ranthambore was one of those original nine, a Rajasthan link RPSC has used. The National Tiger Conservation Authority is not a separate creature: it was given statutory status in 2006 through an amendment to the Wild Life (Protection) Act, 1972. Wetlands are the other half of this section, and Rajasthan's answer here has changed recently, so read the current-affairs box carefully.

- *Project Tiger 1973 (nine reserves at launch, including Ranthambore); NTCA statutory in 2006 under the WPA 1972.*
- *Project Crocodile 1975, Project Elephant 1992, Project Snow Leopard 2009, Project Dolphin 2020.*
- *The all-India tiger estimation of 2022 put the population at about 3,682.*
- *Protected area categories under the WPA - national park, wildlife sanctuary, conservation reserve, community reserve.*
- *Ramsar Convention on Wetlands - signed 1971 at Ramsar in Iran, in force from 1975; a site in trouble goes on the Montreux Record.*

- Keoladeo Ghana is both a Ramsar site and a UNESCO World Heritage Site; Sambhar is India's largest inland saline lake.

Tiger reserves of Rajasthan – district and year

<i>Tiger reserve</i>	<i>District</i>	<i>Note</i>
<i>Ranthambore</i>	<i>Sawai Madhopur</i>	<i>One of the original nine of 1973</i>
<i>Sariska</i>	<i>Alwar</i>	<i>Site of the world's first successful tiger relocation, 2008</i>
<i>Mukundra Hills</i>	<i>Kota</i>	<i>Notified 2013</i>
<i>Ramgarh Vishdhari</i>	<i>Bundi</i>	<i>Notified 2022, India's 52nd</i>
<i>Dholpur-Karauli</i>	<i>Dholpur and Karauli</i>	<i>Approved 2023, India's 54th; Rajasthan's fifth</i>

ASKED IN RAS

Asked in RAS Pre 2024 as a matching question – Ranthambore with Sawai Madhopur, Sariska with Alwar, Mukundra Hills with Kota, Ramgarh Vishdhari with Bundi. And asked in RAS Pre 2016 and again in 2023 – which pair were Rajasthan's FIRST TWO Ramsar sites? Answer: Keoladeo and Sambhar. Learn the district against every reserve, and learn 'first two' as a phrase, because the total has since changed.

CURRENT LINK

CURRENT LINK, as of September 2026 - Rajasthan now has FIVE Ramsar sites: Keoladeo Ghana National Park (1981), Sambhar Lake (1990), and three added in 2025 - Menar in Udaipur, Khichan in Phalodi and Siliserh Lake in Alwar. India's national tally crossed a hundred and stood at 101 in August 2026. Two more datelines worth carrying into the hall: the India State of Forest Report 2023 (released December 2024) is still the latest assessment - forest and tree cover 8,27,357 sq km or 25.17 per cent of the geographical area, forest cover alone 21.76 per cent, mangroves 4,992 sq km, Madhya Pradesh largest by area and Lakshadweep highest by percentage; and Madhav in Madhya Pradesh became India's newest tiger reserve in 2025.

India's environmental laws - the table you must be able to write from memory

This section is pure marks. Every year RPSC asks either an Act-to-year match or one clause of a recent amendment. Learn the table first, then the three amendments that have changed the standard answers since 2022.

Environmental legislation of India - Act and year

Act	Year	The point that gets asked
Wild Life (Protection) Act	1972	The 2022 amendment cut the schedules from six to four, in force 1 April 2023
Water (Prevention and Control of Pollution) Act	1974	Created the CPCB and the State Pollution Control Boards
Forest (Conservation) Act	1980	Renamed by the 2023 amendment
Air (Prevention and Control of Pollution) Act	1981	Enacted to give effect to the Stockholm decisions

Act	Year	The point that gets asked
Environment (Protection) Act	1986	The umbrella Act; passed under Article 253 after Stockholm 1972 and the 1984 Bhopal disaster
Biological Diversity Act	2002	Gives effect to the Convention on Biological Diversity, 1992; amended 2023
Forest Rights Act (Scheduled Tribes and Other Traditional Forest Dwellers)	2006	Ministry of Tribal Affairs is nodal; Gram Sabha is the authority; cut-off 13 December 2005
National Green Tribunal Act	2010	NGT set up on 18 October 2010
Compensatory Afforestation Fund Act	2016	The CAF Fund has its OWN Act - not the 2023 forest amendment

- Wild Life (Protection) Amendment Act 2022 - six schedules cut to four. Schedules I and II are animals, Schedule III is plants, Schedule IV lists CITES specimens.
- Biological Diversity (Amendment) Act 2023 - exempts registered AYUSH practitioners from benefit sharing for codified traditional knowledge, and decriminalises offences, replacing imprisonment with monetary penalties.
- It did NOT abolish Biodiversity Management Committees; they continue and still maintain People's Biodiversity Registers.
- Forest (Conservation) Amendment Act 2023 - renamed the parent Act as the Van (Sanrakshan Evam Samvardhan) Adhiniyam, 1980.
- Its two headline exemptions - forest land within 100 km of an international border for strategic linear projects of national importance, and up to 10 hectares for security-related infrastructure. Zoos, safaris and eco-tourism facilities were exempted too.
- Biological Diversity Act structure is three-tier: National Biodiversity Authority at Chennai, State Biodiversity Boards, Biodiversity Management Committees at local body level.

- NGT - principal bench New Delhi; zonal benches Bhopal, Pune, Kolkata and Chennai; cases to be disposed of within six months.
- India was the third country in the world to set up a dedicated environmental court, after Australia and New Zealand.
- The Environment (Protection) Act 1986 is the parent statute for almost every environmental rule you will meet - EIA, plastic, e-waste, noise.
- Environment Audit Rules were notified on 29 August 2025 under the same 1986 Act.

TRAP

Do not mix up the two amendments of 2023. The BIOLOGICAL Diversity amendment is about AYUSH practitioners and decriminalisation. The FOREST Conservation amendment is about the 100 km border exemption and the Hindi renaming. And the Compensatory Afforestation Fund has its own statute of 2016 - RPSC has floated 'the 2023 amendment gave statutory backing to the CAF' as a deliberately wrong statement.

Pollution - air, water, noise, plastic and e-waste

Air pollution questions come in three shapes: the AQI band table, the list of pollutants inside the index, and the acid rain pair. Take the AQI first. It was launched in 2014, is computed by the CPCB, has six categories and eight pollutants, and the AQI you see reported is the WORST sub-index among those eight - not their average.

National Air Quality Index - the six bands

AQI value	Category	AQI value	Category
0-50	Good	201-300	Poor
51-100	Satisfactory	301-400	Very Poor

AQI value	Category	AQI value	Category
101-200	Moderate	401-500	Severe

- The eight AQI pollutants - PM₁₀, PM_{2.5}, NO₂, SO₂, CO, ground-level ozone (O₃), NH₃ and Pb.
- Methane and carbon dioxide are NOT in the AQI. This is a favourite negative-framed question.
- National Ambient Air Quality Standards were notified in 2009 and cover twelve pollutants.
- National Clean Air Programme - launched January 2019 by the MoEFCC through the CPCB for identified non-attainment cities.
- NCAP's revised target - a 40 per cent cut in particulate matter concentration by 2025-26 over the 2017 level; the original was 20-30 per cent by 2024.
- Acid rain forms when SO₂ and NO_x convert in the atmosphere to sulphuric and nitric acid; rain with a pH below 5.6 counts as acidic.
- Stubble burning in Punjab and Haryana is the seasonal driver of the north Indian winter smog; indoor pollution from biomass chulhas is tackled through the LPG route.

ASKED IN RAS

Asked in RAS Pre 2016 and again word for word in RAS Pre 2018 - acid rain is caused mainly by which pair of pollutants? Answer: sulphur dioxide and oxides of nitrogen. Two consecutive papers, the identical question. If you learn one pollution fact, learn this one.

Water pollution turns on two abbreviations. BOD is the oxygen micro-organisms need to break down the organic matter in a sample, measured over five days at 20 degrees C; a high BOD means dirty water. COD measures everything oxidisable, biodegradable or not, so for the same sample COD is always higher than BOD. Eutrophication is the other standing question - nutrient enrichment feeds an algal bloom, and it is the DECOMPOSITION of that bloom that strips

the oxygen out of the water and kills the fish. The bloom does not raise dissolved oxygen; students tick that wrong statement every year.

Pollutant matched to the disease it causes

Pollutant	Disease
Mercury	Minamata disease
Cadmium	Itai-itai disease
Fluoride	Skeletal fluorosis - a real problem in several western Rajasthan districts
Nitrate in drinking water	Blue baby syndrome (methaemoglobinaemia)
Arsenic	Black foot disease and skin lesions

- Eutrophication is driven chiefly by nitrates and phosphates from fertiliser runoff and sewage.
- Namami Gange - launched 2014 under the National Mission for Clean Ganga.
- Single-use plastic ban on identified items - effective 1 July 2022; carry bag thickness raised to 120 microns from 31 December 2022.
- EPR = Extended Producer Responsibility - the producer stays responsible for the product after the consumer has finished with it.
- E-Waste (Management) Rules 2022 came into force on 1 April 2023.
- Noise limits, day time, in dB(A) Leq - industrial 75, commercial 65, residential 55, silence zone 50; night values 70, 55, 45 and 40.
- Solid Waste Management Rules 2016 made source segregation into wet, dry and hazardous waste mandatory; hazardous waste has its own rules of 2016.
- Soil pollution comes from pesticide and fertiliser residue, industrial effluent and dumped waste; thermal pollution lowers dissolved oxygen; radioactive pollution is measured in becquerel and sievert.

Climate change - the gases, the treaties and India's promises

Two different rankings of greenhouse gases exist and RPSC has asked both, so keep them apart. The first is by CONTRIBUTION to the enhanced greenhouse effect - how much warming each gas is actually delivering, which depends on how much of it is up there. The second is by GLOBAL WARMING POTENTIAL - how strong one molecule is, regardless of quantity. The two orders run in roughly opposite directions.

- By contribution: carbon dioxide (about 60%) > methane (about 20%) > CFCs (about 14%) > nitrous oxide (about 6%).
- By 100-year global warming potential per molecule, sulphur hexafluoride (SF₆) is the highest of all.
- Water vapour is the most abundant greenhouse gas overall but is not counted among the anthropogenic gases whose contribution is ranked.
- The Kyoto basket - CO₂, CH₄, N₂O, HFCs, PFCs and SF₆; nitrogen trifluoride was added later.
- Methane comes largely from paddy fields, enteric fermentation in ruminants and landfills; nitrous oxide largely from nitrogenous fertiliser.
- The IPCC does not do research - it assesses published science. Its Sixth Assessment Report cycle (AR6) ran from 2021 to 2023.

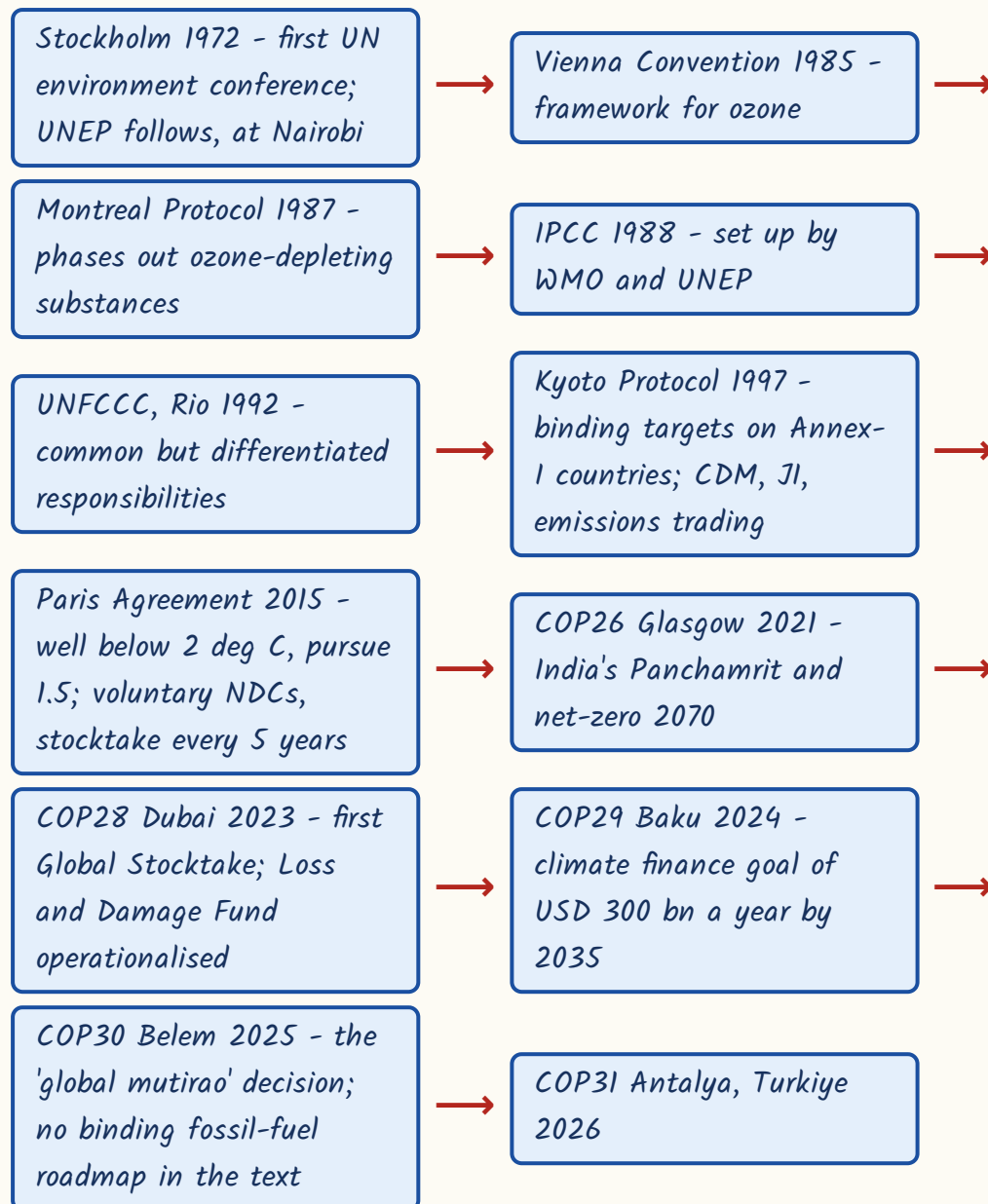
ASKED IN RAS

Asked in RAS Pre 2018 - which gas makes the largest contribution to the enhanced greenhouse effect? Answer: carbon dioxide. Asked in RAS Pre 2016 as an ordering question - arrange methane, carbon dioxide, nitrous oxide and CFCs in descending order of contribution. Answer: CO₂, methane, CFCs, nitrous oxide. Same fact, two papers, two formats - so learn it both as a list and as a rank.

YAAD RAKHO

Mnemonic for the contribution order - 'Car Meets Cool Nights': Carbon dioxide, Methane, CFCs, Nitrous oxide. For potency, one word is enough - SF₆ sits at the top of the GWP table.

The climate and environment chronology - with the one outcome each is asked for



NAPCC 2008 - the eight missions and their nodal ministries

Mission	Nodal ministry
National Solar Mission	New and Renewable Energy
National Mission for Enhanced Energy Efficiency	Power

Mission	Nodal ministry
National Mission on Sustainable Habitat	Housing and Urban Affairs
National Water Mission	Jal Shakti
National Mission for Sustaining the Himalayan Ecosystem	Science and Technology
National Mission for a Green India	Environment, Forest and Climate Change
National Mission for Sustainable Agriculture	Agriculture and Farmers Welfare
National Mission on Strategic Knowledge for Climate Change	Science and Technology

CURRENT LINK

CURRENT LINK, as of September 2026 - India's new Nationally Determined Contribution for 2031-35 was approved by the Union Cabinet on 25 March 2026: a 47 per cent cut in the emissions intensity of GDP by 2035 over the 2005 level, about 60 per cent of cumulative electric power capacity from non-fossil sources by 2035, and an additional carbon sink of 3.5 to 4.0 billion tonnes of CO₂ equivalent. Net-zero stays at 2070. The older 2030 targets - 45 per cent intensity cut and 50 per cent non-fossil capacity - are still asked, so hold both sets. COP31 is scheduled at Antalya, Turkiye, in 2026. Every state also prepares a State Action Plan on Climate Change under the NAPCC umbrella, Rajasthan included. Remember also that there is NO 'National Clean Air Mission' inside the NAPCC's eight - NCAP is a separate 2019 programme, and that distractor has been used. India's other climate vehicles: the International Solar Alliance (2015, India and France, headquarters Gurugram - the first treaty-based international organisation headquartered in India), CDRI (2019, New Delhi), Mission LiFE (October 2022, Kevadia) and the Carbon Credit Trading Scheme notified in 2023 under the Energy Conservation (Amendment) Act, 2022.

Ozone, sustainable development and the SDGs

Keep the ozone story separate from the climate story - students merge them and lose the mark. Ozone depletion is chlorofluorocarbons and other halogenated compounds destroying stratospheric ozone; global warming is greenhouse gases trapping heat. Different problem, different treaty line. Sustainable development then has one definition that is asked almost verbatim: development that meets the needs of the present without compromising the ability of future generations to meet their own needs. It comes from the Brundtland Commission report *Our Common Future*, 1987. Learn the report name and the year as one unit.

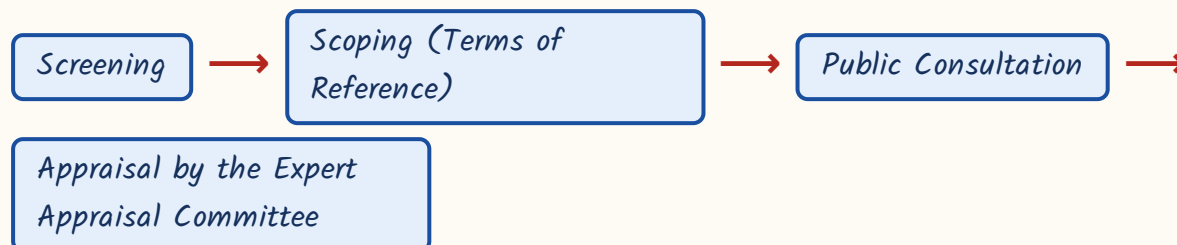
- Vienna Convention for the Protection of the Ozone Layer - 1985, the framework convention.
 - Montreal Protocol - 1987, the operational protocol phasing out ozone-depleting substances. World Ozone Day is 16 September.
 - Kigali Amendment - 2016, phases DOWN hydrofluorocarbons, which harm the climate but not ozone; in force 1 January 2019; India ratified in 2021.
 - The ozone hole forms over Antarctica in the southern spring; ozone thickness is measured in Dobson units.
 - The Montreal Protocol is the only environmental treaty with universal ratification - often called the most successful of them all.
-
- Sustainable Development Goals - 17 goals and 169 targets, adopted in September 2015 as part of the 2030 Agenda.
 - They succeeded the eight Millennium Development Goals (2000-2015).
 - SDG 13 is Climate Action, SDG 14 Life Below Water, SDG 15 Life on Land.
 - In India the SDG India Index is published by NITI Aayog.
 - Ecological footprint - the biologically productive area needed to support a person's consumption; when demand exceeds supply the region runs an ecological deficit.

- Green accounting adjusts national income for the depletion of natural capital and environmental damage.
- Convention on Biological Diversity - Rio 1992; the Nagoya Protocol (2010) covers access and benefit sharing; the Kunming-Montreal Global Biodiversity Framework was adopted in 2022.

Environmental Impact Assessment and disaster management

EIA is the process by which a project is examined for its environmental consequences BEFORE it is allowed to be built. In India it runs under the EIA Notification of 2006, issued under the Environment (Protection) Act, 1986. There are four stages in a fixed order, and RPSC's standard question is which stage does NOT belong.

The four stages of environmental clearance under the EIA Notification 2006



- Category A projects - appraised centrally by an Expert Appraisal Committee and cleared by the MoEFCC.
- Category B projects - appraised by a State Expert Appraisal Committee and cleared by the State Environment Impact Assessment Authority.
- Within Category B, B1 projects need a full EIA report; B2 projects do not.
- The public hearing is conducted by the State Pollution Control Board or Pollution Control Committee.
- 'Cost-benefit certification' and 'post-project audit' are NOT among the four stages - both have been used as distractors.

- The draft EIA Notification 2020 drew heavy criticism for allowing post-facto clearance and shortening the public notice period; applications are filed on the PARIVESH portal.
- Strategic EIA assesses a whole policy, plan or programme rather than one project; a carrying capacity study fixes the limit an area can absorb.

TRAP

Two clean traps here. First: the public hearing is conducted by the STATE POLLUTION CONTROL BOARD - not by the ministry, not by the SEIAA. Second: for Category B the SEAC only APPRAISES while the SEIAA GRANTS the clearance; RPSC swaps the two bodies and most students do not notice.

Bodies under the Disaster Management Act, 2005 - the head of government heads the authority

Body	Ex-officio head
National Disaster Management Authority (NDMA)	Prime Minister
State Disaster Management Authority (SDMA)	Chief Minister
District Disaster Management Authority (DDMA)	District Collector; the Zila Parishad chairperson is co-chairperson
National Executive Committee (NEC)	Union Home Secretary

- The DM Act 2005 also created the National Disaster Response Force and the National Institute of Disaster Management; NDMA has up to nine members besides the Prime Minister.
- India's first National Disaster Management Plan, aligned with the Sendai Framework, was released in 2016.
- Sendai Framework for Disaster Risk Reduction - adopted 2015, runs 2015-2030, has four priorities and seven global targets; it replaced the Hyogo

Framework (2005-2015).

- Disaster types - natural (earthquake, flood, cyclone, drought, landslide) and man-made (industrial, chemical, nuclear, stampede).
- Rajasthan's recurring hazards are drought, heat wave, flash flood in the Hadoti belt and lightning; the state runs an SDMA under the same Act.

REVISION RECAP

1. Ecosystem - term by A.G. Tansley, 1935. Ladder: individual, population, community, ecosystem, landscape, biome, biosphere.
2. 10 per cent law of energy transfer - Raymond Lindeman, 1942; the ecological pyramid concept - Charles Elton.
3. Pyramid of energy always upright; pyramids of number and biomass can be inverted.
4. Grazing chain starts with green plants, detritus chain with dead matter - and on land the detritus chain carries more energy; gaseous cycles are carbon, nitrogen and oxygen, sedimentary are phosphorus and sulphur, and phosphorus has no gaseous phase.
5. Rhizobium-legume, Frankia-Casuarina, Azotobacter free-living aerobic, Anabaena-Azolla.
6. Lichens pioneer bare rock; ecotone is the transition zone and the edge effect its extra richness.
7. Alpha within a habitat, beta turnover along a gradient, gamma the whole region - gamma is never less than alpha.
8. Hotspots - Myers 1988; 1,500 endemic vascular plants plus 70 per cent primary vegetation lost. India's four: Himalaya, Indo-Burma, Western Ghats-Sri Lanka, Sundaland (Nicobar).
9. IUCN 1948, Gland; nine Red List categories; threatened = Critically Endangered + Endangered + Vulnerable.
10. Biosphere reserve = core, buffer, transition; India's first Nilgiri 1986; Rajasthan has none.
11. Project Tiger 1973, NTCA statutory 2006; Rajasthan's five reserves - Ranthambore, Sariska, Mukundra Hills, Ramgarh Vishdhari, Dholpur-Karauli.
12. Rajasthan's five Ramsar sites - Keoladeo 1981, Sambhar 1990, and Menar, Khichan and Siliserh all in 2025 (as of September 2026).
13. Act years - WPA 1972 (four schedules after 2022), Water 1974, FCA 1980 (renamed 2023), Air 1981, EPA 1986, BD Act 2002 (amended 2023), FRA 2006, NGT Act 2010, CAF Act 2016.

14. AQI 2014 - six bands from Good 0-50 to Severe 401-500, eight pollutants; methane and CO₂ are excluded.
15. Acid rain = SO₂ + NO_x, pH below 5.6; BOD over five days at 20 deg C; COD always exceeds BOD.
16. Greenhouse contribution order CO₂ > CH₄ > CFC > N₂O; the highest GWP belongs to SF₆.
17. Stockholm 1972, Vienna 1985, Montreal 1987, IPCC 1988, UNFCCC 1992, Kyoto 1997, Paris 2015, Belem COP30 2025, Antalya COP31 2026.
18. India - net-zero 2070; the 2035 NDC approved 25 March 2026 sets a 47 per cent intensity cut and 60 per cent non-fossil capacity; NAPCC 2008 has eight missions.
19. EIA 2006 - screening, scoping, public consultation, appraisal; public hearing by the SPCB; Category A central, B state, B1 needs an EIA report.
20. DM Act 2005 - NDMA under the PM, SDMA under the CM, DDMA under the Collector, NEC under the Union Home Secretary; Sendai 2015-2030 with four priorities and seven targets.

ST13

Agriculture, Horticulture, Forestry, Animal Husbandry (Rajasthan focus); S&T Policy and Indian Scientists

Worth 3-5 questions in a typical paper. The hardest-working repeat here is the research-institution cluster - National Research Centre on Seed Spices at Ajmer, NRC on Camel at Bikaner, Central Institute for Arid Horticulture at Bikaner (2016, 2018, 2023) and the Udaipur Solar Observatory (2018 and 2021 - the identical question). Crop leadership, breeds by home tract and scientist-to-contribution matching fill the rest.

Two chapters bolted together, and both are Rajasthan-heavy. The first half is the applied science of a desert state - what grows in sand, how you save water, which breed belongs to which tract. The second half is institutional: the research centres sitting in Rajasthan, and the scientists whose names RPSC matches to their contributions. Almost none of it needs reasoning; all of it needs clean, district-level memory.

Agriculture, Livestock & S&T

Soil & nutrients

- 17 essential nutrients
- N chlorosis, P purple, K scorch
- Zn - khaira of rice
- Soil Health Card 19 Feb 2015, Suratgarh
- Neem-coated urea 2015

Seeds

- Breeder - golden yellow
- Foundation - white
- Certified - azure blue
- Truthfully labelled - opal green
- HHB-67 Improved (bajra)

Rajasthan crops

- No.1 in bajra, mustard, guar
- No.1 in coriander, cumin, methi
- Isabgol, henna, moth bean
- Guar is KHARIF
- Dryland below 750 mm

Water

- PMKSY 2015
- Har Khet Ko Pani
- Per Drop More Crop
- Drip ~90%, sprinkler 70-80%
- Khadin, tanka, nadi, johad

Horticulture

- Kinnow - Sri Ganganagar
- Orange - Jhalawar
- Olive - Lunkaransar, Bassi
- Date palm - Jaisalmer, Bikaner
- Sojat henna GI

Livestock

- Rathi best milch
- Chokla finest wool
- Magra carpet wool
- Sirohi goat, Jakhrana milk
- Camel state animal 2014

Institutions

- CAZRI Jodhpur 1959
- CSWRI Avikanagar 1962
- NRC Camel Bikaner 1984
- CIAH Bikaner 1993
- NRC Seed Spices Ajmer 2000
- Udaipur Solar Observatory 1975

S&T policy

- CSIR 1942, DST 1971
- STIP 2020 - fifth policy
- ANRF Act 2023
- NSD 28 Feb, NTD 11 May
- Space Day 23 Aug, Maths 22 Dec

Soil fertility and the plant nutrients

A plant needs seventeen essential nutrients. Three come free from air and water - carbon, hydrogen, oxygen. The rest must come from the soil, and they split into two classes by QUANTITY needed, not by importance. Primary macronutrients are nitrogen, phosphorus and potassium; secondary macronutrients are calcium, magnesium and sulphur; everything else is a micronutrient, needed

in traces. RPSC's standard question is to slip calcium into a list of micronutrients and see if you notice.

- Primary macronutrients - nitrogen (N), phosphorus (P), potassium (K). The NPK you buy in a fertiliser bag.
- Secondary macronutrients - calcium, magnesium, sulphur.
- Micronutrients - iron, manganese, zinc, copper, boron, molybdenum, chlorine, nickel. Eight of them.
- Nitrogen builds protein and chlorophyll; phosphorus builds roots and grain; potassium regulates water and disease resistance.
- Zinc deficiency is widespread in the calcareous and sandy soils of Rajasthan - remember this as a state-specific fact.

Deficiency symptoms - a ready-made matching question

Nutrient	Classic symptom
Nitrogen	General chlorosis (yellowing) of the OLDER leaves
Phosphorus	Purplish tinge on leaves and stems
Potassium	Scorching or burning of the leaf margins
Zinc	Khaira disease of rice
Iron	Chlorosis of the YOUNG leaves, between the veins

- Soil Health Card Scheme - launched by the Prime Minister on 19 February 2015 from Suratgarh in Sri Ganganagar district, Rajasthan. A Rajasthan launch site, so learn the place, not just the year.
- Neem-coated urea was made mandatory in 2015 to slow nitrogen release and improve use efficiency.
- Manure adds bulk organic matter and improves soil structure; fertiliser adds concentrated nutrients and nothing else.
- Integrated Nutrient Management combines fertiliser, organic manure and biofertiliser so that soil health is not mined away.

- Rajasthan's soils - desert and arid sandy soils in the west, alluvial in the east, medium black in the Hadoti plateau, and saline-alkaline patches around the salt lakes.

TRAP

Calcium is NOT a micronutrient. Calcium, magnesium and sulphur are SECONDARY MACRONutrients. RPSC asked exactly this as a 'which one is not a micronutrient' question, with calcium as the answer. Second trap in the same area: nitrogen chlorosis shows on the OLDER leaves, iron chlorosis on the YOUNGER ones.

Seeds - the classes and the multiplication chain

Seed is not one thing; it moves down a certified ladder, and each rung has its own tag colour. The breeder produces a tiny quantity of breeder seed under his own supervision. That is multiplied into foundation seed, and foundation seed into certified seed, which is what actually reaches the farmer. The order and the colours are both asked.

Seed multiplication chain - order and tag colour



- Truthfully labelled seed carries an OPAL GREEN label; it is labelled by the producer, not certified by an agency.
- HYV = high-yielding variety, a pure line that can be saved and resown. Hybrid = the F1 cross of two parents; its yield collapses in the next generation, so seed must be bought afresh every year.

- That single difference – hybrid seed cannot be saved – is why hybrid adoption depends on a working seed supply chain.
- Genetically modified seed carries an inserted gene; in India only Bt cotton is approved for commercial cultivation.
- Seed replacement rate measures how much of the area is sown with fresh certified seed each season.

YAAD RAKHO

Fix the ladder as B-F-C - Breeder, Foundation, Certified - and hang the colours on it in that order: GOLDEN yellow, WHITE, azure BLUE. Say it as 'gold, white, blue' going down the ladder. Opal green sits outside the ladder altogether, on truthfully labelled seed.

Rajasthan's crops and the varieties bred for them

Rajasthan's agriculture is defined by scarcity, so the state leads India in exactly the crops that tolerate it - coarse millets, oilseeds, gums and seed spices. Learn the leadership list as a block, because RPSC asks it as a statement question with one wrong crop slipped in (sugarcane is the usual planted error - that is Uttar Pradesh).

- Rajasthan is India's LARGEST producer of: bajra, rapeseed-mustard, guar, coriander, cumin, fenugreek (methi), isabgol, henna and moth bean.
- It is also India's largest wool producer and holds the country's largest camel population.
- Kharif crops of Rajasthan - bajra, jowar, maize, moth, guar, cotton, groundnut, soybean, til.
- Rabi crops - wheat, barley, gram, mustard, taramira, isabgol. Barley is RABI; it has been used as the odd one out in a kharif list.

- Guar is a Kharif crop, sown with the onset of the monsoon; its gum is exported for use in the oil and gas industry. Main districts - Sri Ganganagar, Hanumangarh, Bikaner, Churu, Nagaur.
- The ICAR Directorate of Rapeseed-Mustard Research is at Sear, Bharatpur - the national mustard institute sits in Rajasthan.
- Crop rotation with a legume restores fertility because root-nodule bacteria fix atmospheric nitrogen. Mixed cropping sows two crops together as insurance; intercropping sows them in a defined row pattern.

Varieties and hybrids grown in Rajasthan - learn crop against name

Crop	Varieties / hybrids	Why it matters here
Bajra (pearl millet)	HHB-67 Improved is the flagship; the HHB and RHB hybrid series dominate the state	HHB-67 Improved is early-maturing and downy-mildew resistant - it escapes drought by finishing fast
Maize	Mahi Kanchan and Mahi Dhawal (Banswara centre); the Pratap Makka series from MPUAT Udaipur	Bred for the southern Rajasthan maize belt - Banswara, Dungarpur, Udaipur, Chittorgarh
Mustard (rapeseed-mustard)	Pusa Bold and Bio-902	Rajasthan is the country's largest producer; the national institute is at Bharatpur
Guar (cluster bean)	RGC-936 and RGC-1003	Bred for the north-western guar tract; guar gum is a major export earner
Wheat	The 'Raj' series of varieties bred in the state	Rabi crop of the canal-irrigated north and the eastern plain; not a Rajasthan leadership crop

TRAP

Three regular traps. One: guar is Kharif, not rabi. Two: Rajasthan does NOT lead in sugarcane - that is Uttar Pradesh, and it is the wrong statement RPSC plants in the leadership list. Three: HHB-67 Improved is a BAJRA hybrid, not a wheat or maize one; it is offered against all three.

Dryland farming, water-use efficiency, drip and sprinkler

Dryland farming is agriculture in areas that get less than about 750 mm of rain a year, without assured irrigation. Its whole objective is not raising yield but **CONSERVING SOIL MOISTURE** - every technique follows from that one aim. Most of western Rajasthan falls in this category, which is why khadin cultivation, mulching and short-duration drought-escaping varieties are standard here rather than exotic.

- Dryland farming - under about 750 mm annual rainfall; rainfed farming can extend above that where rain is more reliable.
- Moisture conservation techniques - mulching (cuts evaporation from the soil surface), contour bunding, deep ploughing before the monsoon, weed control, and wind-breaks.
- Khadin, of the Jaisalmer region, is a runoff-harvesting cropping system: a bund across a slope impounds runoff so a rabi crop can be grown on stored moisture.
- Other traditional structures - tanka, nadi, johad, baori and toba. These recur across the Rajasthan Geography paper too.
- Water-use efficiency, roughly: surface or flood irrigation 30-50 per cent, sprinkler 70-80 per cent, drip about 90 per cent.
- Pradhan Mantri Krishi Sinchayee Yojana (2015) carries two slogans you must attach correctly - 'Har Khet Ko Pani' for coverage and 'Per Drop More Crop' for micro-irrigation.

Drip vs sprinkler - which one for which field

Drip irrigation	Sprinkler irrigation
Water delivered drop by drop at the root zone through emitters	Water sprayed over the crop like rainfall through nozzles
Efficiency about 90 per cent - the highest of all methods	Efficiency about 70-80 per cent
Best for wide-spaced row crops and orchards - kinnow, pomegranate, date palm, vegetables	Best for close-spaced crops on the sandy soils of western Rajasthan - wheat, groundnut, mustard, fodder
Allows fertigation; keeps the leaf dry, so less fungal disease	Cuts conveyance and deep-percolation losses that ruin flood irrigation on sand

- Protected cultivation structures - greenhouse, poly-house, shade-net house, walk-in tunnel. Khadin is NOT one of them; it is a water-harvesting field system, and that is a standard trick option.
- Precision farming uses soil maps, sensors and variable-rate application to match input to need.
- Namo Drone Didi supplies agricultural drones to women's self-help groups for spraying and monitoring.
- Post-harvest technology - grading, cold storage, controlled-atmosphere storage; post-harvest loss is the silent tax on horticulture.

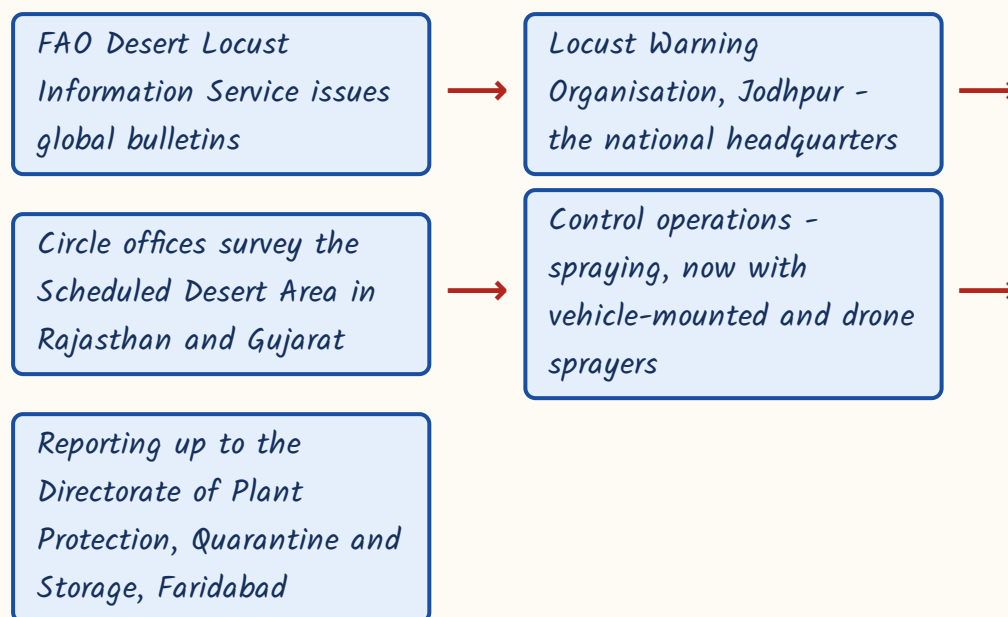
CURRENT LINK

CURRENT LINK, as of September 2026 - the Rajasthan Budget 2026-27 was presented on 11 February 2026 by Deputy Chief Minister Diya Kumari, who also holds Finance. Its farm announcements ran along familiar lines: PM-KUSUM solar pumps with a target of about 50,000 installations, micro-irrigation coverage of roughly three lakh hectares, farm electricity and interest subsidy, crop and livestock insurance, and gaushala support, framed within the 'Viksit Rajasthan @2047' vision. Rajasthan has presented a separate agriculture budget since 2022-23 - that practice itself is an examinable fact.

Pests, the locust threat and integrated pest management

Every arid-zone syllabus in India carries the locust because western Rajasthan is on its invasion path. The species is the desert locust, *Schistocerca gregaria*. Its danger lies in a phase change: in the solitary phase it is a harmless grasshopper, but when rain and vegetation let numbers build, it switches to the gregarious phase, changes colour, and moves in swarms that strip a field bare in hours. India's answer is a permanent watch, and the headquarters of that watch is in Rajasthan.

How a locust warning travels - the chain to learn



- Desert locust = *Schistocerca gregaria*. Solitary phase harmless, gregarious phase destructive.
- The Locust Warning Organisation is headquartered at JODHPUR, under the Directorate of Plant Protection, Quarantine and Storage at Faridabad.
- The Scheduled Desert Area for locust control lies in Rajasthan and Gujarat.
- The last major upsurge reached Rajasthan in 2019-20; drones were deployed for control spraying in 2020.

- Other important pests of Rajasthan crops - white grub in groundnut, termite in the sandy tract, pod borer in gram, and downy mildew in bajra.
- Integrated Pest Management puts cultural, mechanical and biological methods FIRST and treats chemical pesticide as the last resort.
- Its objective is to hold the pest below the Economic Threshold Level - NOT to eradicate the species. That distinction is the standard wrong statement.
- Trichogramma is an egg parasitoid used against lepidopteran (caterpillar) pests; NPV and neem-based formulations are other biocontrol tools.
- Biofertilisers - Rhizobium, Azotobacter, Azospirillum, blue-green algae, mycorrhiza. Biopesticides - Bacillus thuringiensis, Trichoderma.
- Pesticide residue and resistance are the two costs of a chemical-first approach; IPM exists to avoid both.

CURRENT LINK

CURRENT LINK, as of September 2026 - no major locust upsurge has reached Rajasthan since the 2019-20 invasion. The Locust Warning Organisation at Jodhpur continues routine survey of the Scheduled Desert Area and FAO continues to publish desert locust bulletins for the region. If a swarm year does occur before the exam, the examinable facts stay the same: Schistocerca gregaria, LWO Jodhpur, Rajasthan and Gujarat.

Horticulture - fruit belts, the seed spice bowl and medicinal plants

Horticulture in Rajasthan means arid horticulture - fruit that survives heat, sand and salinity. Learn each fruit against its belt, because this is asked as a 'which

pair is NOT correctly matched' question. The classic wrong pair offered is olive with Jhalawar; Jhalawar is oranges, and olive is Lunkaransar and Bassi.

Fruit and its growing belt in Rajasthan

Crop	Belt
Kinnow	Sri Ganganagar and Hanumangarh
Orange	Jhalawar
Olive	Lunkaransar (Bikaner) and Bassi (Jaipur) - the state's olive plantation and refinery project
Date palm	Jaisalmer and Bikaner, through tissue-culture plantations
Pomegranate	Barmer and Jalore
Ber and aonla	The arid tract generally - the core mandate crops of the Bikaner institute

- Seed spices of Rajasthan - coriander, cumin, fennel, fenugreek, ajwain, dill, celery and nigella. Rajasthan and Gujarat together dominate India's seed-spice output.
- Coriander belongs mainly to the Kota-Baran-Jhalawar-Chittorgarh belt; cumin to the western districts around Jodhpur, Barmer, Jalore and Nagaur; fenugreek to Nagaur and Sikar.
- Rajasthan is called India's seed spice bowl, and its dedicated institute is the National Research Centre on Seed Spices at Tabiji, Ajmer.
- Sojat in Pali district is the henna capital of India and its henna carries a Geographical Indication tag.
- Ashwagandha grown around Nagaur is known locally as asgandh; guggul is a threatened resin-yielding shrub of the Aravalli and desert tracts.
- Floriculture and vegetable cultivation cluster around the peri-urban belts of Jaipur, Ajmer and Kota; the national horticulture programme is MIDH.

Medicinal and aromatic plants - botanical names get matched

Common name	Botanical name
Henna (mehndi)	<i>Lawsonia inermis</i>
Isabgol	<i>Plantago ovata</i>
Ashwagandha	<i>Withania somnifera</i>
Safed musli	<i>Chlorophytum borivillianum</i>
Guggul	<i>Commiphora wightii</i>

ASKED IN RAS

Asked in RAS Pre 2016 - the Central Institute for Arid Horticulture at Bikaner works on arid fruits such as ber, date palm, aonla and pomegranate, and was established in 1993 as the National Research Centre for Arid Horticulture. The statement RPSC marks WRONG is that it works under the Ministry of Environment - like every ICAR institute it sits under the Department of Agricultural Research and Education in the Ministry of Agriculture and Farmers Welfare.

Forestry and agroforestry - the khejri system

Agroforestry in Rajasthan has one central tree, and its name is khejri. *Prosopis cineraria* is the state tree and is called the kalpavriksha of the desert because it does everything at once - it fixes nitrogen, its light canopy lets the crop below get sun, its leaves (loong) feed livestock, and its pods, sangri, are a food. A bajra or moth field with khejri standing in it is the classic agri-silviculture system of the desert, and it is the reason the Bishnoi tradition of protecting this tree - Khejarli, 1730, Amrita Devi - runs so deep.

- Khejri (*Prosopis cineraria*) - state tree; pods called sangri; nitrogen-fixing; the backbone of desert agroforestry.

- Rohida (*Tecomella undulata*) - state flower, called Marwar teak or the teak of the desert.
 - *Prosopis juliflora* is the INVASIVE 'vilayati babool'. One word apart from khejri (*Prosopis CINERARIA*) and the opposite answer - never let the two merge.
 - Agroforestry systems - agri-silviculture (crops plus trees), silvipasture (grass plus trees for livestock), horti-pasture.
 - India was the first country in the world to adopt a dedicated National Agroforestry Policy, in 2014.
 - Oran and gochar are the community sacred groves and pastures of Rajasthan - in-situ conservation without any statute behind it.
-
- National Forest Policy 1988 - target of 33 per cent of the geographical area under forest and tree cover, 60 per cent in hills and 20 per cent in the plains.
 - That policy shifted the object of forestry from REVENUE to ecological stability and the needs of forest-dependent communities. 'Maximising revenue' is the wrong statement RPSC uses.
 - Joint Forest Management was formally introduced by a Government of India circular in 1990 - village committees share protection duties and usufruct.
 - Van Dhan Vikas Kendras add value to minor forest produce and are run by TRIFED under the Ministry of Tribal Affairs.
 - Non-timber forest produce of Rajasthan - tendu leaf, gum from khair and kumat, honey, sangri, aonla and medicinal roots.

Animal husbandry - breeds by tract, and the diseases

Breed questions are pure matching, and the trick is to learn the HOME TRACT, not the animal's looks. Fix one anchor fact per breed - Rathii is the best milch cow, Chokla gives the finest wool, Magra gives lustrous carpet wool, Sirohi is the

meat goat, Jakhrana of Alwar is the milk goat. With those anchors the tables fall into place.

Cattle breeds of Rajasthan and their home tracts

Breed	Home tract	Anchor fact
Rathi	Bikaner and Sri Ganganagar	The best MILCH breed of the state
Tharparkar	Jaisalmer and Barmer	Dual purpose - milk and draught
Kankrej	Jalore and Sirohi	Prized draught breed, also good milk
Nagauri	Nagaur	Pure draught breed
Malvi	Jhalawar	Draught, from the Malwa side
Hariana	Bharatpur and Alwar	Comes in from the Haryana tract

Sheep breeds - wool is the anchor

Breed	Home tract	Anchor fact
Chokla	Sikar and Jhunjhunu (Shekhawati)	FINEST wool among Indian sheep breeds - the 'jewel of Rajasthan'
Magra	Bikaner and Jaisalmer	Lustrous wool, prized for carpets
Marwari	Jodhpur and Pali	Hardy, coarse wool, widely kept
Nali	Sri Ganganagar and Churu	Carpet wool
Sonadi	Udaipur and Dungarpur	Southern tract, also gives milk
Pugal, Jaisalmeri, Malpuri	Western and eastern tracts	Names to recognise in an option set

- Goat breeds - Sirohi (meat), Jakhrana of ALWAR (milk), Marwari, Barbari.
- Camel breeds - Bikaneri, Jaisalmeri (the riding breed), Mewari (the milk breed), Kachchhi.
- The camel was declared Rajasthan's state animal (domestic) in 2014 and protected by a state Act of 2015; the chinkara is the state animal (wild). Rajasthan has India's largest camel population.

- Buffalo breeds seen in the state - Murrah, Surti, Badhawari, Jafarabadi.
Horse - Marwari and Kathiawari.
 - Breeding technology - artificial insemination and embryo transfer are the standard routes for spreading superior germplasm.
 - Dairy in Rajasthan runs through RCDF and the Saras brand on the Amul co-operative model; RAJUVAS at Bikaner is the state veterinary university.
 - The state also runs livestock breeding farms - the Chandan sheep breeding farm in Jaisalmer is the best known - and supports gaushalas and Nandi shalas through departmental schemes.
-
- Foot-and-mouth disease - VIRAL, affects cloven-hoofed animals. PPR (peste des petits ruminants) - VIRAL, of sheep and goat.
 - Brucellosis - BACTERIAL and zoonotic, caused by Brucella. This is the odd one out in a 'which is not viral' question.
 - Lumpy skin disease - caused by a capripoxvirus, spread mainly by biting insects, affects cattle and buffalo. It does NOT infect humans through milk; that is the standard wrong statement.
 - The 2022 lumpy skin outbreak caused heavy cattle mortality in Rajasthan; the indigenous Lumpi-ProVacInd vaccine followed.
 - National Animal Disease Control Programme (2019) targets foot-and-mouth disease and brucellosis specifically.
 - Rashtriya Gokul Mission (2014) develops and conserves indigenous bovine breeds; the National Livestock Mission covers feed, fodder and small ruminants.

ASKED IN RAS

Asked in RAS Pre 2018 - where is the National Research Centre on Camel located? Answer: Bikaner, established in 1984. The Central Institute for Arid Horticulture is also at Bikaner, so the city carries two ICAR institutes and RPSC uses that overlap in matching questions.

The research institutions of Rajasthan – the most repeated block in this chapter

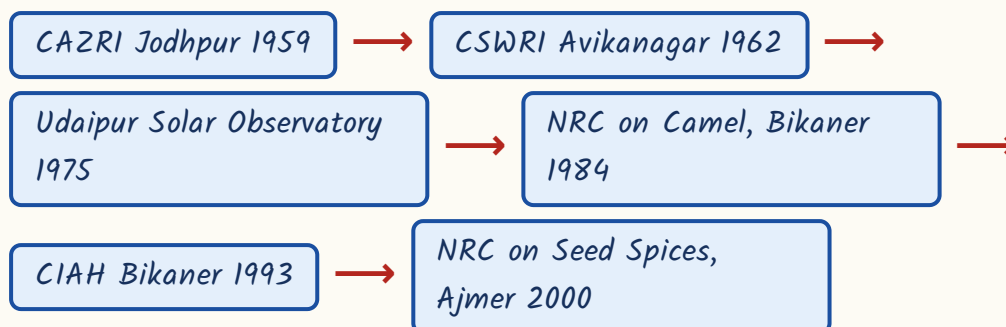
Read this section twice. It is the densest repeat zone in the whole of Science and Technology for Rajasthan – the institution cluster has been asked in 2016, 2018, 2021 and 2023, sometimes as a straight location question, sometimes as a four-pair match, sometimes as a 'which one is NOT in Rajasthan'. Learn the institute, the town, the district where the town is not obvious, and the year.

Research institutions located in Rajasthan

Institution	Place	Year / note
ICAR-Central Arid Zone Research Institute (CAZRI)	Jodhpur	1959; India's premier arid-zone research institute
ICAR-Central Sheep and Wool Research Institute (CSWRI)	Avikanagar, Malpura tehsil, TONK district	1962
ICAR-National Research Centre on Camel	Bikaner	1984
ICAR-Central Institute for Arid Horticulture (CIAH)	Bikaner	1993, as the NRC for Arid Horticulture
ICAR-National Research Centre on Seed Spices	Tabiji, Ajmer	2000
ICAR-Directorate of Rapeseed-Mustard Research	Sewar, Bharatpur	The national mustard institute
Arid Forest Research Institute (AFRI)	Jodhpur	Under ICFRE, not ICAR
CSIR-Central Electronics Engineering Research Institute (CEERI)	Pilani	The ONLY CSIR laboratory in Rajasthan
Udaipur Solar Observatory	An island in Fateh Sagar Lake, Udaipur	1975, founded by Dr. Arvind Bhatnagar; run by PRL Ahmedabad; hosts MAST

Institution	Place	Year / note
Rajasthan Remote Sensing Centre	Jodhpur	State remote sensing agency
Ch. Charan Singh National Institute of Agricultural Marketing (NIAM)	Jaipur	Agricultural marketing training
Rajasthan Agricultural Research Institute (RARI)	Durgapura, Jaipur	State research institute

The institutes in order of establishment - RPSC has asked this as a chronology



ASKED IN RAS

Asked in RAS Pre 2023 - where is the ICAR National Research Centre on Seed Spices? Answer: Tabiji, Ajmer. And asked in RAS Pre 2018 and again in 2023 as a four-pair match - Seed Spices with Ajmer, Arid Horticulture with Bikaner, CAZRI with Jodhpur, CSWRI with Avikanagar (Tonk). This exact matching set is the highest-probability question in the chapter.

ASKED IN RAS

Asked in RAS Pre 2018 and again in RAS Pre 2021 - the identical question - the Udaipur Solar Observatory on the island in Fateh Sagar Lake is at present operated by which body? Answer: the Physical Research Laboratory, Ahmedabad. Twice in consecutive cycles; do not lose this mark. And in RAS Pre 2016 the negative version was asked - which institute is NOT in Rajasthan? Answer: the Central Institute for Research on Goats, which is at Makhdoom in Mathura district, Uttar Pradesh.

Universities of Rajasthan you must place

University	Headquarters
Maharana Pratap University of Agriculture and Technology (MPUAT)	Udaipur
Swami Keshwanand Rajasthan Agricultural University (SKRAU)	Bikaner
Sri Karan Narendra Agriculture University (SKNAU)	Jobner (Jaipur)
Agriculture University	Jodhpur, and a second at Kota
Rajasthan University of Veterinary and Animal Sciences (RAJUVAS)	Bikaner
Birla Institute of Technology and Science (BITS)	Pilani - the state's leading private deemed university

YAAD RAKHO

A city-wise trick. BIKANER holds two ICAR institutes - Camel and Arid Horticulture - plus SKRAU and RAJUVAS. JODHPUR holds three - CAZRI, AFRI and the Remote Sensing Centre - plus the Locust Warning Organisation. AJMER has Seed Spices, BHARATPUR has mustard, TONK has sheep and wool, PILANI has CEERI, UDAIPUR has the Solar Observatory and MPUAT. Learn the CITY first and the institutes hang off it.

S&T policy, the science days, and the scientists RPSC matches

The last block is institutional memory. Two dates matter most - CSIR was founded in 1942, before independence, and DST in 1971. On top of them sits a stack of five policy documents, of which the Science, Technology and Innovation Policy of 2020 is the fifth. Then learn the four national science days, and

finally the scientist-to-contribution table, which is where the marks actually are.

- CSIR founded 1942; DST 1971; DSIR and DBT in the 1980s.
- India's five S&T policy documents - Scientific Policy Resolution 1958, Technology Policy Statement 1983, Science and Technology Policy 2003, STI Policy 2013, and STIP 2020 (the fifth).
- Anusandhan National Research Foundation - created by an Act of 2023, with the Prime Minister as ex-officio President of its Governing Board. It subsumed the Science and Engineering Research Board; it did NOT replace CSIR, which continues.
- Shanti Swarup Bhatnagar Prize - awarded by CSIR; INSPIRE stands for Innovation in Science Pursuit for Inspired Research, run by DST; Vigyan Jyoti encourages girls into STEM.
- The Physical Research Laboratory at Ahmedabad was founded in 1947 by Vikram Sarabhai - and it is PRL that runs the Udaipur Solar Observatory, which ties this section back to the last one.

The four national science days

Day	Date	What it commemorates
National Science Day	28 February	The Raman Effect, 1928
National Technology Day	11 May	The Pokhran-II tests, 1998
National Space Day	23 August	The Chandrayaan-3 landing, 2023
National Mathematics Day	22 December	Srinivasa Ramanujan's birth anniversary

Indian scientists matched to their contributions

Scientist	Contribution
C.V. Raman	Raman Effect - scattering of light; Nobel Prize in PHYSICS, 1930
Srinivasa Ramanujan	Number theory, partitions, infinite series; self-taught mathematician

Scientist	Contribution
Jagdish Chandra Bose	The crescograph, and work on plant response and radio waves
Meghnad Saha	The thermal ionisation equation
Satyendra Nath Bose	Bose-Einstein statistics; the boson is named after him
Homi J. Bhabha	Bhabha scattering; the three-stage nuclear programme; founded TIFR in 1945
Vikram Sarabhai	Founded PRL in 1947; father of India's space programme
Har Gobind Khorana	Genetic code and protein synthesis; Nobel in PHYSIOLOGY OR MEDICINE, 1968
Subrahmanyan Chandrasekhar	Chandrasekhar limit, stellar structure; Nobel in Physics, 1983
Venkatraman Ramakrishnan	Structure and function of the RIBOSOME; Nobel in Chemistry, 2009
M.S. Swaminathan	Father of the Green Revolution in India
Verghese Kurien	Operation Flood and the White Revolution; Amul and NDDB
Salim Ali	The 'Birdman of India'; his ornithological survey of Keoladeo Ghana, Bharatpur
Birbal Sahni	Palaeobotany; the institute at Lucknow carries his name

- Also learn: A.P.J. Abdul Kalam - the missile programme and Pokhran-II, later President of India.
- Daulat Singh Kothari - BORN AT UDAIPUR; chaired the Education Commission of 1964-66 and was the first Scientific Adviser to the Ministry of Defence. A Rajasthan-born scientist, so RPSC likes him.
- India's four science Nobel laureates by birth - C.V. Raman (Physics 1930), Har Gobind Khorana (Medicine 1968), S. Chandrasekhar (Physics 1983), Venkatraman Ramakrishnan (Chemistry 2009).
- Colour revolutions - Green for foodgrains, White for milk, Blue for fish and aquaculture, Yellow for oilseeds, Golden for horticulture and honey, Silver for eggs and poultry, Pink for onion and prawn, Grey for fertiliser.

- The pair RPSC plants as wrong is 'Blue Revolution - oilseeds'. Blue is FISH; oilseeds are YELLOW.
- Other awards to recognise - the Ramanujan Prize for young mathematicians and the Infosys Prize; in the state, science popularisation runs through Regional Science Centres and science parks under the Rajasthan Council of Science and Technology.

TRAP

The most-missed scientist fact: Har Gobind Khorana's 1968 Nobel was in **PHYSIOLOGY OR MEDICINE**, not Chemistry. RPSC has used that exact substitution. Two more in the same family - Chandrasekhar's prize was Physics 1983 and Ramakrishnan's Chemistry 2009, and Raman's 1930 prize was for the **SCATTERING** of light, not for spectroscopy or fluorescence.

REVISION RECAP

1. Seventeen essential nutrients; NPK primary, Ca-Mg-S secondary macronutrients, and eight micronutrients. Calcium is NOT a micronutrient.
2. N - chlorosis of older leaves; P - purplish tinge; K - scorched leaf margins; Zn - khaira of rice; Fe - chlorosis of young leaves.
3. Soil Health Card launched 19 February 2015 from Suratgarh, Sri Ganganagar; neem-coated urea mandatory the same year.
4. Seed chain - breeder (golden yellow), foundation (white), certified (azure blue); truthfully labelled is opal green.
5. Rajasthan leads India in bajra, rapeseed-mustard, guar, coriander, cumin, fenugreek, isabgol, henna and moth bean, plus wool and camels.
6. HHB-67 Improved - bajra, early maturing, downy-mildew resistant. RGC-936 and RGC-1003 - guar. Pusa Bold and Bio-902 - mustard. Mahi Kanchan, Mahi Dhawal and Pratap Makka - maize.
7. Guar is a kharif crop; barley is rabi. The mustard directorate (DRMR) is at Sewar, Bharatpur.
8. Dryland farming - below about 750 mm rain; the aim is moisture conservation, and khadin, mulching and short-duration varieties are the tools.
9. Water-use efficiency - flood 30-50 per cent, sprinkler 70-80 per cent, drip about 90 per cent. PMKSY 2015 = Har Khet Ko Pani + Per Drop More Crop.

10. Desert locust = *Schistocerca gregaria*; Locust Warning Organisation at Jodhpur under DPPQS Faridabad; Scheduled Desert Area in Rajasthan and Gujarat.
11. IPM keeps pests below the Economic Threshold Level, not at zero; *Trichogramma* is the egg parasitoid. Kinnow - Sri Ganganagar; orange - Jhalawar; olive - Lunkaransar and Bassi; date palm - Jaisalmer and Bikaner; pomegranate - Barmer and Jalore.
12. Sojat (Pali) is the henna capital with a GI tag; isabgol is *Plantago ovata*, ashwagandha *Withania somnifera*, safed musli *Chlorophytum borivilianum*, guggul *Commiphora wightii*.
13. Khejri is *Prosopis cineraria*, the state tree and desert agroforestry backbone; Rohida is *Tecomella undulata*, the state flower; *Prosopis juliflora* is the invasive vilayati babool.
14. National Forest Policy 1988 - 33 per cent target, ecology over revenue; JFM circular 1990; Van Dhan Kendras run by TRIFED.
15. Cattle - Rathi (best milch), Tharparkar, Kankrej, Nagauri. Sheep - Chokla (finest wool), Magra (carpet wool), Marwari, Nali, Sonadi. Goat - Sirohi (meat), Jakhrana of Alwar (milk).
16. Camel - state animal (domestic) 2014, state Act 2015; breeds Bikaneri, Jaisalmeri, Mewari, Kachchhi.
17. FMD, PPR and lumpy skin disease are viral; brucellosis is bacterial and zoonotic. NADCP 2019 targets FMD and brucellosis; Rashtriya Gokul Mission 2014 is for indigenous bovines.
18. Institutes - CAZRI Jodhpur 1959, CSWRI Avikanagar (Tonk) 1962, Udaipur Solar Observatory 1975, NRC Camel Bikaner 1984, CIAH Bikaner 1993, NRC Seed Spices Ajmer 2000; CEERI Pilani is the only CSIR lab in the state.
19. CSIR 1942, DST 1971, STIP 2020 the fifth policy, ANRF Act 2023 with the PM as ex-officio President.
20. Science days - 28 February, 11 May, 23 August, 22 December. Khorana's Nobel was in Medicine; Ramakrishnan's in Chemistry for the ribosome; Kothari of Udaipur chaired the 1964-66 Education Commission.